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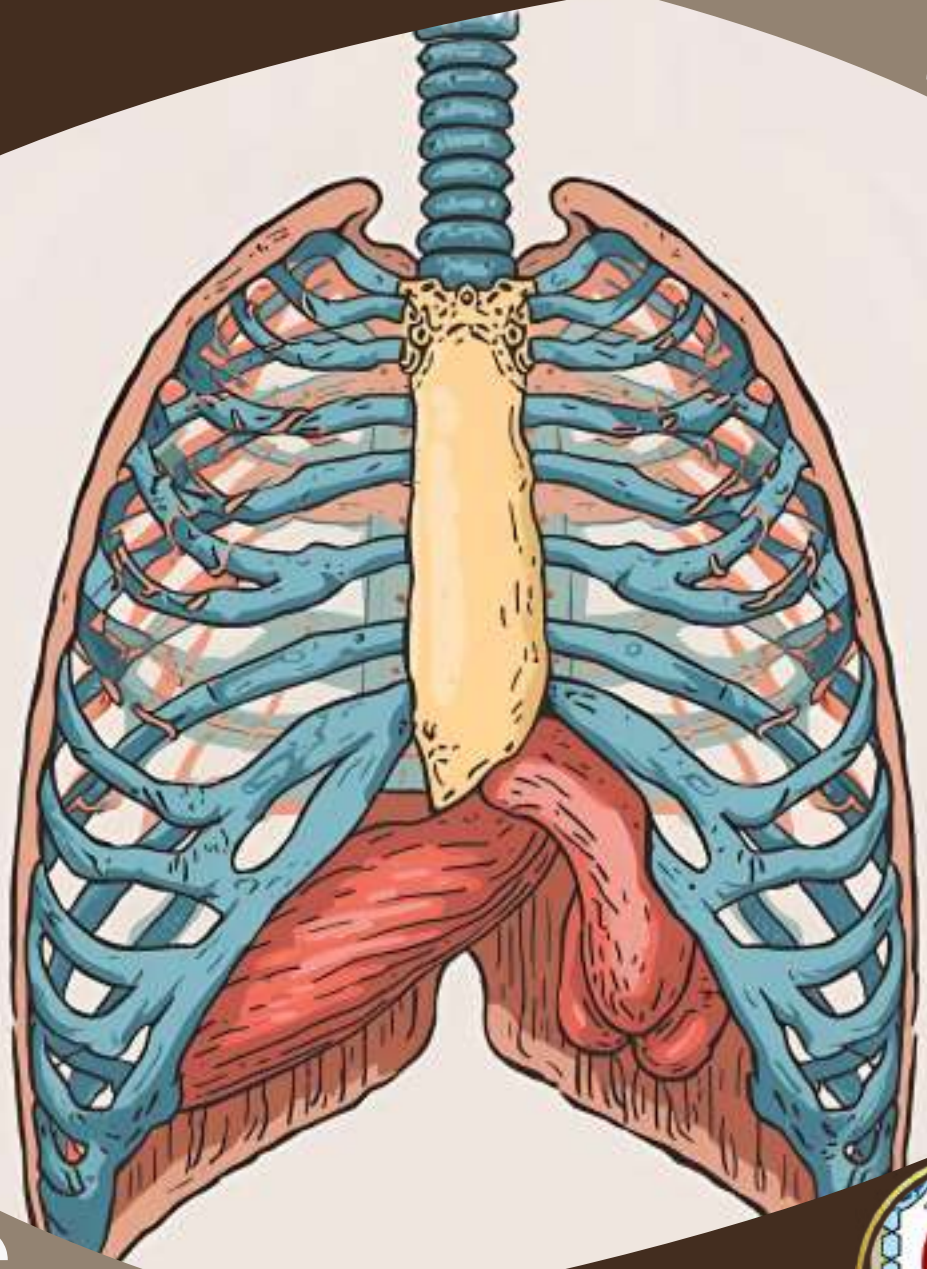


RESPIRATORY SYSTEM

RPS- 314

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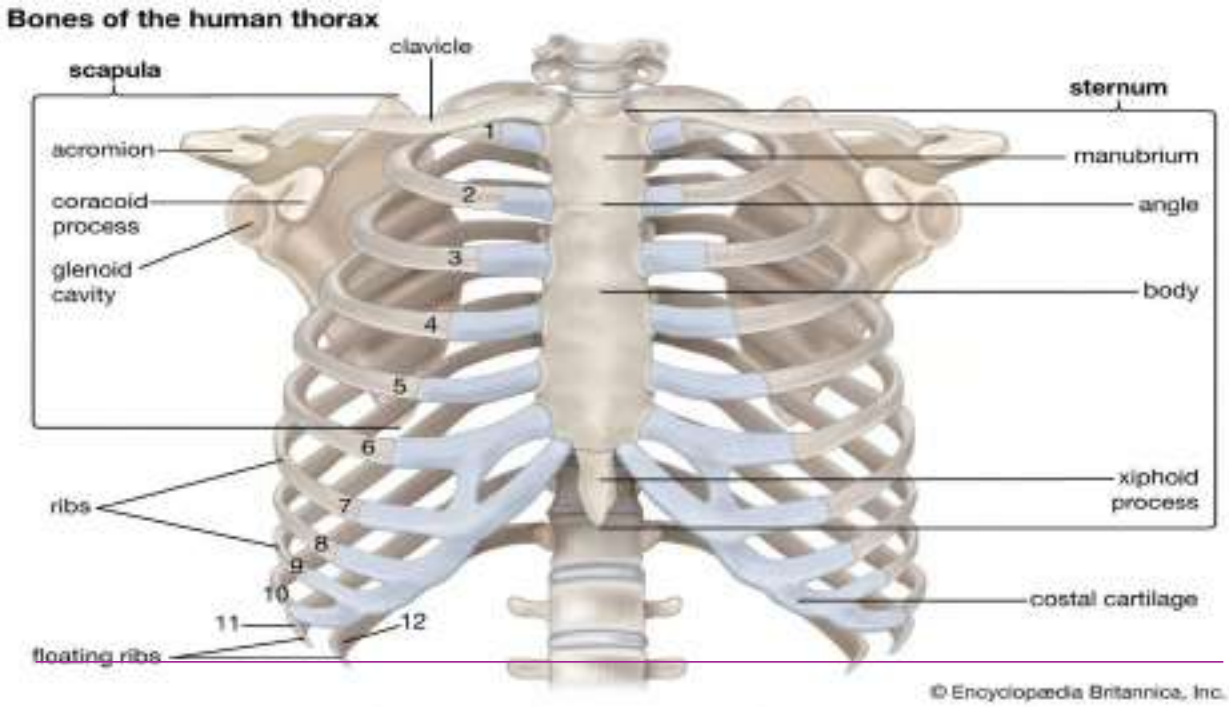
Respiratory system

Anatomy & embryology development

RPS-314

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1 The Thoracic Cage:



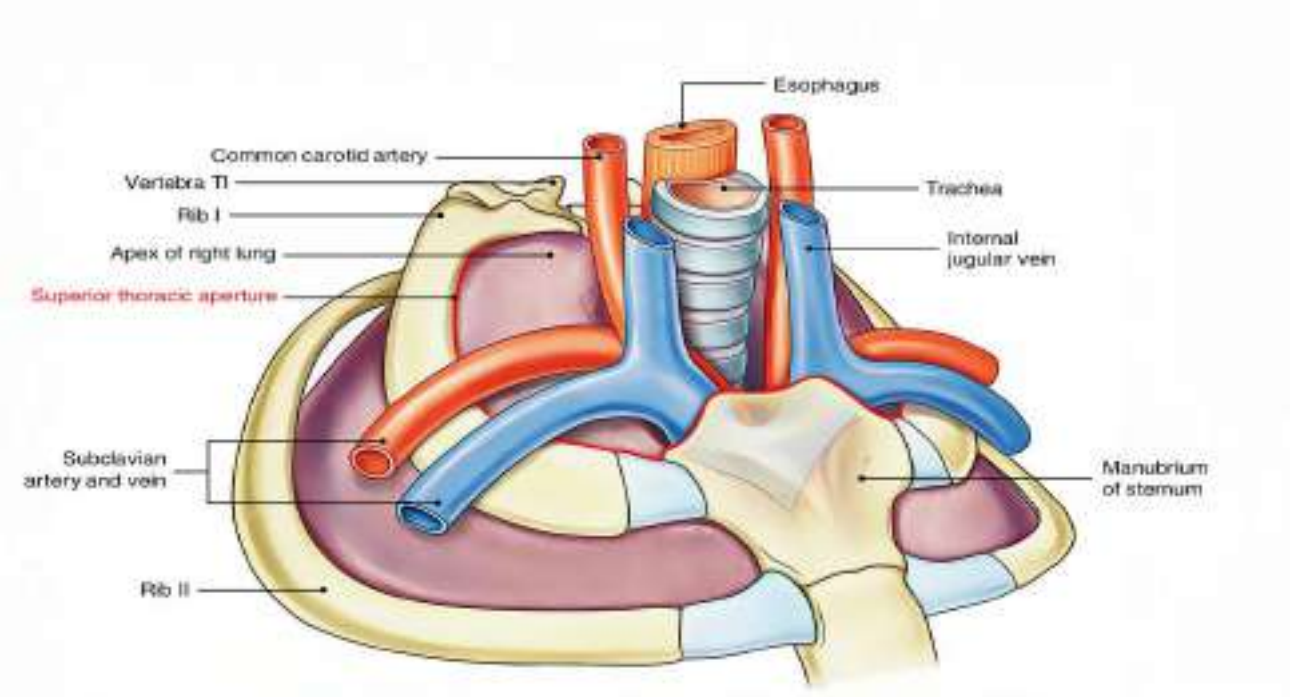
- The thoracic cage is a bony and cartilaginous structure
- It is narrower at the top (thoracic inlet) and wider at the bottom (thoracic outlet).
- The cage protects important organs such as the heart, lungs, and great vessels, and also provides attachment for muscles of respiration and upper limb.

Boundaries:

- Posteriorly: 12 thoracic vertebrae and their intervertebral discs.
- Anteriorly: the sternum (manubrium, body, and xiphoid process).
- Laterally: 12 pairs of ribs with their costal cartilages.

Thoracic Inlet (Superior Thoracic Aperture):

The smaller upper opening of the thorax.



Boundaries:

- Posterior: body of T1 vertebra.
- Laterally: first rib and its costal cartilage.
- Anterior: upper border of the manubrium sterni.
- Contents passing through: trachea, esophagus, great vessels (subclavian arteries & veins), nerves (vagus, phrenic, sympathetic trunk).

Clinical note: Compression of structures at the inlet may cause thoracic outlet syndrome (although the true "outlet" is below, this syndrome involves the upperopening).

Thoracic Outlet (Inferior Thoracic Aperture):

The larger lower opening of the thorax.

Boundaries:

•Posterior: body of T12 vertebra.

•Posterolateral: 11th and 12th ribs.

•Anterolateral: costal cartilages of ribs 7–10 (forming costal margin).

•Anterior: xiphisternal joint.

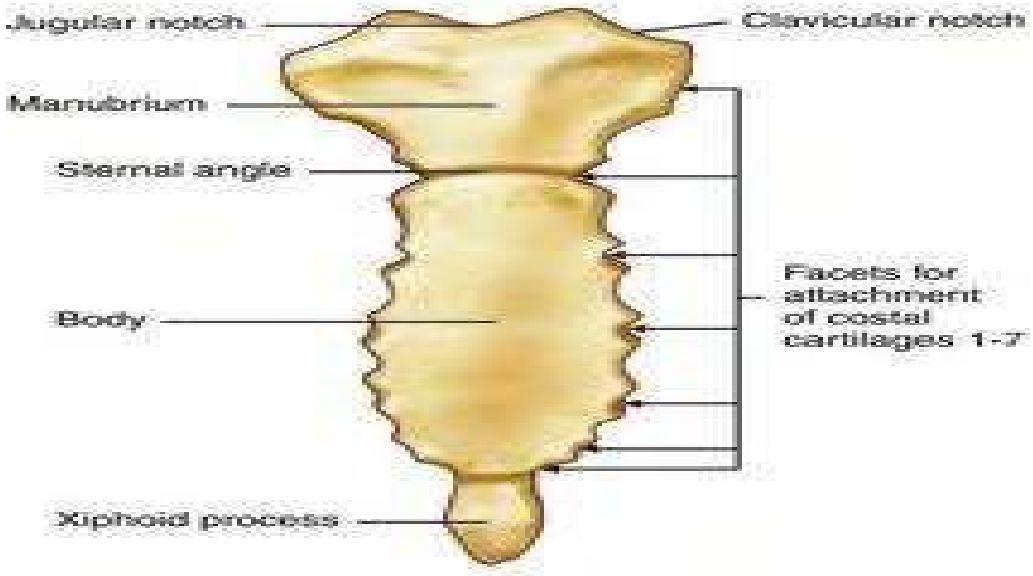
•Closed by: the diaphragm.

Structures Making the Thoracic Wall:

The thoracic wall is formed by bones, muscles, fascia, vessels, and nerves

- **Skeletal Framework:**

Sternum:



The sternum is a flat, elongated bone that lies in the midline of the anterior thoracic wall.it is formed of three parts: manubrium, body &xiphoid process

1. Manubrium: The upper triangular part of sternum.

Landmarks:

1. Jugular (suprasternal) notch → depression at its upper border, between the clavicles.
2. Clavicular notches → lateral articulations with the clavicles
3. Sternal angle (Angle of Louis): junction of manubrium and body → at the level of 2nd costal cartilage and vertebrae T4–T5.

2. Body of Sternum: Longest part of the sternum which articulates with costal cartilages of ribs 2–7.

Surface features: Hascostal notches on each side &Flat anterior surface.

Landmarks: Helpsin rib counting (start from 2nd rib at sternal angle).

3. Xiphoid Process: Small, cartilaginous part at lower end (may ossify in adults).

Landmarks:

•Xiphisternal joint → opposite T9 vertebra.

•Used in CPR (cardiopulmonary resuscitation) → important to avoid fracturing it (can damage liver).

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Ribs :

The ribs are curved, flat bones that form most of the thoracic cage. There are **12 pairs** of ribs which articulate with the thoracic vertebrae posteriorly and most with the sternum anteriorly (via costal cartilages).

Types of Ribs:

1.True ribs (1–7): attach directly to sternum via their own costal cartilages.

2.False ribs (8–10): attach indirectly to sternum (through cartilage of rib above).

3.Floating ribs (11–12): no anterior attachment.

Typical ribs (3rd–9th ribs): share common anatomical features.

Main Parts:

•**Head:** Wedge-shaped, with two articular facets separated by a crest, Superior facet articulates with vertebra above &Inferior facet articulates with numerically corresponding vertebra.

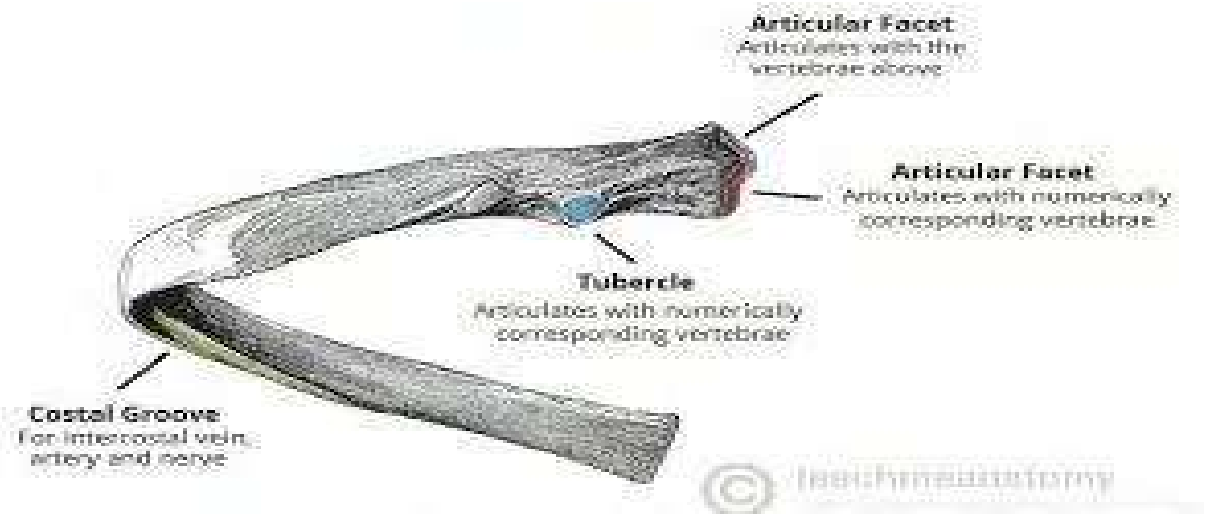
•**Neck:** short, flat portion connecting head to tubercle.

•**Tubercle:** has an it consists of an articular and a non-articular portion. The *articular portion* for transverse process of corresponding vertebra. The *non- articular portion* affords attachment to the ligament of the tubercle.

•**Shaft** (body): thin, curved part.

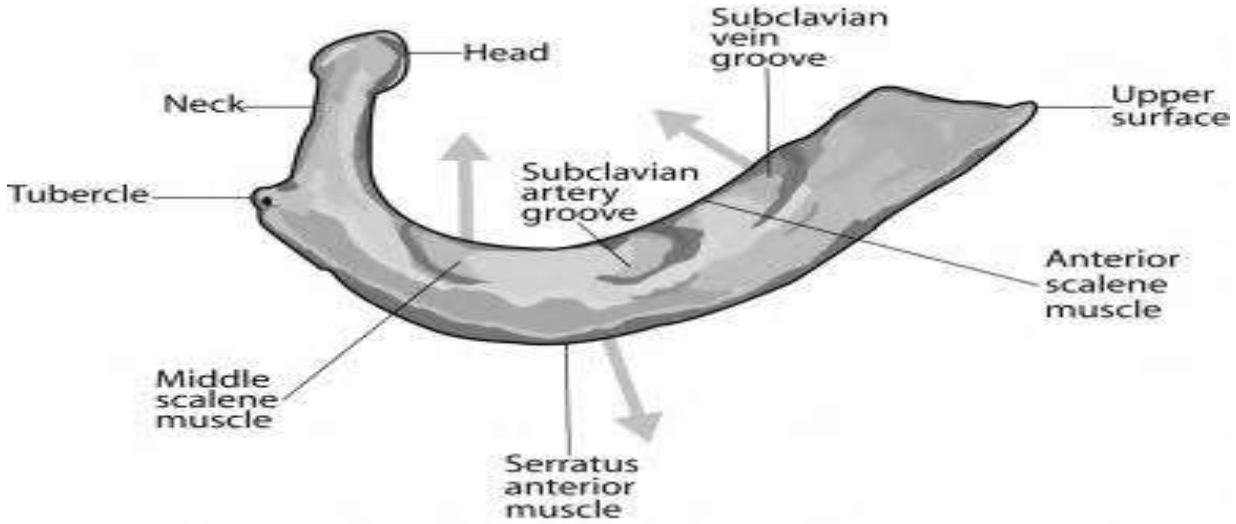
•**Angle:** point of greatest curvature (weakest part, prone to fracture).

•**Costal groove** (on inner inferior border): shelters intercostal vein, artery, and nerve (VAN).



Atypical Ribs: ribs differ from the typical pattern.

1st Rib



•Shortest and broadest, Single facet on head (articulates only with T1). Scalene tubercle on superior surface (for scalenus anterior muscle). Grooves for subclavian vein and artery.

2nd Rib

•Thinner, longer and less curved than 1st rib, Head has two facets (T1 & T2),

Has rough area on upper surface for serratus anterior.

10th Rib

•Head has one facet articulates only with T10.

11th and 12th Ribs (Floating ribs)

•Short, no neck or tubercle, head has one facet (T11 or T12), do not attach to sternum.

NOTE

•Fractures: common in ribs 3–10 at the angle; 1st rib fracture is rare but serious (can damage subclavian vessels, brachial plexus).
•Neurovascular bundle: lies in costal groove → reason chest drains and needles are inserted just above the upper border of rib.
•Floating rib injury: can cause damage to upper abdominal organs.

Thoracic vertebrae

form the posterior part of the thoracic cage. There are 12 thoracic vertebrae (T1– T12), they increase in size from above down ward. Each thoracic vertebra articulates with ribs, giving this region unique features.

Typical Thoracic Vertebra (T2–T8): share common structures:

Body: Heart-shaped, medium sized, has two costal facets (demifacets) on each side → articulate with heads of ribs.

Facets: Superior demifacet articulates with head of rib of same number &Inferior demifacet articulates with head of rib below.

Vertebral Arch: formed of

- Pedicles: two in number, are the connection between the body and the transverse process.
- Laminae: two in number, broad and overlap each other. They are the connection between the transverse process and the spine.
- Vertebral foramen: circular, smaller than cervical & lumbar.

Processes:

- Spinous process: long, slender, directed downward
- Transverse processes: thick, strong, with costal facet at tip for tubercle of rib.
- Superior articular processes& facet→ face backward, slightly lateral for articulation with inferior articular processes of vertebrae above.
- Inferior articular processes& facet → face forward, slightly medial for articulation with superior articular processes of vertebrae below.

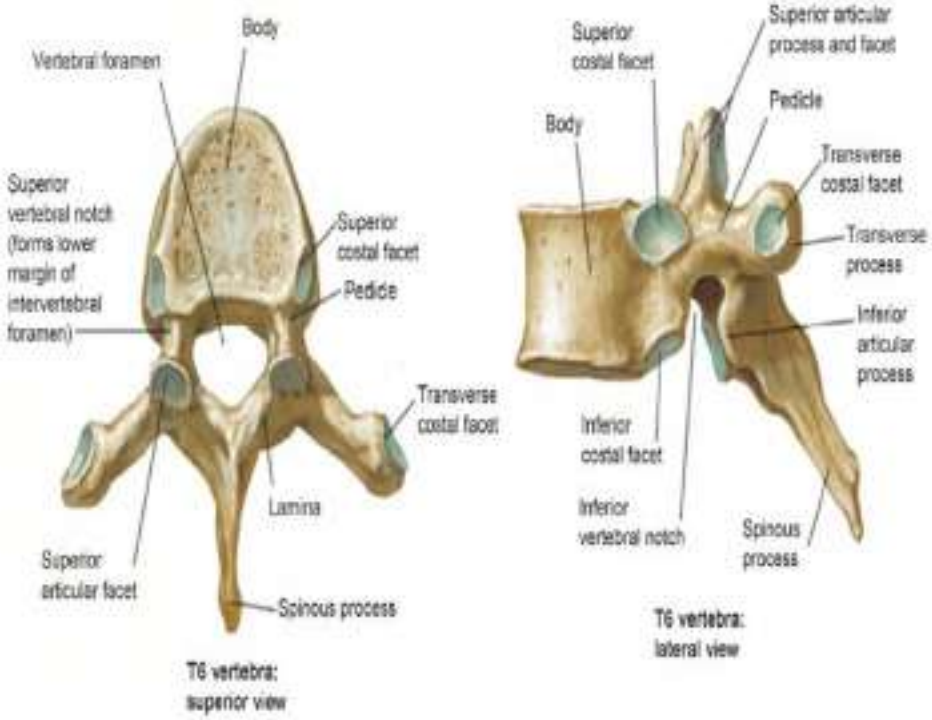
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Typical thoracic vertebra



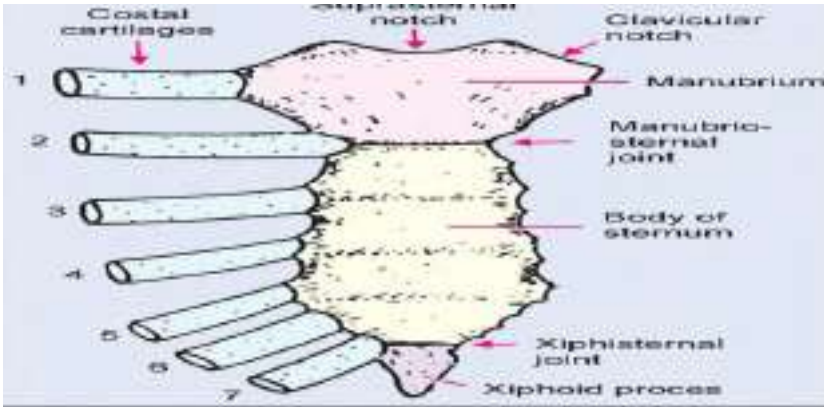
Atypical Thoracic Vertebrae: Some vertebrae differ from the typical pattern. They include T1, T9, T10, T11&T12.

Joints of the Thoracic Wall

The thoracic wall joints connect the ribs, sternum, and vertebrae. They provide flexibility for breathing while maintaining stability of the thoracic cage.

They are grouped into:

Joints of the Sternum



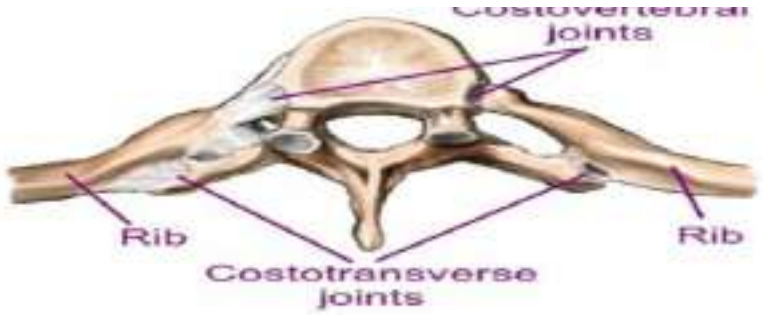
Manubriosternaljoint

- Between manubrium & body of sternum. Type: Secondary
- cartilaginous (symphysis). Note: Forms the sternal angle (Angle of
- Louis, important landmark for 2nd rib).

Xiphisternal joint

- Between body of sternum & xiphoid process.
- Type: secondary cartilaginous.

2-Joints between Ribs and Vertebrae:



Each rib articulates with the vertebral column by two joints:

- *Costovertebral joint (head of rib with vertebral body):*
  - o Rib head articulates with the bodies of its own vertebra and the one above.
  - o Type: Synovial, plane.
- *Costotransverse joint (tubercle of rib with transverse process):*
  - o Present in ribs 1–10.
  - o Type: Synovial, plane.

3-Joints between Ribs and Sternum (Sternocostal Joints)

- Rib 1: Primary cartilaginous (synchondrosis).
- Ribs 2–7: Synovial, plane joints.

4- Costochondral Joints

- Between ribs and their costal cartilages.
- Type: Primary cartilaginous (synchondrosis) no movement.

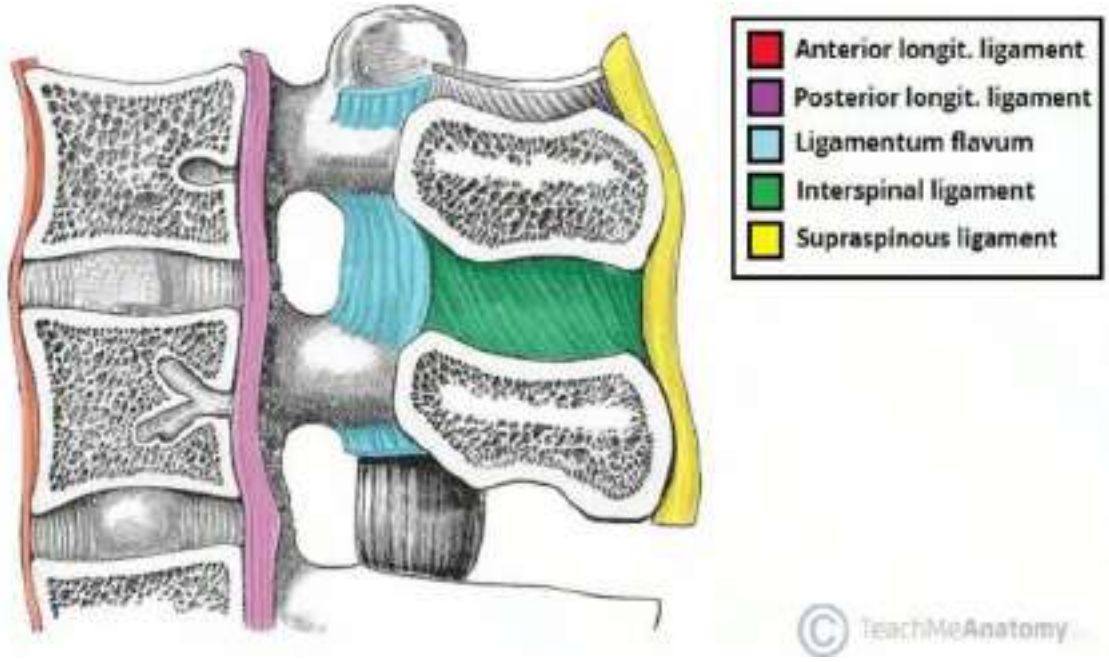
Interchondral Joints

- Between cartilages of ribs 7–10.
- Type: Synovial, plane
- Provide flexibility to costal margin

6-Joints between vertebrae

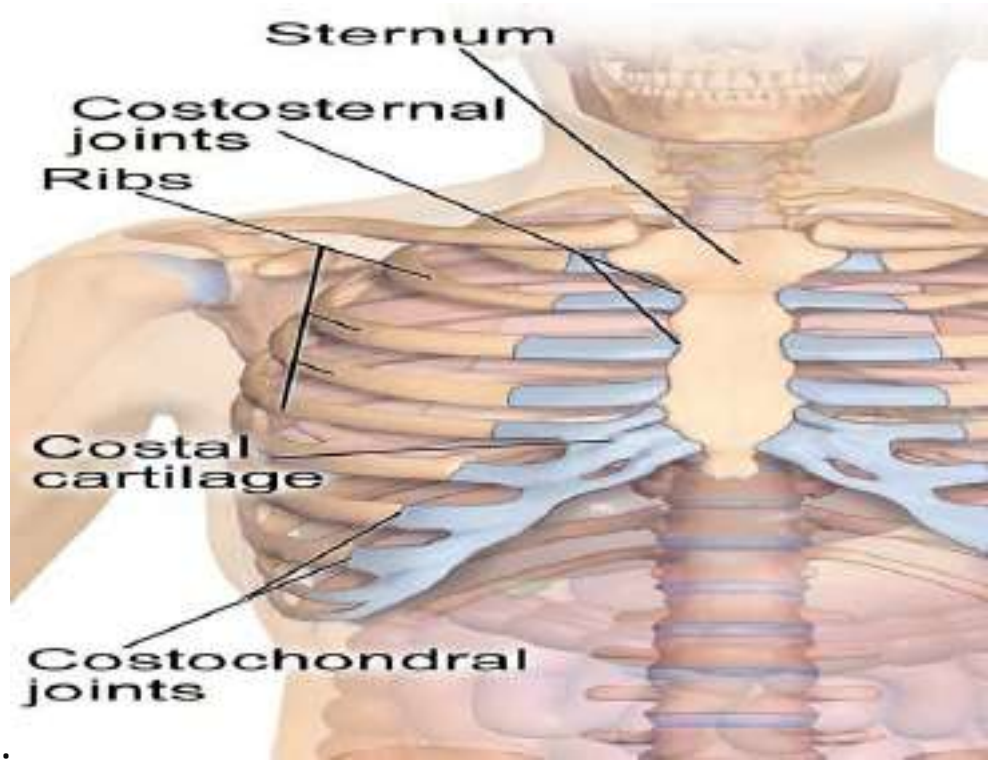
- Intervertebral disc: between bodies of adjacent vertebrae (2ry cartilagenous joints)
- The superior articular processes for articulation with inferior articular processes of vertebrae above (Plane synovial).
- The inferior articular processes for articulation with superior articular processes of vertebrae below (Plane synovial).

Ligaments between vertebrae



- Anterior longitudinal ligament: ligament runs along the anterior surface of all of the vertebral bodies and intervertebral discs.

- Posterior longitudinal ligament (within the vertebral canal) extends along the posterior surfaces of the bodies of the vertebræ. The ligamentum flavum:
- connect the laminae of adjacent vertebrae. The interspinous ligaments
- connect adjoining spinous processes of the vertebra The supraspinous
- ligaments connect the tips of the spinous processes



- .

NOTES:

- Cartilaginous joints → manubriosternal, xiphisternal, 1st sternocostal, costochondral. Synovial joints → 2nd–7th sternocostal,
- interchondral, costovertebral, costotransverse

The intercostal spaces (the spaces between the ribs)

The space is formed of skin, superficial fascia, deep fascia and intercostal muscles, nerves& vessels.

1-Intercostal Muscles There are three layers between the ribs:

(a)ExternalIntercostal Muscles: there are eleven pairs of external intercostal muscles

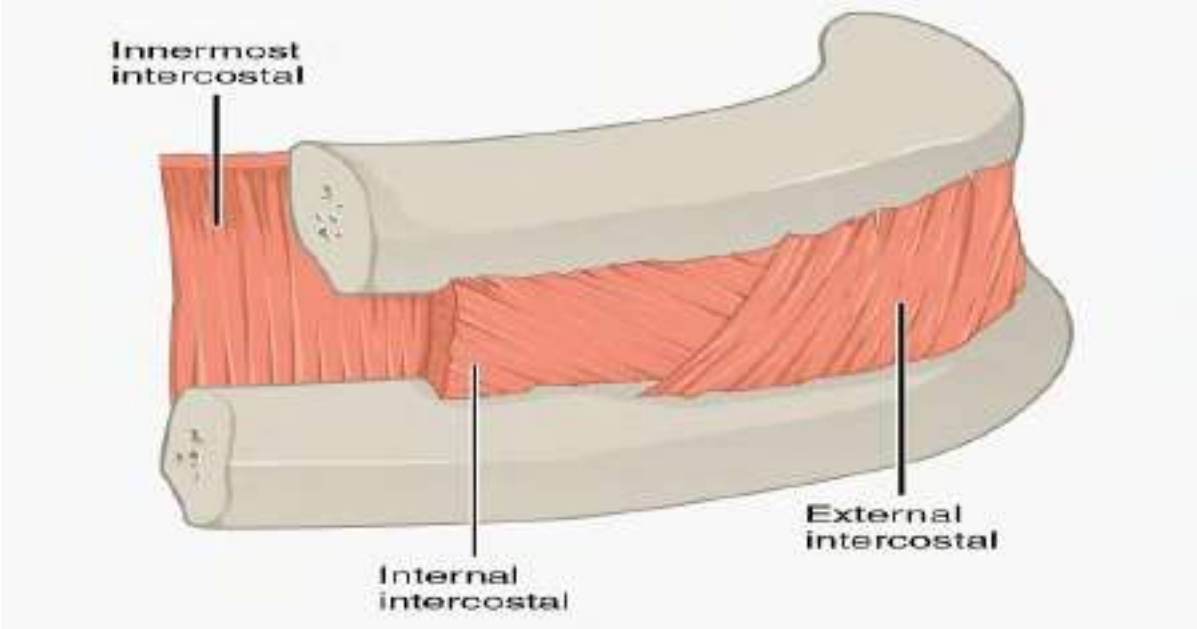
- Origin: Lower border of rib above.
- Insertion: Upper border of rib below.
- Direction: Fibers run downward and forward (like hands in pockets).
- Action: Elevate ribs → inspiration.
- Nerve supply: Intercostal nerves (T1–T11).

(b) Internal Intercostal Muscles: (eleven pairs)

- Origin: Inner surface of rib above.
- Insertion: Upper border of rib below.
- Direction: Fibers run downward and backward (opposite to external).
- Action: Depress ribs → expiration.
- Nerve supply: Intercostal nerves.

(c) Innermost Intercostal Muscles

- Same as internal intercostal but separated by the intercostal vessels and nerves.
- Function: Assist internal intercostals in expiration.



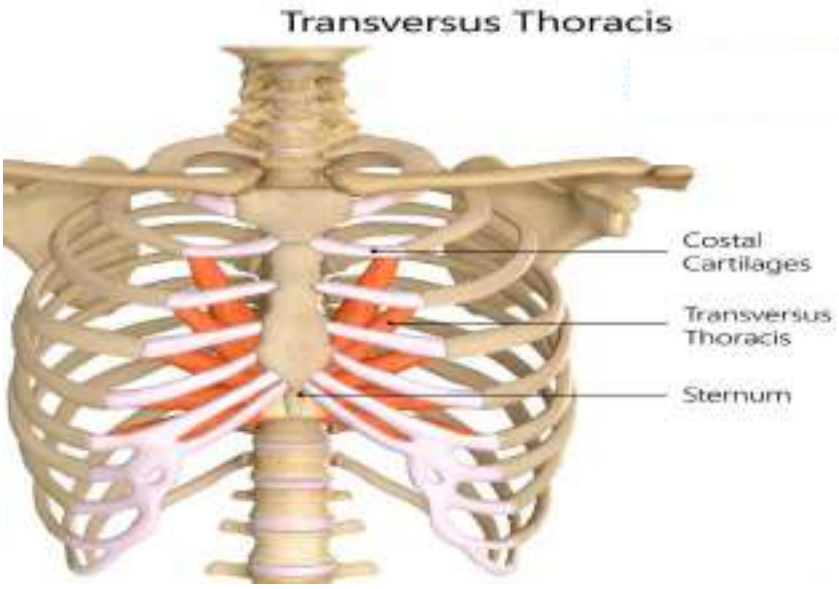
Muscle	Direction of Fibers	Function	Notes
External intercostal	Downward & forward (like putting hands in pockets)	Elevates ribs → inspiration	Most active during inspiration
Internal intercostal	Downward & backward	Depresses ribs → expiration	Most active during expiration
Innermost intercostal	Same direction as internal	Helps internal intercostals	Separated by intercostal vessels & nerves

2- Subcostal Muscles

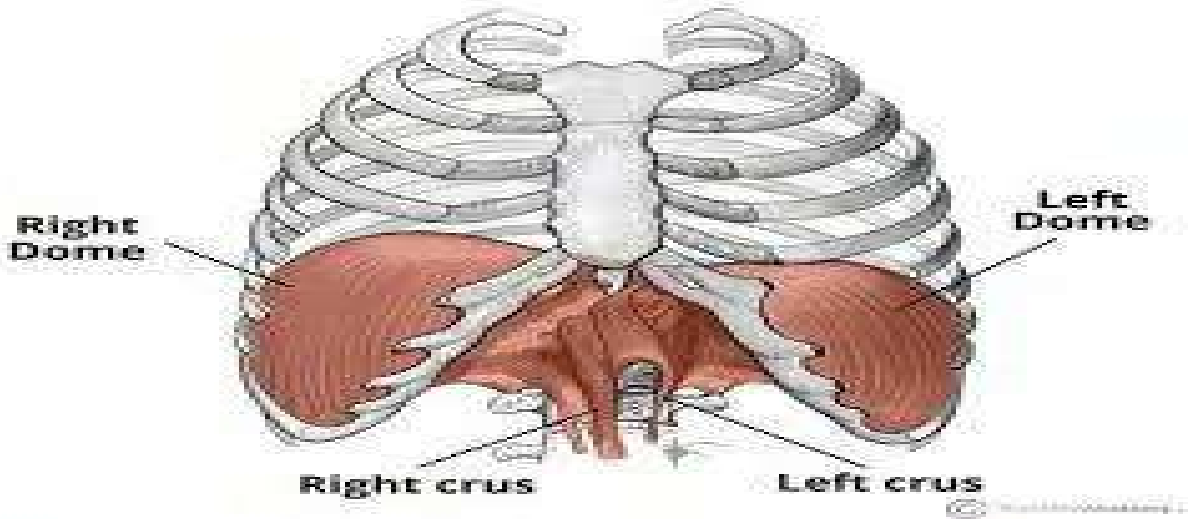
- Location: Inner surface of lower ribs (posteriorly).
- Fibers: Run in same direction as internal intercostals.
- Action: Depress ribs → aid expiration.

3-Transversus Thoracis (Sterno-costalis)

- Origin: Posterior surface of sternum and xiphoid.
- Insertion: Costal cartilages of ribs 2–6.
- Action: Depress ribs during forced expiration.



The Diaphragm



- ***Definition***:

The diaphragm is a large, dome-shaped, musculotendinous sheet that

separates the thoracic cavity (above) from the abdominal cavity (below). It is the primary muscle of respiration.

- ***Shape*:**
  - o Dome-shaped, convex upwards into the thorax. o Right dome is slightly higher than the left (because of the liver). o Central tendon lies at the top, where the muscle fibers converge.
- ***Parts (muscular origins):***
  - a. Sternal part – arises from the back of the xiphoid process.
  - b. Costal part – arises from the inner surfaces of lower six ribs and costal cartilages.
  - c. Lumbar part – arises from right and left crura and arcuate ligaments attached to the lumbar vertebrae.

Insertion:

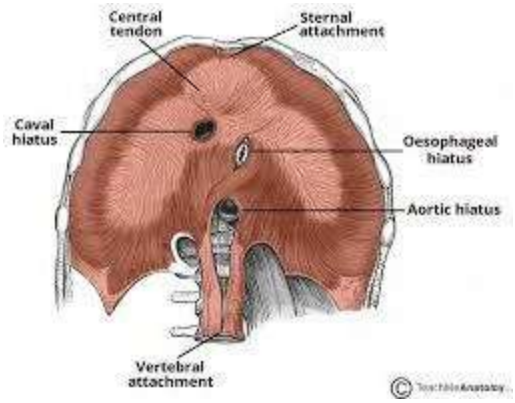
All fibersconverge to insert into the central tendon (a strong aponeurotic sheet, fused withthe fibrous pericardium).

Openingsin the diaphragm: (important for structures passing between thorax and abdomen)

Caval opening (T8) – for inferior vena cava and right phrenic nerve.

Esophageal hiatus(T10) – for esophagus, vagus nerves.

Aortic hiatus (T12) – for aorta, thoracic duct, azygos vein.

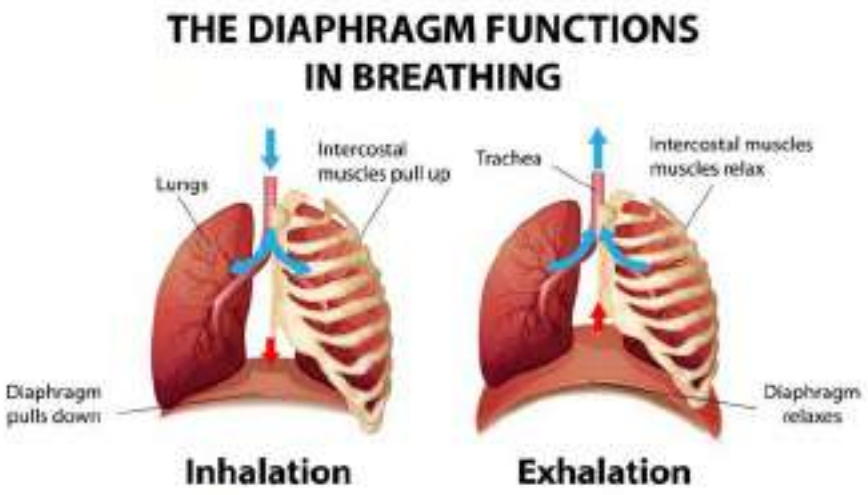


Nerve supply:

- o Motor: Phrenic nerves (C3, C4, C5 – "keeps the diaphragm alive").
- o Sensory: Phrenic nerves (central part) and lower intercostal nerves (peripheral part).

Function in Breathing:

- *Inspiration (active process):*
  - When thediaphragmcontracts, it flattens and moves downward.
  - This increases the vertical dimension of the thoracic cavity.
  - The intrathoracic pressure falls → air is drawn into the lungs.
- *Expiration (mainly passive):*
  - During relaxation, the diaphragm returns to its dome shape (elastic recoil).
  - This decreases thoracic volume, increases intrathoracic pressure → air is expelled.
- *Accessory functions:*
  - Helps in coughing, sneezing, vomiting, defecation, micturition, childbirth (by increasing intra-abdominal pressure).
  - Acts as a barrier between thoracic and abdominal organs.



Intercostal Nerves

The intercostal nerves are the anterior rami of the thoracic spinal nerves (T1–T11). They run in the intercostal spaces and supply muscles, skin, pleura, and (in lower nerves) the peritoneum. They are classified into typical and atypical intercostal nerves.

1. Typical Intercostal Nerves (T3–T6)

:

These represent the “standard” pattern.

Course

- Run in the intercostal space between internal intercostal and innermost intercostal muscles.
- Lie in the costal groove with artery and vein (order = Vein, Artery, Nerve: VAN).
- Near the sternum, pierce muscles to become anterior cutaneous nerves.

Branches

1. Muscular branches → intercostal muscles.
2. Collateral branch → runs along upper border of rib below.
3. Pleural branches → supply parietal pleura.
4. Lateral cutaneous branch → emerges near midaxillary line, divides into anterior and posterior divisions to supply skin.
5. Anterior cutaneous branch → emerges near sternum, supplies skin of anterior thoracic wall.

2. Atypical Intercostal Nerves

These differ from the typical pattern either in course or area of supply.

T1

- Large upper part joins brachial plexus.
- Small branch continues in first intercostal space, supplying intercostal muscles.
- Has no anterior cutaneous branch.

T2

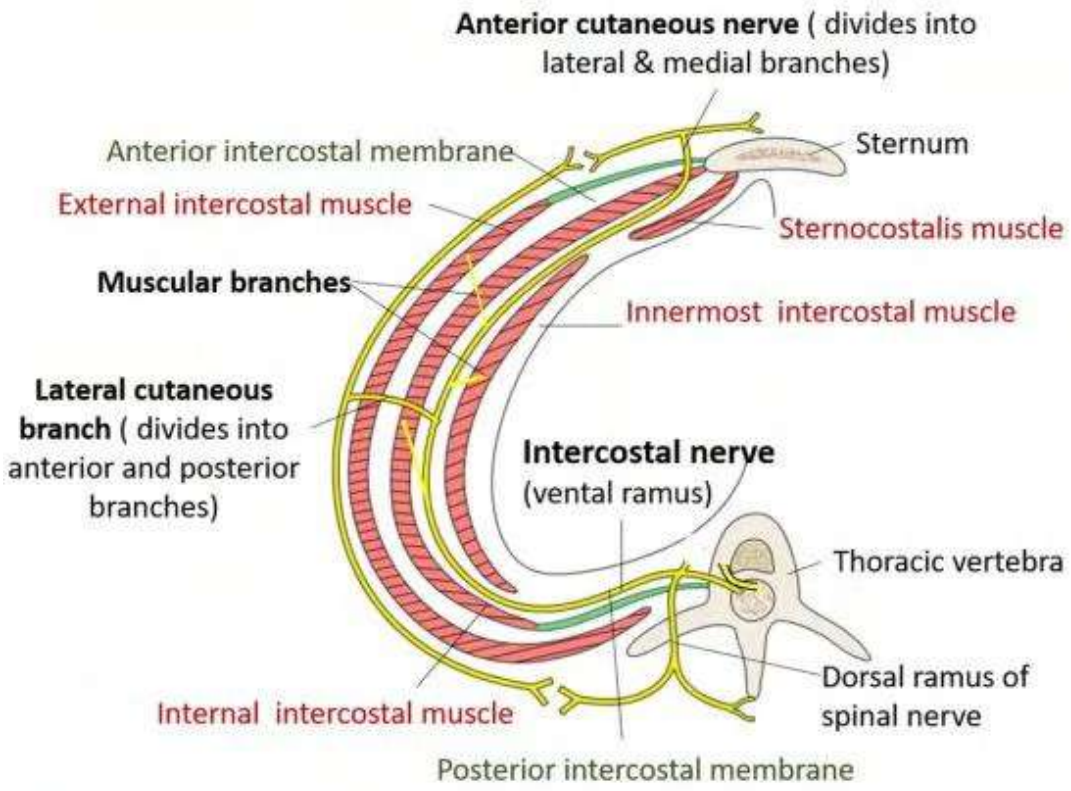
- Gives off the intercostobrachial nerve→ crosses axilla to supply skin of medial upper arm.
- Remaining part is similar to typical.

T7–T11

- Runforward beyondthoracic wall to supply anterior abdominal wall musclesand overlyingskin.
- Called thoracoabdominal nerves

T12 (Subcostal Nerve)

- Runs below the 12th rib (not in an intercostal space).
- Supplies abdominal wall muscles and skin of lower abdominal wall, gluteal region, and upper thigh.

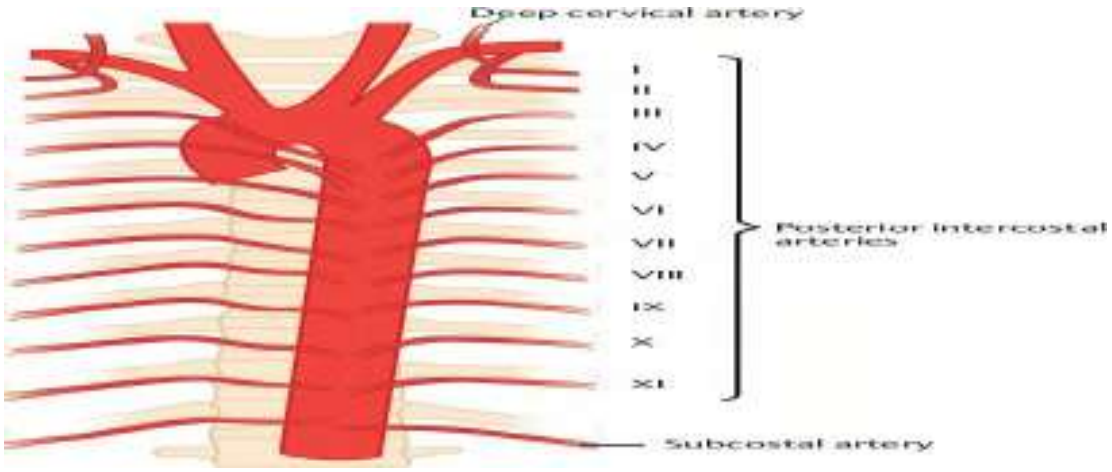


Intercostal Vessels

The intercostal vessels are the arteries and veins that run within the intercostal spaces between the ribs. They supply blood to the thoracic wall, overlying muscles, skin, and parietal pleura.

Arteries

1-Posterior Intercostal Arteries:



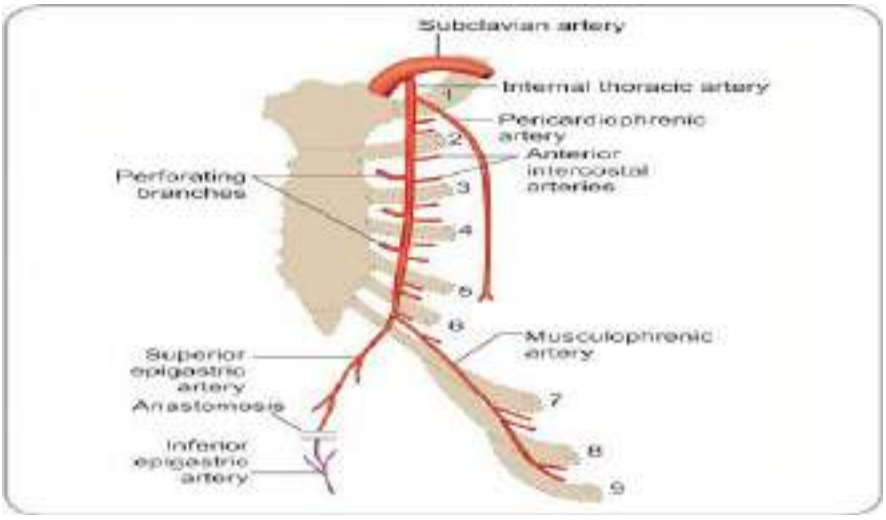
Origin

- :
  - o The first and secondposterior intercostal arteries arise from the superior intercostal artery(a branch of the costocervical trunk of the subclavian artery).
  - o The 3rd to 11thposterior intercostal arteries arise directly from the descending thoracic aorta
- Course: Each runs along the corresponding intercostal space, lying between the internal intercostal and innermost intercostal muscles.
- :

Branches

- o Dorsal branch → supplies the spinal cord, vertebrae, andback muscles.
- o Collateralbranch → runs along the upper border of theribbelow.
- o Lateral cutaneousbranch → supplies skin of lateral thorax
- o End:by anastomosiswithanteriorIntercostalArteries.

2-Anterior Intercostal Arteries:

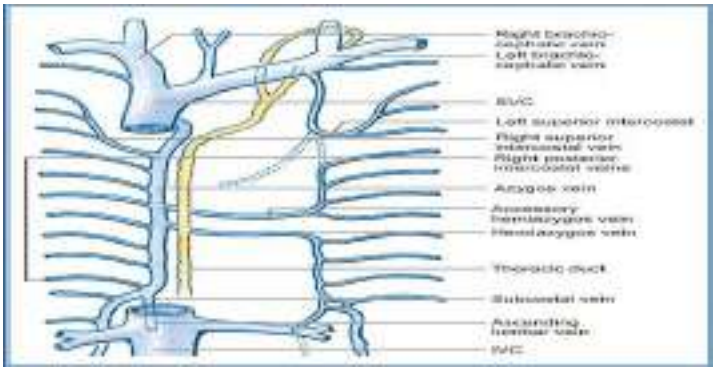


Origin

- :
  - The first six spaces: from the internal thoracic artery (branch of subclavian).
  - o The 7th to 9th spaces: from the musculophrenic artery (terminal branch of internal thoracic).
  - o 2 arteries in each space
  - o No anterior intercostal arteries in the last two spaces (10th & 11th).
- Course: Run in the same plane as posterior intercostal arteries and anastomose with them.

Veins:

1-Posterior Intercostal Veins:



In all the 11 spaces

A- on the right side:

- The 1st → into the right brachiocephalic vein
- The 2nd and 3rd → unite to form the right superior intercostal vein that drains into the arch of azygos
- From the 4th to the 11th → into azygos vein

B- on the left side:

- The 1st → into the left brachiocephalic vein
- The 2nd and 3rd → unite to form the left superior intercostal vein that drains into the left brachiocephalic vein
- From the 4th to 8th → into the superior hemiazygos vein
- From the 9th to 11th → into the inferior hemiazygos vein

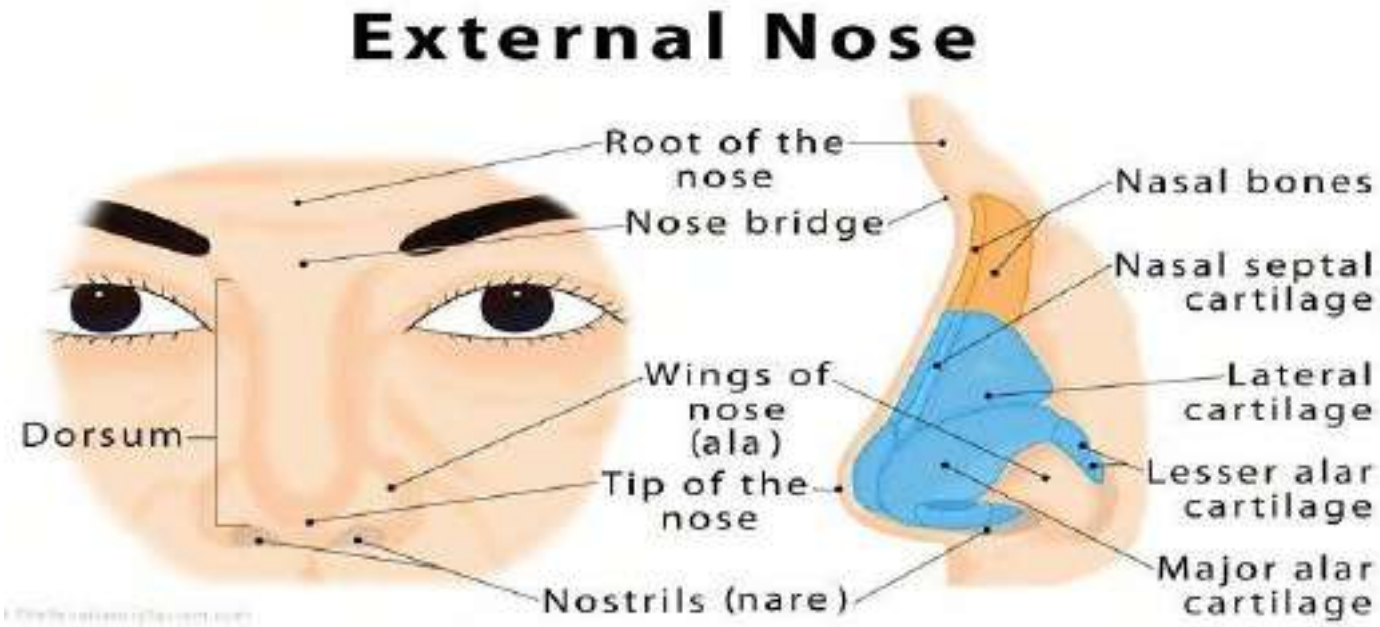
NOTE

The azygous and hemizygous or brachiocephalic veins drain in the superior vena cava then the right atrium.

-Anterior Intercostal Veins:

Alongside the anterior intercostal arteries:

- The first six pairs of anterior intercostal veins drain directly into the internal thoracic vein.
- The seventh to ninth pairs first empty into the musculophrenic vein, which then drains into the internal thoracic vein.



The external nose (pyramidal shape):

The nasal rootsuperiorly, apexof the nose ends inferiorly in a rounded 'tip'. between the root and apex is thedorsum of the nose located immediately inferiorly to the apex are the Anterior nares (Nostrils =Anterior nasal openings) which open into the nasal cavity. The nares are bounded medially by the nasal septum, and laterally by the ala nasi (the lateral cartilaginous wings of the nose).

Anatomy OF Nasal Cavity

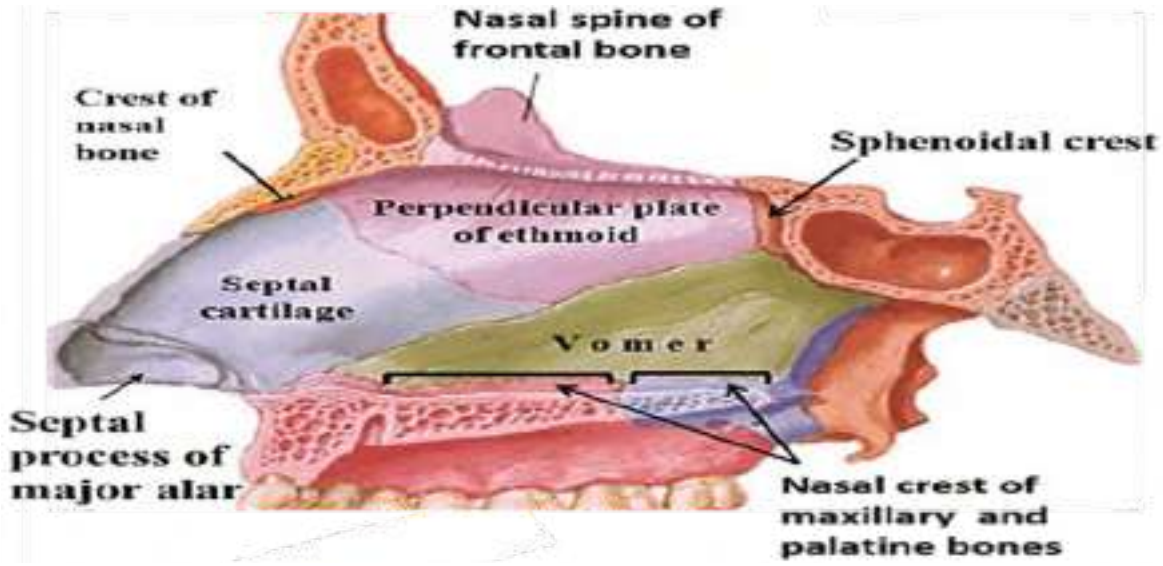
Location and Extent:

The nasal cavity is a large, air-filled space within the skull, it extends from the anterior nasal aperture (nostrils) to the posterior nasal apertures (choanae) , which open into the nasopharynx. It is divided into right and left halves by the nasal septum.

The nasal cavity has three divisions:

- **Vestibule** – the area surrounding the anterior external opening to the nasal cavity.
- **Olfactory region** – located at roof of each nasal cavity.
- **Respiratory region** – most of each nasal cavity.

Boundaries of the Nasal Cavity:

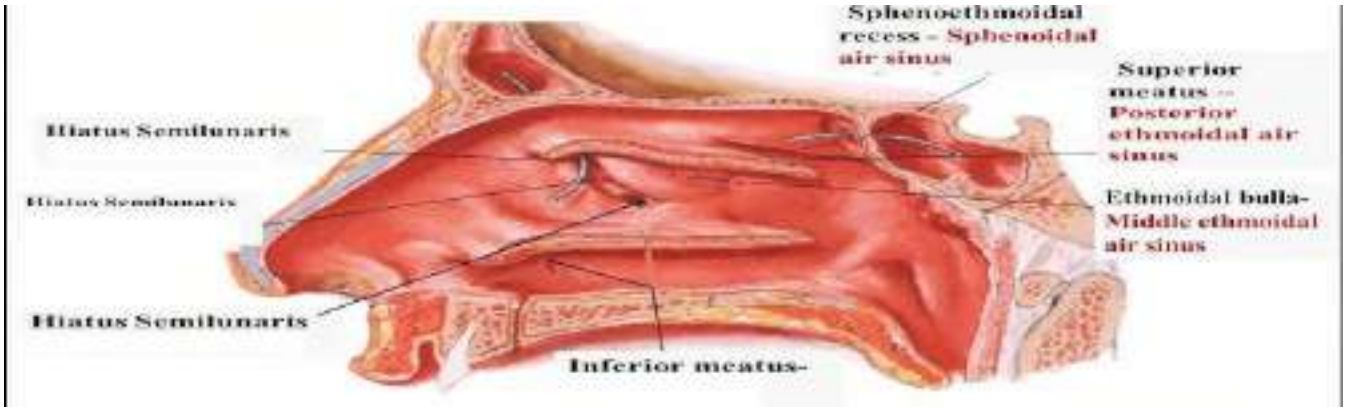


- ***Roof*** **: Formed by**
  - **Fronto nasal bone (anterior)**
  - **Cribriform plate of ethmoid (middle)**
  - **Body of sphenoid (posterior)**

- ***Floor***:
  - Formed by the hard palate → palatine process of maxilla + horizontal plate of palatine bone.
- ***Medial wall (Nasal septum):***
  - Formed by the perpendicular plate of ethmoid, vomer, and septal cartilage.
- ***Lateral wall:***
  - Irregular, formed by maxilla, lacrimal, ethmoid, sphenoid bones.
  - Bears three bony projections → Superior, Middle, and Inferior nasal conchae.

Nasal Conchae and Meatuses:

Each concha overhangs a groove (meatus) where paranasal sinuses open:

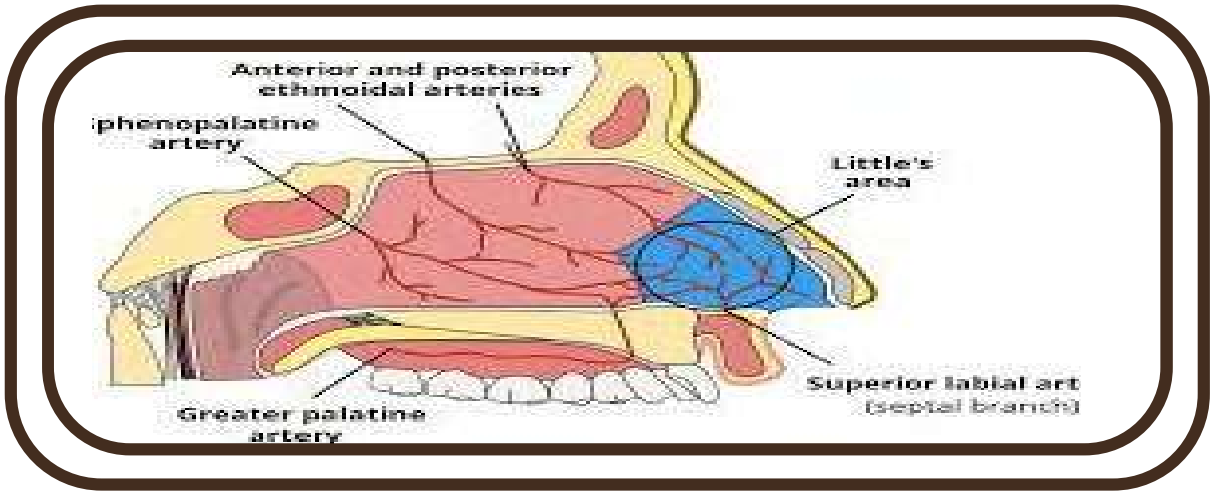


- Superior meatus: Opening of posterior ethmoidal sinuses. Middle meatus:
- Contains the hiatus semilunaris → drainage of maxillary sinus, frontal sinus, and anterior ethmoidal sinus. Bulla ethmoidalis drainage of middle
- ethmoidal air sinus. Inferior meatus: Opening of nasolacrimal duct.
- Sphenoethmoidal recess (above superior concha): Opening of sphenoidal sinus.

Mucosa:

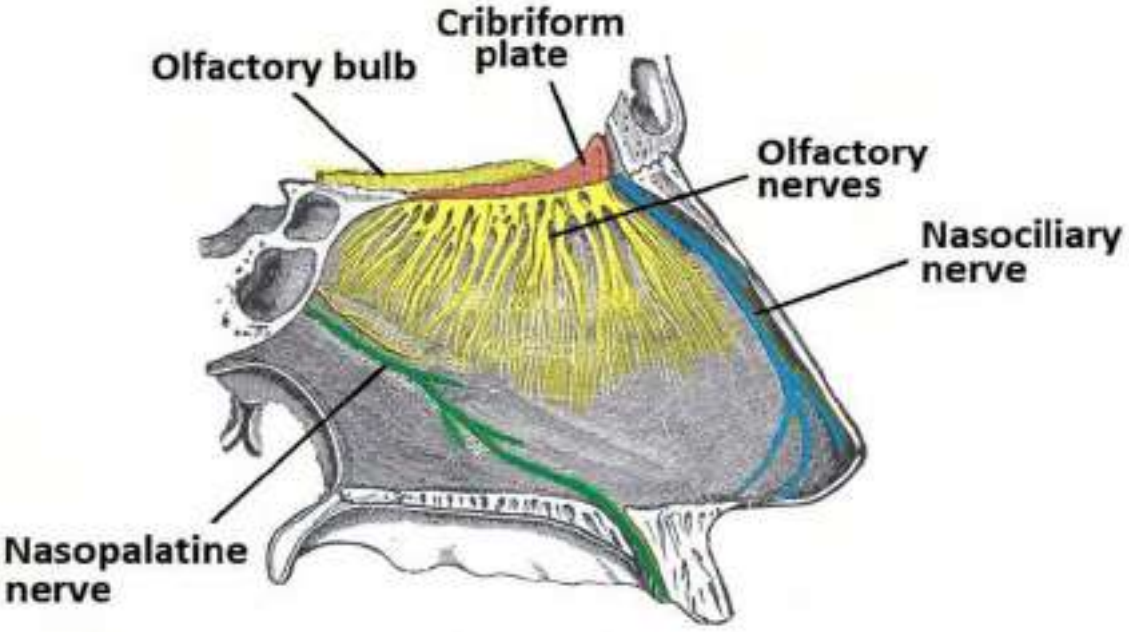
- **Olfactory mucosa** (upper part near roof, superior concha, and upper septum):contain solfactory receptor cells.
- **Respiratory mucosa** (rest of cavity): ciliated pseudostratified columnar epitheliumwithgoblet cells → warms, humidifies, filters inspired air.

Blood Supply:



- Rich vascular supply → warms incoming air.
- **Arteries:** branches from
  - Ophthalmic artery (anterior & posterior ethmoidal)
  - Maxillary artery (sphenopalatine)
  - Facial artery (superior labial)
- Kiesselbach's plexus (Little's area): common site of epistaxis.
- The veins of the nose tend to follow the arteries. They drain into the pterygoid plexus, facial vein or cavernous sinus.

Nerve Supply:



- ***Sensory***
  - Ophthalmic (V1) → anterosuperior part.
  - Maxillary(V2)→ posteroinferior part.
- *Olfactory nerve (CN I)*→ olfactory mucosa.
- *Autonomic* :parasympathetic fibers to glands (via pterygopalatine ganglion).

Functions :

1. Passage of air. Warming, humidifying, and filtering inspired air.
2. Sense of smell(olfaction).
3. Resonance forspeech.

Clinical Notes:

Nasal obstruction: Commonly due to deviated septum, polyps, enlarged turbinates, or foreign body (especially in children).

another cliniical notes

- Cribriform Plate Fracture: Due to nose trauma, a fractured cribriform plate can penetrate the meningeal linings of the brain, causing leakage of cerebro-spinal fluid and meningitis. Epistaxis (nosebleed):
  - Most frequent from *(Little's area)* in the anterior septum.
  - Posterior epistaxis (from sphenopalatine artery) is less common but may be life-threatening.
- Sinusitis: Inflammation of paranasal sinuses spreads easily from the nasal cavity, producing facial pain, nasal blockage, and fever.
- Spread of infection: Venous connections (facial vein ↔ ophthalmic vein ↔ cavernous sinus) allow infection to spread to the brain → cavernous sinus thrombosis.
- Anosmia: Loss of smell due to damage of olfactory mucosa/nerve (trauma, infections, tumors, or neurodegenerative disease).
  CSF rhinorrhea: Leakage of cerebrospinal fluid through the nose after fracture of the cribriform plate of the ethmoid.
- Neoplasms: Tumors (e.g., nasopharyngeal carcinoma) may present with nasal obstruction, bleeding, or hearing problems.

paranasal sinuses

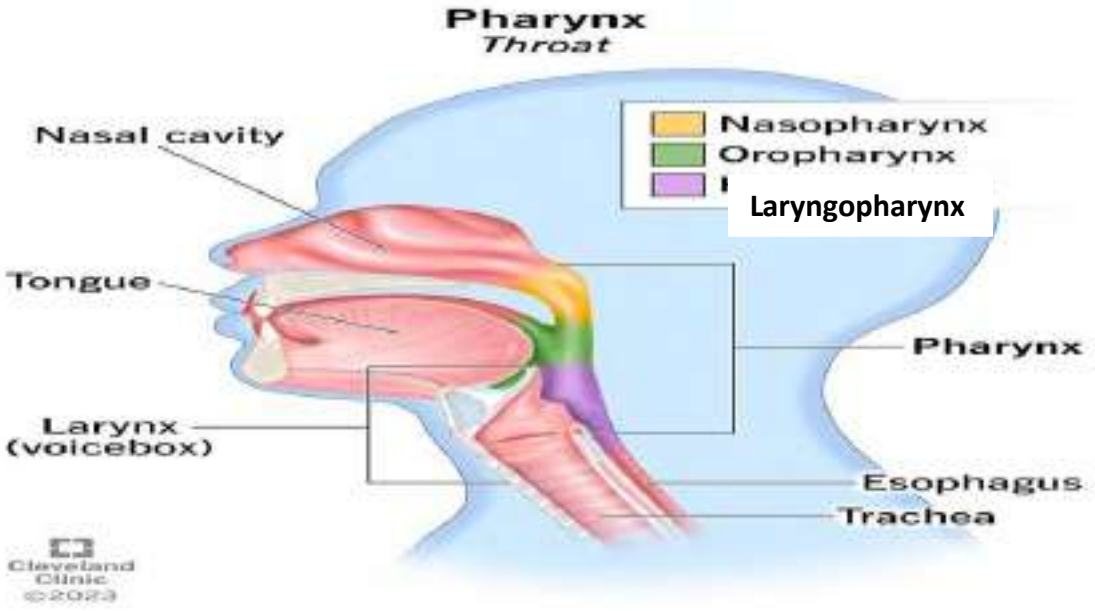


Air-filled spaces located within the bones of **the skull and facial bones** around the nose within the frontal, ethmoidal, sphenoidal, and maxillary bones. They are outgrowths from the nasal cavity. All of them drain into the **superior or lateral aspect of the nose.**

There are 4 pairs of the paranasal sinuses:

- The pyramid-shaped maxillary sinuses are located within the body of the maxilla.
- The frontal sinuses, which are separated from each other by a thin bony lamella, lie within the frontal bone.
- The ethmoidal sinuses are number of small cavities within the ethmoid bone. The ethmoid sinuses are divided into 3 groups anterior, middle and posterior groups
- The sphenoidal sinuses lie within the body of the sphenoid bone. They are located at the most posterior (farthest toward the back of the head) of the paranasal sinuses.
- **The functions of the paranasal sinuses:**

1. Lightening the weight of the head.
2. Supporting immune defense of the nasal cavity.
3. Humidifying inspired air.
4. Increasing resonance of the voice.



Pharynx Anatomy

definition

- The pharynx isamuscular tube forming the upper part of the digestive and respiratory systems.
- Actsasa common pathwayfor food and air.

Extent

- **Above:** Base of skull (behind the nasal cavity).
- **Below:** Continuous with the esophagus at the level of the C6 vertebra (lowerborder of cricoid cartilage).
- **Length:** About 12–14 cm.

Parts of the Pharynx

The pharynx is divided into 3 regions:

1. Nasopharynx (respiratory part)
   - o Location: Behind the nasal cavity, above the soft palate.
   - o Features:
     - Opening of Eustachian tube (pharyngotympanic tube).
     - Pharyngeal tonsil (adenoids).
     - **Communicates with nasal cavity via choanae.**
2. Oropharynx (digestive + respiratory part)
   - o Location: Behind the oral cavity, from soft palate → epiglottis.
   - o Features:
     - Palatine tonsils in the tonsillar fossa in the lateral wall of
     - oropharynx between the palato-glossal arch (Anteriorly) &
     - palate-pharyngeal arch. (Posteriorly). Lingual tonsil on tongue base.

     Communicates with oral cavity via oropharyngeal isthmus.
3. Laryngopharynx (Hypopharynx) (digestive part)
   - o Location: Behind the larynx, from epiglottis → cricoid cartilage.
   - o Features:
     - Piriform fossa (recesses on each side of laryngeal inlet – common site for foreign body lodgment).
     - Continuous below with esophagus.

Waldeyer's Ring

Aringof lymphoid tissuein the pharynx guarding entry of respiratory & digestive tracts.

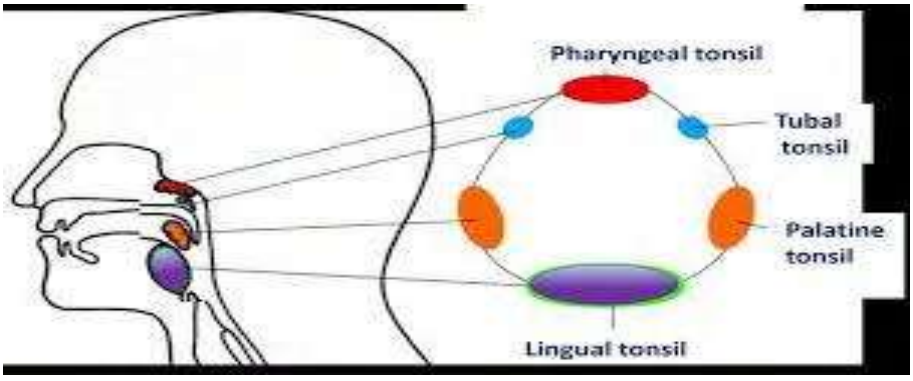
Components:

- o Pharyngeal tonsil (adenoid) – roof of nasopharynx
- o Tubal tonsils – near Eustachian tube opening
- o Palatine tonsils – oropharynx, between arches
- o Lingual tonsil – posterior 1/3 of tongue

Functions: Immune defense (local antibody formation, lymphocyte production).

Clinical relevance:

- Hypertrophy→ adenoids (children), nasal obstruction, otitis
- media Recurrent tonsillitis → tonsillectomy Adenoidectomy if
- persistent obstruction/infection



Walls of the Pharynx

1. *Mucosa*The innermost layer (lining epithelium: respiratory type in
2. nasopharynx, stratified squamous in lower parts).
3. *Pharyngobasilar fascia*(fibrous layer replacing muscle at skull base).

Muscle layer

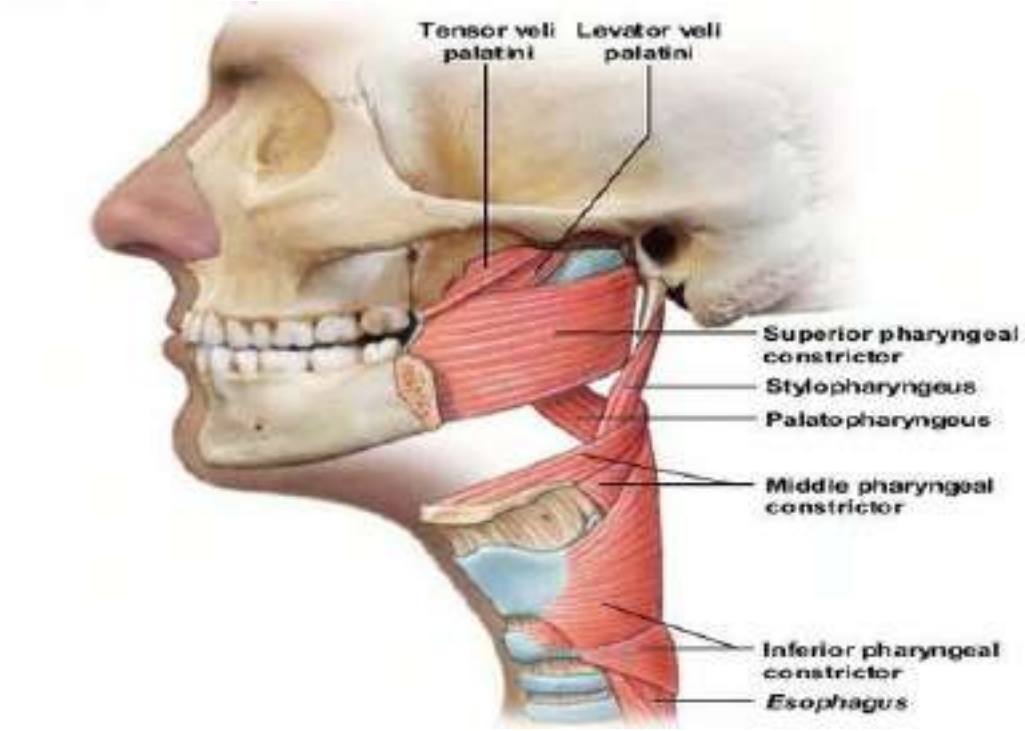
- Outer circular muscles: Superior, middle, and inferior constrictors (help in swallowing).
- Inner longitudinal muscles: stylopharyngeus, salpingopharyngeus, and palatopharyngeus.

- palatopharyngeus (elevate pharynx).

Buccopharyngeal fascia

(outer covering, continuous with pretracheal fascia).

muscle of pharynx



4- The buccopharyngeal fascia: The outermost layer of connective tissue, covering the external surface of the pharyngeal muscles.

Blood Supply

- **Arterial:** Ascending pharyngeal artery, branches from facial, maxillary, lingual, and superior thyroid arteries.
- **Venous:** Pharyngeal venous plexus → internal jugular vein.

Nerve Supply

- Motor: Mostly by pharyngeal plexus(cranial accessory via vagus).
  - Exception: Stylopharyngeus→ supplied by glossopharyngeal nerve (CN IX).
- **Sensory:**
  - Nasopharynx → maxillary nerve (V2).
  - Oropharynx → glossopharyngeal nerve (CN IX).
  - Laryngopharynx → vagus nerve (CN X).

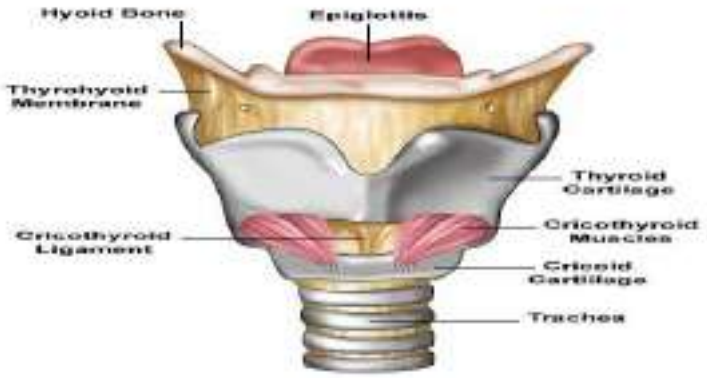
Lymphatic Drainage

- Mainly into retropharyngeal, deep cervical nodes (especially jugulodigastric).

anatomy of larynx

Introduction

- The **larynx** is a hollow organ of voice (voice box) and a passage for air.
- Lies in the anterior neck, **opposite C3–C6 vertebrae**.
- **Functions:**
  1. Air passage between pharynx and trachea.
  2. Phonation (production of voice).
  3. Sphincter – prevents aspiration of food into airway.

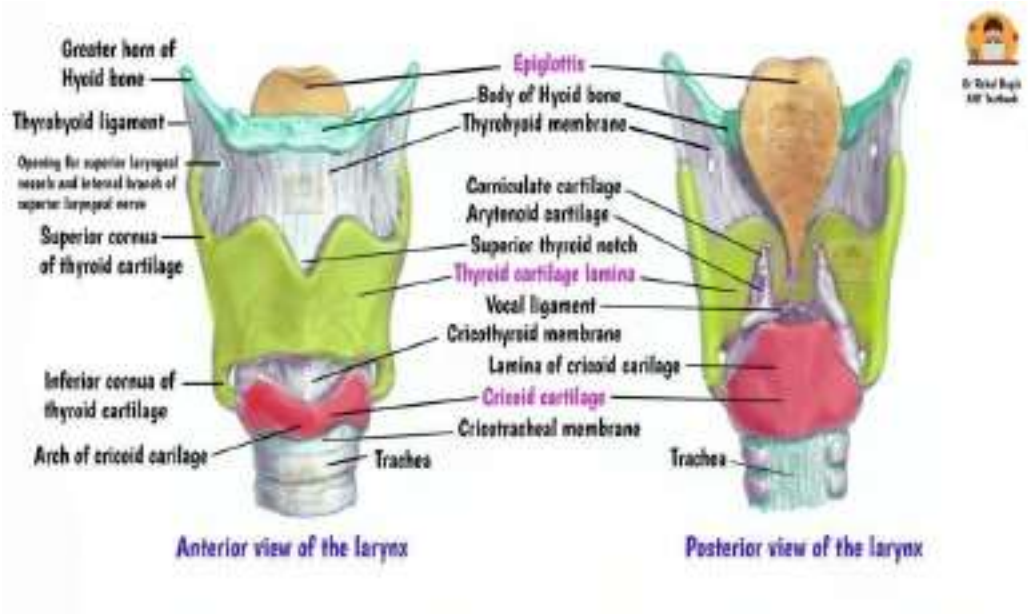


Situation and Relations

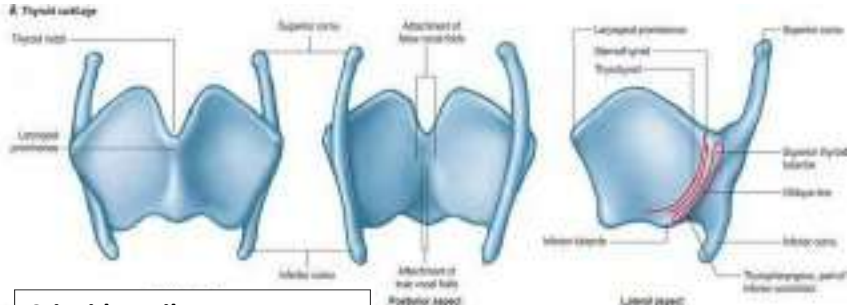
- Above: Pharynx(laryngopharynx).
- Below: Continuouswith trachea.
- Anteriorly: Infrahyoid muscles, skin.
- Posteriorly: Laryngopharynx.
- Laterally: Lobesofthyroid gland, carotid sheath.

Skeleton of the Larynx (Cartilages)

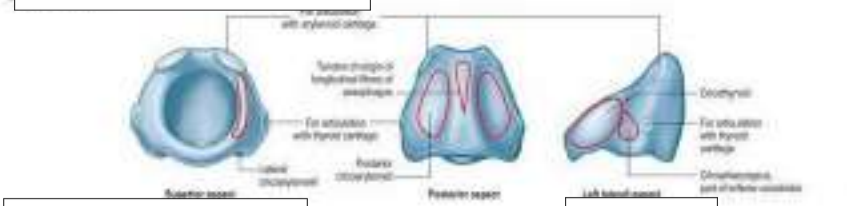
The laryngeal framework is made of 9 cartilages (3 single + 3 paired):



Thyroid cartilage



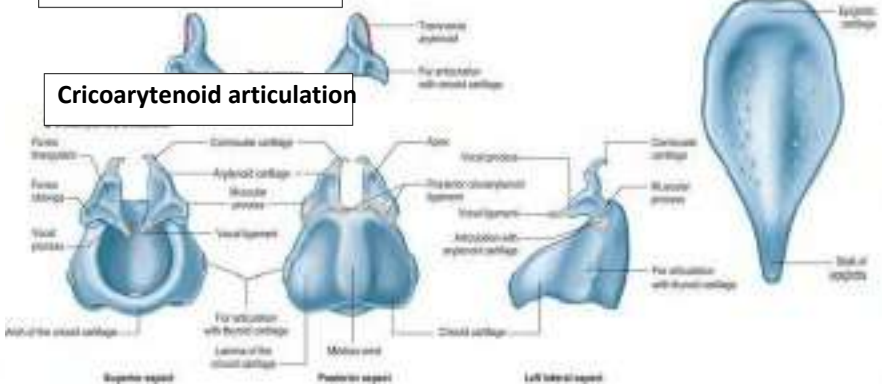
Cricoid cartilage



Arytenoid cartilage

Epiglottis

Cricoarytenoid articulation



Single cartilages:

1. Thyroid cartilage – largest, forms Adam's apple (laryngeal prominence).
2. Cricoid cartilage – complete ring, lies below thyroid.
3. Epiglottis –leaf-shaped, closes laryngeal inlet during swallowing.

Paired cartilages:

1. Arytenoid cartilages – pyramidal, sit on cricoid, control vocal cords.
2. Corniculate cartilages – small, on apex of arytenoids.
3. Cuneiform cartilages – small, in aryepiglottic fold.

Joints of the Larynx

- Cricothyroid joint – between cricoid and thyroid; allows tilting (tension of vocal cords).
- Cricoarytenoid joint – between cricoid and arytenoid; allows rotation, gliding (opening/closing cords).

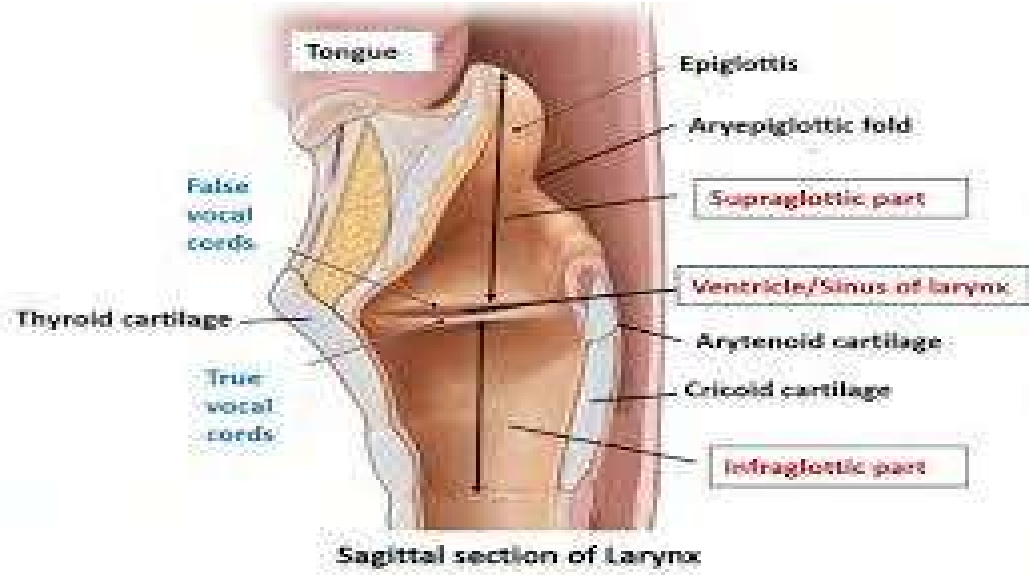
Ligaments and Membranes

- Thyrohyoid membrane– between hyoid bone & thyroid cartilage.
- Cricothyroid ligament – between cricoid & thyroid.
- Quadrangular membrane – forms vestibular fold (false cord).
- Cricothyriod membrane(conuselasticus) – forms vocal fold (true cord).

Cavity of the Larynx *Divided into 3 parts:*

1. Vestibule – from inlet to vestibular folds.
2. Ventricle – between vestibular and vocal folds.
3. Infraglotticcavity – from vocal folds to trachea.

Vocal folds (true cords): contain vocal ligament & vocalis muscle → produce sound.
Vestibular folds (false cords): protective, no role in sound.



Sagittal section of Larynx

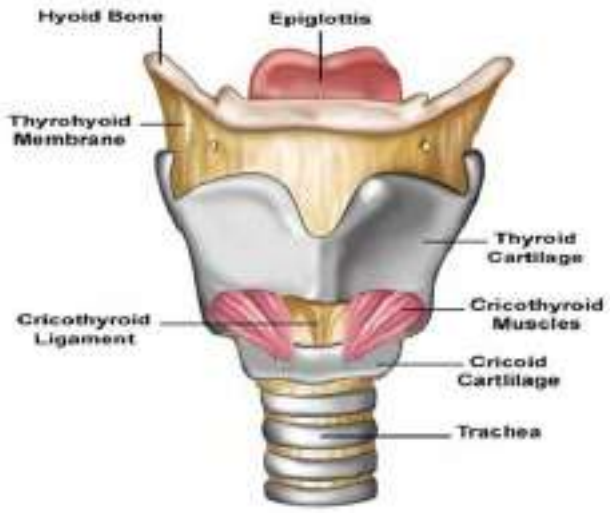
Muscles of the Larynx

A.*Extrinsic muscles* – move larynx as a whole:

- Elevators: suprahyoid muscles.
- Depressors: infrahyoid muscles.

B. *Intrinsic muscles* – control cords and inlet:

- Posterior cricoarytenoid → abducts cords (only opener of rima glottidis).
- Lateral cricoarytenoid → adducts cords.
- Cricothyroid → tenses cords.
- Thyroarytenoid & vocalis → relax cords.
- Arytenoid (transverse & oblique) → adduct aryteniods.
- Aryepiglotti: close the inlet & thyroepiglottic: widen the inlet.





Nerve Supply

- ***Motor:***
  - All intrinsic muscles by recurrent laryngeal nerve (branch of vagus).
  - Exception: Cricothyroid by external branch of superior laryngeal nerve (branch of vagus).
  - .
- ***Sensory:***
  - Above vocal cords → internal branch of superior laryngeal nerve.
  - Below vocal cords → recurrent laryngeal nerve.

Blood Supply

- Above vocalcords→Superiorlaryngealartery (from superior thyroid artery).
- Below vocalcords→Inferiorlaryngealartery (from inferior thyroid artery).
- Venous drainage follows arteries → thyroid veins.

Lymph drainage

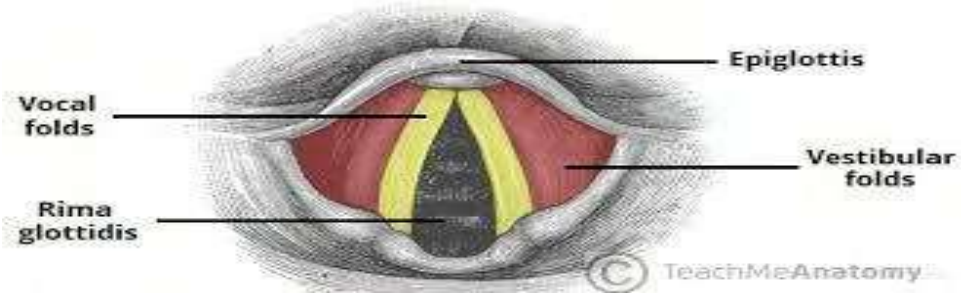
- Above the VCs: Upper deep cervical L.Ns.
- Below the VCs: Lower deep cervical L.Ns.

Clinical Notes

- Tracheostomy:donebelowcricoid.
- Cricothyrotomy:emergencyairway through cricothyroid ligament.
- Recurrent laryngeal nerve injury: hoarseness or loss of voice.
- Epiglottitis:inflammationthatcan block airway.

Rima Glottidis:

- Definition: Narrow opening between the vocal folds inside the
- larynx. Parts:
  - o Intermembranous (anterior 3/5): phonation
  - o Intercartilaginous (posterior 2/5): respiration & cough
- Functions: Respiration, phonation, airway protection, cough.
- Clinical: Laryngospasm, vocal cord paralysis (recurrent laryngeal nerve injury), site of intubation.



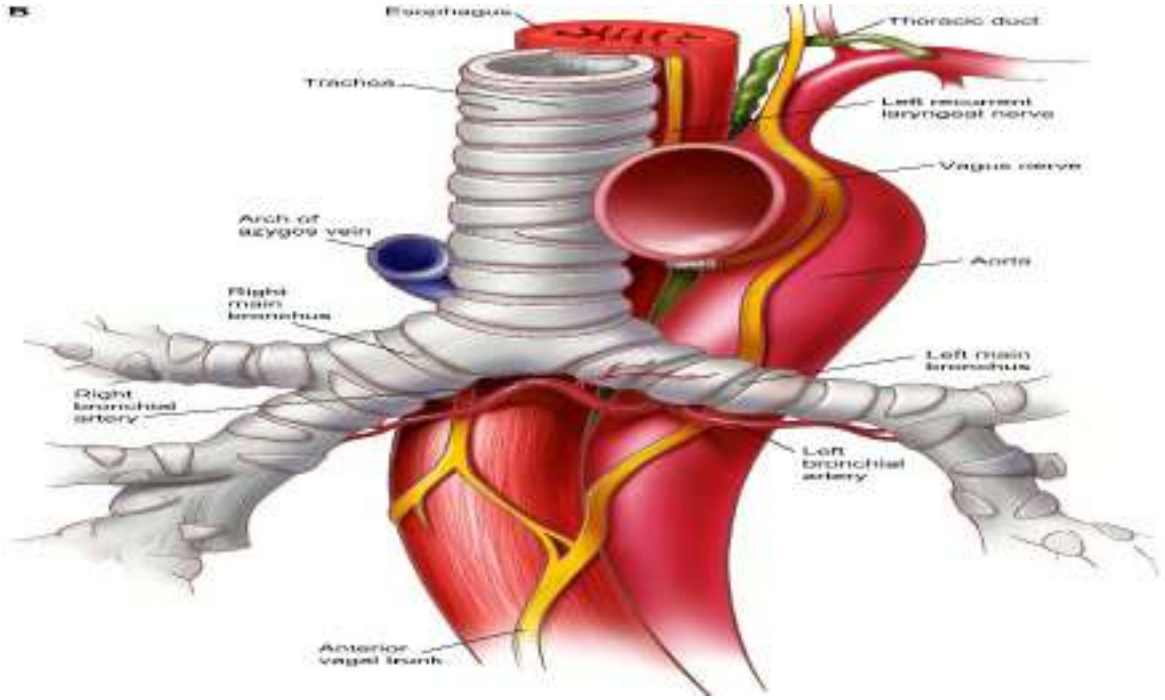
trachea

- **Definition**: The trachea (windpipe) is a fibrocartilaginous tube that forms the lower part of the respiratory tract, connecting the larynx above with the bronchi below.
- **Extent:**
  - o Begins at the lower border of the cricoid cartilage (opposite C6 vertebra).
  - o Ends at the level of the sternal angle (T4–T5 vertebrae) by dividing into the right and left main bronchi.
- **Dimensions:**
  - o Length: about 10–12 cm.
  - o Diameter: about 2 cm in adults (slightly wider in males).
- **Structure:**
  - o Composed of 16–20 C-shaped hyaline cartilages, open posteriorly.
- **Relations:**
  - o Anteriorly: thyroid isthmus, inferior thyroid veins, thymus (in children), sternum.
  - o Posteriorly: esophagus.
  - o Laterally: lobes of thyroid gland, common carotid arteries, recurrent laryngeal nerves.
- **Blood Supply:**
  - o Arteries: inferior thyroid and bronchial arteries.
  - o Veins: drain into inferior thyroid veins.
- **Nerve Supply:**

o Sensory and motor innervation mainly from the recurrent laryngeal nerves.

- Bronchi & Bronchial Tree:
- Trachea divides at the level of T4–T5 (sternal angle) into right and left main (primary) bronchi.
- Right main bronchus: wider, shorter, more vertical → common site for inhaled foreign bodies.
- Left main bronchus: narrower, longer, more horizontal.
- Main bronchi enter the lungs at the hilum and branch into:
- Lobar (secondary) bronchi → one for each lobe (3 right, 2 left).
- Segmental (tertiary) bronchi → supply bronchopulmonary segments (10 right, 8–10 left).
- Beyond segmental bronchi → smaller bronchi → bronchioles.
- Terminal bronchioles: last part of conducting zone.

- **Respiratory bronchioles: beginning of respiratory zone → give rise to alveolar** ducts and alveoli (gas exchange units).
- Together, this branching system from trachea down to alveoli is called the bronchial tree.

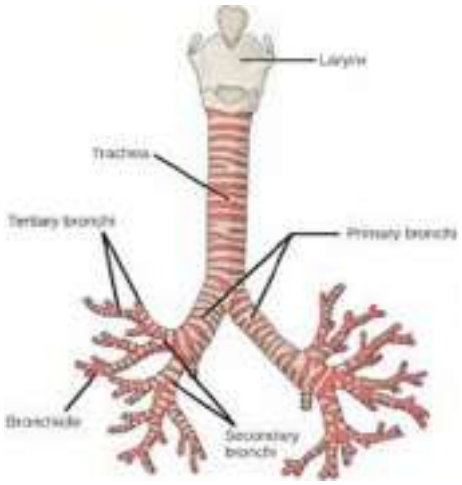


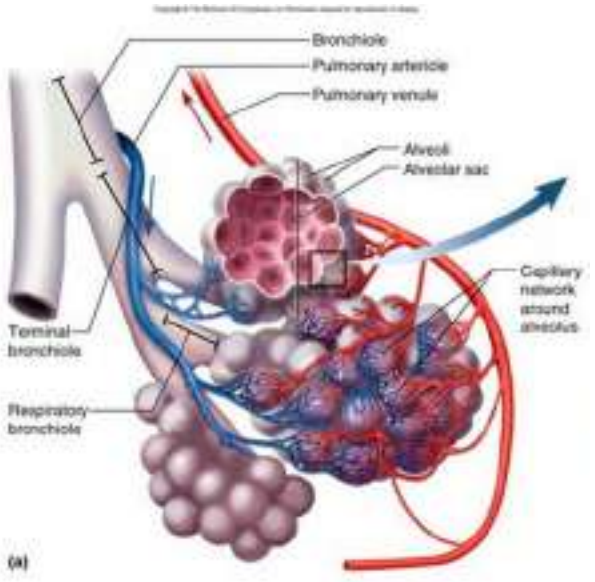
Bronchopulmonary Segments:

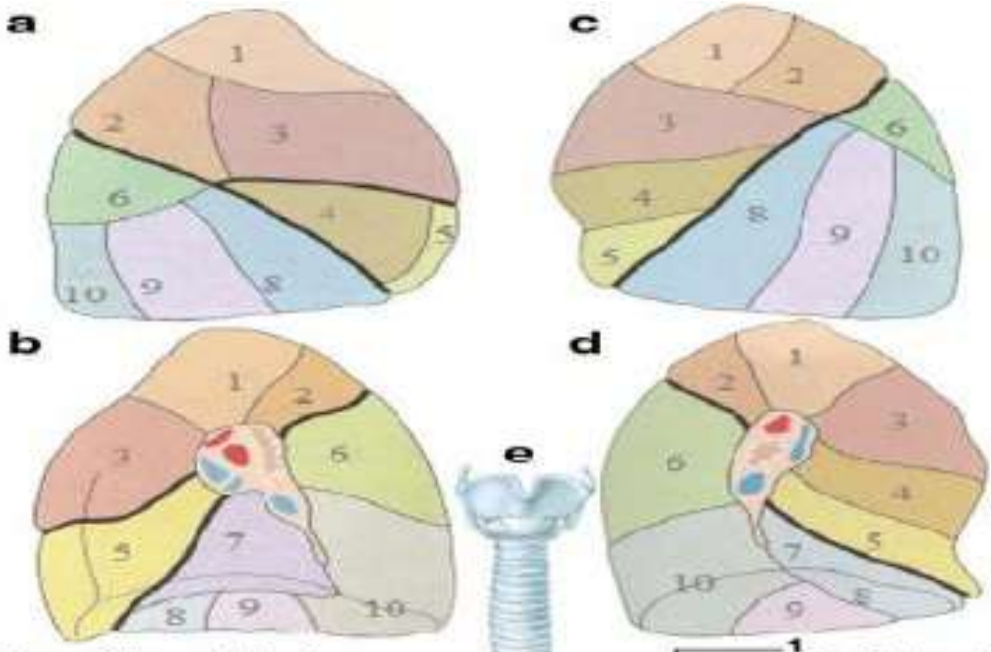
A bronchopulmonarysegmentis the smallestfunctionallyindependentunitof thelung. It is a pyramidal-shaped portion of lung tissue that is aerated by a segmental (tertiary) bronchus and supplied by its own branch of the pulmonary artery & drained by pulmonary veins.

Clinical note

Each segment is separated by connective tissue septa, making it possible to surgically remove a diseased segment without affecting the rest of the lung.







Right lung (Upper lobe):
1. Apical
2. Posterior
3. Anterior

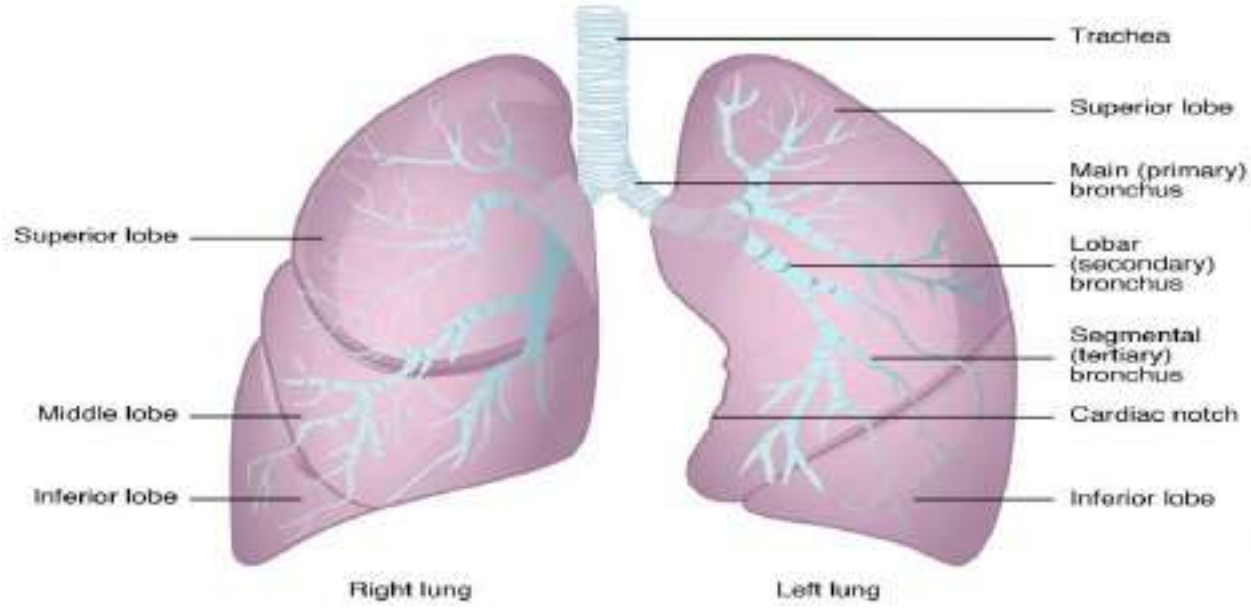
Right lung (Middle lobe):
4. Lateral
5. Medial

Right lung (Lower lobe):
6. Superior
7. Medial basal
8. Anterior basal
9. Lateral basal
10. Posterior basal

Left lung (Upper lobe):
1. Apical
2. Posterior
3. Anterior
4. Lingula-Superior
5. Lingula Inferior

Left lung (lower lobe):
6. Superior
7. Medial basal
8. Anterior basal
9. Lateral basal
10. Posterior basal

LUNG



Shape and Position

- The lungs are paired organs of respiration, lying on either side of the mediastinum.
- Each lung is conical in shape with:
  - ***Apex***: projects above the clavicle into the root of the neck.
  - ***Base*:** rests on the diaphragm.
  - ***Surfaces*:**
    - Costal surface: convex, related to ribs.
    - Mediastinal surface: concave, related to mediastinum and hilum.
    - Diaphragmatic surface: concave, rests on diaphragm.
  - ***Borders*:**
    - Anterior border: sharp, meets the other lung behind sternum.
    - Posterior border: rounded, lies beside vertebral column.
    - Inferior border: separates base from costal surface.

Hilum of Lung

- Medial surface has a hilum through which structures enter/leave.
- Contents form the root of lung:
  - One pulmonary artery.
  - Two pulmonary veins.
  - Main bronchus.
  - Bronchial arteries & veins.
  - Lymphatics & nerves.

Differences between Hilum of left & right lung

Left lung: One bronchus (mostposterior structure). -Pulmonary artery (anterior andabove the bronchus -Twopulmonary veins(most anterior and inferior structures).

Right lung: Two Bronchi (most posterior structure). - Pulmonary artery (anterior and inbetween the two bronchi). - Two pulmonary veins (most anterior and inferiorstructures).

BloodSupply

- Functional supply:pulmonary arteries → alveoli → pulmonary veins.
- Nutritional supply:bronchial arteries.

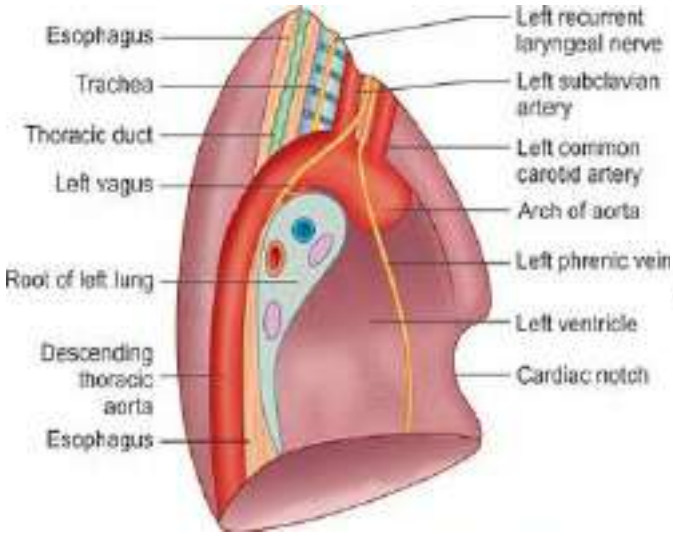
Nerve Supply

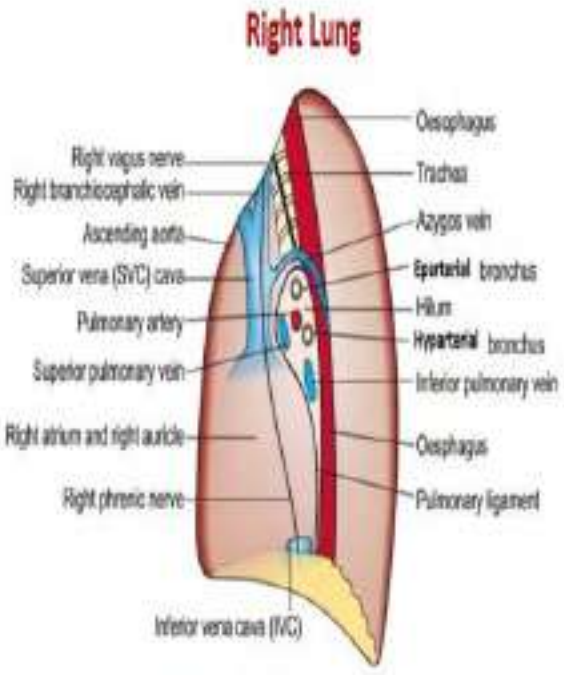
- Parasympathetic: vagus nerve (bronchoconstriction, secretomotor).
- Sympathetic: thoracic sympathetic chain (bronchodilation, vasoconstriction).

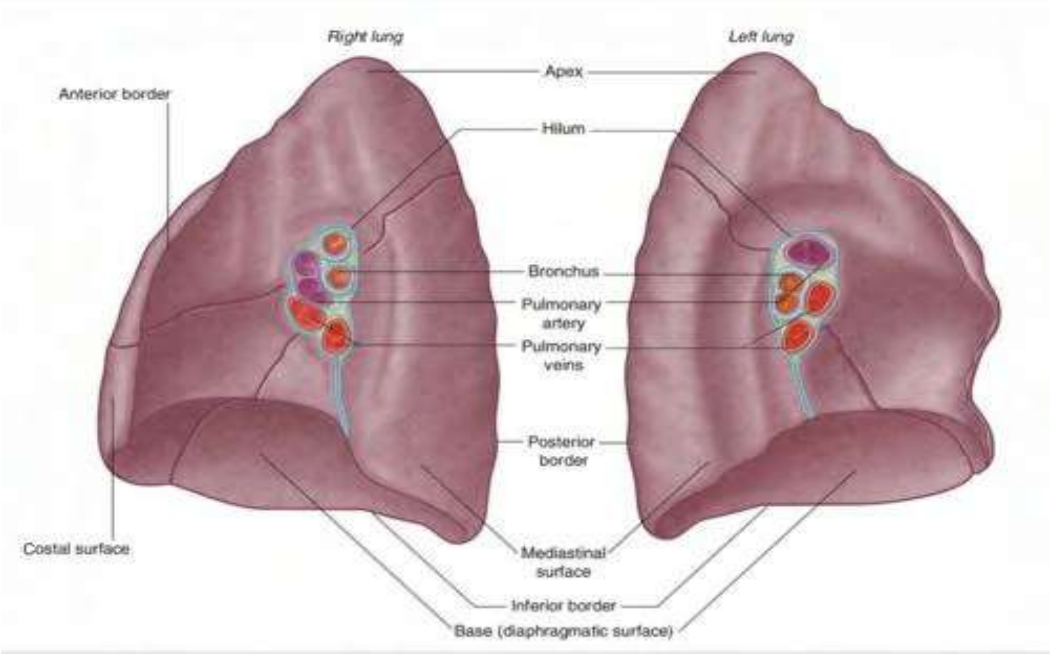
Lymphatic Drainage

- Superficial (subpleural) and deep plexuses → hilar (bronchopulmonary) nodes → tracheobronchial nodes → right & left mediastinal lymph trunks.
- On the right side, drains into the right lymph duct
- On the left side, drains into thoracic duct
- → Brachiocephalic veins

Left lung


Esophagus
Trachea
Thoracic duct
Left vagus
Root of left lung
Descending thoracic aorta
Esophagus
Left recurrent laryngeal nerve
Left subclavian artery
Left common carotid artery
Arch of aorta
Left phrenic vein
Left ventricle
Cardiac notch

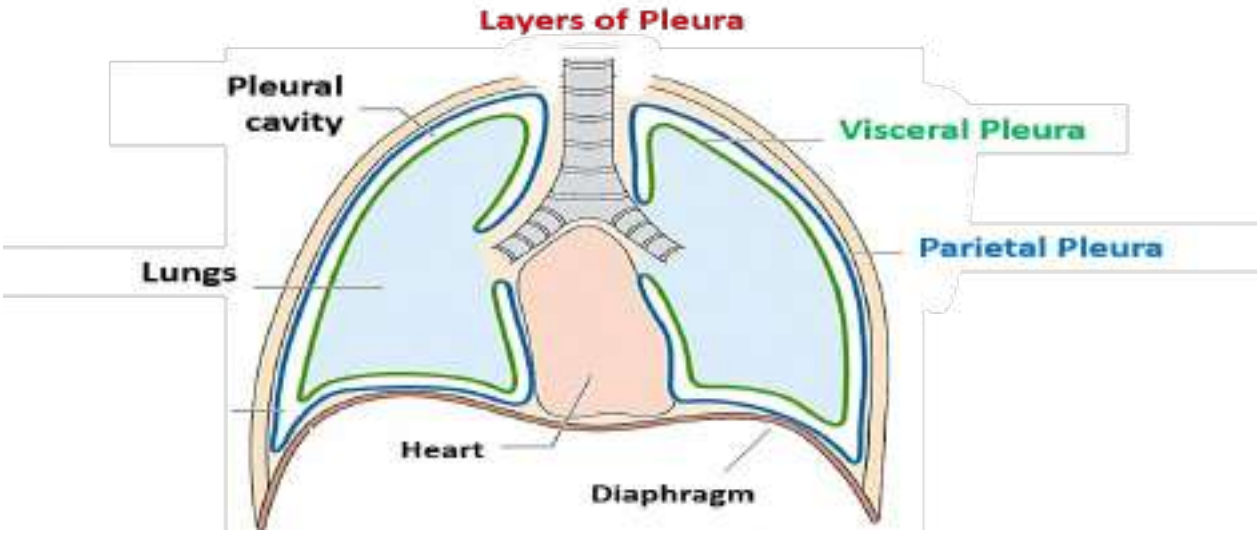

Right Lung
Right vagus nerve
Right brachiocephalic vein
Ascending aorta
Superior vena (SVC) cava
Pulmonary artery
Superior pulmonary vein
Right atrium and right auricle
Right phrenic nerve
Inferior vena cava (IVC)
Oesophagus
Trachea
Azygos vein
Eparterial bronchus
Hilum
Hyparterial bronchus
Inferior pulmonary vein
Oesophagus
Pulmonary ligament


Right lung
Left lung
Anterior border
Apex
Hilum
Bronchus
Pulmonary artery
Pulmonary veins
Posterior border
Costal surface
Mediastinal surface
Inferior border
Base (diaphragmatic surface)

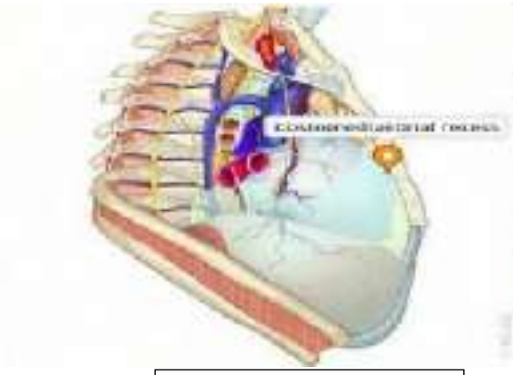
Comparison Between Right and Left Lung

Feature	Right Lung	Left Lung
Number of lobes	3(superior,middle, inferior)	2 (superior, inferior)
Fissures	2fissures: oblique& horizontal	1 fissure: oblique
Size	Larger,heavier(≈ 700 g)	Smaller, lighter (≈ 600 g)
Length	Shorter(duetohigher right dome ofdiaphragm by liver) Broader	Longer
Width Anterior border	Straight	Narrower (due to heart bulge) Shows cardiac notch
Special features	Has a middle lobe	Has lingula (tongue like projection of the upper lobe)
Relations (mediastinal surface)	Related to superior vena cava, right atrium, azygos vein	Related to arch of aorta, descending aorta, left ventricle
Hilum arrangement (from before backward)	Vein – Artery – Bronchus	Artery – Bronchus – Vein
-		

pleura



- ***Definition*** :Serous membrane surrounding the lungs, forming a closed sac.
- ***Parts*:**
  - Parietal pleura: lines thoracic wall, diaphragm, and mediastinum.
    - Divisions: costal, diaphragmatic, mediastinal, cervical.
  - Visceral pleura: covers lungs and dips into fissures.
- ***Pleural Cavity***
  - Potential space between parietal and visceral pleura.
  - Contains a thin film of fluid → lubrication, allows smooth lung movement.
- ***Recesses*** (clinically important spaces)
  - Costodiaphragmatic recess: between costal and diaphragmatic pleura; lowest part where fluid collects.
  - Costomediastinal recess: near anterior border, left side deeper due to cardiac notch.



Costmediastinal recess



Costodiaphragmatic recess

Clinical Notes

- Pleural effusion: accumulation of excess fluid in pleural cavity, usually in costodiaphragmatic recess.
- Pneumothorax: entry of air into pleural cavity → lung collapse.
- Pleurisy (pleuritis): inflammation of pleura causing sharp chest pain during breathing.
- Thoracentesis: needle insertion in 9th intercostal space at midaxillary line (above rib) to aspirate fluid from pleural cavity.

Surface anatomy of the lung:

1-Apex: 1 inch above the medial 3rd of the clavicle

2- *Anterior border:*

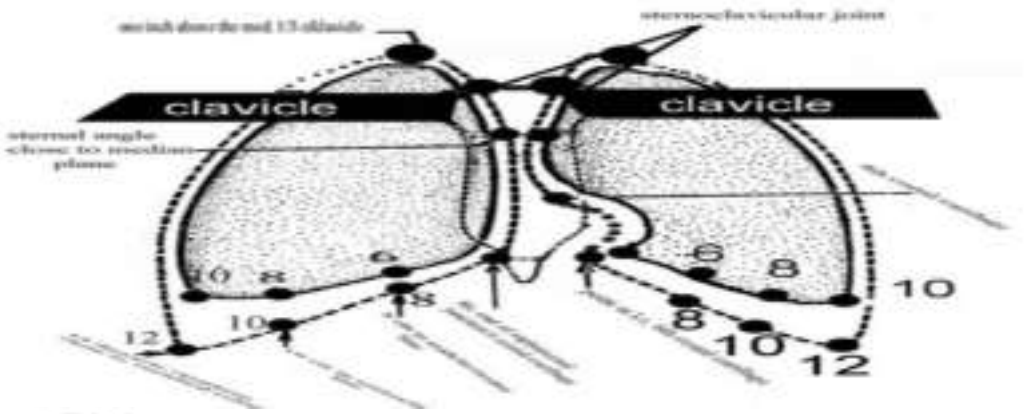
1. Rt. Lung: extend from the apex to the sterno-clavicular joint, sternal angle then vertically to the level of 6th costal cartilage.
2. Lt. lung: extend from the apex to the sterno-clavicular joint, sternal angle then vertically to the level of 4th costal cartilage, deviates to the Lt. & descend again 1inch lateral to sternal margin down to the 6th costal cartilage.

3- Inferior border:

6th rib (in the mid clavicular plane)
8th rib (in the mid axillary line).
10th rib (close to the vertebral column)

Surface anatomy of pleura: (the same as lung but it differs from it in the following):

- -The anterior border descends down to the 7th costal cartilage
- - The inferior border at 8th,10th,12th ribs instead of 6th ,8th ,10th ribs



Development of the Respiratory System

1. Origin

- The respiratory system develops from the endoderm of the foregut and splanchnic mesoderm.
- The endoderm gives rise to:
  - Epithelial lining of larynx, trachea, bronchi, and alveoli.
- The splanchnic mesoderm forms:
  - Cartilage, muscle, and connective tissue of airways.
  - Visceral pleura.
- Somatic mesoderm → parietal pleura.

2. Formation of Respiratory Diverticulum (lung bud):

- Around week 4 of embryonic life, a laryngotracheal diverticulum (lung bud) appears as an outgrowth from the ventral wall of foregut .
- This separates from the foregut by tracheoesophageal ridges , which fuse to form the tracheoesophageal septum.
- This septum divides the foregut into:
  - Dorsal part → esophagus.
  - Ventral part → larynx, trachea, and lung buds.

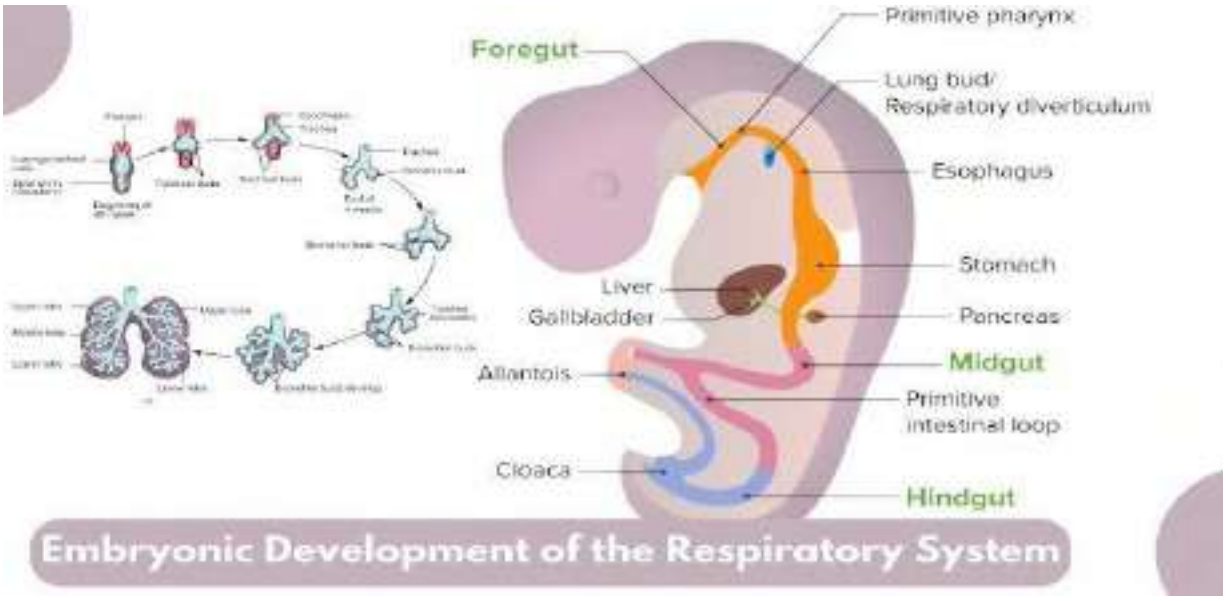
3. Development of Larynx, Trachea, and Bronchi

- Larynx :developsfrom thecranialend of the diverticulum; cartilages from 4th and 6th pharyngeal arches
- Trachea:elongation of thediverticulum.
- Bronchi:
  - By week 5, lung buddividesinto two bronchial buds (right and left primarybronchi).
  - Right forms 3 secondary bronchi → 3 lobes. Left forms 2 secondary
  - bronchi → 2 lobes. Further branching forms tertiary bronchi
  - (bronchopulmonary segments).

4. Development of Lungs (Histological Stages)

Lung development occurs in five stages

1- Embryonic stage (weeks 4-7):
Formationoftrachea,bronchi, andinitial lung buds.
2- Pseudoglandular stage (weeks 5–17):
o Branchingforms terminal bronchioles.
o No gas exchange yet (not viable).
3- Canalicular stage (weeks 16–25):
o Bronchioles divide into respiratory bronchioles.
o Primitive alveolar ducts form.
o Some vascularization appears (beginning of possible viability ~24 weeks).
4- Saccular stage (weeks 24–birth):
o Formation of terminal sacs (primitive alveoli).
o Thinning of epithelium → closer contact with capillaries.
o Type I pneumocytes (gas exchange) and Type II pneumocytes (surfactant secretion) appear.
5- Alveolar stage (late fetal period – 8 years):
o True alveoli form.
o Postnatal alveolar multiplication: ~150 million at birth → ~300 million in adults.



Congenital Anomalies of the Respiratory System

1-Nasal and Upper Airway Anomalies

- Choanalatresia : failure of posterior nasal aperture to open → newborn presents with respiratory distress relieved by crying.
- Congenital nasal stenosis : narrowing of nasal passages.
- Congenital laryngeal anomalieṡ
  - *Laryngomalacia*: collapse of supraglottic structures on inspiration (stridor in infants).
  - *Laryngeal web or atresia*: abnormal membrane partially or completely obstructing the larynx.

2-Tracheal and Bronchial Anomalies

- Tracheal stenosis or atresia: narrowing or absence of tracheal lumen. Tracheoesophageal fistula (TEF): abnormal communication between trachea and esophagus (often with esophageal atresia).

- Tracheomalacia : weakness of tracheal wall → collapse during breathing.
- Bronchial atresia : blind-ending bronchus, distal lung may be hyperinflated.
- Accessory bronchi : e.g., tracheal bronchus (extra bronchus directly from trachea).

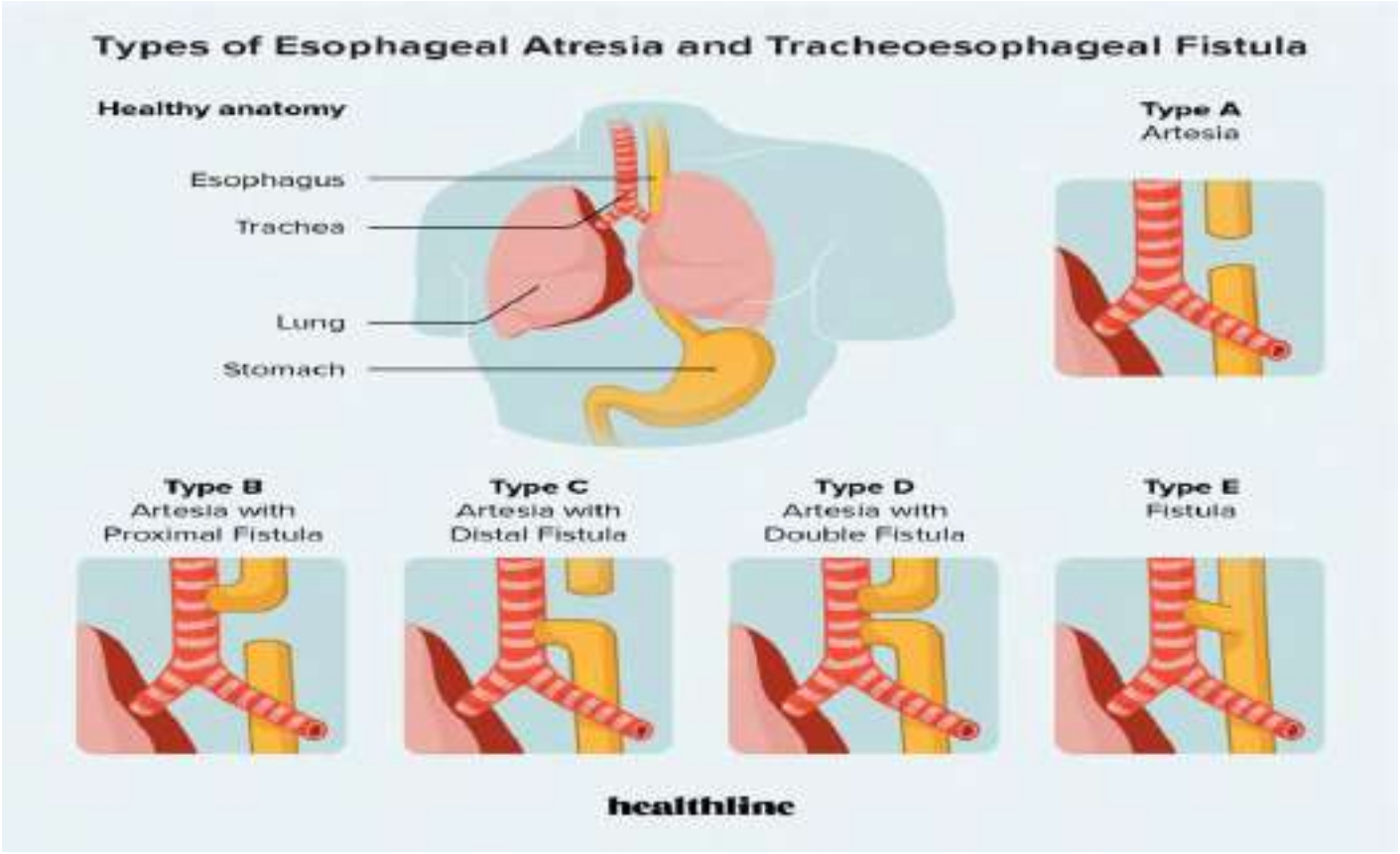
3-Lung Anomalies

- Congenital Respiratory Distress Syndrome is due to surfactant deficiency in premature infants, causing alveolar collapse, hypoxemia, and formation of hyaline membranes. It presents with rapid breathing, grunting, and cyanosis soon after birth.
- Pulmonary hypoplasia: underdeveloped lung (seen with congenital diaphragmatic hernia, oligohydramnios).
- Pulmonary agenesis: complete absence of one lung (rare).

- Development of the diaphragm:
- Thediaphragm develops from fourmain components:
- Septum transversum → central tendon.
- Pleuroperitoneal membranes → close pericardioperitoneal canals.
- Dorsal mesentery of esophagus → crura.
  Body wall musculature → peripheral muscular part.

4-Congenital anomalies of the diaphragm:

- Congenital diaphragmatic hernia (CDH): failure of pleuroperitoneal membranes to close (most commonly left posterolateral → Bochdalek hernia).
- Hiatal hernia: due to enlargement of esophageal hiatus.
- Eventration of diaphragm: defective muscle development, diaphragm thin and elevated.
- Supernumerary lobes: extra lobe of lung.


Types of Esophageal Atresia and Tracheoesophageal Fistula
Healthy anatomy
Esophagus
Trachea
Lung
Stomach
Type A
Artesia
Type B
Artesia with
Proximal Fistula
Type C
Artesia with
Distal Fistula
Type D
Artesia with
Double Fistula
Type E
Fistula
healthline

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Respiratory System

Histology

RPS_314

page	lecture
1	Conductive part of respiratory
12	Respiratory portion of respiratory

1 Conductive part of respiratory

The Respiratory System:

- Air enters the nose through two openings called the external nares (nostrils) to the nasal cavity, and then travels through the pharynx, which is a Produce mucus that bathes the exposed muscular tube which carries both food surfaces

and air throughout most of its length Air
- then enters the larynx to the trachea that bifurcates into a right and a left primary bronchus, which carry the air into the lungs
- The **diaphragm** and the **intercostal muscles**, although they are not apart of the respiratory system, play important roles in breathing
- The upper respiratory system, above the larynx
- The lower respiratory system, from the larynx down
- **The Respiratory Tract Consists of :**
  - Conductive portion
  - Respiratory portion

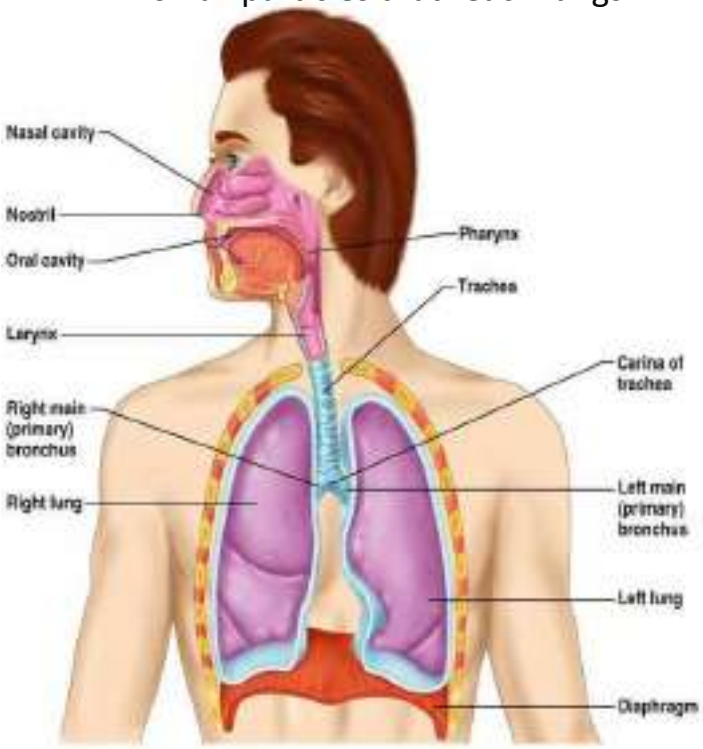
- **Components of the Respiratory Defense System :**

➢ Goblet cells and mucous glands

➢ Cilia:

Sweep debris trapped in mucus toward the pharynx

➢ Alveolar macrophages engulf small particles that reach lungs



Conductive tract	Respiratory tract
• Nose & nasal cavity	• Respiratory bronchioles
• Nasopharynx &oropharynx	• Alveolar duct
• Larynx, Trachea, Bronchi	• Alveolar sac
• Bronchioles till terminal	• Alveoli
• bronchioles	
•	

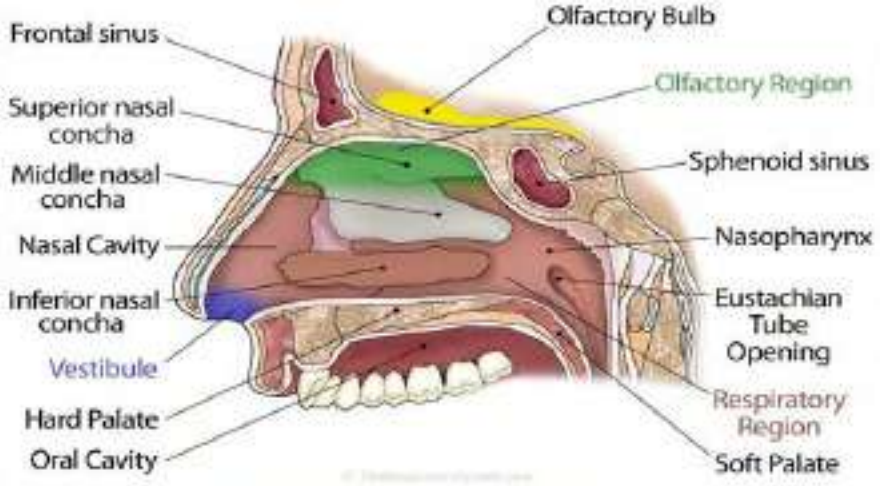
1) THE NASAL CAVITIES

They include one cavity on each side separated by nasal septa, formed of:

1-Vestibule of the nose
- Lined by stratified squamous epithelium, keratinized (thin skin) contains thick short hair sebaceous and sweat glands

2-olfactory area
- The roof of the nasal cavity, the superior aspect of the nasal septum and the superior concha

3-The respiratory area

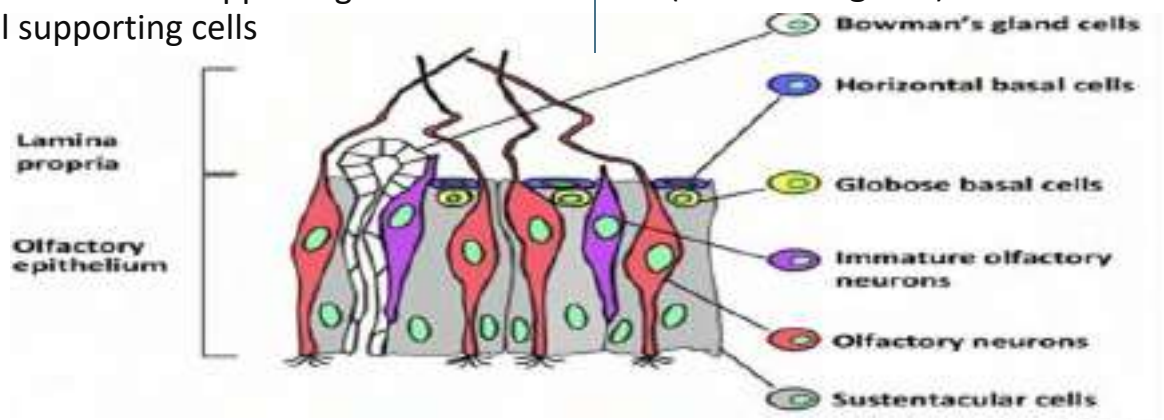


OLFACTORY MUCOSA

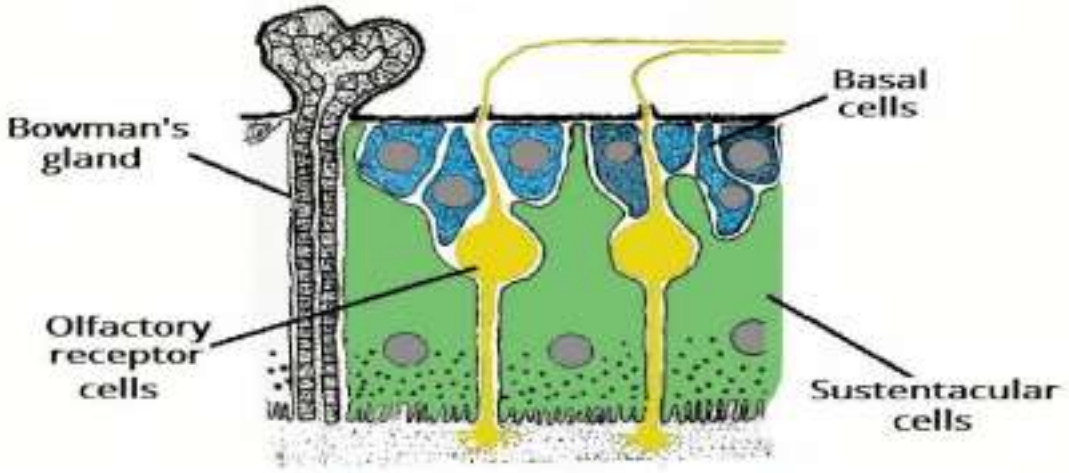
- **The olfactory epithelium :**
  - Is thick yellowish-brown color due to the presence of pigments in its covering epithelium and in the glands
  - Is modified type of pseudo stratified columnar epithelium without goblet composed of 4 types of cells :

1.Olfactory sensory cells
2. Sustentacular or supporting cells
3. Basal supporting cells

- **Lamina propria :**
  - Highly vascularized, loose to dense, irregular collagenous connective tissue that is firmly attached to the underlying periosteum
  - Contains lymphoid elements, collection of axons that arise from the olfactory cells and serous olfactory glands (**Bowman's gland**).



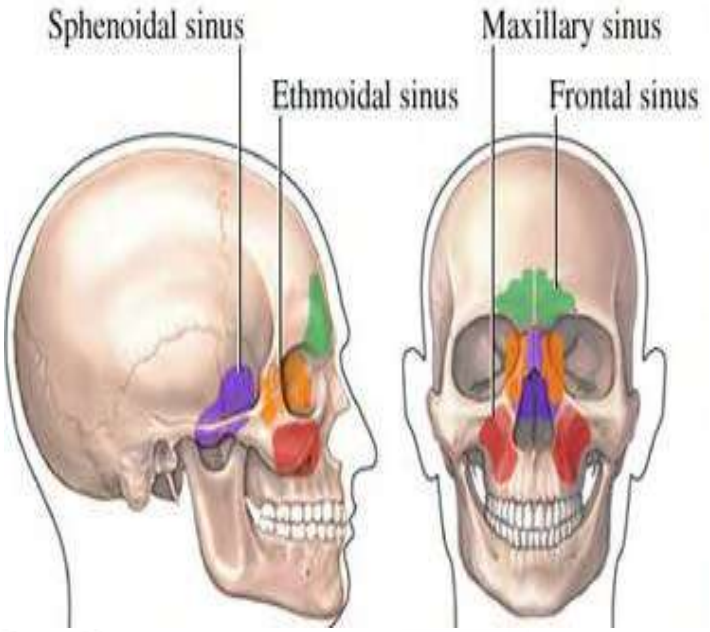
1- OLFACTORY SENSORY (RECEPTOR CELLS)	*2- SUSTENTACULAR (SUPPORTING) CELLS:*	*3- BASAL CELLS*
• Olfactorycells are bipolar nerve cells specialized for smell	• They are the most numerous cells; provide physical support, nourishment, and electrical insulation for the olfactory cells	• **They are of two types** o Horizontal cells o Globose cells
• The apical end of the dendrite is modified to form a bulb (**olfactory vesicle**), which projects above the surface of the sustentacular cells	• Long columnar cells in between the sensory cells, wide striated apical surface and narrow near the basement membrane	• **Horizontal cells**: flat and lie against the basement membrane • **Globose cells:** short, basophilic, pyramid-shaped cells
• The base of the olfactory cells has unmyelinated axons that penetrate the basal lamina to form bundles of olfactory nerve fibers	• Have pale oval nuclei are in the apical third of the cell	• The horizontal basal cells replicate to replace the globose cells
• Olfactory cells live for 3 months, and if injured, they are replaced	• Have junctional complexes with the olfactory vesicle of olfactory cells and with other sustentacular cells	• The globose cells are stem cells and have considerable proliferative capacity and can replace both sustentacular and olfactory cells



Paranasal Sinuses

- Four large, mucoperiosteum-lined spaces in the skull called **ethmoid**, **sphenoid**, **frontal**, and **maxilla**, which communicate with the nasal cavity through narrow opening in the respiratory mucosa

- Lined by **the respiratory epithelium** with few goblet cells.
- **Lamina propria** contains seromucous glands and lymphoid elements and fused with the periosteum. Humidifiy the nasal
- cavity and produce clear voice



2) THE NASOPHARYNX

- It is the first portion of the pharynx that acts a passageway for air and food, lies below the base of the skull, behind the nose and above the soft palate
- **Lined by Pseudostratified columnar ciliated with goblet cells**.
  Its lamina propria contains seromucous glands, lymphoid tissues called pharyngeal tonsil (adenoid).

3) THE OROPHARYNX

- It lies behind the oral cavity and posterior surface of the tongue, **lined by stratified squamous non-keratinized epithelium**

4) THE LARYNX

- The larynx (voice box) situated between the oropharynx and the trachea

- It is a rigid, short, cylindrical tube 4 cm length and 4 cm width formed of :
  - ***Mucous membrane:***
    1) 2 pairs of mucosal folds
    - **Vestibular folds**: covered by respiratory epithelium
    - **The vocal fold (True vocal cords)**: covered by non-keratinized stratified squamous epithelium

    2) **Lining epithelium**: pseudostratified columnar ciliated with goblet cells

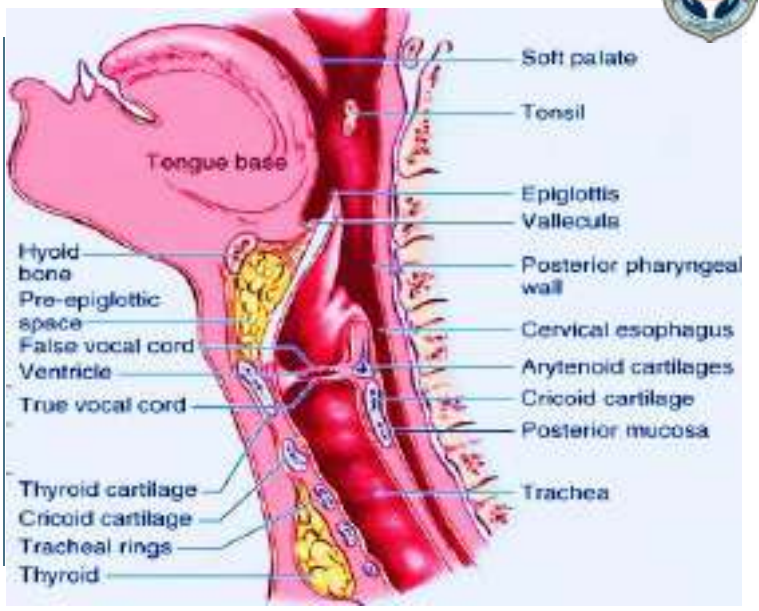
    3) **Lamina propria:** loose C.T. contains glands &lymphatic nodules

Cartilages:

1)Hyaline cartilage: The thyroid and cricoid and the inferior part of arytenoids

2) Elastic cartilage:

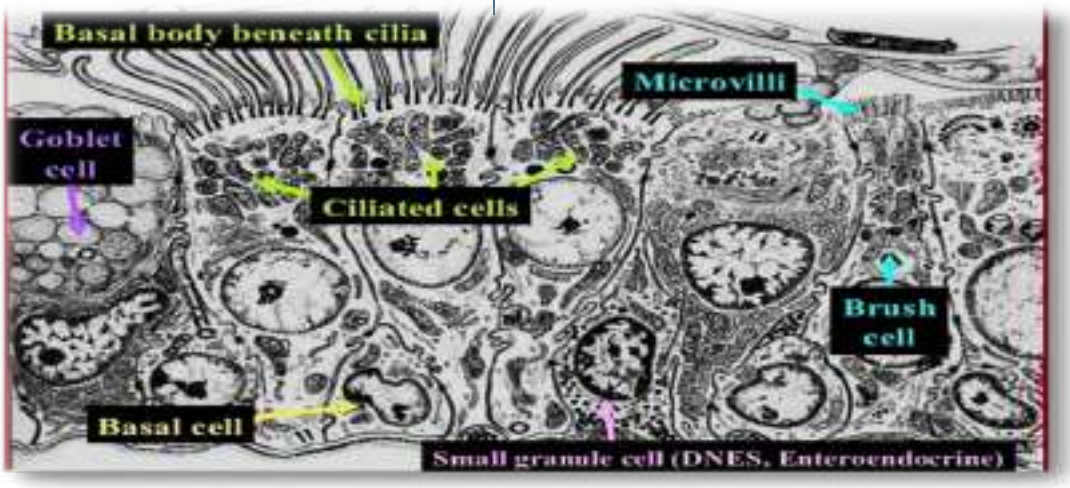
epiglottis, corniculate and cuneiform, and tip of the arytenoids



Respiratory Epithelium

- **Pseudostratified ciliated columnar epithelium with goblet cells**
- rests on basement membrane and lamina propria Lines the conducting portion of respiratory system ***except:***
  - o The vestibule of the nasal cavity.
  - o The olfactory area of the nasal cavity.
  - o The oro-pharynx.
  - o The true vocal cords of the larynx.
  - o The anterior surface and the upper posterior surface of the epiglottis

- The epithelium is composed of 6 cell types :
  1) Ciliated columnar cells,
  2) Mucous Goblet cells,
  3) Brush cells,
  4) Basal cells,
  5) Small granule cells (Cells of the diffuse neuroendocrine system) (DNES)



1) Ciliated columnar cells

- It is the most common cells; tall, slender cells with a basally located nucleus and possess 300 cilia and short microvilli on their apical cell membrane
- The cytoplasm contains mitochondria, Golgi complex, some RER and a few ribosomes
- Their cilia move the mucus and particles toward the nasopharynx

2) Small granule cells

- (Cells of the diffuse neuroendocrine system) (DNES cells)
- Known as **Kulchitsky cells:** basally located and stained with **silver**
- They monitor the O2 and Co2 levels in the lumen of the airway
- They are closely associated with naked sensory nerve endings
- DNES cells contain numerous granules with dense core (100 – 300 nm) in their basal cytoplasm

3) Mucous Goblet cells

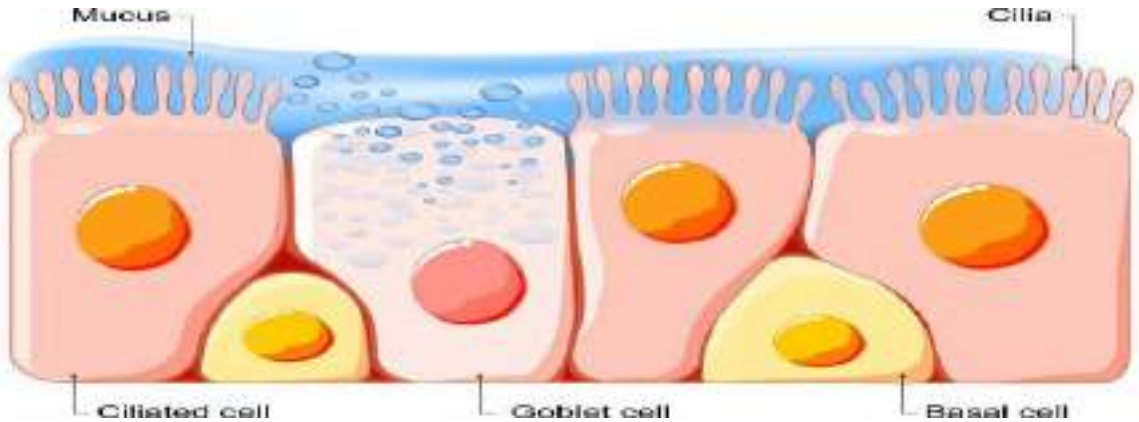
- It is the next common cells
- have a narrow base
- contain flat nucleus and an expanded apical portion containing abundant , secretory granules contain mucin

4) Short Basal cells

- They are undifferentiated cells located on the basement membrane, but their apical surfaces do not reach the luminal surface
- They proliferate to replace other cell types in the respiratory epithelium

5) Brush cells (small-granule mucous cells)

- Columnar cells with numerous tall blunt microvilli that exhibit dendritic synapses with an afferent nerve ending at their basal surfaces
- They have a sensory role (the receptors of sneezing and cough reflexes). Others believe that they are goblet cells that have released their contents



5) TRACHEA

- The trachea is a flexible air tube concerned with the conduction and conditioning of air
- 20 cm in length and 2cm in diameter
- It begins at the larynx cricoid cartilage, ends at the point of its bifurcation into the
- **2 primarybronchi**

The tracheal wall is formed of the following layers from inside outwards :

A. Mucosa :

- The respiratory epithelium lies on a basement membrane
- Lamina Propria: Loose,fibroelastic Connective tissue contains lymphoid elements, mucous, and seromucous glands. Elastic membrane: between the lamina propria and submucosa

- **B. Submucosa** composed of**:** • Areolar C.T. contains blood vessels, nerves, numerous mucous and mucoserous glands open onto the epithelial surface and diffuse Lymphoid elements and lymphatic nodules

C. Supporting layer (fibro- cartilaginous coat of the trachea):

- It is formed of 16 to 20 horseshoe-) shaped **(C-rings hyaline cartilage rings)** embedded in **dense,irregular fibroelastic C.T**
- The rings ends opened posteriorly and are connected to each other by smooth muscle, **the trachealis muscle**

D. Adventitia (External fibrous membrane) :

Fibroelastic C.T. contains blood vessels, nerves and lymphatics

THE LUNGS

Gross and microscopic Structure of the Lungs and Bronchial tree;

- The lung is the main respiratory composed of the bronchial tree, interalveolar septa and blood vessels, covered with the pleura
- lung has a medial indentation, **the hilum (lung root)** receives :
  - ✓ One main primary bronchus Bronchiolar arteries and veins
  - ✓ pulmonary arteries and veins
  - ✓ Lymphatic vessels

- It is subdivided by connective tissue septa into lung lobes :
  - ✓ the right lung has 3 lobes (10 bronchopulmonary segments),
  - ✓ the left has 2 lobes (8 bronchopulmonary segments).

6) THE BRONCHIAL TREE

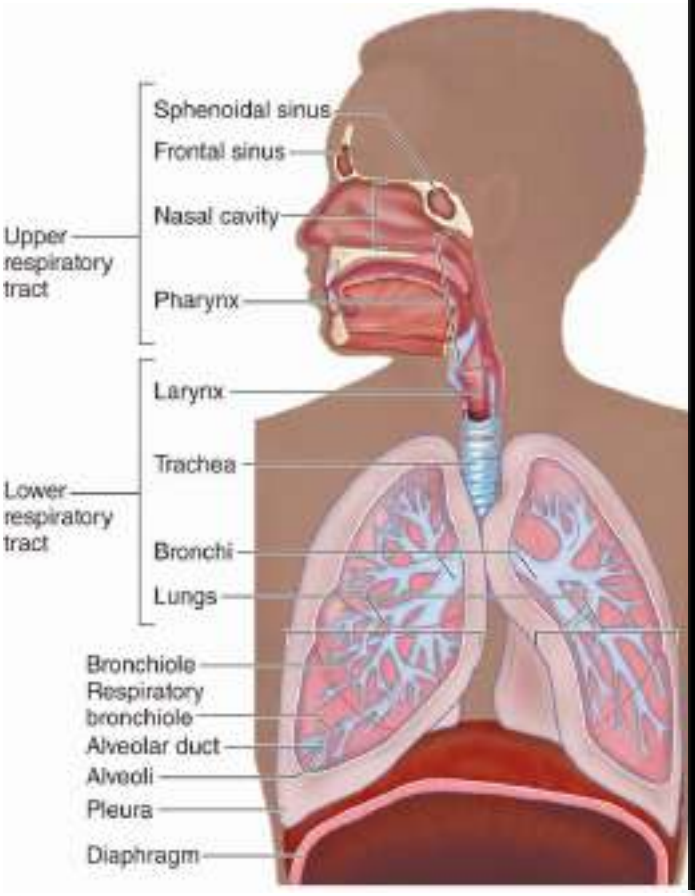
The bronchial tree begins at the bifurcationof the trachea, which branches gradually and decrease in size, it is composed of :

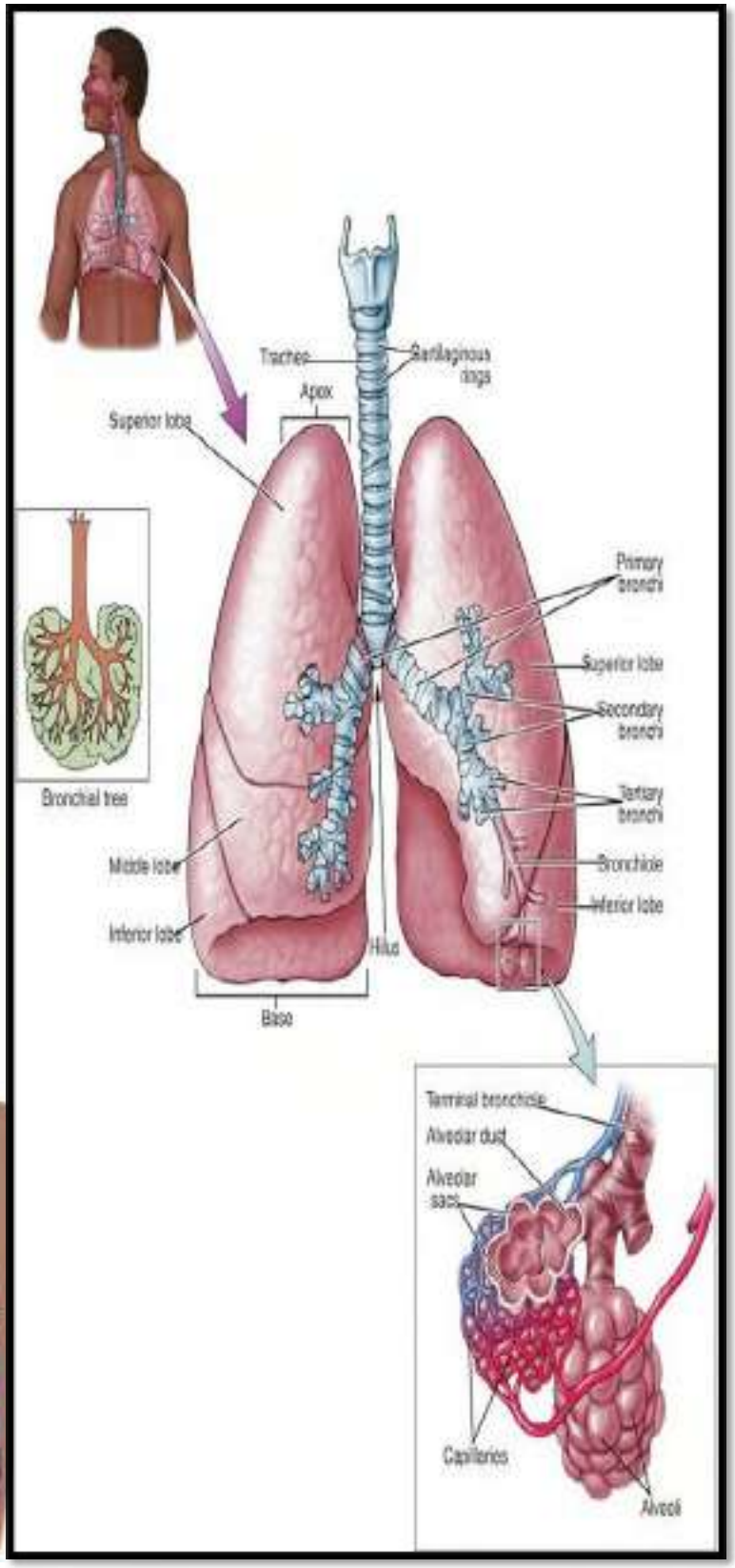
- **Airways located outside the lungs:**

1) Right and left primary bronchi (extrapulmonary bronchi)

- **entering the lung hilum (airways located inside the lungs):**

1) Intrapulmonary bronchi (secondary bronchi)
2) Intrapulmonary bronchi (tertiary bronchi)
3) Bronchioles
4) Terminal bronchioles
5) Respiratory bronchioles
6) Alveolar ducts
7) The alveolar sacs and alveoli

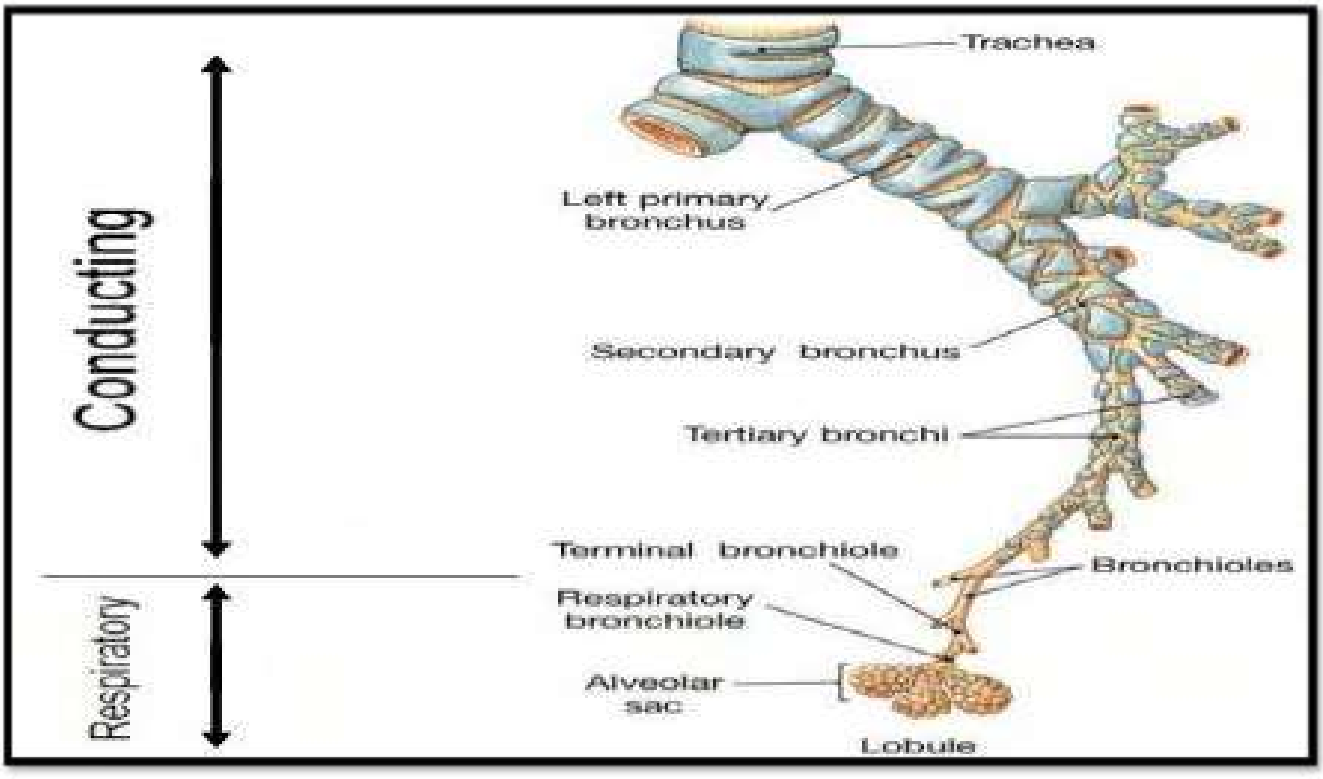




1) Primary Bronchi	2) secondary bronchi	3) tertiary bronchi
• Their structure is identical to the trachea, but smaller in diameter and with a thinner wall.	• Three on the rightand Two on the left	• Ten on the right and eight on the left
• Each primary bronchus pierces the root of the lung. Their branches enter the substance of the lungs as intrapulmonary bronchi	• supply each lung lobe	• supply each bronchopulmonary segments

4) Bronchioles	5) Terminal bronchioles	6) Respiratory bronchioles
• Less than 1mmin diameter	• Less than 0.5mmin diameter	• Supply the primarylung lobules (the respiratory units) of the lung
• supply each lung secondary lobules	• supply each lung (pulmonary) acinus	•

7) Alveolar ducts	8) The alveolar sacs and alveoli
• Supply the Ductal unit	• As the airways branch progressively,
•	• they **decrease** in size, decrease in the amount of cartilage, and the number of glands and goblet cells. The height of the epithelium **increases** in
•	• smooth muscle and elastic tissue



Extra-Pulmonary Bronchus	*Intra-Pulmonary Bronchus*
• it has larger diameter	• It has a smaller diameter
• Its mucosa is **less folded**	• Its mucosa is **highly folded**
• Lined by: respiratory epithelium with too many goblet cells.	• Respiratory epithelium, with little number of goble cells
• Well-developed submucosa	• Absence of submucosa
• Well-developed C- shaped cartilaginous rings in its wall	• Multiple plates of cartilage are connected by C.T. in its wall
• The smooth muscle is located in the posterior wall of the bronchus at the ends of the C- shaped cartilaginous rings	• The smooth muscles are arranged spirally and form a complete ring directly under the mucosal epithelium, encircling the lumen.
• Elastic membrane in C.T. of the lamina propria	• Elastic fibers between cartilage plates and in adventitia
• There is **no** external fibrous elastic membrane.	• There is an external fibrous elastic membrane.
• Mucosal glands are located in the submucosal layer	• Mucous glands, lymph nodules and cartilage plates in adventitia

Bronchioles

- Branching of the bronchi results in different types of bronchioles
- The walls of bronchioles and their branches have no cartilage
- The wall of each bronchiole consists of:
  - ✓ **Mucosa**
  - ✓ **Muscolosa**
  - ✓ **Connective tissue fibrosa**
- The epithelium of :
  large bronchioles: ciliated simple columnar with occasional goblet cells and non-ciliated Clara cells
- **smaller bronchioles: simple cuboidal with occasional non-ciliated Clara cells and no goblet cells**
- The epithelium of the bronchioles rests on a lamina propria, which is surrounded by a continuous well- developed smooth muscle layer ensheathed in elastic fibers.
- Outside the muscle layer, there is a connective tissue layer, which does not contain cartilage, lymph follicles or glands.
- The smallest or the finest branches of bronchioles are called the **terminal bronchioles**

Clara cells

❖ ***LM :***

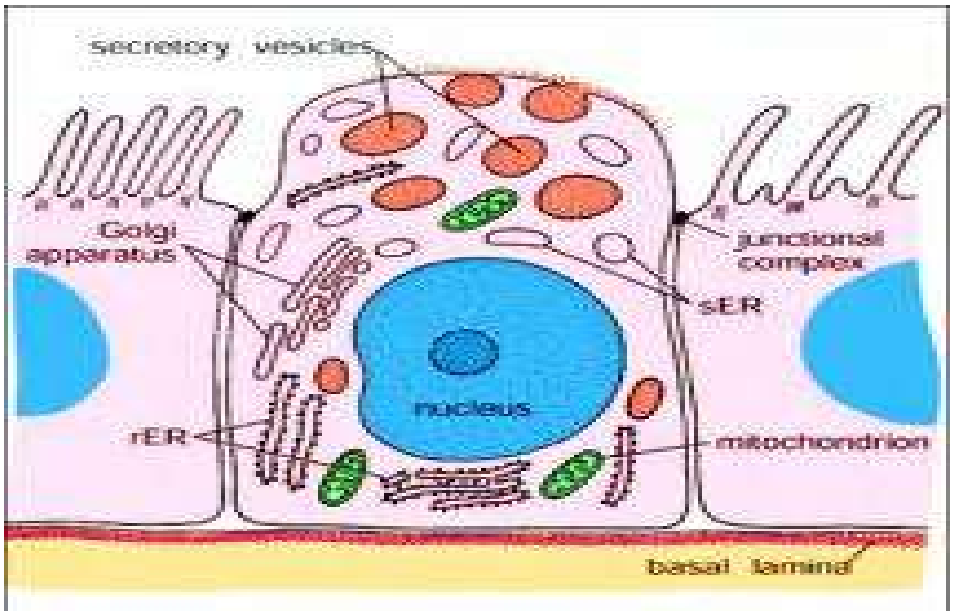
- Columnar cells with dome-shaped apices protrude into the lumen

❖ ***EM :***

- Have short, blunt microvilli, free of cilia with numerous secretory granules containing glycoproteins in apical cytoplasm
- The basally located nucleus is ovoid or slightly indented
- Abundant long cisternae of well-developed basal RER , Lateral or supranuclear Golgi apparatus and Scattered elongated mitochondria

❖ ***Function :***

- Protect the bronchiolar epithelium with their secretory product
- produce a surfactant-like material that
- reduces the surface tension of bronchioles and facilitates the maintenance of their patency.
- They divide to regenerate the
- bronchiolar epithelium



2 Respiratory portion of respiratory

The respiratory portion of the respiratory system is composed of :

1-Respiratory bronchioles	2-Alveolar ducts	3- Alveolar sacs	4-Alveoli

1) *Respiratory Bronchioles*

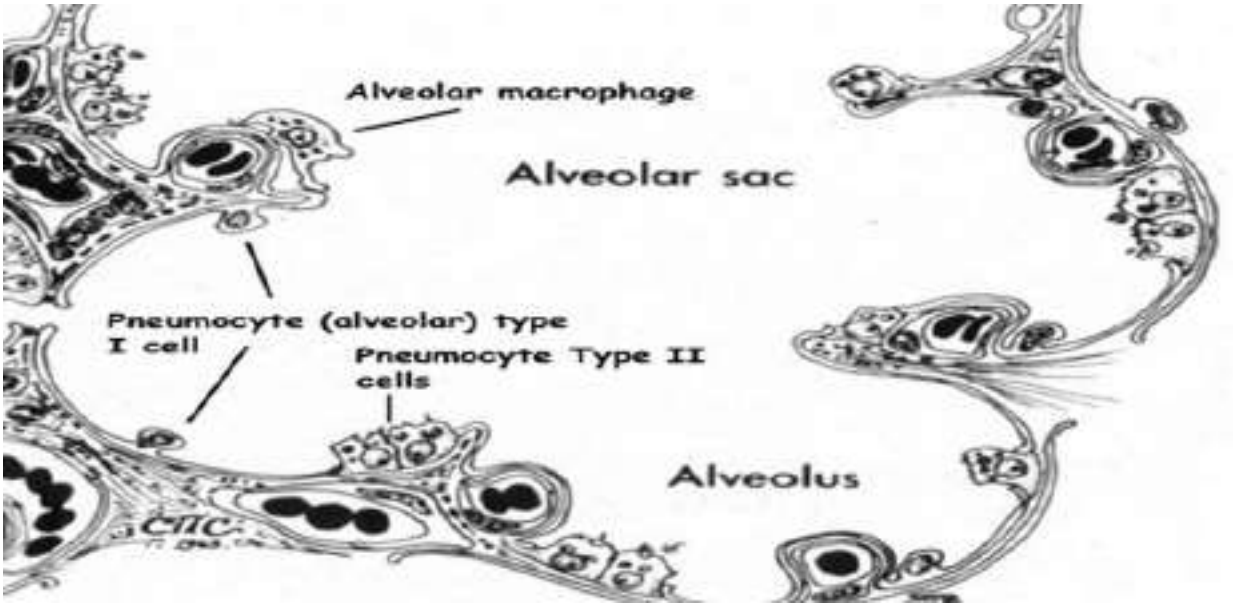
- Respiratory bronchioles are the first region of the respiratory system and are similar in structure to terminal bronchioles, but their wall is interrupted by the presence of thin-walled, pouch-like structures known as **alveoli**, where gaseous exchange (O2 for CO2) can occur
- As respiratory bronchioles branch, they become narrower in diameter and their population of alveoli increases, terminating in an alveolar duct

2) *Alveolar Duct, Atrium Alveolar Sac*

- They do not have walls of their own; they are merely linear arrangements of alveoli Lined with simple cubical cells surrounded by **elastic fibers** and **smooth muscle fibers**
- Arises from a respiratory bronchiole branch and ends as a blind outpouching composed of two or more small clusters of alveoli (**alveolar sac**).
- These alveolar sacs thus open into a common space (atrium)

3) *alveoli*

- Alveoli form the primary structural and functional unit of the respiratory system Their thin walls permit exchange of CO2 for O2 between the air in their lumina and blood in adjacent capillaries.
  The alveoli are lined with alveolar epithelium.
- The region between the two adjacent alveoli is known as the **interalveolar septum** and is lined on both sides by alveolar epithelium.
- These epithelial cells are invested by a prominent basal lamina.
- The capillaries and connective tissue matrix are called **Interstitium**
- The air spaces of the two alveoli may communicate with each other through an **alveolar pore** to equalize the pressure inside the alveoli or making possible collaterals when the bronchioles are obstructed



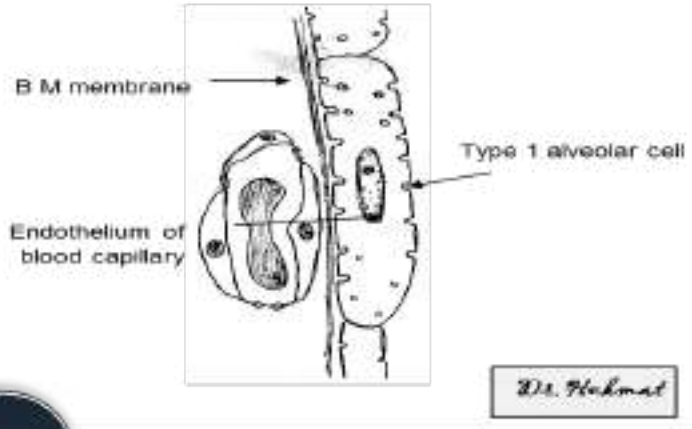
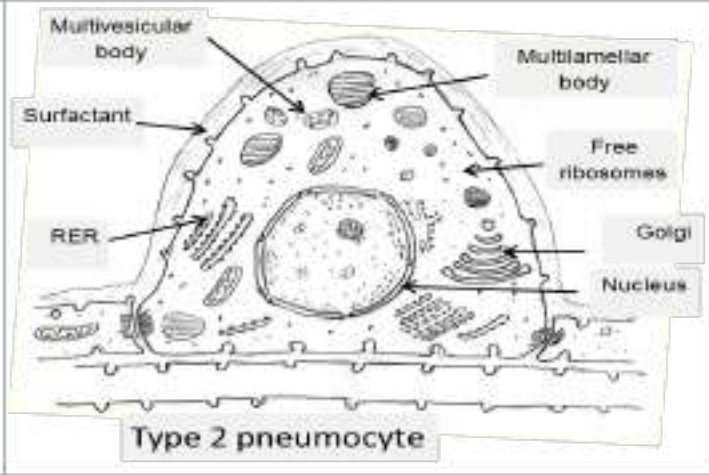
Alveolar duct, alveolar sac and alveolar cells and macrophages

Interalveolar septum

- Connective tissue elements between alveoli containing elastic fibers reticular fibers
- and pulmonary alveolar phagocytic cells as well as blood capillaries The interalveolar septum is composed of five main cell types:

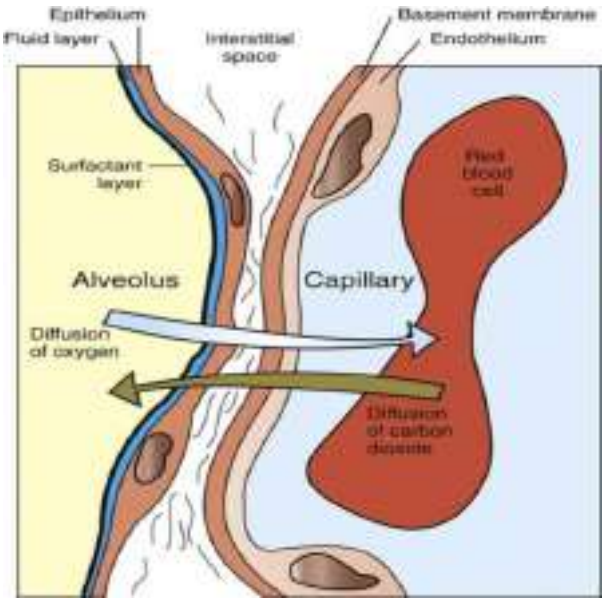
a) Capillary endothelial cells (30%). Type I (squamous alveolar)
b) epithelial cells (8%). Type II (septal, great alveolar) epithelial
c) cells (16%). Interstitial cells, including fibroblast and mast cells
d) (36%). Alveolar macrophages (10%)

The alveolar epithelium consists of 2 Types cells:

Type I pneumocytes (type I alveolar cells, squamous cells) They	Type II Pneumocytes (Secretory cells)
• are the most numerouscells in the alveolar cells (Approximately 97% of thealveolar surface)).	• Also known as **great alveolar cells, septal cells** , and **type II alveolar cells)** are more numerous than type I pneumocytes
❖ ***LM:*** • Theyare flat squamous cells with smallamount of cytoplasm	❖ ***LM:*** • These cells are of cubical shape with dome-shaped apical surface project into the lumen of the alveolus • A centrally placed nucleus, and rest on the basal lamina with large amount of vacuolated cytoplasm
❖ ***EM:*** • The cell is full of pinocytic vesicles and contains a small number of mitochondria, a few profiles of RER, and a Golgi apparatus • They have short microvilli at their alveolar surfaces • Occluding junctions with each other, thus preventing the leakage of extracellular fluid (tissue fluid) into the alveolar lumen • Basally covered by a well-developed basal lamina while apically they are lined by surfactant. • They are important in gas exchange • They are differentiated cells, they cannot divide	❖ ***EM:*** • It contains well, developed mitochondria, ribosomes, RER, extensive Golgi, numerous multivesicular bodies and multilamellar bodies • These cells are interspersed among, and form occluding junctions with, type I pneumocytes • Its apical surface contains microvilli • The most distinguishing feature of these cells is the presence of membrane-bound lamellar bodies (cytosome) that contain pulmonary surfactant • They also undergo mitosis and act as progenitor cells for type I and type II pneumocytes
	

Blood air barrier

- The blood air barrier is present between air in the alveoli and the blood in the pulmonary capillaries. It
- is composed of the :
  - ✓ type I alveolar cells
  - ✓ basement membrane
  - ✓ endothelial cells of capillaries



Alveolar phagocytes (Pulmonary alveolar macrophages)

- These phagocytic cells are present inside the alveoli and in the interalveolar septa.They have an important role in the defense mechanism of the respiratory system

❖ **Function:**

- They phagocytose dust particles (dust cells) or haemosidrin particles in cases of heart failure (**heart failure cells**).

❖ **E.M:**

- Their cytoplasm often contains phagocytosed particles, which are of variable sizes and shapes
- They have indented eccentric nuclei
- They originate from cells in the interalveolar septum

Pleura

- It is a **thin layer** of fibrous connective tissue with:

collagenous elastic fibers few cells, mainly fibroblasts and macrophages. It is **covered** with a layer of mesothelium It contains:

- numerous lymph
  blood capillaries
  small nerve fibers.

- It **secretes** the pleural fluid, which permits free movement between parietal andvisceral pleura

2025



2026

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Respiratory system

Physiology

RPS-314

page	lecture
1	introduction of the respiratory system
10	Elastic properties of the lung
17	Airway resistance and alveolar ventilation
25	physiology of Respiration
52	Acid/Base balance & Respiratory disorders

1 The function of the respiratory system

Exchange of oxygen and carbon dioxide between the environment and body cell

1) The four major components of respiration are the following:

1-pulmonary ventilation, which means the inflow and outflow of air between the atmosphere and the lung alveoli.

2-diffusion of oxygen (O 2) and carbon dioxide (CO 2) between the alveoli and the blood.

3-transport of oxygen and carbon dioxide in the blood and body fluids to and from the body's tissue cells.

4-regulation of ventilation and other aspects of respiration.

2) Structure of the respiratory system

The respiratory system includes the lungs and a series of airways that connect the lungs to the external environment. It subdivided into:

Conducting zone	Respiratory zone
Includes the nose, nasopharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles. • Brings air into and out of the respiratory zone for gas exchange.	Includes the structures that are lined with alveoli, the respiratory bronchioles, alveolar ducts, and alveolar sacs. • Gas exchange

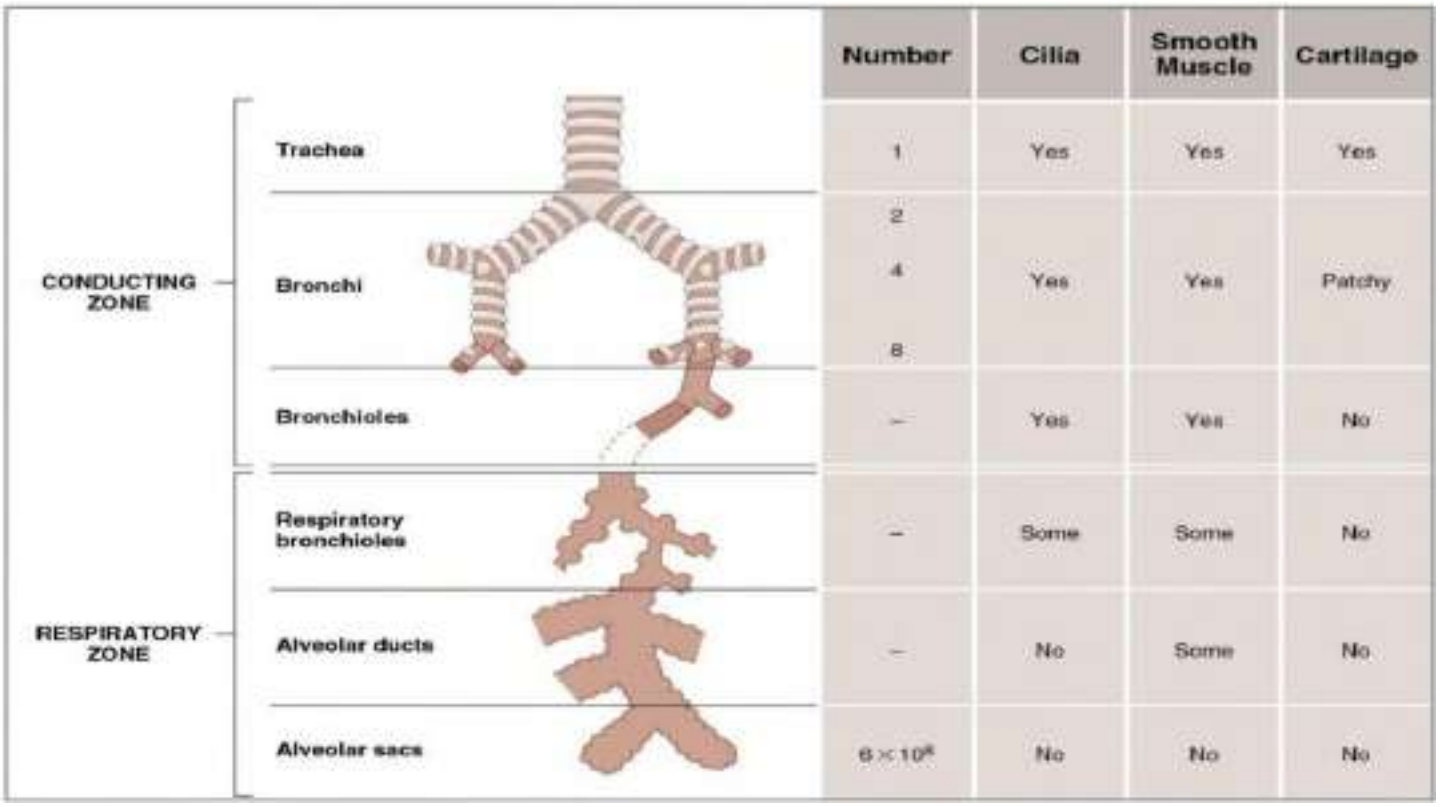
• warm, humidify, and filter the air before it reaches the critical gas exchange region

-The progressively bifurcating airways are referred to by their generation number

-The trachea, which is **the zeroth generation**, is **the main conducting airway.**

-The trachea divides into the right and left mainstem bronchi **the first generation**, which divide into two smaller bronchi, which divide again.

-Ultimately, there are 23 such divisions into increasingly smaller airways, culminating in the airways of the 23rd generation.



	Number	Cilia	Smooth Muscle	Cartilage
Trachea	1	Yes	Yes	Yes
Bronchi	2 4 8	Yes	Yes	Patchy
Bronchioles	–	Yes	Yes	No
Respiratory bronchioles	–	Some	Some	No
Alveolar ducts	–	No	Some	No
Alveolar sacs	6×10^8	No	No	No

Costanzo Physiology

Figure 1: Structure of the airways. The number of the various structures is reported for two lungs.

The alveoli

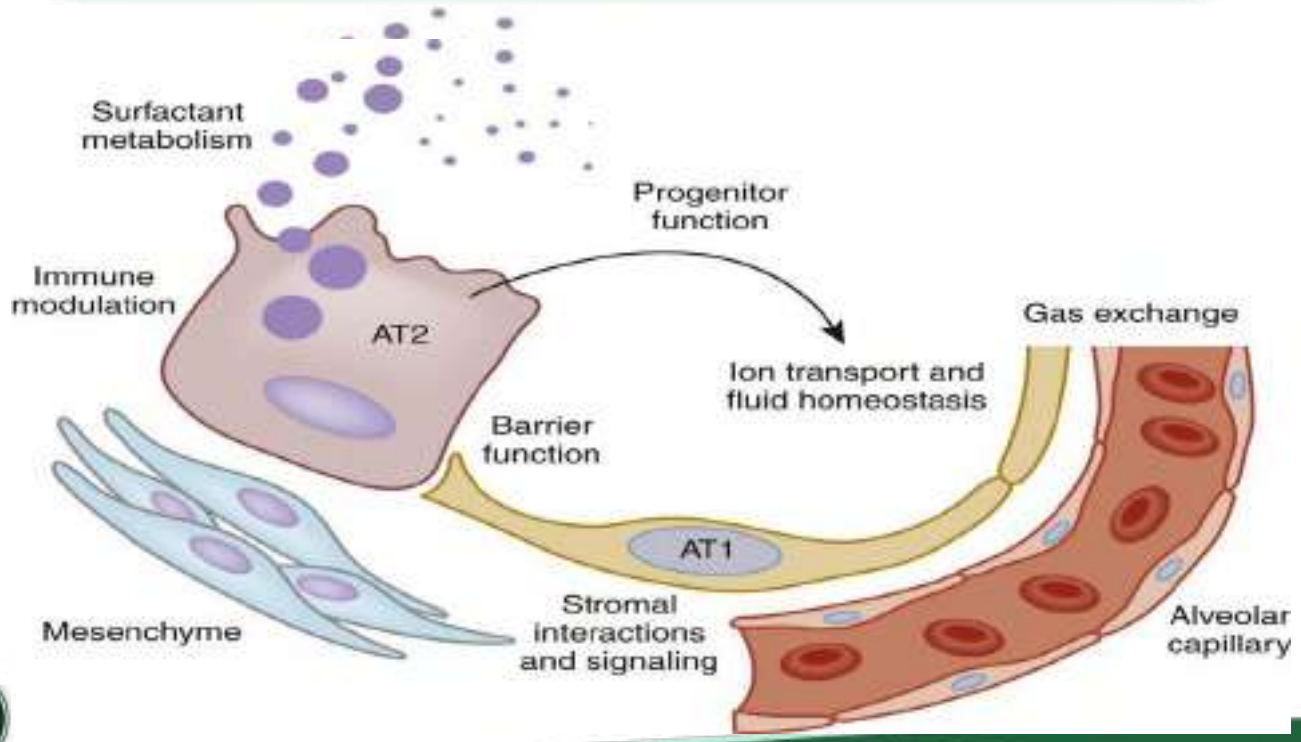
1-Are pouch like evaginations of the walls of the respiratory bronchiole the Alveolar ducts, and the alveolar sacs
- 300 million alveoli for each lung 2-Alveolus is approximately **200 micrometers** in diameter 3-Alveolar walls are thin and have a large surface area for diffusion of gases

2-The alveolar walls are lined with epithelial cells called **type I and type II pneumocytes** (or alveolar cells)

-AT1 Cells: facilitate gas exchan ge **-AT2 Cells:** synthesize pulmonary surfactant components and act as

progenitor cells for repair and regeneration of the alveolus 5-The alveoli contain phagocytic cells called alveolar macrophages keep The alveoli free of dust and debris

3-Macrophages fill with debris and migrate to the bronchioles, where ,the beating cilia carry debris to the upper airways and the pharynx where it can be swallowed or expectorated



Non-respiratory function of the respiratory system

-Breathing movement affect the rates of venous return and lymph flow
-Regulation of acid- base balance
-Regulation of body temperature
-Excretion of volatile waste products
-Lungs perform several and endocrine functions
- Secretion of heparin
- Angiotensin I converted to angiotensin II in the lungs by ACE activity

-Conducting zone
- Perception of smell sensation
- Phonation or vocalization
- Protective reflexes

3) *Mechanics of Pulmonary Ventilation*

Muscles That Cause Lung Expansion and Contraction

The lungs can be expanded and contracted in two ways:
1- By downward or upward movement of the diaphragm to lengthen or shorten the chest cavity
2- By elevation or depression of the ribs to increase or decrease the anteroposterior diameter of the chest cavity.

Normal quiet breathing is accomplished almost entirely by movement of the diaphragm (supplied by the phrenic nerve which originates from the C3-C5 spinal nerves).

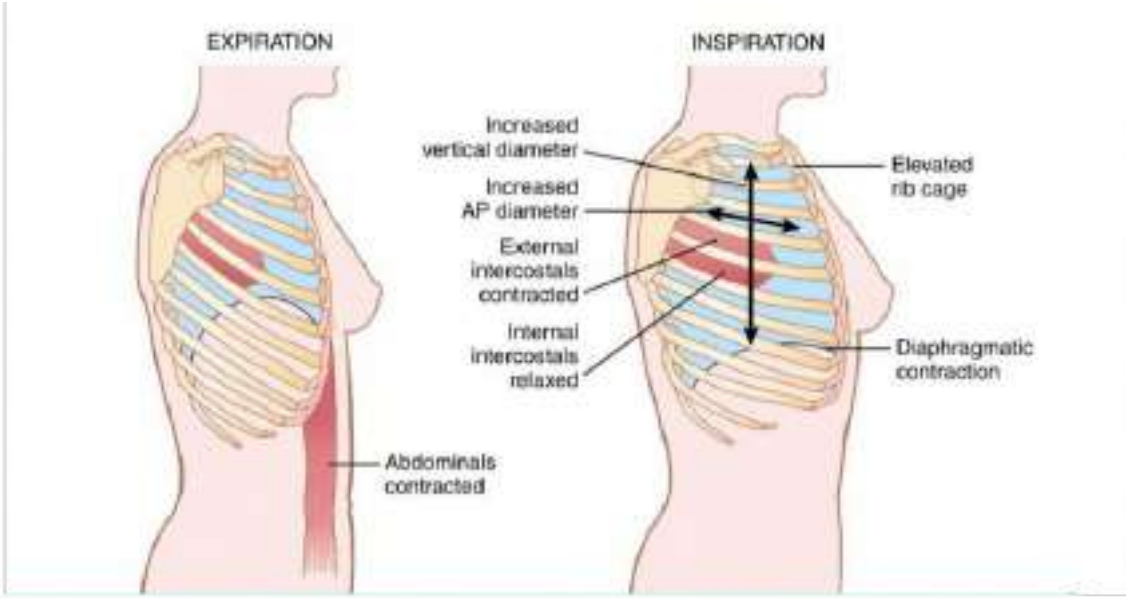


Figure 2: Contraction and expansion of the thoracic cage during expiration and inspiration, demonstrating diaphragmatic contraction, function of the interc ostal muscles, and elevation and depression of the rib cage. AP, Anteroposterior. (Guyton and Hall).

Inspiration	Expiration
An active process **During eupnea** ↳ Thediaphragm is themost important muscle for inspiration. When the diaphragm contracts, the abdominal contents are pushed downward, and the ribs are lifted upward and outward. These changes produce an increase in intrathoracic volume, which lowers intrathoracic pressure and initiates the flow of air into the lungs. **During exercise,↑ breathing frequency and tidal volume)** -The external intercostal muscles, and the accessory muscles of inspiration	**A passive process** The diaphragm simply relaxes, and the elastic recoil of the lungs, chest wall, and abdominal structures compresses the lungs and expels the air. Muscles of expiration contract only during Forced expiration as occur in cases of •During muscular exercise •Increased air way resistance (asthma) •Diminished elasticity of the lung (emphysema)

-sternocleidomastoid muscles - anterior serrati -scaleni may also be used for more vigorous inspiration.	**Muscles of expiration** -Anterior abdominal wall muscles - The internal intercostal muscles

Pressures That Cause Movement of air in and out of the Lungs

-The lung is an elastic structure that collapses like a balloon and expels all its air through the trachea whenever there is no force to keep it inflated. - The lung "floats" in the thoracic cavity, surrounded by a thin layer of pleural fluid that lubricates movement of the lungs within the cavity.

- Continual suction of excess fluid into lymphatic channels maintains a slight suction between the visceral surface of the lung pleura and the parietal pleural surface of the thoracic cavity. - The lungs are held to the thoracic wall as if glued there, except that they are well lubricated and can slide freely as the chest expands and contracts.

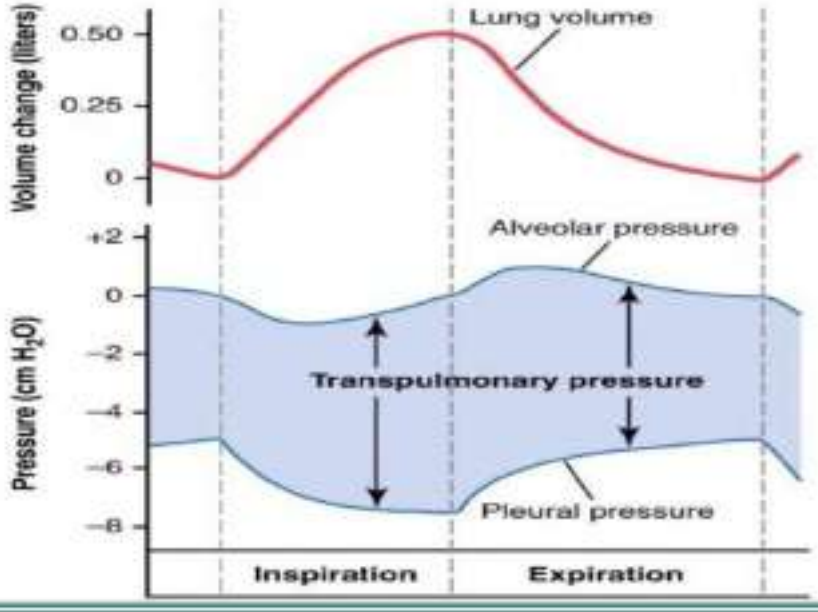


Figure 3: Changes in lung volume, alveolar pressure, pleural pressure, and transpulmonary pressure during normal breathing. (Guyton and Hall)

Pleural Pressure and Its Changes During Respiration

- Pleural pressure is the pressure of the fluid in the thin space between the lung pleura and chest wall pleura.

-This pressure is normally slightly negative pressure due to elastic properties of both the lungs and chest wall

Elastic properties of the lung:

tend to decrease the lungs volume, pulling the visceral pleura inward this is due to : - Elastic element of the lung. - Surface tension of the fluid layer that lines the alveoli.

Elastic properties of the chest wall:

this tends to expand the thoracic cage, pulling the parietal pleura outward. These opposing forces →continuous pulling of the 2 pleural layer apart from each other→ create an equal –ve pressure in both the pleura and chest cavities.

• **At the beginning of inspiration**: The normal pleural pressure is about −5 (cmH2O), which is the amount of suction required to hold the lungs open to theirresting level. • **During normal inspiration:** The pleural pressure becomes more negative toanaverage of about −7.5 cm H 2 O as expansion of the chest cage pulls outwardon the lungs with greater force.

• **During forced inspiratio**n: The IPP decreases to about -40 cm H20. Muller'smaneuver (forced inspiration against closed glottis)

• **During forced expiration:** The IPP increases and becomes +ve •

Valsalva's maneuver: Forced expiration against closed glottis: theIPPBECOME more +ve

Significance of IPP negativity:

1-Itprevents collapse of thelungs and keeps them inflated.

2-Helps lung expansion during inspiration, decrease the work of breathing

3-Help the venous returnand lymph drainage.

Alveolar Pressure—Air Pressure Inside the Lung Alveoli

When the glottis is open and no air is flowing into or out of the lungs, the pressures in all parts of the respiratory tree, all the way to the alveoli, are equal to atmospheric pressure, which is considered to be zero reference pressure in the airways—that is, 0 cm H2O pressure

- **During normal inspiration:** Alveolar pressure decreases to about −1 cm H 2O due to lung expansion→ pull 0.5 liter of air into the lungs in the 2 seconds required for normal quiet inspiration, which causes the alveolar pressure to increase gradually till becoming atmospheric again by the end of inspiration.

- **During expiration:** Alveolar pressure rises to about +1 cm H2O → forces the 0.5 liter of inspired air out of the lungs during the 2 to 3 seconds of expiration. The alveolar pressure decreases gradually till becoming atmospheric again by the end of expiration

Transpulmonary Pressure—Difference between Alveolar and Pleural Pressures

It is the pressure difference between that in the alveoli and that on the outer surfaces of the lungs (pleural pressure); it is a measure of the elastic forces in the lungs that tend to collapse the lungs at each instant of respiration, called the recoil pressure.

Pneumothorax

-This is the entry of air inside the pleural cavity.

-The IPP increases leading to collapse of the lung on the affected side and shift of the mediastinum to the normal side. **Causes:**

1- Injury of the thoracic wall 2-Rupture of the alveoli in the pleural sac

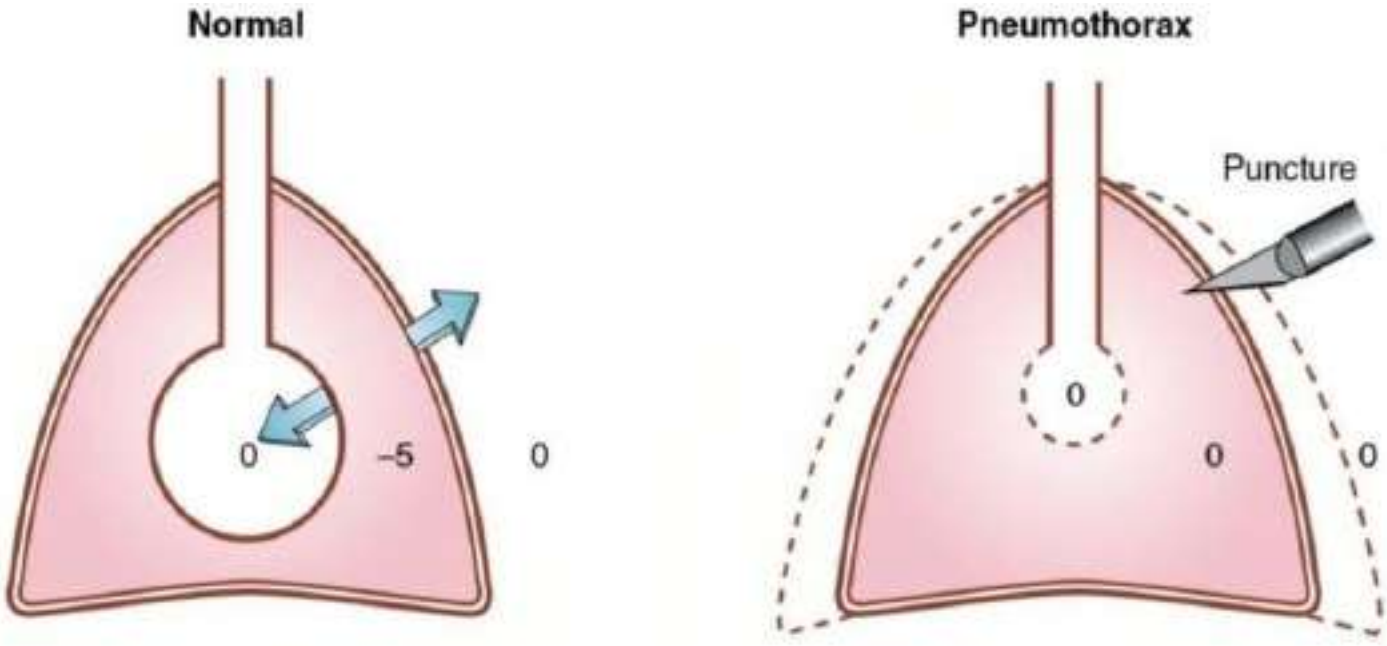


Figure 4: Intrapleural pressure in a normal person and in a person with a pneumothorax. The numbers are pressures in centimeters of water. Pressures are referred to atmospheric pressure; thus, zero pressure means equal to atmospheric pressure. The arrows show expanding or collapsing elastic forces. Normally, at rest, intrapleural pressure is −5 cm H2O because of equal and opposite forces trying to collapse the lungs and expand the chest wall. With a pneumothorax, the intrapleural pressure becomes equal to atmospheric pressure, causing the lungs to collapse and the chest wall

2 Elastic properties of the lung

1 Surfactant, surface tension, and collapse of the alveoli

Principle of Surface Tension

- When water forms a surface with air, the water molecules on the surface of the water have an especially strong attraction for one another. As a result, the water surface is always attempting to contract.
- On the inner surfaces of the alveoli, the water surface is also attempting to contract. This tends to force air out of the alveoli through the bronchi → causes the alveoli to try to collapse.
- The net effect is to cause an elastic contractile force of the entire lungs, which is called the **surface tension elastic force.**

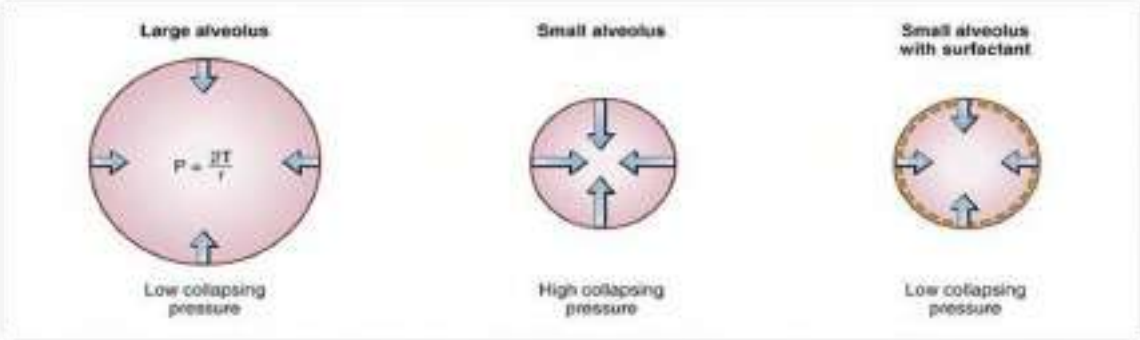


Figure 5: Effect of alveolar size and surfactant on collapsing pressure.
The length of the arrows shows the relative magnitude of the collapsing pressure.

Surfactant and its effect on surface tension

Surfactant is a surface-active agent in water, which greatly reduces the surface tension of water, from one-twelfth to one-half the surface tension of a pure water surface tension. Secreted by **type II** alveolar epithelial cells, these cells are granular ,containing lipid inclusions that are secreted in the surfactant into the alveoli. Surfactant is a complex mixture of several phospholipids, proteins, and ions. The most important components are:

✓ Phospholipid dipalmitoyl phosphatidylcholine.
✓Surfactant apoproteins .
✓ Calcium ions.

Surfactant decreases the surface tension of the fluid film that lines the alveoli .
Its effect is.

- **Directly** proportional to Its concentration
- **Inversely** proportional to Surface area of the fluid

Factors that affect the formation of the surfactant:

Stim ulate
- Thyroid hormone
- Glucocorticoids

Inhibit
- Smoking
- Insulin
- Prolonged oxygen inhalation
- Prolonged cessation of pulmonary circulation

Functions of the surfactant

- Itincreases lungcompliance
- It helps keepthealveoli dry.
- It prevents alveolar collapse during expiration.
- Plays a major role in maintaining alveolar stability.
- It decreases the work of breathing.

Alveolar stability:stability of the size of different alveoli.

Surfactant main tains alveolarst ability by keeping a nearly equal pressure in various alveoli (p=2T/r).

The respiratory distress syndrome(RDS)

This is a serious lung disease due to deficient formation of the surfactant.
RDS may occur in adult,more common in newly born infant particularly:

1. Premature infants

Many of whom have:
- oAlveoli with radii less than 25% than of an adult person.
- oLittle orno surfactant in the alveoli when they are born.

Surfactant normally begin stop secreted between 6th and 7th months of gestation and in some cases even later . Their lungs have an extreme tendency to collapse as much 6 to 8 times that in normal adult person.

2. Hypothyr oid infant.

3. Infant born by d iabetic mothers.

2 Compliance of the Lungs

- Compliance is the extent to which the lungs will expand for each unit increase in transpulmonary pressure if enough time is allowed to reach equilibrium.
- ΔV / Δ P
- Compliance is a measure of distensibility (or expansibility) of elastic structure.
- The compliance of the lungs and chest wall is inversely correlated with their elastic properties or elastance.
- The total compliance of both lungs together in the normal adult averages about 200 ml of air/cm H 2 O transpulmonary pressure.

Compliance Diagram of the Lungs

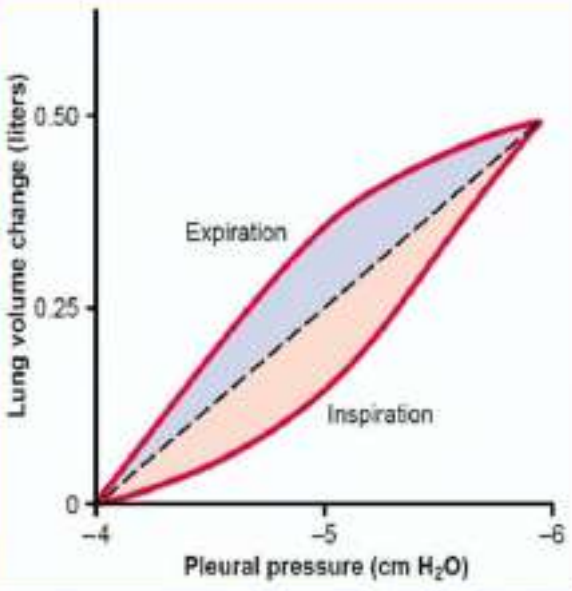
Relating lung volume changes to changes in pleural pressure.

The relationship is different for inspiration and expiration.

Each curve is recorded by changing the pleural pressure in small steps and allowing the lung volume to come to a steady level between successive steps.

Inspiratory compliance curve.

Expiratory compliance curve.



The characteristics of the compliance diagram are determined by the **elastic forces** of the lungs.
Elasticforces can be divided into two parts:

1. Elastic forces of the lung tissue:

- Determined mainly by elastin and collagen fibers intertwined among the lung
- parenchyma.
- Represent only about **one-third** of the total lung elasticity.
- Indeflated lungs, these fibers are in an elastically contracted and kinked state. Inexpanded lungs, the fibers become stretched and unkinked, thereby elongating, and exerting even more elastic force.

2. Elastic forces caused bysurface tension of the fluid:

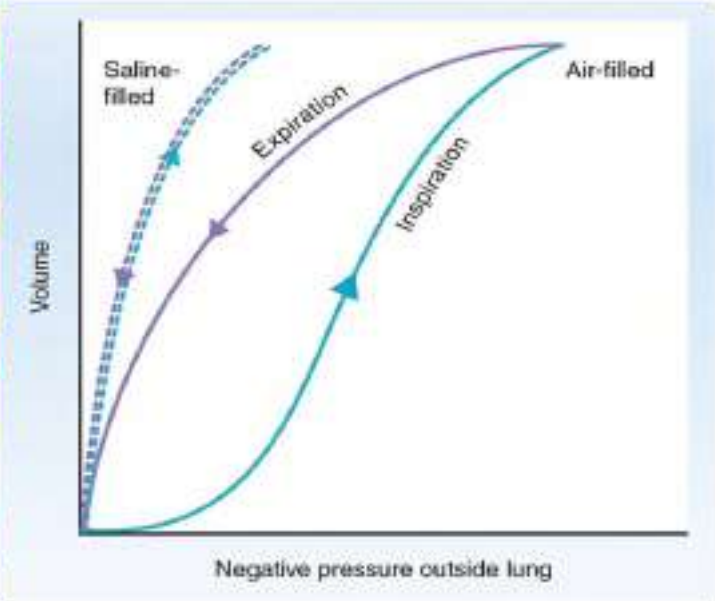
oCaused by surface tension of the fluidthat lines the inside walls of the alveoli and other lung air spaces.
oRepresent about **two-thirds** of the total lung elastic force.
oCompliance diagram of the lungs when filled with saline solution, there is no air-fluid interface and, therefore, the surface tension effect is not present.
oCompliance diagram of the lungs when filled with air, there is an alveolar fluid air interface in the alveoli → which causes surface tension.
- Transpleural pressures required to expand air-filled lungs are about three times as great as those required to expand lungs filled with saline solution.

Figure6: Compliance of the lung (Costanzo Physiology) .

The relationship between lung volume and lung pressure is obtained by inflating and deflating an isolated lung.

The slope of each curve is the compliance.

In theair-filled lung ,inspiration (inflation) and expiration (deflation) follow different curves,which is known as hysteresis.



Feature of the pressure-volume loop for the air-filled lung

Hysteresis: the slopes of the relationships for inspiration and expiration are different.

Foragiven outside pressure, the volume of the lung is greater during expiration thanduring inspiration.

Compliance is measured on the expiration limb of the pressure-volume loop because the inspiration limb is complicated by the decrease in compliance at maximal expanding pressures.

Why aretheinspirationand expiration limbs ofth elungcompliancecurve different?

Surface tension at the liquid-air interface of the air-filled lung.

Inspiratory compliance curve:
One begins at low lung volume → liquid molecules are closest together and intermolecular forces are highest → to inflate the lung, one must first break up these intermolecular forces→ surfactant, which is newly produced by typeII alveolar cells,enters the liquid layer lining the alveoli and breaks up these intermolecular forces to reduce surface tension.

The initial part of the inspiration curve:
the lung surface area is increasing faster than surfactant can be added to the liquidlayer
→ surfactant density is low, surface tension is high,compliance is low, and the curve is flat.
Asinflationproceeds:the surfactant density increases,which decreases surface tension,increases compliance, and increases the slope of the curve.

Expiratory compliance curve:
One begins at high lungvolume, where intermolecular forces between liquid molecules are low; one does not need to break up intermolecular forces to deflate the lung.
During deflation of the lung, lung surface area decreases faster than surfactant can be removed from the liquid lining and the density of surfactant molecules rapidly increases → decreases surface tension → increases compliance.

Compliance of Thorax and Lungs Together

To inflate the entire pulmonary system (the lung and thoracic cage together) ,almost twice as much pressure is required compared with the same lungs after removal from the chest cage.

The compliance of the combined lung-thorax system is almost exactly half that of the lungs alone → 110 ml/cm H2O pressure for the combined system ,compared with 200 ml/cm H2O for the lungs alone.

Factors affecting lung compliance

Decreased compliance

- Decrease size of the lung.
- ↑Surface tension.
- Diseases as ARDS.
- Diseases that cause lung stiffness e.g., Pulmonary fibrosis, edema, congestion.
- Thoracic compliance decreased in cases of deformities of the vertebral column , arthritis, and obesity.

Increased c ompliance

- Saline filled lung
- Old age
- Emphysema

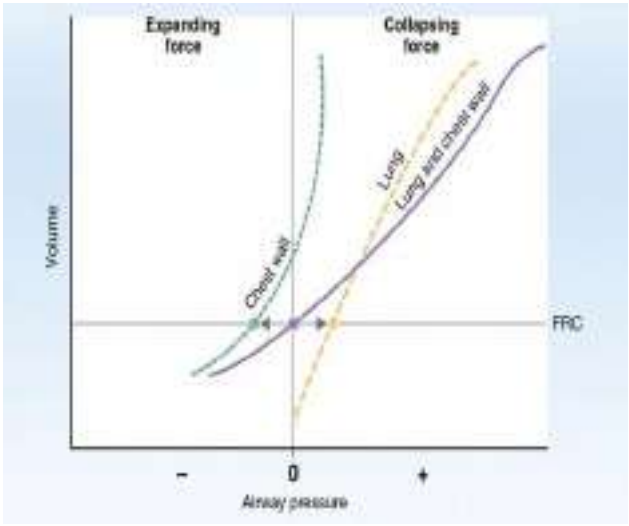


Figure 7: Compliance of the lungs, chest wall, and combined lung and chest-wall system.

The equilibrium position is at functional residual capacity (FRC), where the expanding force on the chest wall is exactly equal to the collapsing force on the lungs.

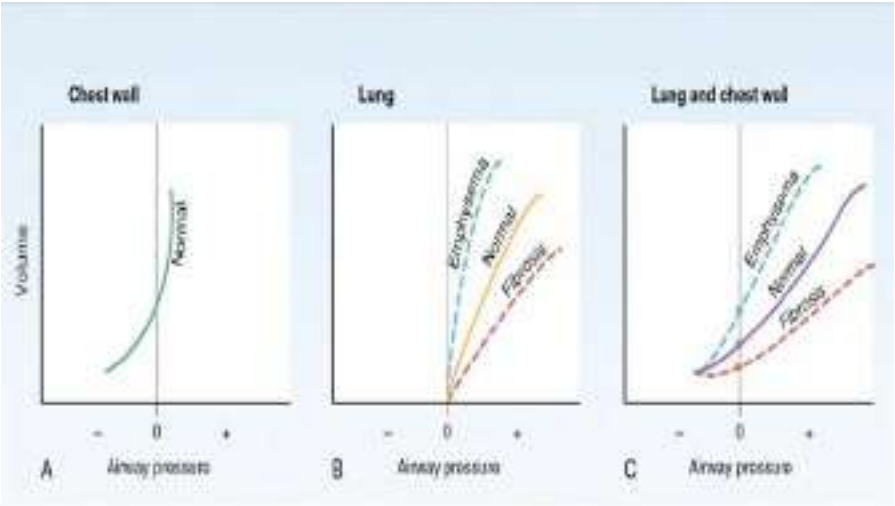


Figure 8: Changes in compliance of the chest wall (A), lungs (B), and combined lung and chest-wall system (C) in emphysema and fibrosis. The equilibrium point, functional residual capacity (FRC), is increased in emphysema and decreased in fibrosis.

3 Work o f Breathing

Almost all the work of breathing is Performed by the inspiratory muscles as expiration is normally a passive process.
It includes:

1. Elastic (compliance component):

- This is about 65% of the total work .
- Used for stretching the elastic tissues of the lungs and chest wall.

2. Non elastic (resistive) work:

- This is about 35% of the total work .
- Used for resisting the frictional resistance to air movement in the lungs, it includes:

a) Air way resistance work (28%):

Used to overcome resistance to airflow through the respiratory passages.

b) Tissue resistance work (7%):

Used to overcome the viscous resistance in the lung.

Factors that increase the work of breathing

Hyperpnea.
Reduction of the respiratory compliance.
Deficient secretion of surfactant.
Diseases that increase the resistance to air flow e.g., asthma ,emphysema.

3 Airway resistance and alveolar ventilation

1) Alveolar Ventilation



•The rate at which new air reaches the gas exchange areas of the lungs.

Dead space air:

•The air that fills respiratory passages, such as the nose, pharynx, and trachea where gas exchange does not occur.

•On expiration, the air in the dead space is expired firs t before any of the air from the alveoli reaches the atmosphere

•the dead space is very disadvantageous for removing the expiratory gases from the lungs

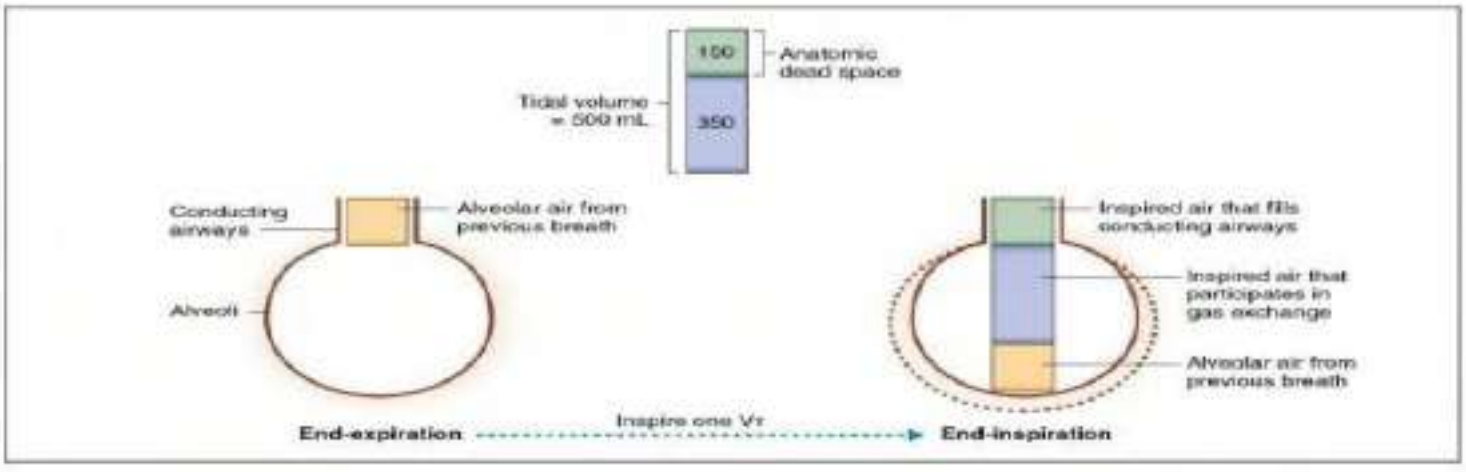


Figure 9: Anatomic dead space.

One-third of each tidal volume (V t) fills the anatomic dead space

Typ es of dead space:

Anatomical dead space	Alveolar dead space	Physiologic dead space
1-is the volume of the conducting airways including the nose (and/or mouth), trachea, bronchi, and bronchioles. 2- It does not include the respiratory bronchioles and alveoli 3- The volume: 150 ml increasing slightly with age.	Ventilated alveoli that do not participate in gas exchange because of a mismatch of ventilation and perfusion, or so-called ventilation/perfusion defect, in which ventilated alveoli are not perfused by pulmonary capillary blood .	Is the total volume of the lungs that does not participate in gas exhange . = Anatomic dead space of the conducting airways + Afunctional dead space in alveoloi (alveolar dead space)

- In normal person: The physiologic dead space is nearly equal to the anatomic dead space→ alveolar ventilation and perfusion are normally well matched and functional dead space is small.

Measurement of Dead Space Volume:

The anatomical dead space (DS) is measured by **fowler's method**:

- the subject suddenly takes a deep breath of 100% O 2, which fills the entire dead space with pure O 2.
- Some oxygen also mixes with the alveolar air but does not completely replace this air.
- Then the person expires through a rapidly recording nitrogen meter.

The first portion of the expired air:

1- only O 2 appears, and the nitrogen concentration is 0

- This air comes from the dead space regions, where the air has been completely replaced by O 2.

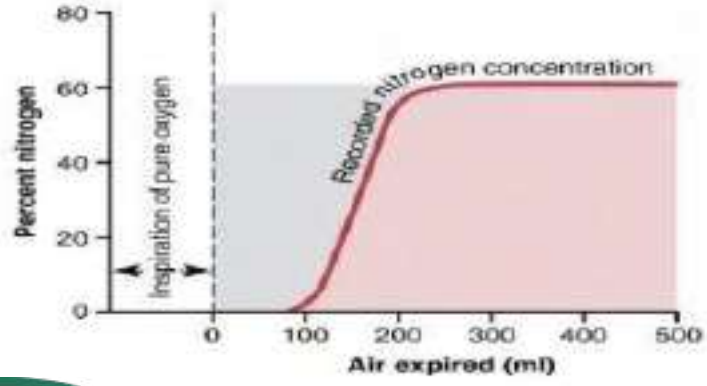
2. When alveolar air begins to reach the nitrogen meter:
the nitrogen concentration rises rapidly because alveolar air containing large amounts of nitrogen begins to mix with the dead space air

•The recorded nitrogen concentration reaches a plateau level equal to its concentration in the alveol

• all the dead space air has been washed from the passages and only alveolar air remains

The volume of dead space air =The gray area represents the air that has nonitrogen

$$V_D = \frac{\text{Gray area} \times V_E}{\text{Pink area} + \text{Gray area}}$$



Measurement of the physiological dead space

• The physiological dead space is measured by **Bohr's method**.

This method is based on the fact that CO2 in the expired air is derived mainly from the alveolar air and not from the dead space (because after inspirationthedead space is filled with atmospheric air, which normally contains a negligible amount of CO2) Therefore, the amount of CO2 in the whole expiredairvolume as well as in its part that is derived from alveolar air are equal.

1- Each of them =its volume X CO2 concentration

2- Expired air volume X CO2 % =alveolar air volume X its CO2%

3- Expired air volume= Tidal volume (TV)

4- Alveolar air volume = expired air volume – DS= TV- DS

5-TV (CO2% in expired air) = (TV-DS) CO2% in alveolar air= TV (CO2 alveolar air)-DS (CO2% in alveolar air) by shifting

6- DS (CO2% in alveolar air) =TV (CO2% in alveolar air) - TV (CO2% in expired air)= TV(CO2% in alveolar air−CO2 % in expired air)

$$\frac{Ds}{TV} = \frac{CO2\% \text{ in alveolar air} - CO2\% \text{ in expired air}}{CO2\% \text{ in alveolar air}}$$

$$Ds = TV \times \frac{CO2\% \text{ in alveolar air} - CO2\% \text{ in expired air}}{CO2\% \text{ in alveolar air}}$$

•Since the partial pressure (p) of a gas is proportionate to its concentration the PCO2 can be used instead of the CO2.

Bohr's equation $$Ds = TV \times \frac{CO2\% \text{ in alveolar air} - CO2\% \text{ in expired air}}{PCO2\% \text{ in alveolar air}}$$

Rate of Alveolar Ventilation

• Alveolar ventilation per minute is the total volume of new air entering the alveoli and adjacent gas exchange areas each minute.

• the respiratory rate times the amount of new air that enters these areas with each = breath: VA = Freq × (VT – VD)

• Alveolar ventilation is one of the major factors determining the concentrations of oxygen and carbon dioxide in the alveoli

The alveolar air

• Is that air undergoes gas exchange with blood in the pulmonary capillaries After a quite expiration. It continuously loses oxygen and gain CO2. Its O2 content is less, and its CO2 content is more than the amount of these gases in both the inspired and the expired air

Resistance to Airflow in the Bronchial Tree

Under normal respiratory conditions:

•Air flows through the respiratory passageways so easily that less than 1cmH2O pressure gradient from the alveoli to the atmosphere is sufficient to cause enough airflow for quiet breathing

•The greatest amount of resistance to airflow occurs in some of the larger bronchioles and bronchi near the trachea

Because:

•There are relatively few of these larger bronchi in comparison with the approximately 65,000 parallel terminal bronchioles, through each of which only a minute amount of air must pass.

In some disease conditions:

The smaller bronchioles play a far greater role in determining airflow resistance because of their small size and because they are easily occluded by the following:

- muscle contraction in their walls
- mucus collecting in the lumens of the bronchioles
- edema in the walls

• Cartilage is present in the walls of the zeroth to 10th generations of conducting airways; it functions, structurally, to keep those airways open Starting with the 11th generation, cartilage disappears; to remain open, those airways with no cartilage depend on the presence of a favorable transmural pressure

The walls of the conducting airways contain smooth muscle. This smooth muscle has both sympathetic and parasympathetic innervations

1-Sympathetic adrenergic neurons

activate β2 receptors on bronchial smooth muscle, which leads to relaxation and dilation of the airways.
B2 receptors are activated by circulating epinephrine released from the adrenal medulla and by β2 adrenergic agonists such as isoproterenol

2-Parasympathetic cholinergic neurons

activate muscarinic receptors, which leads to contraction and constriction of the airway

• Changes in diameter of the conducting airways result in changes in their resistance, which produce changes in air flow. the autonomic nervous system affects airway diameter, so have predictable effects on airway resistance and , .air flow. The most notable effects are those of β 2 -adrenergic agonists (e.g epinephrine, isoproterenol, albuterol), which are used to dilate the airways in . the treatment of asthma

• Many obstructive diseases of the lung result from narrowing of the smaller bronchi and larger bronchioles, often because of excessive contraction of the smooth muscle.

Nervous and Local Control of the Bronchiolar Musculature

• Sympathetic Dilation of the Bronchioles

• Parasympathetic Constriction of the Bronchioles

• These nerves secrete acetylcholine and, when activated, cause mild to moderate constriction of the bronchioles. When a disease process such as asthma has already caused some bronchiolar constriction, superimposed parasympathetic nervous stimulation often worsens the condition Administration of anticholinergic as atropine improve the condition

.

Parasympathetic nerves are activated by

• Reflexes that originate in the lungs, stimulated by irritation of the epithelial membrane of the respiratory passageways e.g noxious gases dust, cigarette smoke, or bronchial infection

• Microemboli occlude small pulmonary arteries cause bronchiolar constrictor reflex

Local Secretory Factors May Cause Bronchiolar Constriction

• Histamine

•Slow reactive substance of anaphylaxis

- Released in the lung tissues by mast cells during allergic reactions .especially those caused by pollen in the air

- They play key roles in causing airway obstruction in allergic asthma this is especially true of the slow reactive substance of anaphylaxis

• the same irritants that cause parasympathetic constrictor reflexes of the Airways as smoke, dust, sulfur dioxide, and some of the acidic elements in Smog

may also act directly on the lung tissues to initiate local, non nervous reactions that cause obstructive constriction of the airways.

Obstructive and restrictive lung diseases

Obstructive lung disease	Restrictive lung disease
Characterized by an increase in resistance to airflow due to diffuse airway disease, which may affect any level of the respiratory tract	Characterized by reduced expansion of lung parenchyma and decreased total lung capacity

Forced expiratory volume

Forced vital capacity (FVC)

- The total volume of air that can be forcibly expired after a maximal inspiration

The volume of air that can be forcibly expired in the first second is called FEV1	the cumulative volume expired in 2 seconds is called FEV2	the cumulative volume expired in 3 seconds is called FEV3

FVC and FEV1 are useful indices of lung disease. Specifically, the fraction of the vital capacity that can be expired in the first second

FEV1, FVC can be used to differentiate among disease

In a normal person:

• FEV1, FVC is approximately 0.8, meaning that 80% of the vital capacity can be expired in the first second of forced expiration

In patients with obstructive lung diseases such as asthma and chronic) obstructive pulmonary disease (COPD):

• Both FVC and FEV1 decreased, but FEV 1 is decreased more than FVC is Thus FEV1, FVC is also decreased, which is typical of airway obstruction with its increased resistance to expiratory air flow

In a patient with a restrictive lung disease such as fibrosis:

• Both FVC and FEV1 are decreased but FEV1 is decreased less than FVC is, so in fibrosis, FEV1 FVC is actually increased

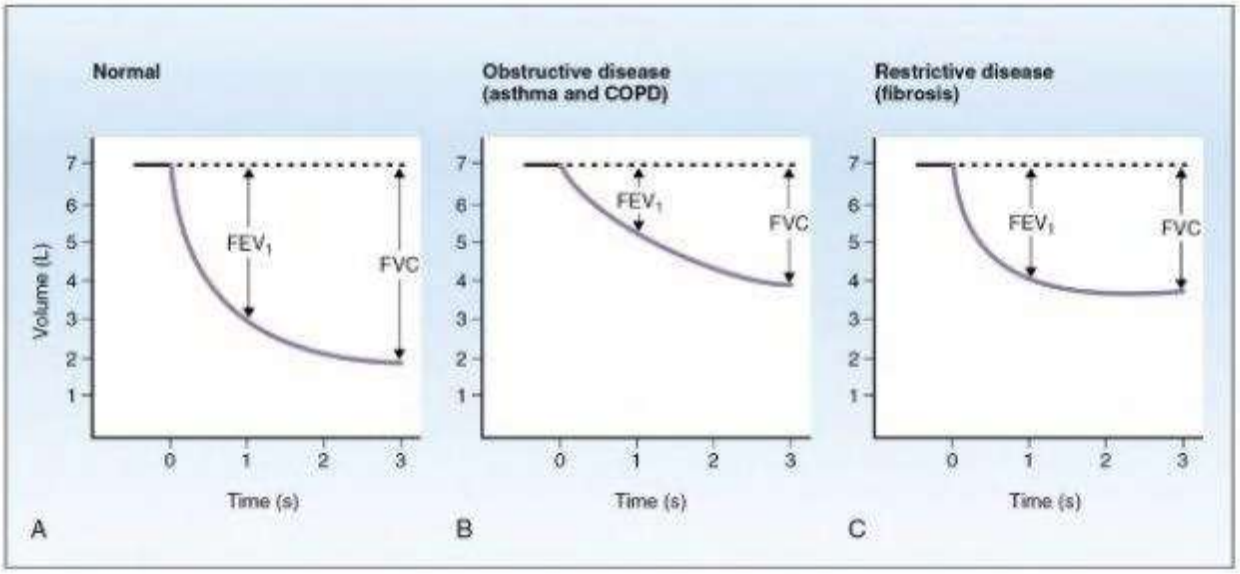


Figure 10: FVC and FEV 1 in normal subjects and patients with lung disease. Subjects inspired maximally and then expired forcibly. A–C, the graphs show the phase of forced expiration. The total volume that is forcibly expired is called the forced vital capacity (FVC). The volume expired in the first second is called FEV 1. COPD, Chronic obstructive pulmonary disease.

4 physiology of Respiration

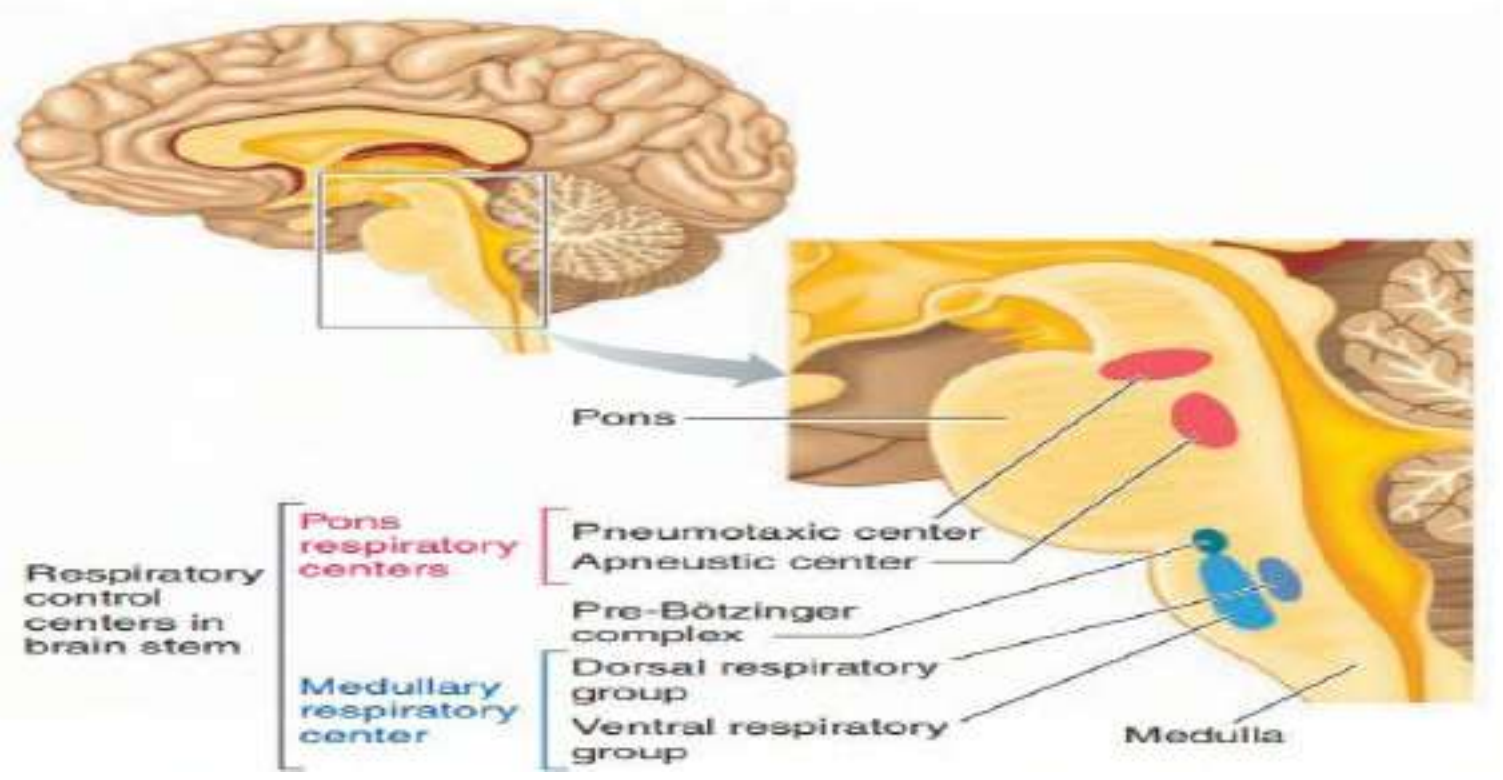
Control of respiration

- Respiration is a reflex process.
- But it can be controlled voluntarily for a short period.
- Normally, quiet regular breathing occurs because of two regulatory mechanisms:
  - o Nervous or neural mechanism
  - o Chemical mechanism

❑ Nervous mechanism

- **Respiratory centers**
- **Afferent nerves**
- **Efferent nerves**

❑ Respiratory centers



Respiratory centers

Respiratory centers

- Respiratory centers are groups of neurons, which control the rate, rhythm and force of respiration.
- These centers are bilaterally situated **in the brainstem**. Depending upon the situationin brainstem, the respiratory centers are classified into two groups:

❑ Medullarycenters
- **Dorsal respiratory group of neurons**
- **Ventral respiratory group of neurons**

❑ Pontine centers
- **Apneustic center**
- **Pneumotaxic center**

Medullary centers

1) Dorsal Respiratory Group of Neurons	2) Ventral Respiratory Group of Neurons
• Theyare presentin the **upper part of the medulla oblongata,** and are called the **inspiratory center.** • They generate inspiratory ramp (pacemaker of respiration) by the their autorhythmic property. They are responsible for basic rhythm of respiration.	• They are present **anteriorand lateralto the dorsal respiratory** group of neurons. • They are called the **expiratory center**. • They have both inspiratory and expiratory neurons. • Normally, ventral group of neurons are inactive during quiet breathing and become active during forced breathing. • During forced breathing, these neurons stimulate both inspiratory muscles and expiratory muscles.

pontine centers

1) Apneustic Center	2) *Pneumotaxic Center*
• Apneusticcenterissituated**in lower pons.** • It **increases depth of inspiration** by acting directly on dorsal group neurons.	• Pneumotaxiccenter is situated**inthe upper pons.** • Its primary function is to controlthe medullary respiratory centers, particularlythe dorsal group neurons. • It acts **through apneustic center**

N.B

- **Pneumotaxic center inhibits the apneustic center** so that the **dorsal group neurons** are **inhibited**. Because of this, **inspiration stops and expiration starts**.

- Thus, pneumotaxic center influences the switching between inspiration and expiration.
- Pneumotaxic center increases respiratory rate by reducing the duration of inspiration.

Connections of respiratory

❑ EfferentrespiratoryfibersPathway

o Phrenic nerve fibers **(C3 to C5),** which supply the **diaphragm**
o Intercostal nerve fibers **(T1 to T11),** which supplythe **external intercostal muscles.**
o Vagus nerve also contains some efferent fibers from the respiratory centers.

❑ Afferent Pathway

- Respiratorycentersreceiveafferentimpulsesfrom:
  - **Peripheral chemoreceptors and baroreceptors** via branches of **glossopharyngeal and vagus nerves.**
  - Stretch receptors of **lungs via vagus nerve.**

- By receiving afferent impulses from these receptors, respiratory centers modulate the movements of thoracic cage and lungs through efferent

Inspiratory ramp signals

- To start with, the amplitude of action potential is low. It is due to the activation of only few neurons. Later, more and more neurons are activated, leading to gradual increase in the amplitude of action potential in a ramp fashion. Impulses of this type discharged from dorsal group of neurons are called inspiratory ramp signals.
- Ramp signals are not produced continuously but only for a period of 2 seconds, during which inspiration occurs. After 2 seconds, ramp signals stop abruptly and do not appear for another 3 seconds. Switching off the ramp signals causes expiration. At the end of 3 seconds, inspiratory ramp signals reappear in the same pattern and the cycle is repeated.

N.B

- **Significance of inspiratory** rampsignals
  - There is a slow and steady inspiration, so that the filling of lungs with air is also steady

Factors affecting respiratory

Impulses from Higher Centers

- Cerebral cortex
  - **voluntary apnea**
  - **Voluntary hyperventilation**
  - **Controlled expiratory effort with singing or talking.**
- Hypothalamus
  - **Moderate pain ,↑ temperature , moderate emotions → ↑ respiratory rate**
  - **Stimulation of anterior hypothalamus → ↓ respiratory rate.**
  - **Stimulation of posterior hypothalamus → ↑ respiratory rate**

Impulses from Stretch Receptors of Lungs

- Hering-Breuer inflation Reflex
  - Hering-Breuerreflex is a **protective reflex** that restricts inspiration and prevents overstretching of lung tissues. It is initiated by the stimulation of stretch receptors of air passage.
  - Stretch receptors are the receptors which give response to stretch of the tissues. These receptors are situated on the wall of the bronchi and bronchioles.
  - Expansion of lungs during inspiration stimulates the stretch receptors. Impulses from stretch receptors reach the dorsal group neurons via vagal afferent fibers and inhibit them. So, inspiration stops and expiration starts. Thus, the overstretching of lung tissues is prevented.
  - However, Hering-Breuer reflex does not operate during quiet breathing. It operates, onlywhenthe tidalvolume increases beyond 1,000 mL.
- Hering-Breuer deflation Reflex
- Reverse of Hering-Breuer inflation Reflex, it takes place during expiration. During expiration, as the stretching of lungs is absent, deflation occurs.

Impulses from Irritant Receptors of Lungs

- Besides stretch receptors, there is another type of receptors in the bronchi and bronchioles of lungs, called irritant receptors.
- Irritant receptors are stimulated by irritant chemical agents such as **ammonia** and **sulfur dioxide.**
- These receptors send afferent impulses to respiratory centers **via vagal nerve fibers.**
- Stimulation of irritant receptors produces **reflex hyperventilation** along with **bronchospasm.** Hyperventilation along with bronchospasm prevents further entry of harmful agents into the alveoli.

Impulses from Baroreceptors

- Baroreceptors or **pressoreceptors** are the receptors which give response to change in blood pressure.

❑ Function

- Baroreceptors **in carotid sinus** and **arch of aorta** give response to **increase in blood pressure**. Whenever arterial blood pressure increases, baroreceptors are activated and send inhibitory impulses to vasomotor center in medulla oblongata. This causes decrease in blood pressure and inhibition of respiration. However, in physiological conditions, the role of baroreceptors in regulation of respiration is insignificant.

❑ Impulses from right atrium (Harrison ' s reflex)

- Distension of **right atrium** → stimulate **atrial stretch receptors** → send impulses along the **vagus nerve** → stimulate the **RC** →
- ↑ **respiratory rate.**
  - This reflex is responsible for dyspnea in cardiac patient (***Rightsided heart failure***).
  - **Aimof the reflex**: oxygenation & removal of CO2 from the excess blood.

Impulses from Chemoreceptors

- Chemoreceptors play an important role **in the chemical regulation of respiration**

Impulses from Proprioceptors

- Proprioceptors are the receptors which give response to change **in the position of body**. These receptors are **situated in joints, tendons and muscles.** Proprioceptors are stimulated during the muscular exercise and send impulses to brain, particularly cerebral cortex, through somatic afferent nerves. Cerebral cortex in turn **causes hyperventilation** by sending impulses to medullary respiratory centers.

Impulses from Thermoreceptors

- Thermoreceptors are cutaneous receptors, which **give response to change in the environmental temperature**. Thermoreceptors are of two types, namely receptors for cold and receptors for warmth. When body is exposed to cold or when cold water is applied over the body, cold receptors are stimulated and send impulses to cerebral cortex via somatic afferent nerves. Cerebral cortex in turn, stimulates the respiratory centers and causes hyper ventilation.

Impulses from Pain Receptors

- Pain receptors are those which **give response to pain stimulus**. Whenever pain receptors are stimulated,the impulses are sent to cerebral cortex via somatic afferent nerves. Cerebralcortex in turn, stimulates the respiratory centers and **causes hyperventilation.**

Chemical mechanism

- Chemical mechanism of **regulation of respiration is operated through the chemoreceptors**. Chemoreceptors are the sensory nerve endings, which give response to changes in chemical constituents of blood.
  - o **Hypoxia(decreased pO2)**
  - o **Hypercapnea (increased pCO2)**
  - o **Increased hydrogen ion concentration**

❑ TypesofChemoreceptors

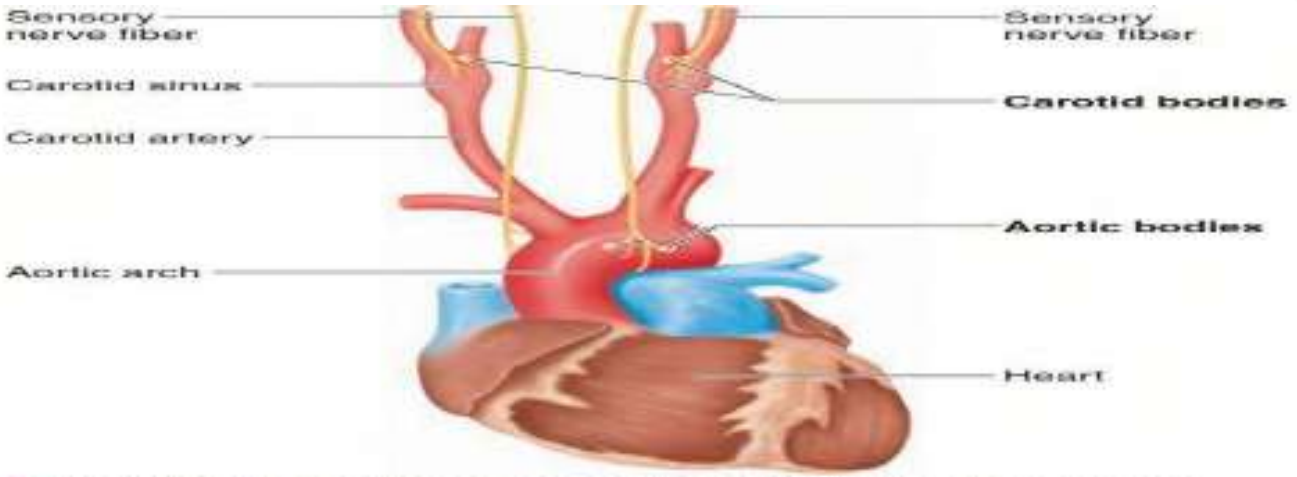
- ○ **Central chemoreceptors**
- ○ **Peripheral chemoreceptors.**

Central chemoreceptors

- ○ Central chemoreceptors are situated **in deeper part of medulla oblongata,** close to the dorsal respiratory group of neurons.
- ○ Chemoreceptors are in **closecontact with blood and cerebrospinal fluid**.

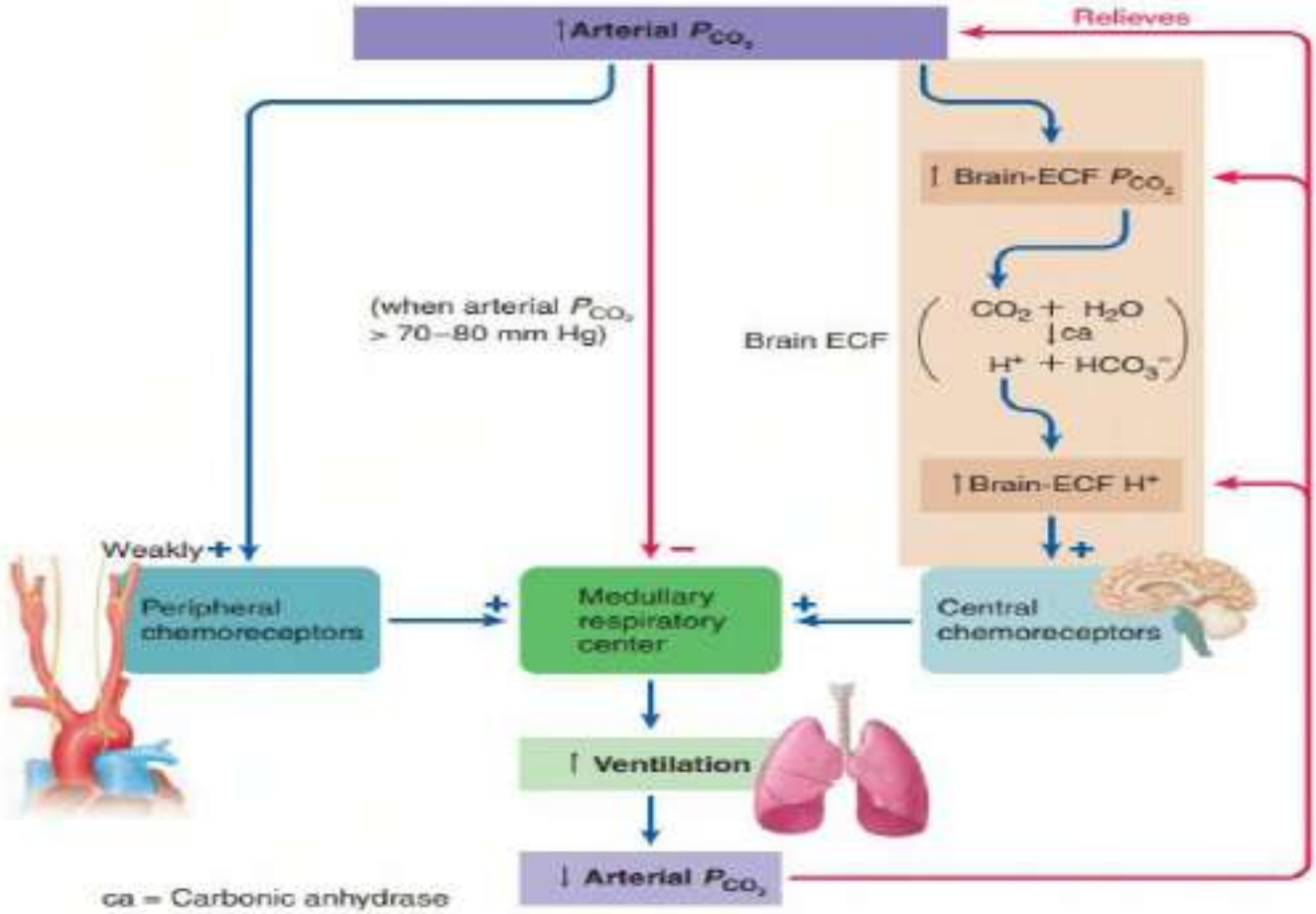
Peripheral chemoreceptors

- ○ Peripheral chemoreceptors are the chemoreceptors **present in carotid and aortic bodies.**
- ○ **Hypoxia is the most potent stimulant** for peripheral chemoreceptors.
- ○ **peripheral chemoreceptors** are also **stimulated by hypercapnia** and **increased hydrogen ion concentration (mild sensitive)**



● **FIGURE 13-34 Location of the peripheral chemoreceptors.** The carotid bodies are located in the carotid sinus, and the aortic bodies are located in the aortic arch.

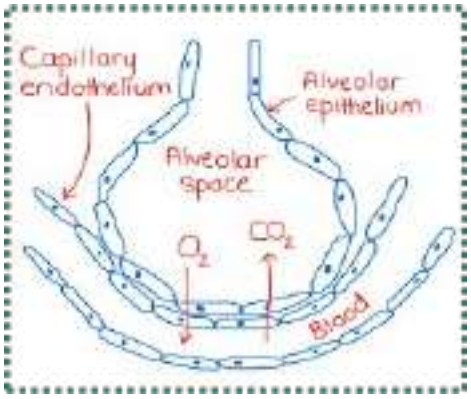
❑ Effect of increased CO2



● **FIGURE 13-35 Effect of increased arterial P_{CO_2} on ventilation.**

- **Giving O2 to patient under deep anesthesia may end their life, why?**

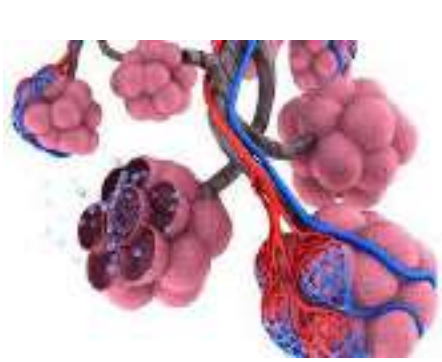
Gas exchange



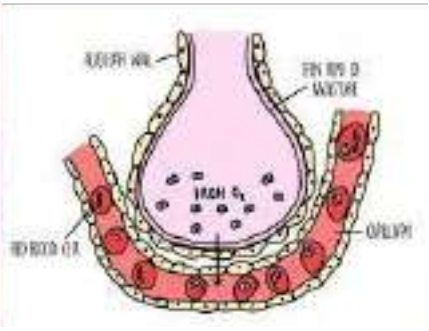
- In the lungs, oxygen enters the blood from alveoli and carbon dioxide is expelled out of blood into alveoli.
- Exchange occurs through **bulk flow diffusion**.
- Exchange of gases between blood and alveoli takes place through **respiratory membrane.**

1) Alveoli

- Respiratory membrane is **very thin** with an average thickness of 0.5 µ.
- Total surface area of the respiratory membrane in both the lungs is about **70 square meters**.
- Average diameter of pulmonary capillary is only 8 µ, which means that the RBCs with a diameter of 7.4 µ actually squeeze through the capillaries.
- Therefore, the membrane of RBCs is in close contact with capillary wall. This facilitates quick exchange of oxygen and carbon dioxide between the blood and alveoli.







2) Diffusing capacity

Diffusing capacity is defined as **the volume of gas that diffuses through the respiratory membrane each minute for a pressure gradient of 1 mm Hg**.

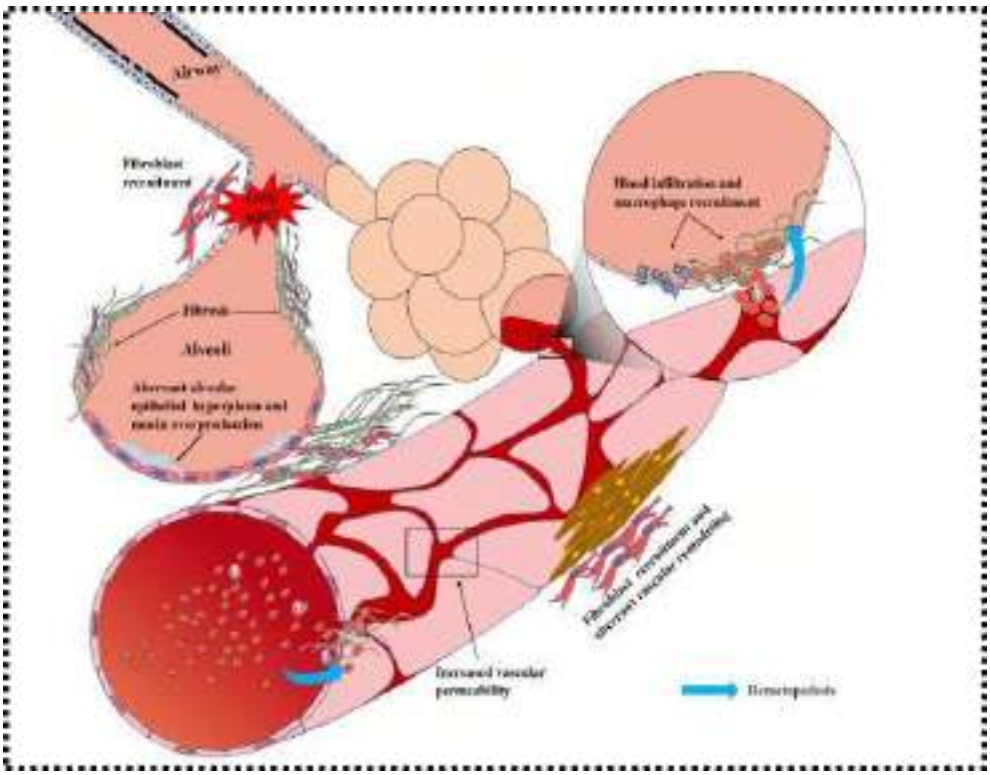
difusing capacity for oxygen is21mL/minute/1mm Hg.

diffusing capacity for carbon dioxideis400mL/minute/1 mmHg .

diffusing capacity for carbon dioxide is about **20 times** more than that of oxygen.

Factors affecting diffusing capacity

1. Pressure gradient
2. Solubility of gas in fluid medium
3. Molecular weight of the gas
4. Total surface area of respiratory membrane
5. Thickness of respiratory membrane

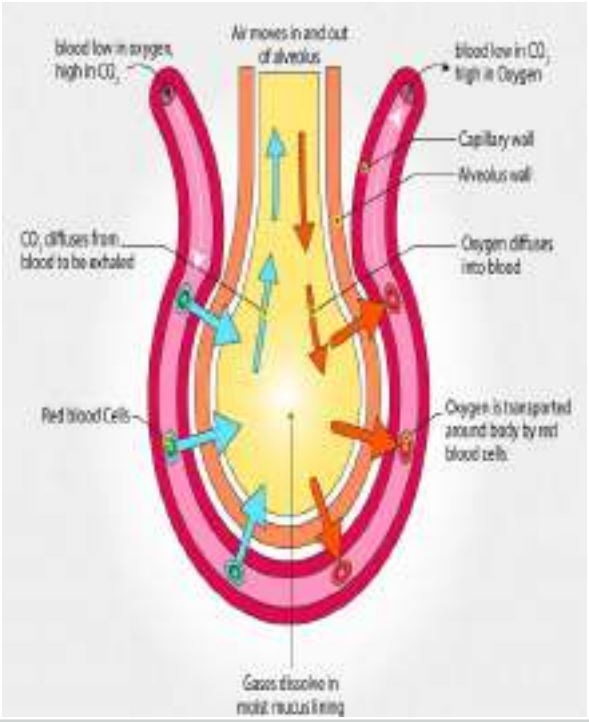
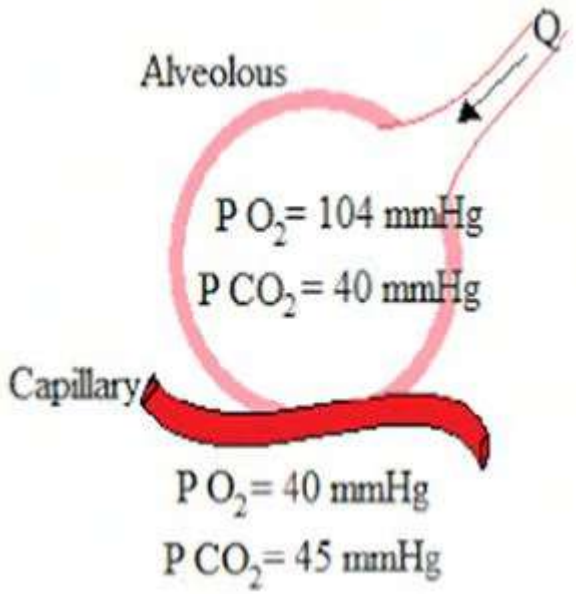


Relation between Diffusing Capacity and its factors

$$DC \propto \frac{Pg \times S \times A}{Mw \times D}$$

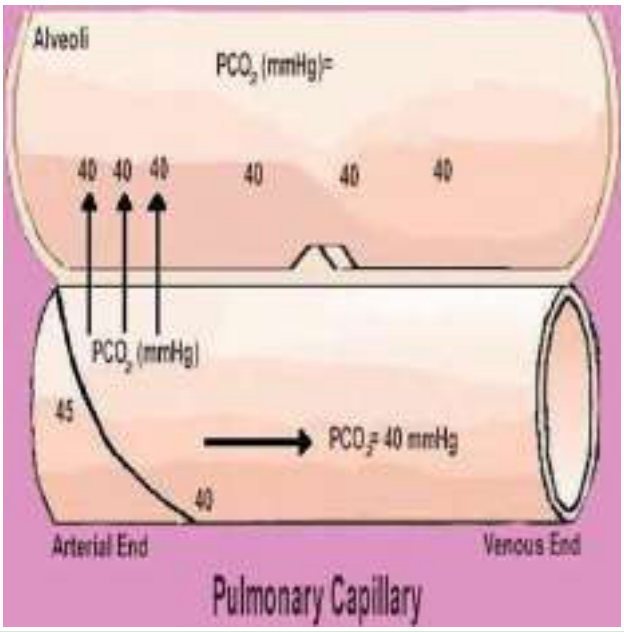
- DC = Diffusing capacity
- Pg = Pressure gradient
- S = Solubility of gas
- A = Surface area of respiratory membrane
- Mw = Molecular weight
- D = Thickness of respiratory membrane

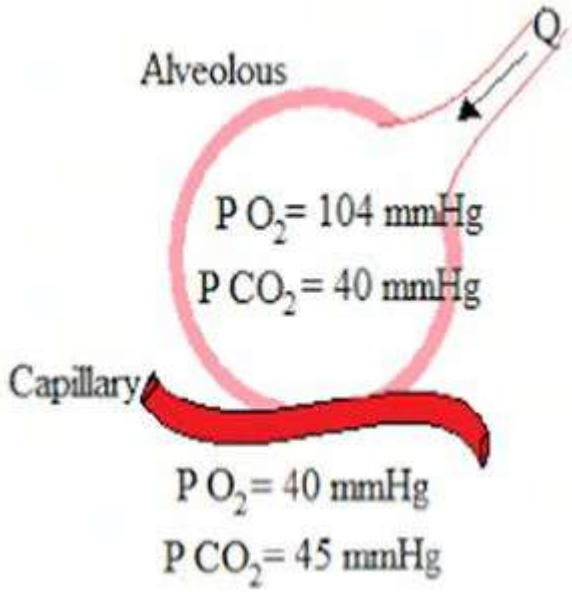
3) Diffusion of Oxygen

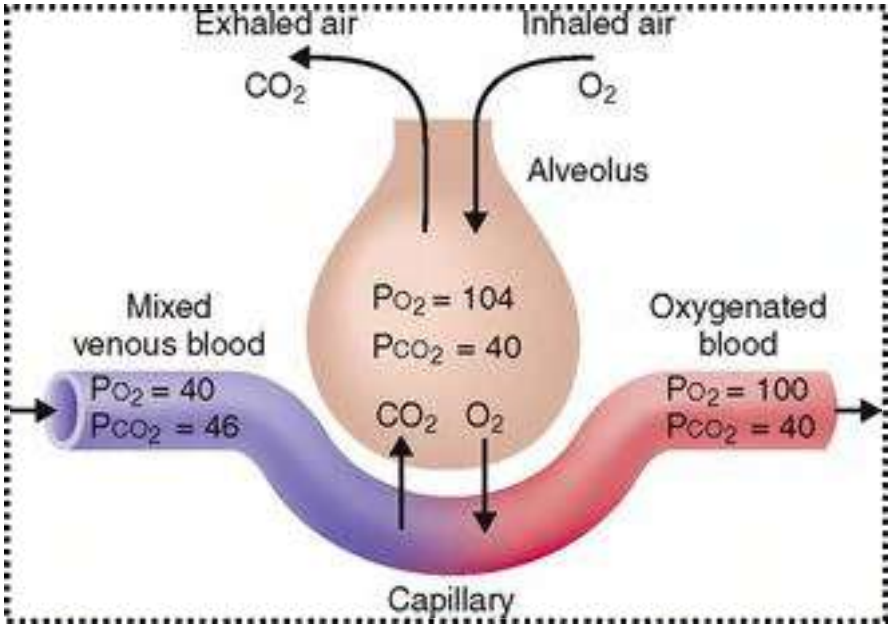
Diffusion of oxygen from atmospheric air into alveoli	Diffusion of oxygen from alveoli into blood
• Partial pressure of oxygeninthe **atmospheric air is 159 mmHg** and, in the **alveoli, it is 104 mm Hg**. • Because of **the pressuregradient of 55 mm Hg**, oxygen easily entersfrom atmospheric air into thealveoli.  	• PO2 in **pulmonary capillary is 40 mm Hg** • and in the **alveoli, it is 104 mm Hg**. **Pressure gradient is 64 mm Hg**. • It facilitates the diffusion of oxygen from alveoli into the blood. 

4) Diffusion of Carbon Dioxide

Diffusion of carbon dioxide from alveoli into atmospheric air	Diffusion of carbon dioxide from blood into alveoli
• In atmospheric air, PCO2 is very insignificant and is only about 0.3 mm Hg whereas, in the alveoli, it is 40 mm Hg. So, carbon dioxide enters passes to atmosphere from alveoli easily.	• PCO2 in alveoli is40mmHgwhereas in the blood it is 46mmHg. • Pressure gradientof6mmHgis responsible for thediffusionofcarbon dioxide from bloodintothealveoli.







Oxygen transport

The oxygen is carried in the blood in two forms :-

1. Physically dissolved (3%)
2. Chemically combined (97%)

1) Physically dissolved

Physically dissolved in the water of plasma or water of RBCs [3%]. It is very small amount but important because of:

a- It **determines partial pressure of O2 (Po2)**, because gases in chemical form have no pressure. So, it determines the **direction** and **rate** of diffusion of gas in chemical combination from or to blood as well as the **saturation** of Hb with O2

b- It is a **stimulant to the peripheral chemoreceptors**.

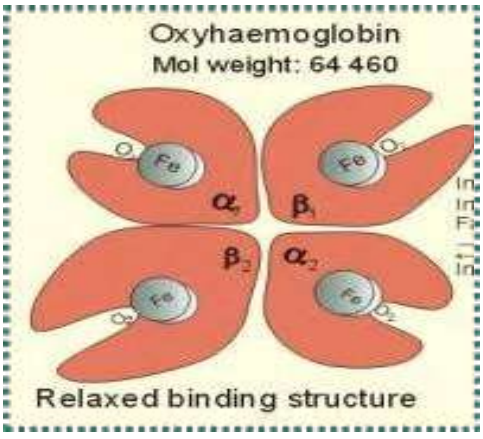
2) Chemically combined

NOTE

Chemically combined with Hb (oxyhemoglobin) [97%].

Characteristics of Hb

- Hb is made up of **four subunits**. Each of these subunits is attached to an iron-containing molecule called heme; the iron in the center of each heme can **reversibly** bind to one oxygen molecule. So, each hemoglobin molecule can associate with between **0** and **4 oxygen molecules** at any one time.
- **Normal adult hemoglobin** (hemoglobin A) makes up about **97%** of adult hemoglobin and is comprised of **2 α** and **2 β** subunits.



O2 content

It is the volume of O2 (in ml or cc.) present in chemical combination with Hb/ 100 ml blood.

It depends on the amount of Hb, amount of O2, O2 affinity and metabolic state of the body organs.

O2 capacity

- It is **the maximal** volume of O2 (in ml or cc.) present in chemical combination with Hb **in 100 ml blood** when the Hb is **fully saturated** with O2.
- Blood in circulation is never fully saturated with O2.
- Hb is 15 gm/dL and each 1 gm can carry 1.33 ml O2
- O2 capacity = 15 X 1.33 = 20 ml (but only 97 % of Hb which carried O2, so it contains 19.5 ml).

O2 s a turation

- It is the % saturation of Hb with O2, and is calculated as follows:
- O2 saturation= $\frac{O2\ content}{O2\ capacity} \times 100$
- In arterial blood, O2 saturation is 97%

O2 utilization

- It is the amount of oxygen that the body takes up and utilizes.
- Every 100 ml (100 cc.) of arterial blood (O2 content 19.5 ml) loses O2 while passing into tissues and change into venous blood (O2 content 14 ml).
- So, 1 liter of arterial blood loses 50 ml of O2.

Coefficient of O^2 utilization

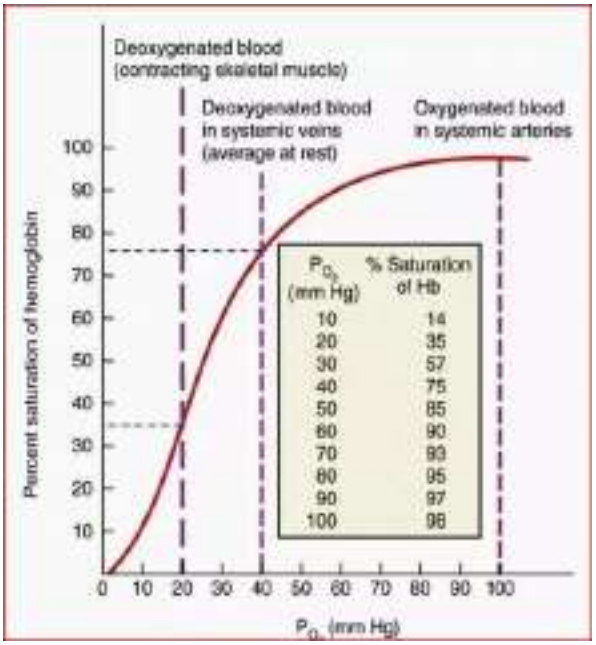
- It is the % of oxygen in arterial blood that is taken by tissues.
- It is calculated as follows:

$$\frac{arterial O_2 content - venous O_2 content}{arterial\ O_2\ content} \times 100$$

- Normally it = 25% at rest, while during exercise it increases to ≥ 75%
- It depends on:

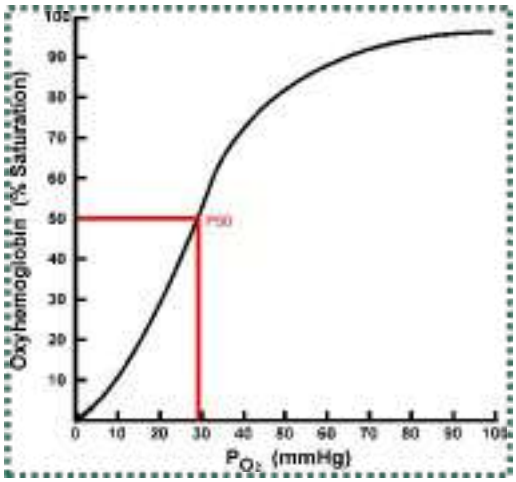
a- **The tissue activity**: It is directly proportional to tissue activity.

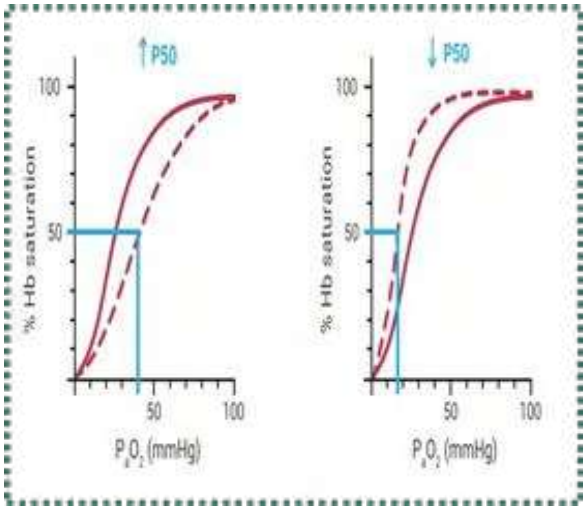
b- **The rate of blood flow**: It is inversely proportional to the rate of blood flow, so tissues will not have enough O2 if the blood flow became high.



P50

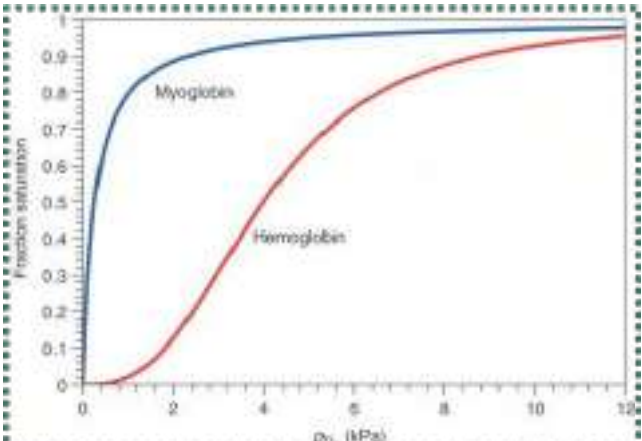
- It is the partial pressure of O2 required for the saturation of 50% of Hb with oxygen.
- It is an index of the Hb-O2 dissociation curve
- The normal P50 at rest is about 27 mmHg
- Increased of P50 indicates decreased affinity of Hg to O2 (shift to right) and vice versa



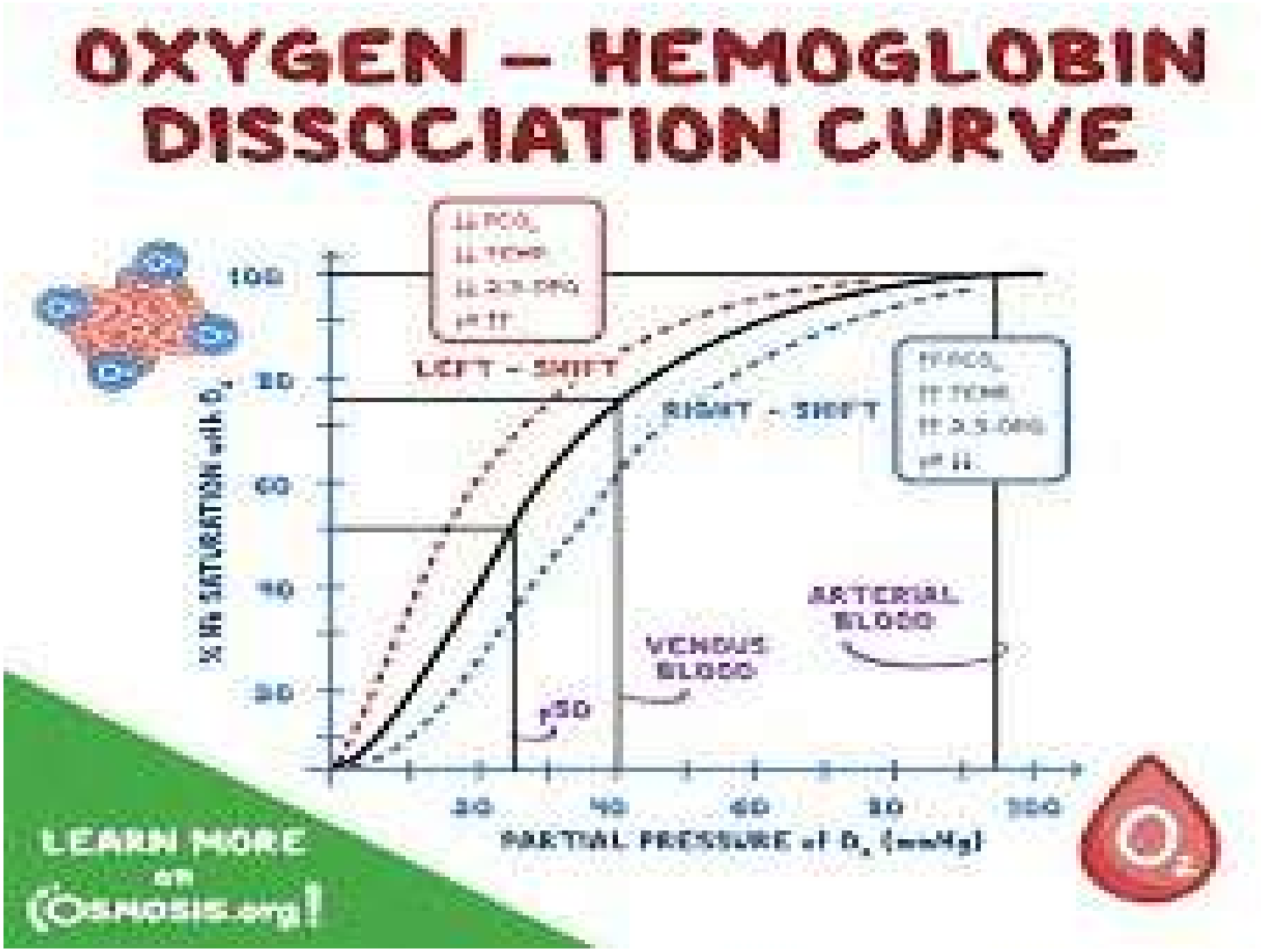


Myoglobin

- It is an iron containing pigment found in skeletal muscles. it binds with one molecule of O2 rather than 4 molecules as in Hb.
- It takes up O2 from the Hb in the blood and store it, then release it only at very low PO2 [as in exercise].



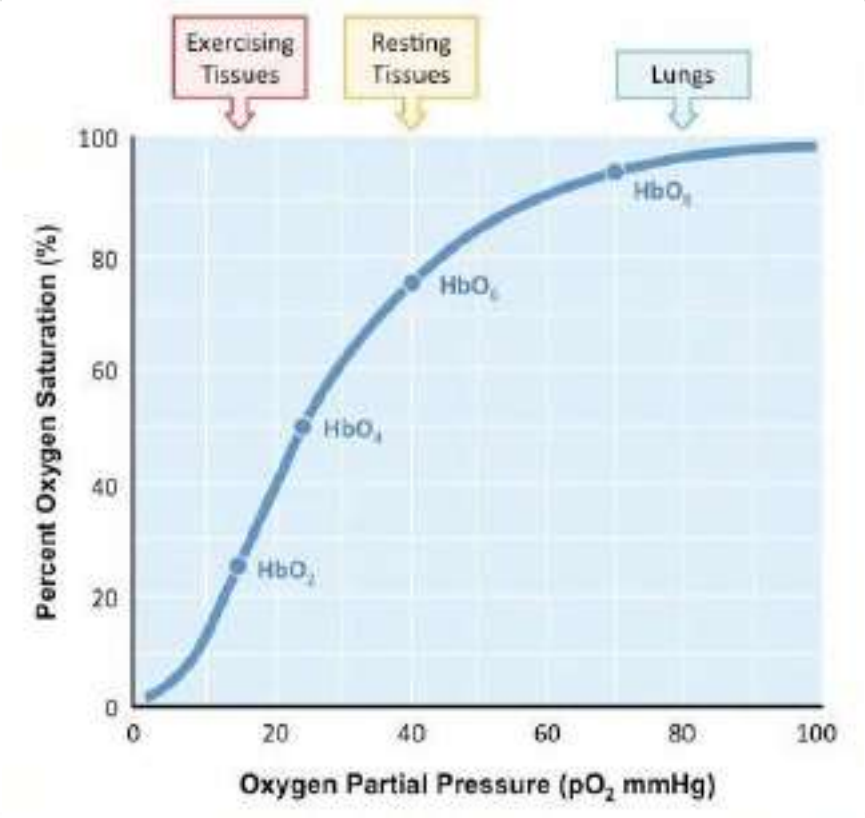
3) *Hemoglobin-oxygen affinity (oxygen-hemoglobin dissociation curve)*



It is a **sigmoid shaped** curve that describes the affinity of Hb for O2 [i.e., describes the relation between the percentage of saturation of Hb by oxygen & PO2 in the blood].

Physiological significance

Normal O2 supply to tissues during rest	Normal O2 supply to tissues during Muscular exercise
• At 100 mmHg. O2 saturation is 97%, this point is arterial blood that is rich with O2 supply to tissues. • At 40 mmHg, O2 saturation is 75%, this point is venous blood. • Tissues take about 25% of arterial blood O2 (97-75 =22%).	Arterial blood, (PO2 is 100 mmHg & O2 saturation is 97 %), venous blood (PO2 is l5-30 mmHg due to excess utilization of O2 & O2 saturation is 25%) (the more activity the less affinity). Tissues take about 72% of arterial blood O2.

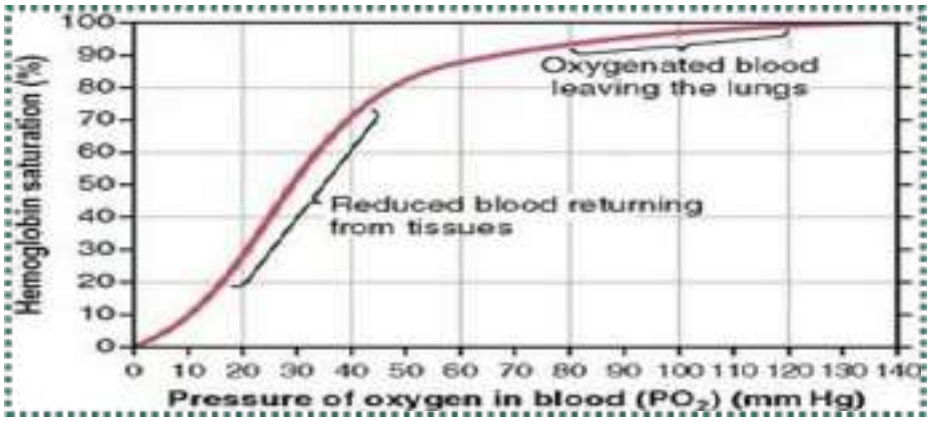


NOTE

The normal arterial O2 saturation is 97% not l00% due to mixing of arterial and venous blood in **physiological shunts**

The shape of the curve

- is **S-shaped**:theupper segment is horizontal and the lowersegmentis vertical.

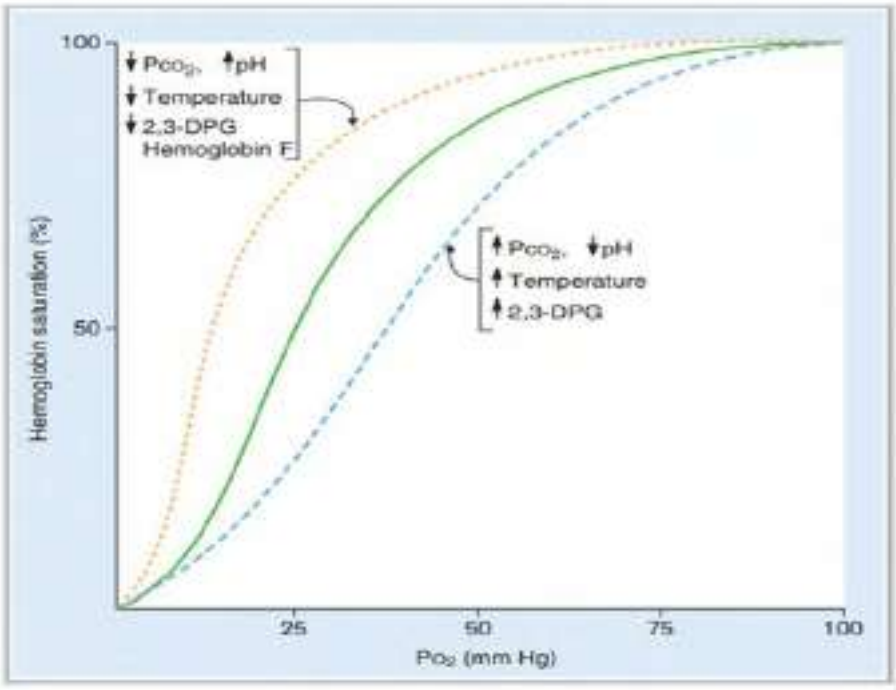


- upper segment is horizontal this means that O2 tension can be reduced to about 60mmHg while the O2 saturation is only decreased to 90%,this means that Hb can be still be saturated even at lower O2 tension as at high altitudes and in extensive lung diseases. Lower segment is vertical this means that slight decrease in O2 tension is accompanied bymarked decrease in O2 saturation. This occurs at tissue level so oxy-Hg givesO2 to the tissue easily.

Deviation

- This curve could be shifted to the right or to the left by many physiological factors.

right	left
• When the oxygen hemoglobin dissociation curve shifts to the right, this means that the affinity of Hb to O2 is ↓ (i.e., ↑ dissociation and liberation of O2 from the Hb → facilitate the delivery of oxygen to the tissues).	• On the other hand,when this curve shift to the **left**, the opposite will occur [↑ affinity of Hb to O²→↓ dissociation & liberation of oxygenfrom Hb → binding of large amount of Oèven at low PO2 level (this occurs at thepulmonary capillaries)].
The factors that shift the curve to the right include: 1- ↑ H+ concentration (acidosis) [↓ PH]. 2- ↑ temperature. 3- ↑ PCO2 (Bohr's effect)]. 4- ↑ 2,3 (DPG) 2, 3 Diphosphoglycerate 5- ↑ Hb concentration.	The factors that shift this curve to the left include: 1- ↓ H+concentration(alkalosis)[↑ PH]. 2- ↓ temperature. 3- ↓ PCO2 (at pulmonary capillaries, during sleep). 4- Presence of HbF [has more affinity for O2]. 5- A decrease in Hb concentration (as in anemia). 6- Presence of carboxy hemoglobin [COHb].

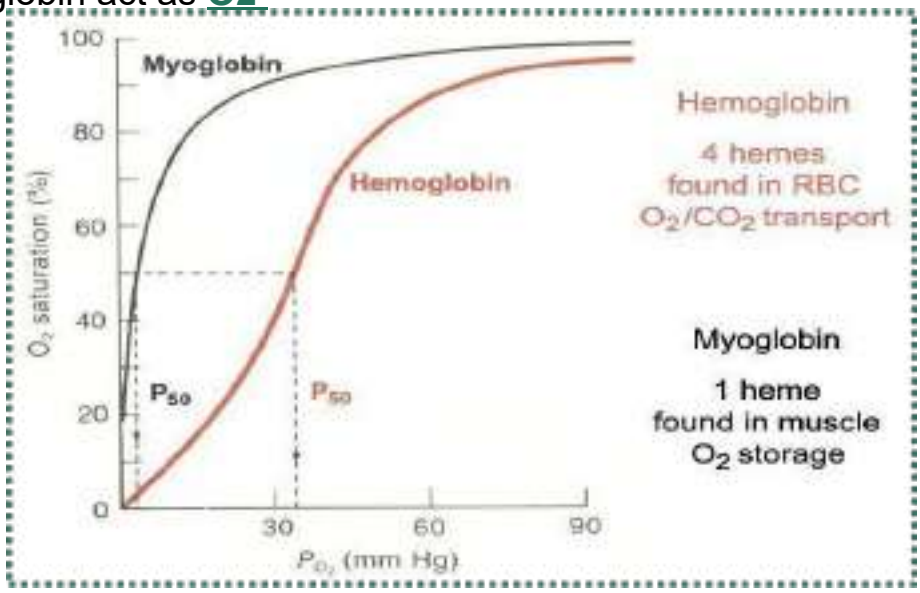


O2 dissociation curve of myoglobin

It is the pigment in **skeletal muscle** (similar to Hb in blood). It is a single chain globular protein containing a hem. So, it binds only with one mole of O2 (while Hb with four O2).

Its O2 dissociation curve is **rectangular hyperbola** i.e. it remains horizontal till very low O2 tension (PO2 near zero), in other words this occurs when **the blood Hb** has given most of its O2. So myoglobin act as **O2 store** to be used by muscles when O2 tension becomes very low in tissues in **severe exercise**.

Myoglobin gives O2 at very low PO2 (near zero) i.e., other source of O2 when the Hb becomes very low. Following exercise, the myo-Hb sucks O2 from the blood.



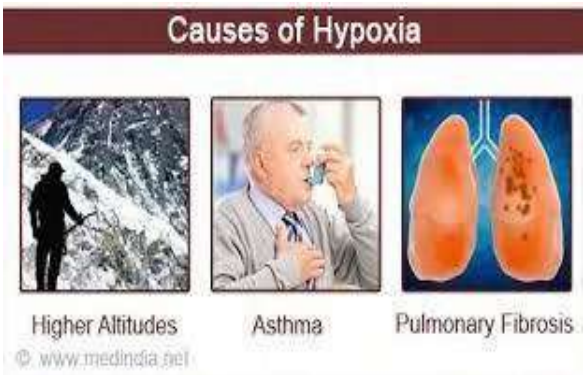
4) *Hypoxia and its types*

Definition: an inadequate supply of oxygen to tissues. The decrease of oxygen in the bloodis called **hypoxemia**.

Hypoxic hypoxia

itiscaused by any condition that interferes with theproper oxygenation of blood in the lungs like incases of:

- Ascending to high altitude [↓PAO2 → ↓ PaO2].
- Breathing air with low PO2.
- Arteriovenous shunt.



Anemic hypoxia

This type is caused by decrease in the oxygen carrying capacity of the blood. It is caused by:

- Hemorrhage & anemia.
- CO poisoning [formation of carboxyhemoglobin].

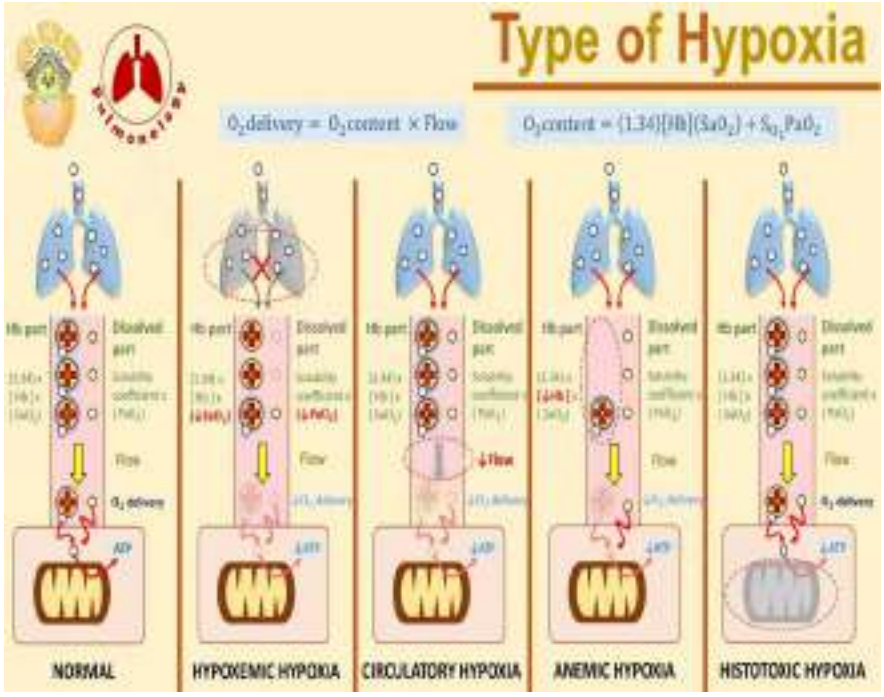
Stagnant hypoxia

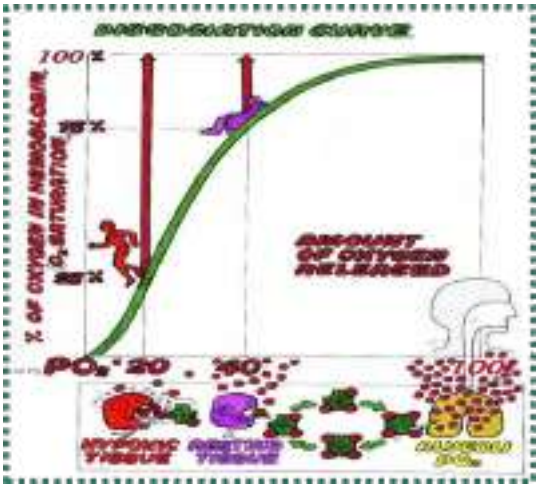
It is caused by decrease in the rate of blood flow to tissues. It is caused by:

- Congestive heart failure [left side].
- Raynaud's disease [vasoconstriction in the hands and feet by cold].
- Hyperviscosity of the blood → ↓ blood flow [e.g. in polycythemia].

Histotoxic hypoxia

Itisdue to a defect in the oxidative enzymes of the cells that is caused by poisons [e.g. potassium cyanide]. In this case the venous blood looks arterial because the cells are unable to utilize the oxygen.





CO2 transport

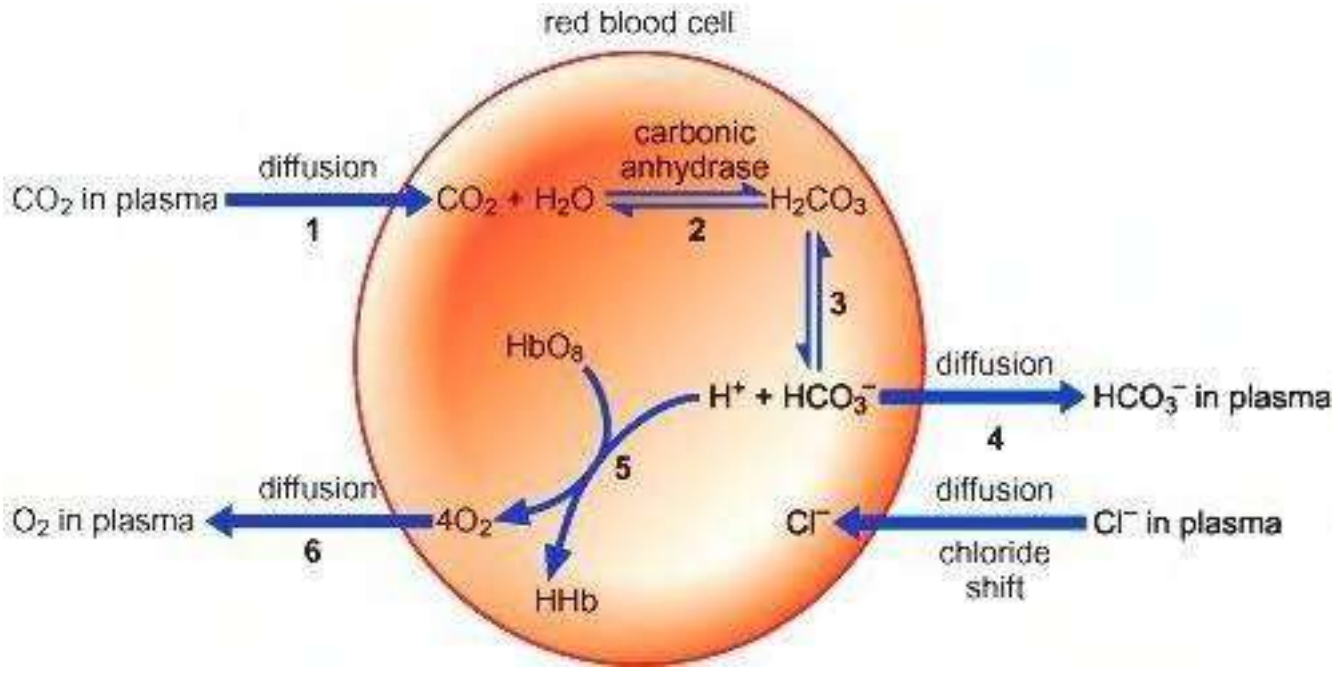
- **CO2** is produced in the tissues and carried to the lungs in the venous blood in the following forms:

1. **Physically Dissolved** in the blood [3ml/100mlblood,about 5%]. It is responsible for **PCO2**.

- In **arterial blood,** PCO2inarterialbloodis **40 mmHg,**
- In **venous blood 46 mmHg.**

1. As **bicarbonate** (HCO3-)[About **89%**].
2. **Carbamino compounds 6% combined** (with plasma proteins and with Hb).

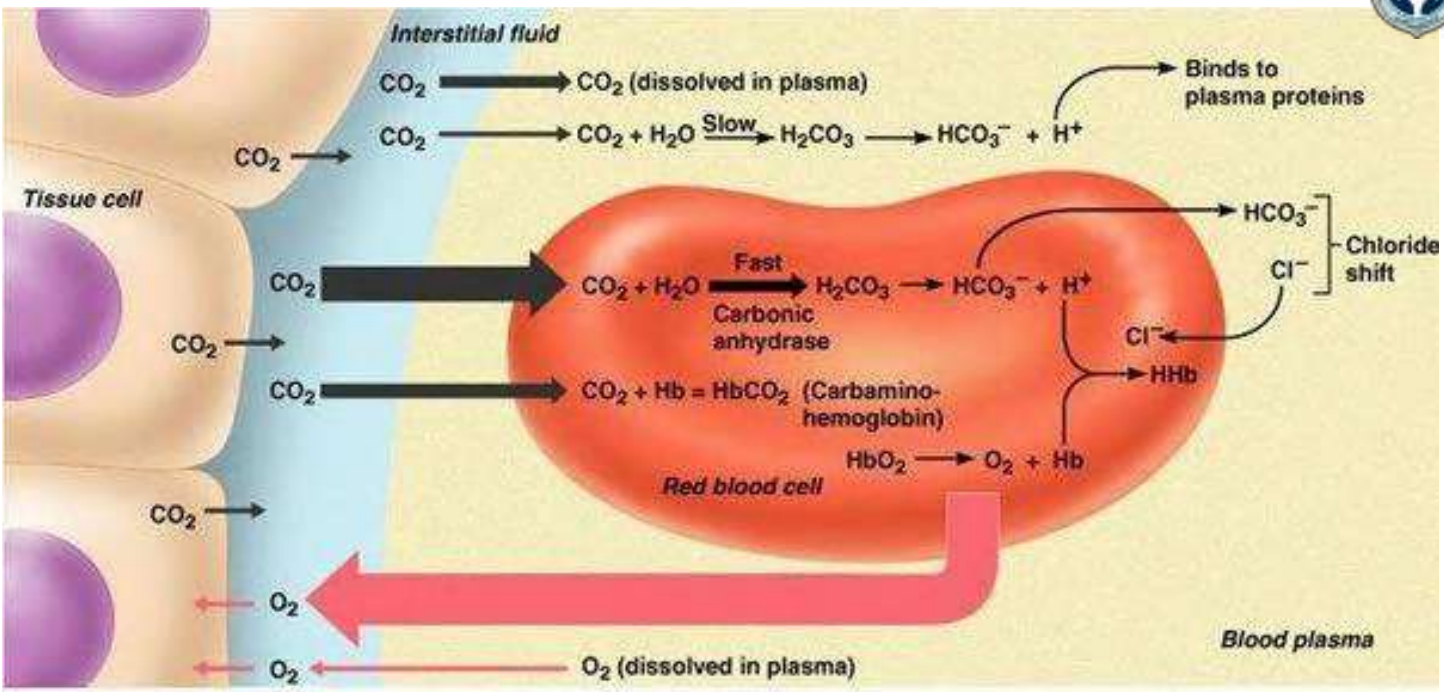
Bicarbonate



Results of chloride shift:

- The **HCO3-** concentration is increasedin both **RBCs and plasma.**
- **Cl-** concentration is decreasedin **plasma** andincreasedin **RBCs**.
- **Osmotic pressureinside** the **red cells** increasesduetoaccumulationof **both Cl- and HCO3-**. This leadsto **water movement** fromtheplasmaintothered cellsby **osmosis**(**water shift**),which **results** inswellingofthe **red cells** (increaseinsize).

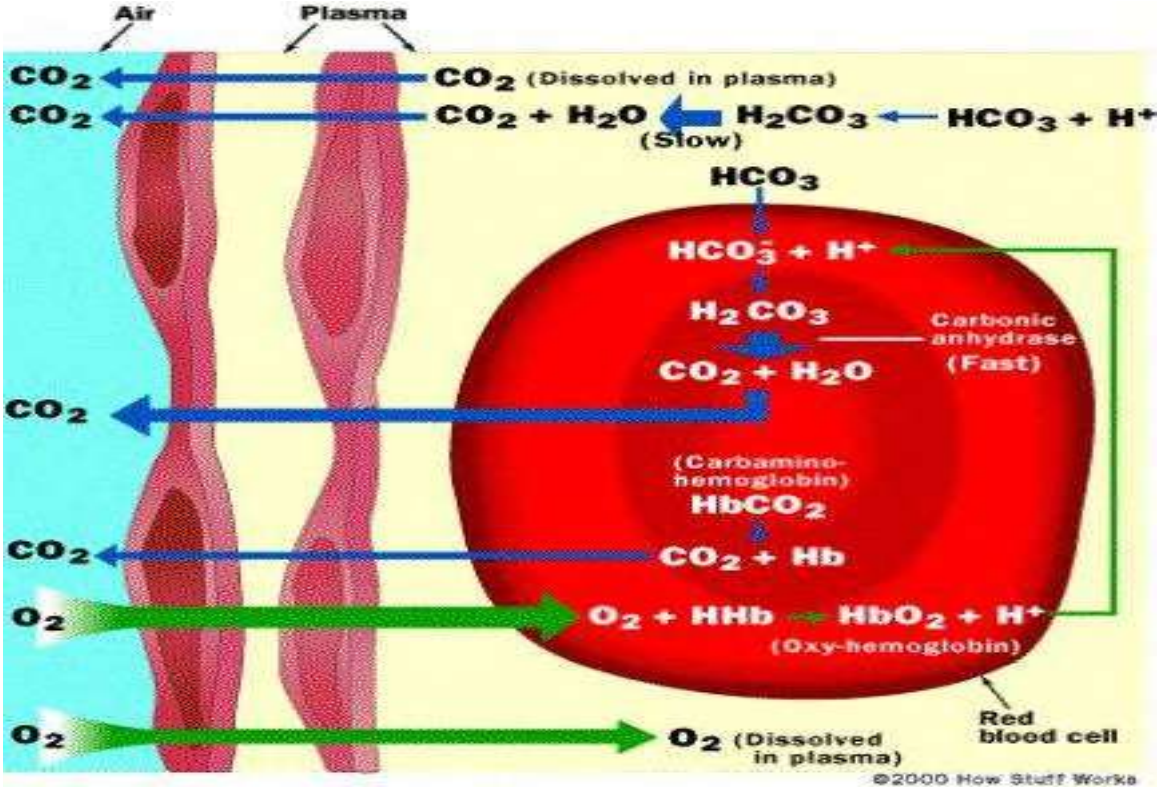
There is a **negligible change** in **H+ concentration** (pH) of both **plasma and RBCs .**

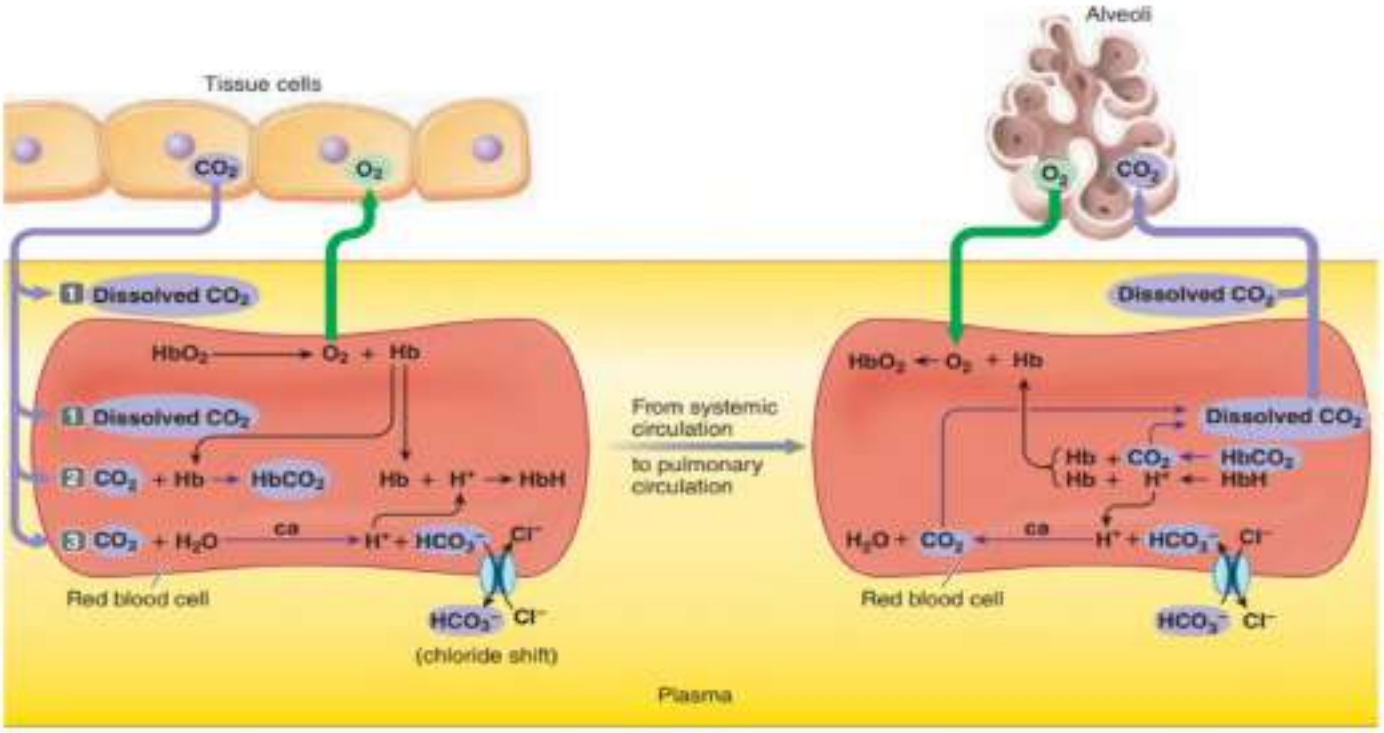


(a) Oxygen release and carbon dioxide pickup at the tissues

In the lungs,

- **HCO3** – enters the **RBCs** in **exchange** for **Cl**–.
- **HCO3** – **recombines** with **H+** to form **H2CO3**, which **decomposes** into **CO2** and **H2O**.
- Thus, **CO2**, originally **generated** in the **tissues**,is **expired** .





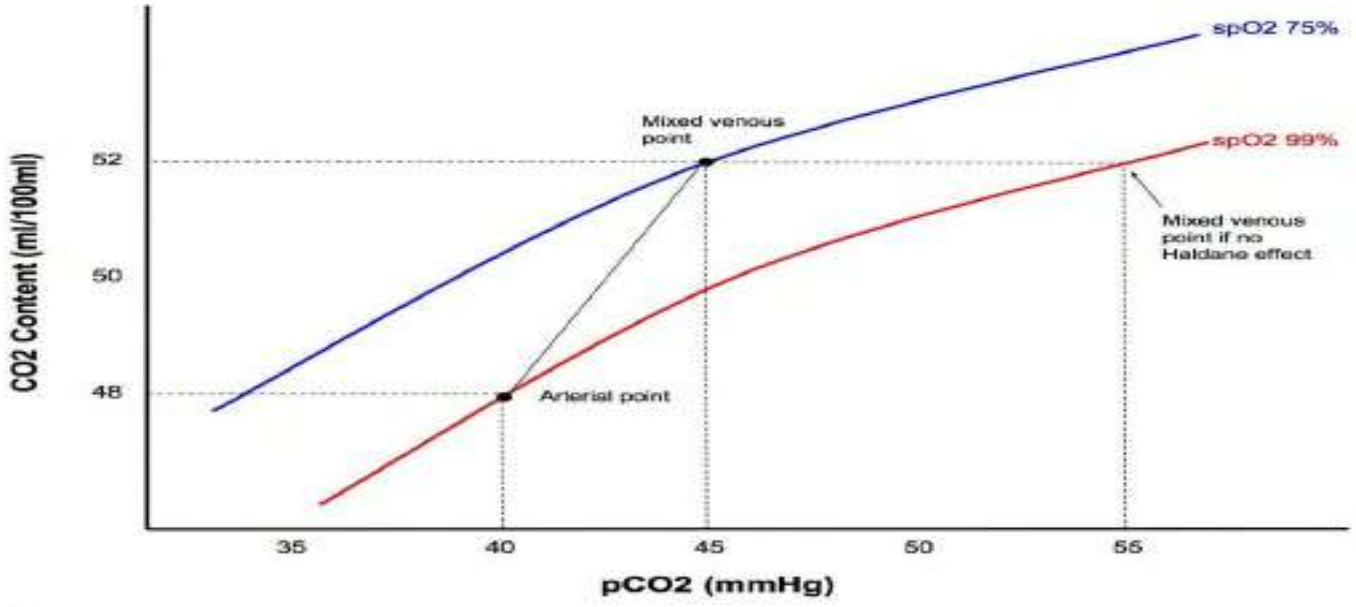
Tidal CO2:

- Is the **CO2** given by the **tissue** to **100 ml blood.**
- Is **4 ml/100 ml** blood .

Normal Carbon Dioxide Dissociation Curve:

Normal carbon dioxide dissociation cu e shows that:

- The **carbon dioxide content** in the **blood** is **48 mL%** when the **partial pressure** of carbon dioxide is **40 mm Hg** (**arterial blood**)
- and it is **52 mL%** when the **partial pressure** of carbon dioxide is **48 mm Hg** (**venous blood**).



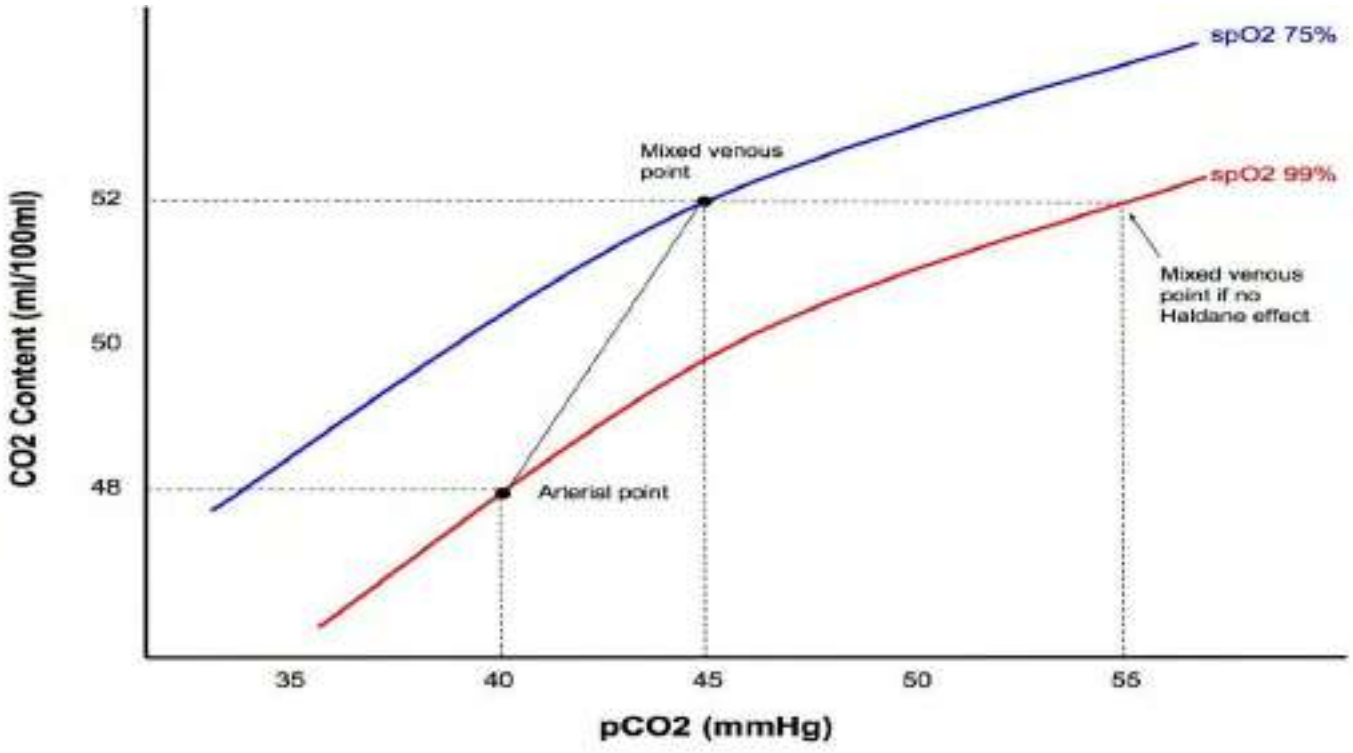
Haldane Effect

Haldane effect is the **effect** by which **combination** of **oxygen with hemoglobin displaces carbon dioxide** from **hemoglobin**.

Causes for Haldane effect:

Duetothe **combination** with **oxygen, hemoglobin** becomes **strongly acidic**.It causes **displacement** of **carbon dioxide** from **hemoglobin** in **two ways**:

1. **Highly acidic hemoglobin** haslowtendency tocombinewith **carbon dioxide.** So, **carbon dioxide** isdisplacedfrom **blood** . .Hydrogen ions
2. Becauseofthe **acidity**,hydrogenions are **released** in **excess** bindwith bicarbonate ionstoform **carbonic acid. Carbonic acid** inturn dissociates into **water and carbon dioxide. Carbon dioxide** is **released** from **blood into alveoli .**



Significance of Haldane effect:

1. **Release** of carbon dioxidefrom **blood** into the **alveoli** of lungs.
2. **Uptake** of oxygen by the **blood.**

Giving **pure O2** to patients under **deep anesthesia** may **end** their life, **why ?**

Cyanosis

- It is a **bluish discoloration** of the **tissues** when the reduced hemoglobin concentration in the **capillaries** is **> 5 gm / dL.**
- **Normal amount o**f reduced hemoglobin are **2.7 gm/ dL.**



Causes:

1. **Allcauses** of **hypoxichypoxia** (decreased **Hb saturation**—> reduced Hb).
2. **Allcauses** of **stagnanthypoxia** (more time for **extraction** of O2).
3. **Allcauses** of **asphyxia**.
4. In**case** of **severemuscular exercise.**
5. In **case** of exposure to **moderate cold** (in **severe cold cyanosis not occurs** due to **depressionofmetabolism** and **shift of O2** dissociation curve to **left** with **lessreductionofHb**).

Types of Hypoxiathatcannotcausecyanosis:

1. **Anemic hypoxia** because the total amount of hemoglobin is decreased and so **reduced hemoglobin** is also decreased.
2. **Histotoxic hypoxia** because all hemoglobin in the **venous** is **saturated** (**no reduced Hb**).

5 Acid/Base balance

- H+ concentration affects the activities of almost all enzyme systems in the body.
- Therefore, changes in H+ concentration effectively alter all cell and body functions.
- The blood H+ concentration is normally maintained within tight limits around a normal value of about 0.00004 mEq/L (40 nEq/L).
- Normal variations are only about 3 to 5 nEq/L.
- Under extreme conditions, the H+ concentration can vary from as low as 10 nEq/L to as high as 160 nEq/L without resulting in death.

1) PH

- Because H+ concentration normally is low, and because these small numbers are heavy, it is usual to express H+ concentration on a logarithm scale using **pH units**.
- pH is related to the actual H+ concentration by the following formula:

$$pH = \log \frac{1}{H+} = -\log (H+)$$

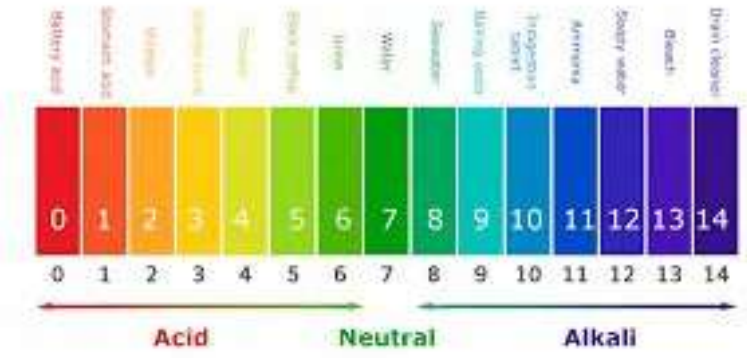
- For example, normal [H+] is 40 nEq/L (0.00000004 Eq/L).
- Therefore, the normal pH is as follows:

$$pH = -\log [0.00000004]$$

$$pH = 7.4$$

- From this formula:
  - o pH is inversely related to the H+ concentration
  - o A low pH corresponds to a high H+ concentration
  - o A high pH corresponds to a low H+ concentration

- The normal pH of arterial blood is 7.4
- The pH of venous blood and interstitial fluids is about 7.35 because of the extra amounts of carbon dioxide (CO2) released from the tissues to form H2CO3 in these fluids.
- The lower limit of pH at which a person can live more than a few hours is about 6.8, and the upper limit is about 8.0
- **Acidemia** the pH falls significantly below the normal pH of arterial blood 7.4
- **Alkalemia** when the pH rises above 7.4



pH and H+ Concentration of Body Fluids

	H+ Concentration (mEq/L)	pH
Extracellular Fluid		
• Arterial blood	• 4.0×10^{-5}	• 7.40
• Venous blood	• 4.5×10^{-5}	• 7.35
• Interstitial fluid	• 4.5×10^{-5}	• 7.35
Intracellular fluid	1×10^{-7} to 4×10^{-7}	6.0–7.4
Urine	3×10^{-2} to 1×10^{-4}	4.5–8.0
Gastric HCl	160	0.8

2) *Defending against Changes in H+ Concentration*

- The hydrogen ion concentration [H^+] in the blood is determined by the relationship between dissolved carbon dioxide, which is dependent on the Pco2, and the bicarbonate ion concentration [HCO_3^-]
- This relationship is expressed in the **Henderson-Hasselbalch equation**:

$$pH = pK + \log \frac{(HCO_3^-)}{0.03 P_{CO2}}$$

- where pK is the dissociation constant (6.1) for the overall reaction between carbon dioxide and H2O and the factor 0.03 is the solubility coefficient for carbon dioxide in the plasma

The H + concentration in the body fluids is regulated by

Chemical acid-base buffer systems of body fluids

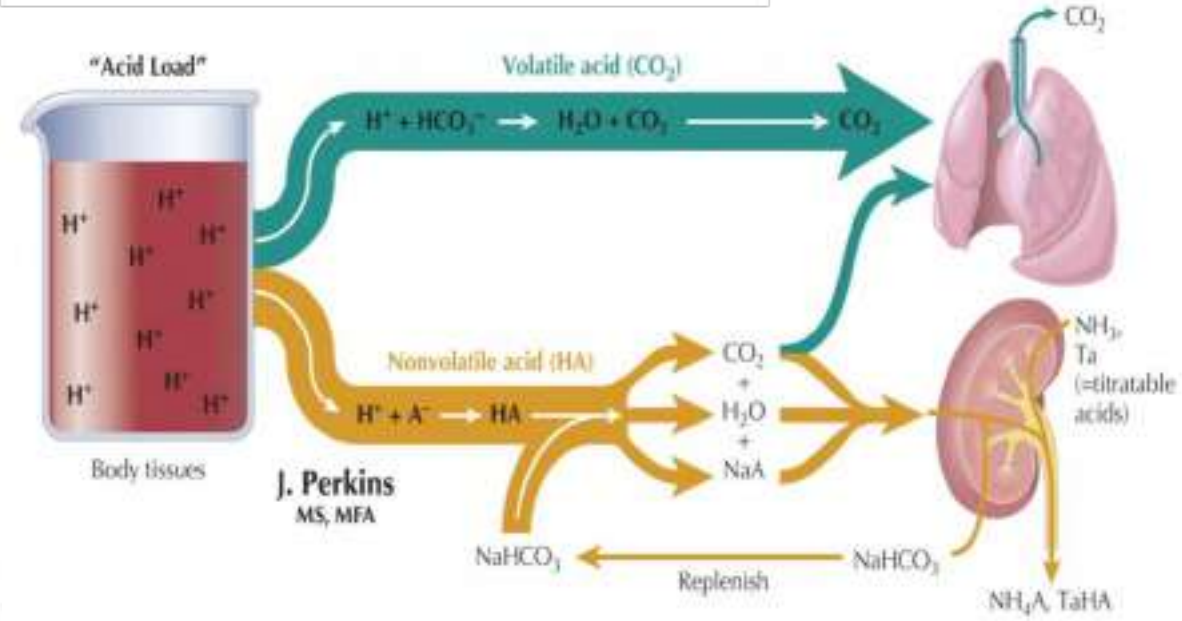
- Combine immediately with an acid or a base to prevent excessive changes in H+ concentration.
- In case of change in H+ concentration, react within seconds to minimize these changes.
- Do not eliminate H+ from or add H+ to the body but only keep them tied up until balance can be re-established.

Respiratory Center

- Acts within a few minutes regulate the removal of CO2 (and, therefore, H2CO3) from the extracellular fluid.

The Kidneys

- Can excrete acidic or alkaline urine, by re-adjusting the extracellular fluid H+ concentration toward normal during acidosis or alkalosis.
- The **most powerful** of the acid–base regulatory systems, although of its slow response over period of hours to several days.



3) Respiratory regulation of acid-base balance

- Control of extracellular fluid CO2 concentration by the lungs is the second line of defense against acid–base disturbances.

⇧ An increase in ventilation → Eliminates CO2 from extracellular fluid → ⇩ Reduces the H+ concentration.

⇩ Decreased ventilation → ⇧ increases CO_2 and H+ concentrations in the extracellular fluid.

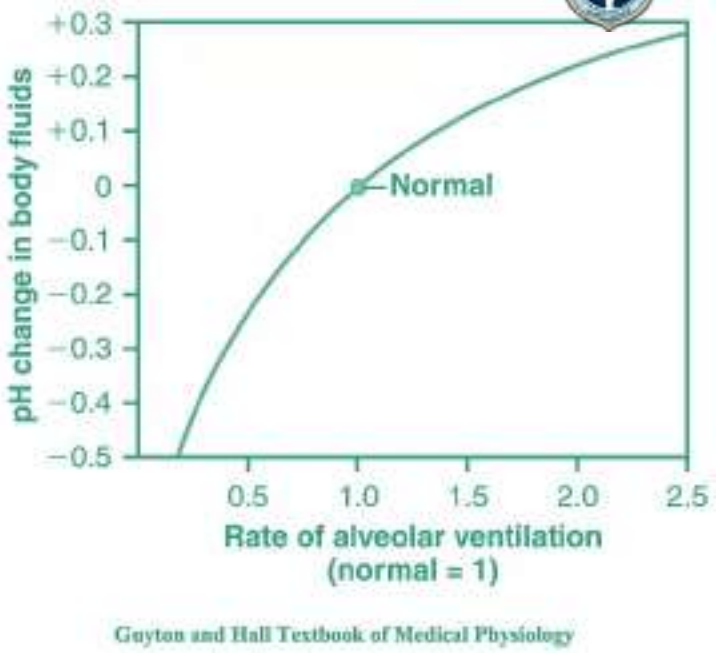
Pulmonary expiration of CO2 balances metabolic formation of CO2

- CO 2 is formed continually in the body by intracellular metabolic processes.
- After it is formed, it diffuses from the cells into the interstitial fluids and blood, and the flowing blood transports it to the lungs, where it diffuses into the alveoli and then is transferred to the atmosphere by pulmonary ventilation.
- About 1.2 mol/L of dissolved CO 2 is normally in the extracellular fluid, corresponding to a Pco2 of 40 mm Hg.
- Changes in pulmonary ventilation or the rate of CO2 formation by the tissues can change the extracellular fluid Pco2

Increasing Alveolar Ventilation Decreases Extracellular Fluid H+ concentration and Raises pH

- If the metabolic formation of CO2 remains constant, the only other factor that affects Pco2 in extracellular fluid is the rate of alveolar ventilation.
- Increasing alveolar ventilation to about twice normal raises extracellular fluid pH by about 0.23 (from pH of 7.4 to 7.63)

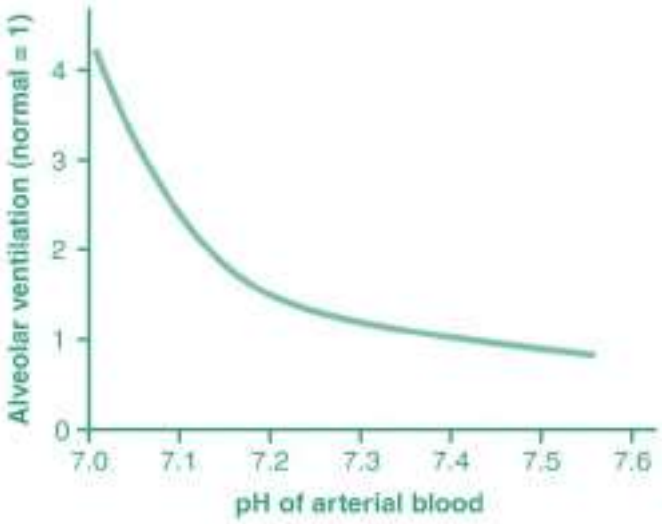
- A decrease in alveolar ventilation to one-fourth normal reduces the pH by 0.45. That is, if the pH is 7.4 at a normal ventilation to one-fourth normal reduces the pH to 6.95 The alveolar ventilation rate can change
- markedly, from as low as 0 to as high as 15 times normal, the pH of the body fluids can be changed markedly by the respiratory system.



Guyton and Hall Textbook of Medical Physiology

The H + concentration affects the rate of alveolar ventilation

- Reduction of the pH from the normal value of 7.4 to the strongly acidic value of 7.0 increases alveolar ventilation rate four to five times normal A rise in plasma pH
- above 7.4 causes a decrease in the ventilation rate
- The change in ventilation rate per unit pH change is much greater at reduced levels of pH compared with increased levels of pH
- Respiratory responses to metabolic alkalosis are limited by **hypoxemia** associated with reduced alveolar ventilation



Guyton and Hall Textbook of Medical Physiology

Feedback Control of H+ Concentration by the Respiratory System

The respiratory system acts as a typical negative feedback controller of H + concentration



Buffering power of the respiratory system

- Respiratory control cannot return the H+ concentration all the way back to normal when a disturbance outside the respiratory system has altered the pH
- The respiratory mechanism for controlling H+ concentration is approximately 50% to 75% effective, that is, if the pH is suddenly decreased by adding acid to the extracellular fluid, and the pH falls from 7.4 to 7.0, the respiratory system can return the pH to a value of about 7.2 to 7.3. This response occurs within 3 to 12 minutes
- The overall buffering power of the respiratory system is one to two times as great as the buffering power of all other chemical buffers in the extracellular fluid combined

4) *Respiratory acid-base disorders*

Impairment of Lung Function Can Cause Respiratory Acidosis , such as severe emphysema causes:

- Decreases the ability of the lungs to eliminate CO2, which causes a buildup of CO2 in the extracellular fluid and a tendency toward respiratory acidosis.
- Impaired ability to respond to metabolic acidosis because the compensatory reductions in Pco2 that would normally occur by means of increased ventilation are blunted.

Respiratory Acidosis

⇩Decreased pulmonary ventilation → ⇧increased Pco2 →⇧increase in H_2CO_3 and H^+concentrations → **Acidosis**

Causes

1. Damage to the respiratory center in the medulla oblongata
2. Obstruction of the passageways of the respiratory tract
3. Pneumonia, emphysema, or decreased pulmonary membrane surface area
4. Any factor that interferes with exchange of gases between the blood and alveolar air

Compensatory responses

1. The buffers of the body fluids
2. The kidneys, which require several days to compensate for the disorder

Respiratory Alkalosis

⇧ Increased pulmonary ventilation → ⇩ Decreased Pco2 → ⇩ Decrease in H_2CO_3 and H^+ concentrations → **Alkalosis**

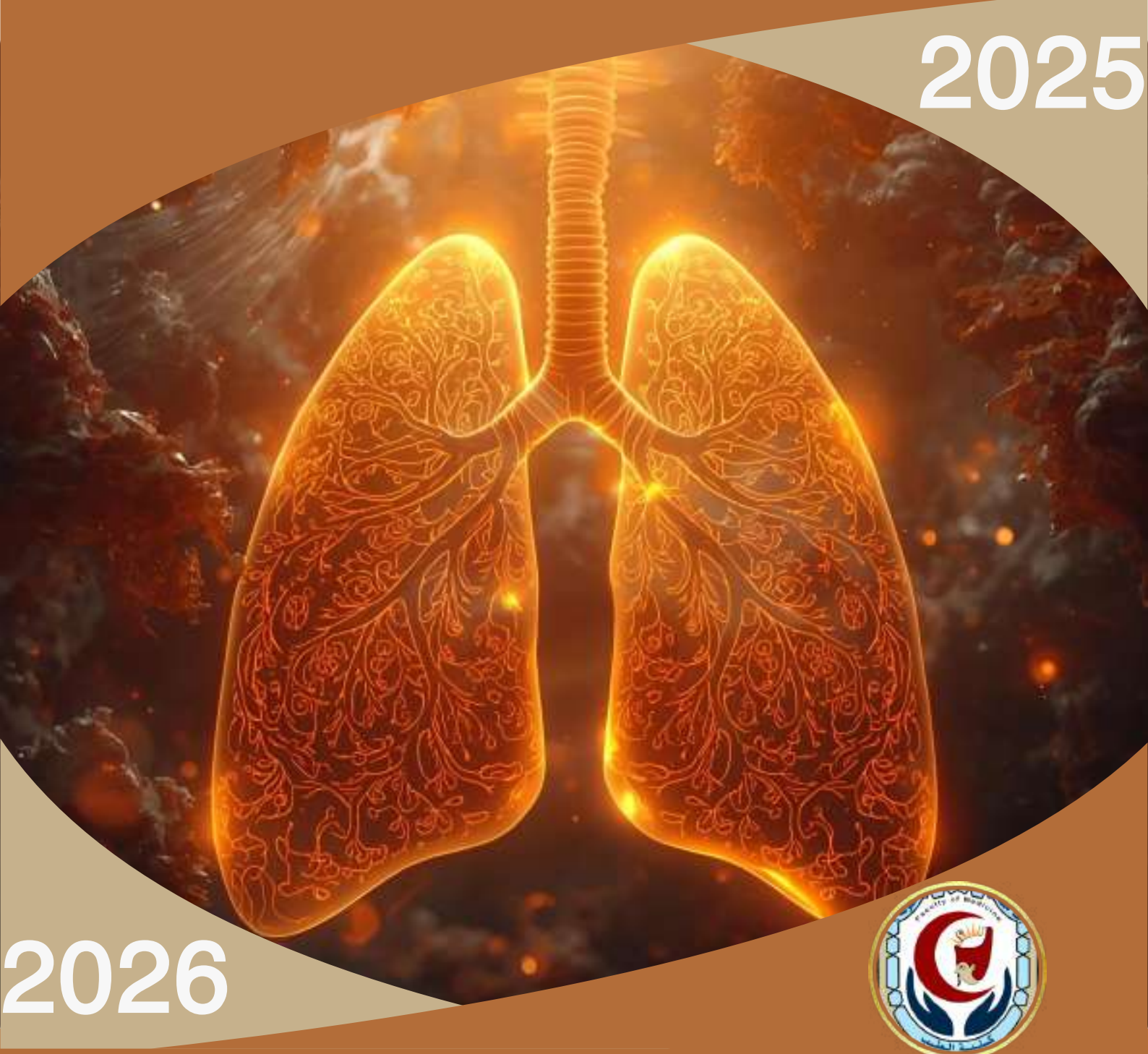
- **Causes**

1. Psychoneurosis
   can sometimes accelerate breathing to the point where an individual becomes alkalotic.
2. High altitude
   The low oxygen content of the air stimulates respiration, which causes loss of CO_2 and development of mild respiratory alkalosis

Compensatory responses

1. The chemical buffers of the body fluids
2. The ability of the kidneys to increase HCO 3excretion.

2025



2026

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MEDICINE

Respiratory system

Biochemistry

RPS-314

page	lecture
1	Arterial blood gas analysis

1 Arterial blood gas analysis

basic principles

➢ Arterial blood gases (ABG) are obtained for two basic purposes:

1.To determine oxygenation. 2.To determine acid-base status.

➢ When is an ABG required?

1.To establish a diagnosis.
2.To assess illness severity.
3.To guide and monitor treatment.

➢ Definitions

- **PH**:itis the negative logarithm to base ten of hydrogen ion concentrations i.e.number of gram ions of H+ in a liter (represents the free hydrogen ions concentration).

•If[H+] is high, the solution is acidic; pH < 7
•If[H+] is low, the solution is basic or alkaline ;pH > 7

- **Acids** are H+ donors
- **Bases** are H+ acceptors, or give up OH- in solution.

➢ Acids and bases can be:

- **Strong** :– dissociate completely in solution **Examples:** HCl, NaOH .
- **Weak** :– dissociate only partially in solution **Examples :** Lactic acid, carbonic acid .
- **Buffer**: a substance that reacts with acids and bases to maintain a neutral environment of stablePH.

➢ The Body and pH

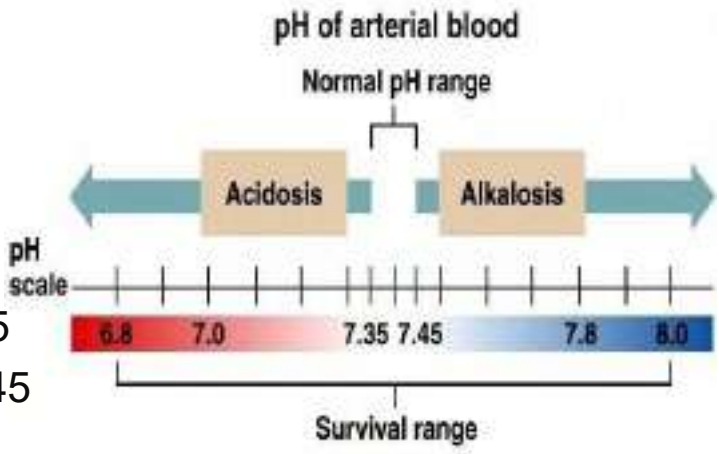
- HomeostasisofpHistightly controlled

>>> Extracellular fluid PH = 7.4

>>> Blood PH = 7.35 – 7.45

✓ Acidosis (academia) below 7.35

✓ Alkalosis (alkalemia) above 7.45



Ventilation is the process of exchanging oxygen and carbon dioxide, which is
essentially breathing. Oxygen comes in to the body via the airway, it's offloaded onto the red blood cells while carbon dioxide diffuses across the membrane into the alveoli and is then exhaled. You breathe in oxygen andexhale CO2.

Diffusion involves substances moving across concentration gradients from areas of higher
concentration to areas of lower concentration. This is the process involved with gas exchange.

Perfusion is the body process of supplying oxygenated blood to the cells and is
reliant on adequate cardiac output in order to be optimal.

Pulmonary gas exchange refers to the transfer of O2 from the atmosphere to the bloodstream (oxygenation) and CO2 from the bloodstream to the atmosphere (CO2 elimination) Arterial blood gases help us to assess the effectiveness of gas exchange by providing measurements of the partial pressure of O2 and CO2 in arterial blood.

Partial pressure – contribution of one individual gas within a gas mixture.Gases move from areas of higher partial pressure to lower partial pressure. At the alveolar- capillary membrane, air in alveoli has a higher PO2 and lower PCO2 . O2 molecules move from alveoli to blood and CO2 move from blood to alveoli.

NOTE

-PO2= partial pressures of O2
-PaO2=partial pressures of O2 **inarterial blood**
-PCO2= partial pressures of CO2
-PaCO2= partial pressures of CO2 **in arterial blood**

➢ Carbon dioxide elimination:

◆ PaCO2 is determined by alveolar ventilation and the level of ventilation is adjusted to maintain PaCO2 within tight limits (ref. range: PaCO2= 35-45 mm Hg).

◆ Increased PaCO2 (hypercapnia) implies reduced alveolar ventilation

Ventilation is regulated by an area in brainstem called respiratory center.This area contains receptors that sense PaCO2 and connect with the muscles involved in breathing. If PaCO2 is abnormal ,the respiratory center adjusts the rate and depth of breathing accordingly.

➢ ✓ Oxygenation:

Almost all O2 molecules in blood are bound to Hb.The amount of O2 in blood depends on the twofactors:

1. **Hb concentration**-how much O2 blood has the capacity to carry.
2. **Saturation of HbwithO2**(SO2)-the percentage of available binding sites on Hb occupied by O2 molecules.

SO2 – saturation in (any) blood .
SaO2 – saturation in arterial blood (80-100 mm Hg) .

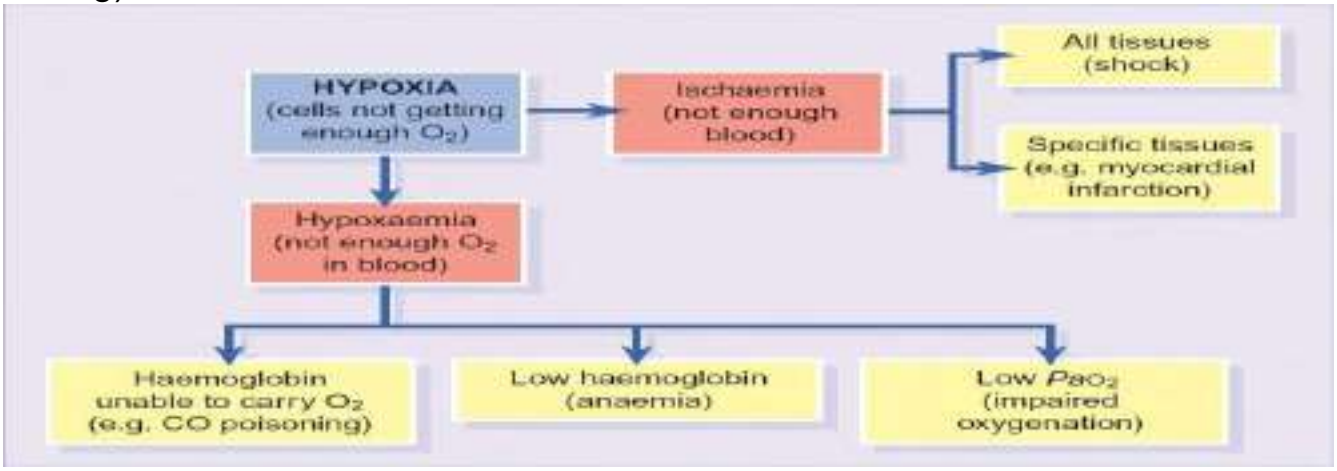
With a normal PaO2 (80-100 mm Hg), Hb is maximally saturated (SaO2>95%). This means blood has used up its O2-carrying capacity and any further increase in PaO2 will not significantly increase arterial O2 content.

✓ The term hypoxia is a reduced O2 delivery to tissues.

✓ the term hypoxemia is a reduced O2 content (PaO2) in arterial blood .It may result from impaired oxygenation, low haemoglobin (anaemia) or reduced affinity of haemoglobin for O2 (e.g. carbon monoxide) .

Impaired oxygenation refers to hypoxaemia resulting from reduced transfer of

O2 from lungs to the bloodstream. It is identified by a low PaO2 (<10.7 kPa; <80 mmHg).



Arterial blood gas analysis:

acid-base balance

✓ pH - measurementof acidity or alkalinity, based on hydrogen (H+) ions present.

✓ Human blood normally has a pH of 7.35–7.45 (H+ = 35–45 nmol/L) and, therefore, it is slightly alkaline. If blood pH is below the normal range (<7.35), there is an acidaemia. If it is above the normal range (>7.45), there is an alkalaemia.

An acidosis is any process that lowers blood pH, whereas an alkalosis is any process that raises blood PH.

Maintaining Acid-base Balance

The respiratory buffer response :

✓The lungs are responsible for removing CO2. Pa CO2,the partial pressure of carbon dioxide in blood, is determined by alveolar ventilation. If CO2 production is altered, we adjust our breathing to exhale more or lessCO2, as necessary, to maintain Pa CO2 within normal limits.

✓CO2 is carried in the blood to the lungs. In blood CO2combines with water to form carbonic acid (H2CO3)The more CO2 is added toblood, the more H2CO3 is produced, which dissociates to release H+.

$$H_2O + CO_2 \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$$

↑ H_2CO_3 level in blood

↓ blood pH

↑ the rate and depth of lung ventilation until the appropriate amount of CO_2 has been re-established

✓Activation of the lungs to compensate foranimbalance starts to occur within 1-3minutes.

The renal buffer response :

Thekidneys are responsible for excreting metabolic acids. They secrete H+ions into urine and reabsorb HCO3− from urine. HCO3− is a base (and thereforeaccepts H+ ions), so it reduces the concentration of H+ ions in blood. The kidneyscan adjust urinary H+ and HCO3− excretion in response to changes in metabolicacid production .

$$H_2O + CO_2 \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$$

↓ blood pH	↑ blood pH
kidneys will compensate by retaining HCO3	kidneys will compensate by excreting HCO3
↑ HCO3 level	↓ HCO3 level

➢ Compensatory response or regulation of pH

-By3mechanisms:

- **Chemical buffers:**
  React instantly to compensate for the addition or subtraction of H+ ions
- **CO2 elimination:**
  Controlled by the respiratory system
  Change in pH result in **>>>** change in PCO2 within minutes
- **HCO3 - elimination:**
  Controlled by the kidneys
  Change in pH result in **>>>** change in HCO3-
  It takes hours to days and full compensation occurs in 2-5 days

➢ Rates of correction :

- Buffers functionalmost instantaneously .
- Respiratory mechanisms take several minutes to hours.
- Renal mechanisms may take several hours to days.

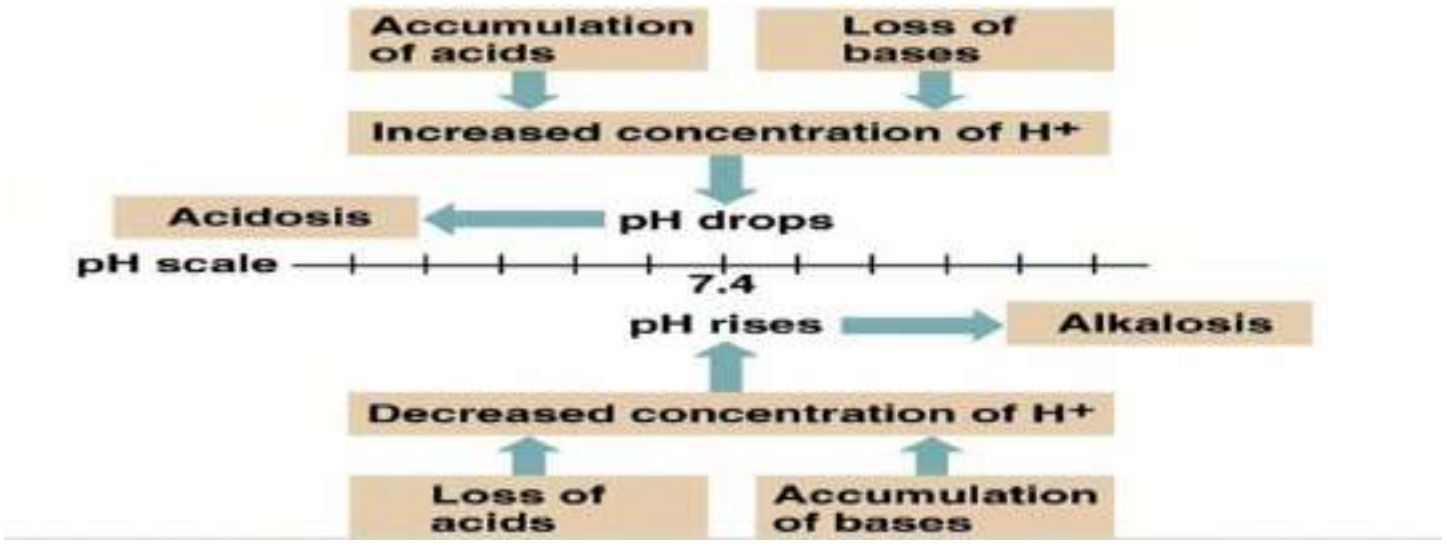
➢ Disturbances of acid-base balance:

1. **Metabolic**
2. **Respiratory**

• **Metabolic processes** are those that primarily alter the HCO3 concentration in the blood. A decrease in serum HCO3 (base) leads to a metabolic acidosis, while an increase in serum HCO3 leads to a metabolic alkalosis.

• **Respiratory processes** alter the pH by changing the CO2 levels.

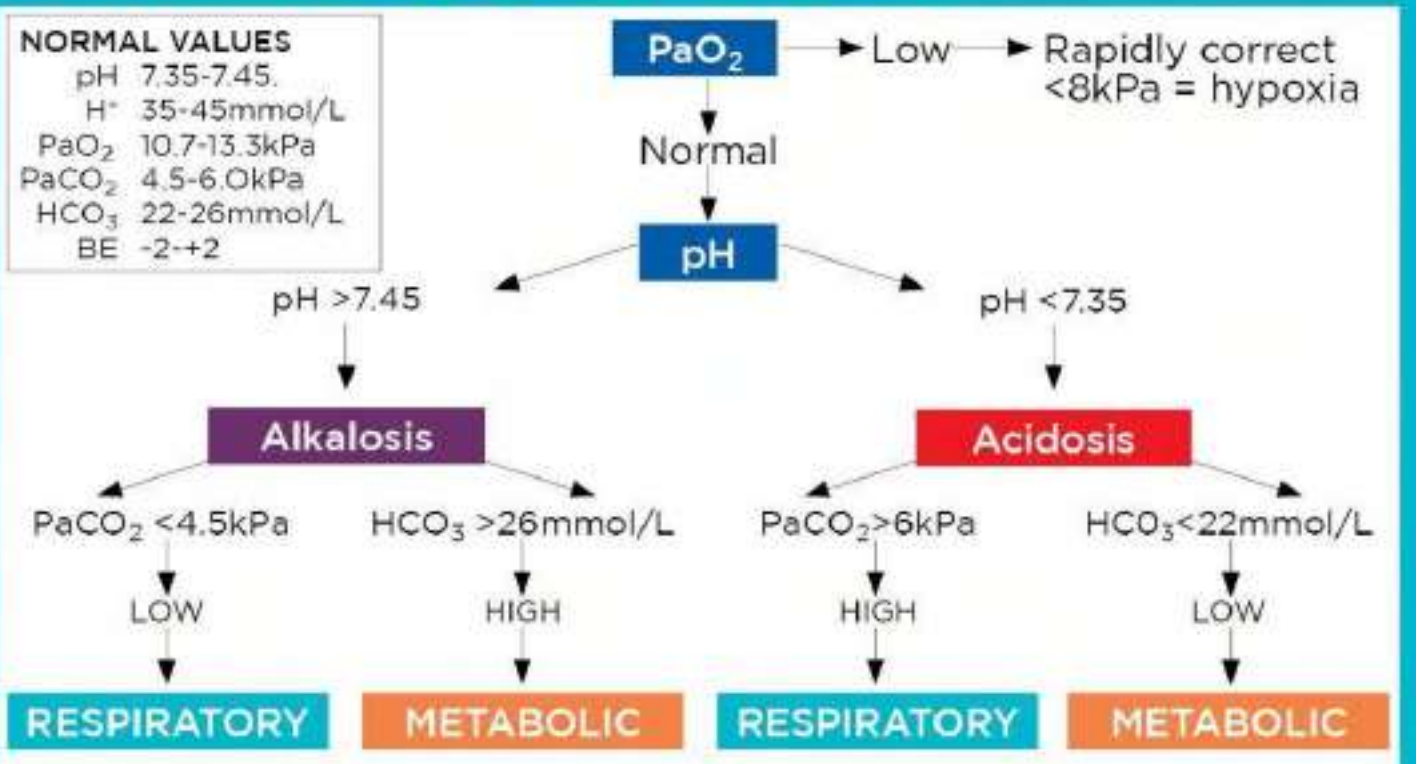
CO2 is a respiratory acid. CO2 accumulation causes an acid state in the blood (through carbonic acid), and as respirations (respiratory rate and/or tidal volume) increase, the body eliminates more CO2 (acid) and is left with a respiratory alkalosis. A decrease in ventilation leads to retention and increased levels of CO_2 and thus a respiratory acidosis .



➢ Steps for ABG analysis :

1. What is the pH? Acidosis or alkalosis ?
2. Respiratory or metabolic(Co2 ,Hco3) 3.
Is there appropriate compensation ?

Fig 1. **Arterial blood gas analysis flowchart**



Maintaining acid-base balance

Compensation

When a patient develops an acid – base imbalance, the body attempts to compensate. Lungs and kidneys are the primary buffer response system in the body. The body tries to overcome either a respiratory or metabolic dysfunction in an attempt to return pH into normal value.

The patient can be

1. uncompensated
2. partially compensated

} pH outside normal range

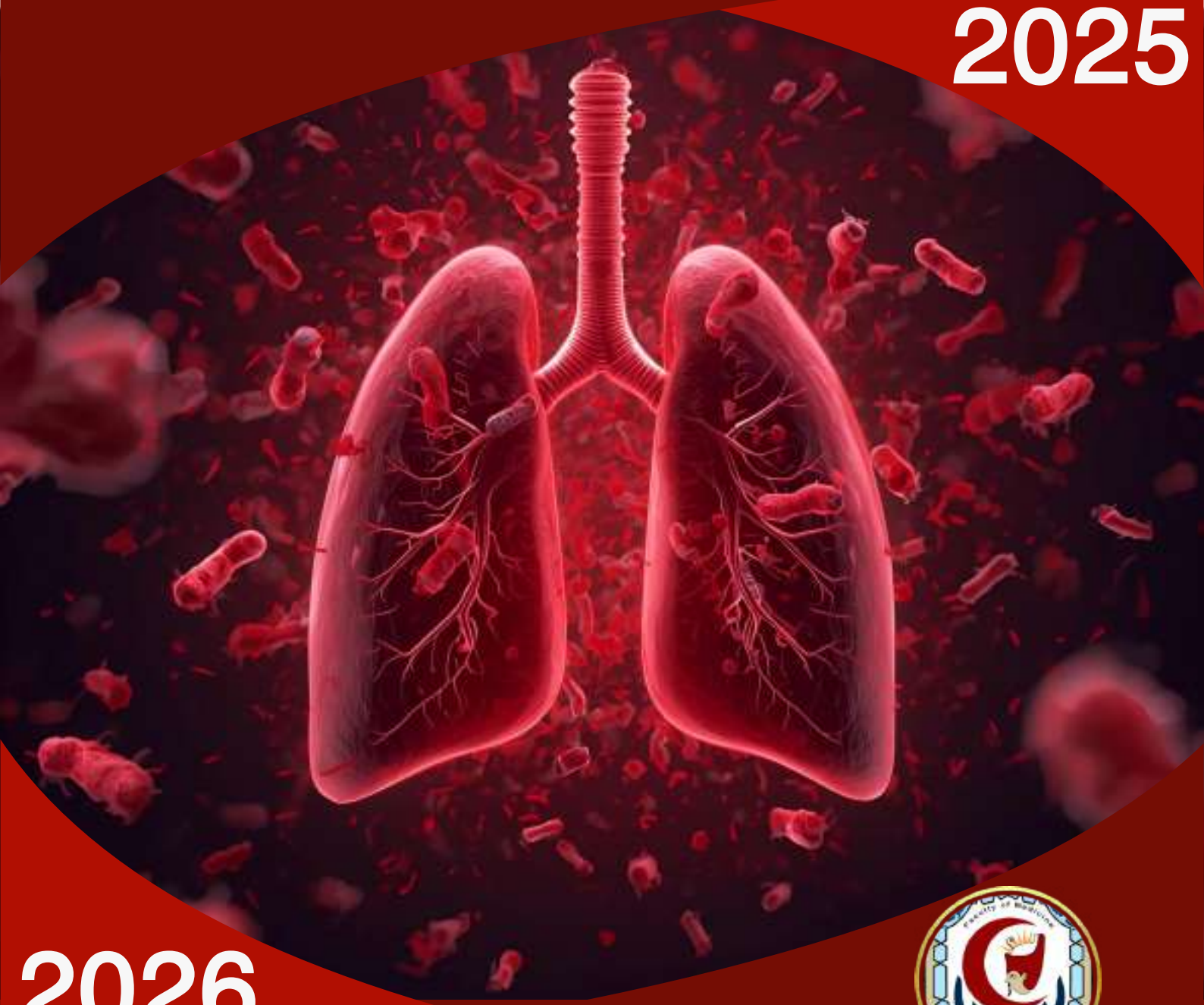
3. fully compensated — pH within normal range

Without compensation	pH	PaCO2	HCO3
Respiratory acidosis	↓	↑	normal
Respiratory alkalosis	↑	↓	normal
Metabolic acidosis	↓	normal	↓
Metabolic alkalosis	↑	normal	↑

Partially compensated status	pH	PaCO2	HCO3
Respiratory acidosis	↓	↑	↑
Respiratory alkalosis	↑	↓	↓
Metabolic acidosis	↓	↓	↓
Metabolic alkalosis	↑	↑	↑

Fully compensated status	pH	PaCO2	HCO3
Respiratory acidosis	normal but <7,40	↑	↑
Respiratory alkalosis	normal but >7,40	↓	↓
Metabolic acidosis	normal but <7,40	↓	↓
Metabolic alkalosis	normal but >7,40	↑	↑

2025



2026

SOHAG UNIVERSITY
FACULITY OF
MEDICINE

Respiratory System

Microbiology & immunity

NEU-316

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1 Bacteria causing upper respiratory tract infections

ILOs:

- Describe the etiology of the common upper respiratory tract infections caused by bacteria.
- Describe the major clinical manifestation.
- Describe the pathogenesis, diagnosis and treatment of infections

Sore throat, Tonsillitis and Pharyngitis

are **caused** by:

✓ Streptococcus pyogenes.
✓ Corynebacterium diphtheriae.
✓ H. influenzae type b. Vincent
✓ angina. Bordetella pertussis

Acute Otitis Media:

✓ 50% of infections are due to viruses, however, it is **difficult clinically** to **differentiate** viral from bacterial causes. **Bacterial causes** are: Str. pneumoniae,
✓ H. influenzae, Pseudomonas aeruginosa, Staph.aureus, Proteus species, Str. pyogenes, E. coli, Klebsiella species, Moraxella catarrhalis. Anaerobic organisms and Fungi.

Corynebacterium

- **Corynebacterium diphtheriae** is the most important member of the group. It produces a powerful exotoxin that causes **diphtheria** in humans.
- **Other corynebacteria species** (termed coryneforms or diphtheroids) **resemble C. diphtheriae** in morphology and occur as **normal flora** on the skin and mucous membranes, **but** some of them are **implicated in opportunistic infections.**

Corynebacterium diphtheriae

C. diphtheriae causes diphtheria which is an exclusively human disease.

Morphology:

- Gram positive rods with a **swollen end** "**club shaped**". They lie in **small** groups at **acute angles** or **parallel** to each other, "**chinese-letter arrangement**".
- The **bacilli** are **beaded** due to the metachromatic granules that contain polyphosphate and are seen better when **stained** with methylene blue.

Cultural characters:

- Facultative anaerobes;growbeston **Loeffler's serum** at37°C, can grow on blood agar.
- On blood tellurite medium(**selective differential medium**),colonies appear grey to black. They are **catalase positive**.

Diphtheria toxin:

- C. diphtheriae is a **powerful exotoxin** producer. The **toxin** is the **virulence factor**. **Only diphtheria** strains **lysogenized** by a prophage are toxigenic and virulent. The **amount of toxin produced** depends on the **extracellular iron concentration.**
- Diphtheria toxin **inhibits protein synthesis**, resulting in the **necrotic and neurotoxic** effects of the toxin.
- The toxin is heat labile, highly toxic, highly antigenic and **stimulates** the **production of antibodies** (antitoxin) that **neutralize the toxin's activity**. Formalin treatment of the toxin produces a **toxoid** that retains the antigenicity but **not the toxicity** of the molecule. **The toxoid is used for immunization.**

Toxigenicity testing of C.diphtheriae isolates:

Itis essential to demonstrate toxin production by diphtheria isolates **obtained** from throat swabs of **cases or carriers;** this is done by:

1. **Elek's test** : precipitation lines formed with the toxigenic strain. Absence of precipitation lines indicates that the strain is non-toxigenic.
2. **PCR** : for detection of diphtheria toxin genes.
3. **ELISA** : to detect diphtheria toxin from clinical isolates.
4. **Immunochromographic strip assay** allows detection of diphtheria toxin in few hours.
5. **Tissue culture cytotoxicity assay**: On incorporation of the test strain into an agar overlay of cell culture monolayer. The produced toxin diffuses into the cells below and kills them.

Diphtheria

- **Diphtheria** has virtually disappeared in **developed** countries following mass immunization but is still endemic in **many regions** of the world. **Tonsillar**
- **diphtheria** is the **commonest type** and is **transmitted** by airborne droplets. **Conjunctival or skin diphtheria** is rare and spreads by contact.
- In **tonsillar diphtheria** the **organism** multiplies locally **releasing the toxin causing inflammation** of the throat, local necrosis with fibrinous exudate resulting in formation of a spreading greyish white, tough, adherent pseudo-membrane. **Any attempt to remove this membrane** causes bleeding.
- The **released exotoxin** diffuses to the blood stream causing toxaemia and affection of the **heart, kidney, liver and nervous tissue.**

Clinical findings:

- The **patient presents** with mild fever, sore throat and general ill health. **Cervical lymph nodes** are enlarged. The **tonsils** are **covered** with a greyish membrane which **may extend** to the **posterior pharyngeal wall or larynx.** This may cause **laryngeal obstruction** and **suffocation**. **Generalized symptoms** occur due to **production and absorption** of the **toxin**. **Nerve involvement** may lead to
- paralysis of muscles of the **soft palate** and **pharynx** causing **difficulties in swallowing and speech**, **peripheral neuritis** or **paralysis of limbs**. Myocarditis accompanied by **arrhythmias and circulatory failure.**

Diagnosis ofa case:

Ifthe **clinical picture** is strongly suggestive of **diphtheria**, **specific treatment** with **antitoxic serum** should be started immediately to **neutralize** the **toxin before irreversible damage occurs**. The **clinician should not wait** for laboratory results which serve to confirm the clinical diagnosis. **Throat swabs** from the membrane are examined as follows:

- **Direct smears** are stained with gram and methylene blue. **Gram positive bacilli** with the **characteristic morphology** of diphtheria may be seen.
- **Direct detection** of diphtheria toxin **genes** by PCR in clinical specimens.
- **Cultures** are made on Loeffler's serum and blood tellurite media. These are examined after 12-18 hrs. Colonies are picked and stained with **gram** and **methylene blue.** The presence of **organisms morphologically** similar to diphtheria is reported as "diphtheria-like" organisms. A **blood agar plate** is also cultured to exclude Str. pyogenes tonsillitis.

The physician should be **notified** of the **high possibility of diphtheria** which is being confirmed further by **toxigenicity** tests as previously described.

Carriers of diphtheria in the **throat or nose** are diagnosed as in a case.

Treatment:

1. **Diphtheria antitoxic serum** should be **given without delay**, when there is a **strong clinical suspicion** of **diphtheria**. The **antitoxin** neutralizes the toxin **before** it causes **irreversible damage.** From 20,000 to 120,000 units are injected **intramuscularly** or **intravenously** after suitable precautions have been taken to rule out hypersensitivity to the animal serum.

To avoid such complications, **tests** for **sensitivity** must be performed before treatment with antitoxin as follows:

- **Very minute amounts of diluted antitoxin** (one drop or 0.02 ml) are **dropped** in the **eye or injected intradermally**. If **reaction occurs** (conjunctivitis or erythema within 20 min) **desensitization** is done by injecting the antitoxin, using **highly diluted slowly increasing doses at 20 minutes intervals**.
- To **prevent anaphylactic shock**, **pretreatment with cortisone** and **anti-histaminics** is recommended. An **epinephrine injection** should be **ready** for use.

2. **Antibiotics** e.g.penicillinanderythromycin aregivenin adequate doses, however,theyare **not a substitute for antitoxins**.They inhibit growth of the **organism** inthe **throat**,thus **reduce toxin production** and **decrease** the **incidence of carriers.**
3. **Respiratory support may be needed.**

Prophylaxis:

- **Active immunization:** Diphtheria can be **prevented** by proper active immunization with **diphtheria toxoid**:
  Such toxoids are **commonly combined** with **tetanus toxoid and pertussis vaccine (DTP)** and given intramuscularly to children at the **age of 2,4, 6 and 18 months.** A **booster dose** is given at **school age**. **Adults** should **receive a booster every 10 years**, but it **should contain diphtheria and tetanus toxoids Td only.**
- **Close contacts of a case** of diphtheria should **receive a booster dose** of **diphtheria toxoid** and a course of erythromycin or an **injection of long acting penicillin**. **Antitoxic serum is no longer indicated for contacts.**

Haem ophilus

Members of this genus **require certain growth factors** present in **blood** (haemophilic) for their growth.

I-Haemophilus influenzae:

Non-capsulated strains are often found in **normal throats**. **Capsulated strains** cause meningitis, epiglottitis and otitis mediain **children**.

Morphology:

Gram negative pleomorphic small cocco-bacilli. **Long filamentous** forms occur. **Many strains are capsulated.**

Cultural characters:

Facultative anaerobes require **heated blood** containing **media** for their growth. They **grow** on chocolate agar which **provides X factor (haematin) and V factor (NAD)** that are **essential for their growth**. **Around Staph, aureus colonies,** the colonies of H.**influenzae** are **much larger and denser**, since **staphylococci produce V factor** This is called **satellitism**.

Serological characters:

Smooth capsulated strains canbe **classified** into 6 types (a-f), **depending** on the capsular polysaccharide. **H. influenzae type b (Hib)** is the **most pathogenic.** The **capsule** is a major virulence factor.

Pathogenesis,virulence factors and diseases:

- The organism **enters** the **upper respiratory tract** by inhalation or droplets. The **polysaccharide capsule** is the **major virulence factor** that **facilitates invasion. Other factors** include **fimbriae** and **IgA protease** that **degrades secretory IgA and facilitates attachment** to **respiratory mucosa**, **resulting in colonization. Outer membrane proteins** and **lipopolysaccharide** contribute to **invasion**.
- **Capsulated strains** mainly Hib cause **invasive diseases** in **children below 5 years** including **meningitis, epiglottitis, septic arthritis, pneumonia, cellulitis and bacteraemia. Non- capsulated strains** cause **non-invasive diseases**; **otitis media** and **upper respiratory infections**, initiated by viral infections.

Diagnosis:

Specimensare pus, sputum or CSF.

1. **Smears** are stained with **gram** When **present** in **large numbers** in specimens, the organism can be **directly detected** by **Quellung reaction, immunofluorescence or by PCR.**
2. **Direct detection** of H. influenzae polysaccharide in **CSF** by **latex agglutination commercial kits.**
3. **Culture** on chocolate agar at 37°C in 5% CO2. Colonies are **identified** by; their **morphology**, **inability to grow** in absence of X-V growth factors, **serologically** with **specific antisera** or by **DNA probes**.

Treatment:

Cefotaxime given **intravenously**,or ceftriaxone are the **drugs of choice** for Hib meningitis.

Prevention:

- **Haemophilus b conjugated (Hib) vaccines containing** the capsular polysaccharide **conjugated** to a carrier protein is recommended for **children at 2, 4, 6 months** and at **12-15 months.** The vaccine has **reduced the incidence of meningitis** caused by this organism by 90% in immunized children. The **vaccine** may be given in **combination with DTP i.e.DTPH. Rifampicin** is giving
- **prophylactically** to non-immune children in **close contact** with patients **particularly** those with **meningitis**.

Bordetella pertussis

Bordetella pertussis is the most importantmember of the **genus Bordetella.** It **causes whooping cough** which is **mainly a disease of children.**

Morphology: Gramnegative capsulated cocco-bacilli.

Cultural characters:

Itgrows on**complexenriched media** e.g.Bordet-Gengoumedium(**potato-blood-glycerol agar**) or Regan Lowe medium (after 3- **charcoal-horse blood agar**)**Growth** occurs 7 days. **Colonies** are greyish-whitewitha **drop" shiny convex surface**, **"mercury** appearance.

Virulence factors and pathogenesis:

1. Filamentoushaemagglutinin(FHA) and fimbriae **mediates adherence** tothe **cilia of epithelial cells** ofthe **trachea and bronchi,** where theorganism **multiplies and colonizes,** interfering with **ciliary action.**
2. Pertussistoxin(PTx): **functions** as **adhesins helping colonization** andpartly **secreted acting** as a toxin, which, with the tracheal cytotoxin and endotoxin, **irritate and damage** the **ciliated** cellsoftherespiratorytractcausingthe **paroxysmal toxaemicstage.**
3. Adenylate cyclase **inhibits** bactericidal action of phagocytes, while **pertactin** (surface protein) **helpadhesion.**

Whooping cough

- **Infection** occurs by inhalation of droplets from **early cases**. After an **incubation period** of 2 weeks, the catarrhal stage develops with **mild coughing and sneezing. During this stage** (colonization), the **patient** is **highly infectious.**
- This is **followed** by **severe paroxysmal cough (toxemic stage)** characterized by a **series of hacking coughs** that end with an inspiratory "whoop" that may be followed by **vomiting**, **cyanosis and convulsions**. It **lasts for 1-4 weeks** and is associated with **marked leucocytosis and absolute lymphocytosis.**
- The disease may be **complicated** by **pneumonia, otitis media, sub-conjunctival haemorrhages** and **rarely encephalopathy or seizures**.
- **B. pertussis** is a **common cause** of prolonged cough in adults.

Diagnosis: **Samples** are nasopharyngeal swabs or saline nasal wash.

- **Immunofluorescence** may be used to **identify** the organism in the samples directly. **PCR** may be used when available.
- **Isolation of the organism** is positive in the catarrhal stage. Samples are inoculated on the media and incubated in a **humid atmosphere** for 7 days. Colonies are **identified** by **immunofluorescence** or **slide agglutination** with specific antisera or PCR
- **Serologic detection of antibodies**, which **start to appear 3 weeks after onset of symptoms** by agglutination or ELISA.

Treatment: **Erythromycin** is the drug of choice for treatment and prophylaxis.

Prophylaxis:

1. A **heat killed vaccine** preparedfrom capsulatedstrains of B.pertussisis **given** to **children** duringthe firstyearof life in **combination** with **diphtheria and tetanus toxoid i.e. DTP vaccine**.**Adults should not receive this pertussis vaccine** sinceit **may cause encephalopathy** if **given after 6 years of age.**
2. **Acellular vaccines** have atleasttwoofthefollowing **antigens**:Inactivated pertussistoxin which isthe **main immunogen in the vaccine**,**FHA, pertactin**

and **2 types of fimbriae.** These **vaccines** have fewer side effects than **the killed** vaccine. The **vaccine** is given to **children with diphtheria and tetanus toxoids (DTaP)** at **2, 4, 6 months** and **boosters at 15-18 months** and **at 4-6 years.** In 2013, it was recommended by the **Advisory Committee** on **Immunization Practices** (**ACIP**) that **all adolescents and adults** as well as **pregnant women should receive a booster dose of DTaP.**

Vincent Angina

- In **certain conditions**which **cause injury or devitalization** of the **mucous membrane** e.g. nutritional deficiency, stasis ulcers, human bites or herpes simplex lesions; the **normal spirochaetes** of the **mouth** together with **anaerobic cigar-shaped fusiform bacilli (fusobacteria) multiply** in the lesions **causing acute ulcerative gingivostomatitis. Ulcers** may become **covered** with a
- **pseudomembrane** containing **pus cells and necrotic tissues,** on the **tonsils and gums**, the condition is called "Vincent angina" and it **should be differentiated from** diphtheria and follicular tonsillitis.

Diagnosis:

Smears prepared from the pseudomembrane show **gram negative spirochaetes** in **large numbers** associated with **fusiformgram-negative bacilli (fusobacteria**). **Pus cells** and other organisms are also found.

2 Viral Infections of the respiratory tract

ILOs:

- Enlist different viruses causing respiratory tract infections.
- Identify the common viral etiological agents causing RT infections.
- Discuss the pathogenesis, signs and symptoms of the viral RT infections.
- Discuss the laboratory diagnosis of common viral RT infections
- Discuss the pathogenesis and clinical presentation of the three types of Influenza & Corona viral infections.

Causes of viral respiratory tract infection

1. **Influenza viruses:** causing influenza, commonly known as "the flu"
2. **Parainfluenza viruses:** major cause of respiratory infections in **infants** and **young adults** as common cold, croup, and bronchitis.
3. **Rhinoviruses:** causing **upper respiratory tract infections especially** common cold.
4. **Coronavirus:** cause illness **ranging** from the common cold to more severe diseases such as **Middle East Respiratory Syndrome** (MERS-CoV) , **Severe Acute Respiratory Syndrome** (SARS-CoV) and COVID-2019
5. **Respiratory Syncytial Virus (RSV):** is the **main cause** of **bronchiolitis** and **pneumonia** in infants and cause **mild upper respiratory tract infection** in older children.
6. **Adenoviruses:** may be asymptomatic or causing acute respiratory disease and pharyngeal infection.

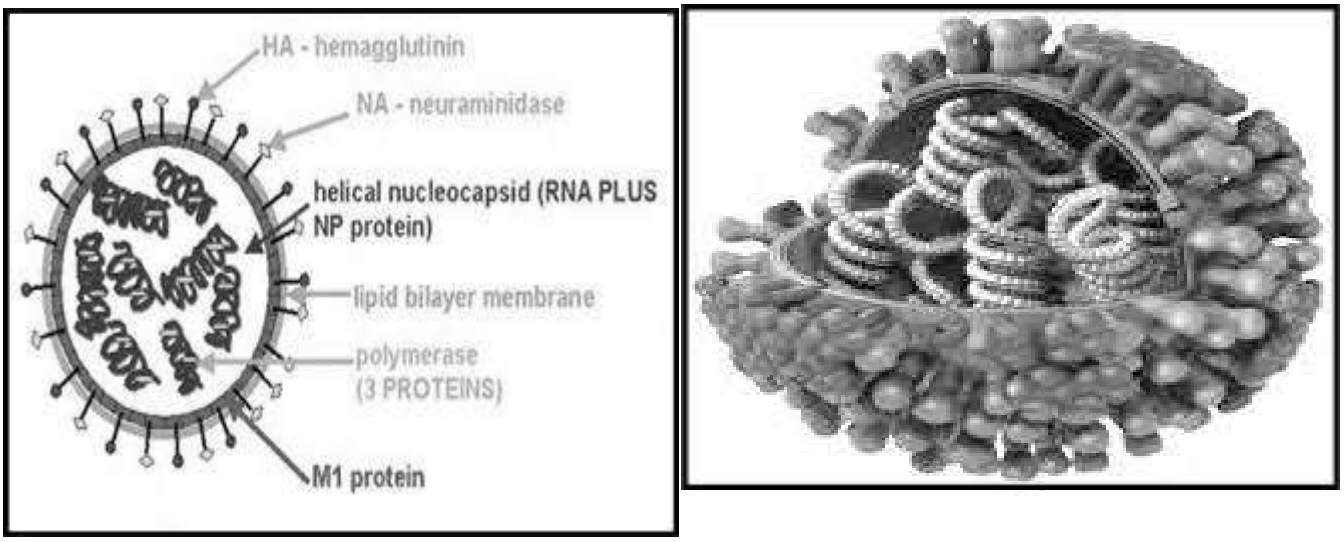
Orthomyxovi ridae (Influenza viruses)

- (**Ortho**= direct and **Myxo**= mucous; viruses in this family has direct affinity for **mucous membranes**).
- They cause INFUENES which is an acute respiratory disease.
- There are **three types** of influenza virus: Influenza A virus, Influenza B virus, Influenza C virus .

1. **Influenza A:** causes worldwide influenza epidemics **every 10-20 years** (pandemics) and **outbreaks every year**. It infects **humans, birds, pigs** and **horses**.
2. **Influenza B:** causes outbreaks, but **less often** than influenza A. (**infectshuman only**)
3. **Influenza C:** causes **mild respiratory disease**. infects human & pigs.

Characters:

- Spherical or filamentous, **80-120 nm** in diameter.
- **-ve sense ssRNA enveloped** viruses with helical symmetry.
- Has segmented genome (**7-8 segments**) that promotes genetic **diversity** caused by reassortment (in influenza A) in addition to **diversity** caused by mutation.
- The **helical nucleocapsid** is surrounded by a lipid envelope which gives it a spherical shape. **Two** glycoproteins are inserted in the envelope:
  a. **Hemagglutinin(HA)** binds to the **cellsurfacereceptor** (sialic acid) to **initiate infection**
  b. **Neuraminidase (NA) facilitates** virus release from infected cells



(Structure of Influenza viruses)

Types of influenza virus:

Influenza viruses areclassified as type A, B, or C by their **nucleoproteins (NP)** and **matrix proteins (M1).**

Subtypes ofinfluenza A viruses:

- **Influenza A viruses** are further classified, **based** on the viral surface proteins hemagglutinin **(HA or H)** and neuraminidase (**NAorN**).
- **16 H subtypes** (or **serotypes**) and **9 N subtypes** of influenza A virus have been identified
- **Currently** the most famous subtypes are:

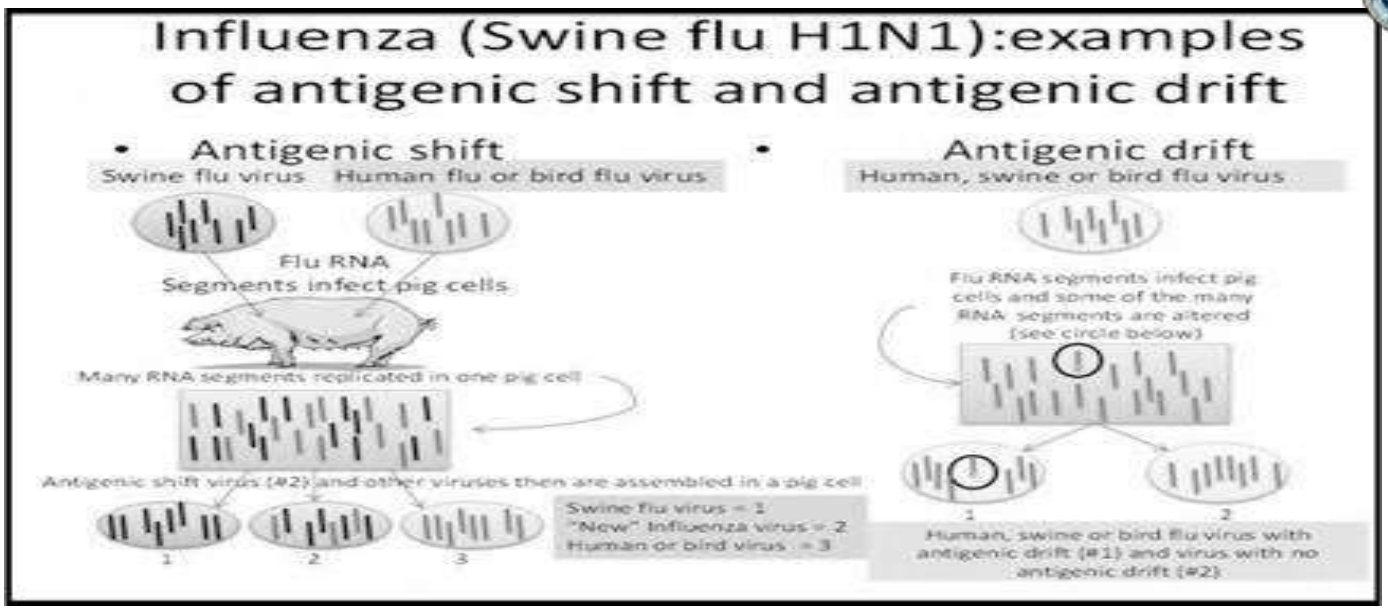
1. A **(H1N1)** circulating in **humans** caused the **2009** influenza **pandemic (swineflu)**.
2. A **(H5N1)** circulating in **birds** causing **avian flu** and **infecting humans** who closely handle infected birds.

Antigenic variations of influenza viruses :

Antigenic drift	Antigenic shift
• Theseare **minor** changesthat occur due to point mutations in **viralgenome** lead to **changes** in surface HAand NA result in: • **Appearance** of new strains that cause yearly outbreaks. The need to change the strains used to produce the annual vaccine . It occurs in **both** influenza A and less commonly B.	• They are **major** changes dueto reassortment of **gene segments** that occurs when **one cell is infected simultaneously** with 2 different influenza **A** viruses (e.g. **avian & human influenza viruses**), mixtures of parental gene segments may be assembled into **progeny virions**, resulting in a **new variant** of influenza A virus with **sudden changes** in **viral surface antigens**. • Occur **only in the case of influenza A** • This phenomenon occurs **less frequently,** about **every 10 years** causing epidemics and pandemics.

Difference between types of Influenza viruses

Influenza A virus	Influenza B virus	Influenza C virus
• Can infect **multiple species** (birds, human, swine, horses)	• **Only** infect human	• caninfect**multiplespecies**
• **Commonly** undergo **genetic variations**drifts or shifts Cause	• Can go **less commonly drifts** but **never shifts**	• **antigenically stable**
• **outbreaks**and **epidemics** (pandemics)	• **Less commonly small outbreaks**	• **Never**
• Can cause **severe respiratory disease**	• Causes **mild respiratory disease**	• Cause **minor respiratory disease**



Influenza (swine flu H1N1: antigenic shift and antigenic drift)

Pathogenesis:

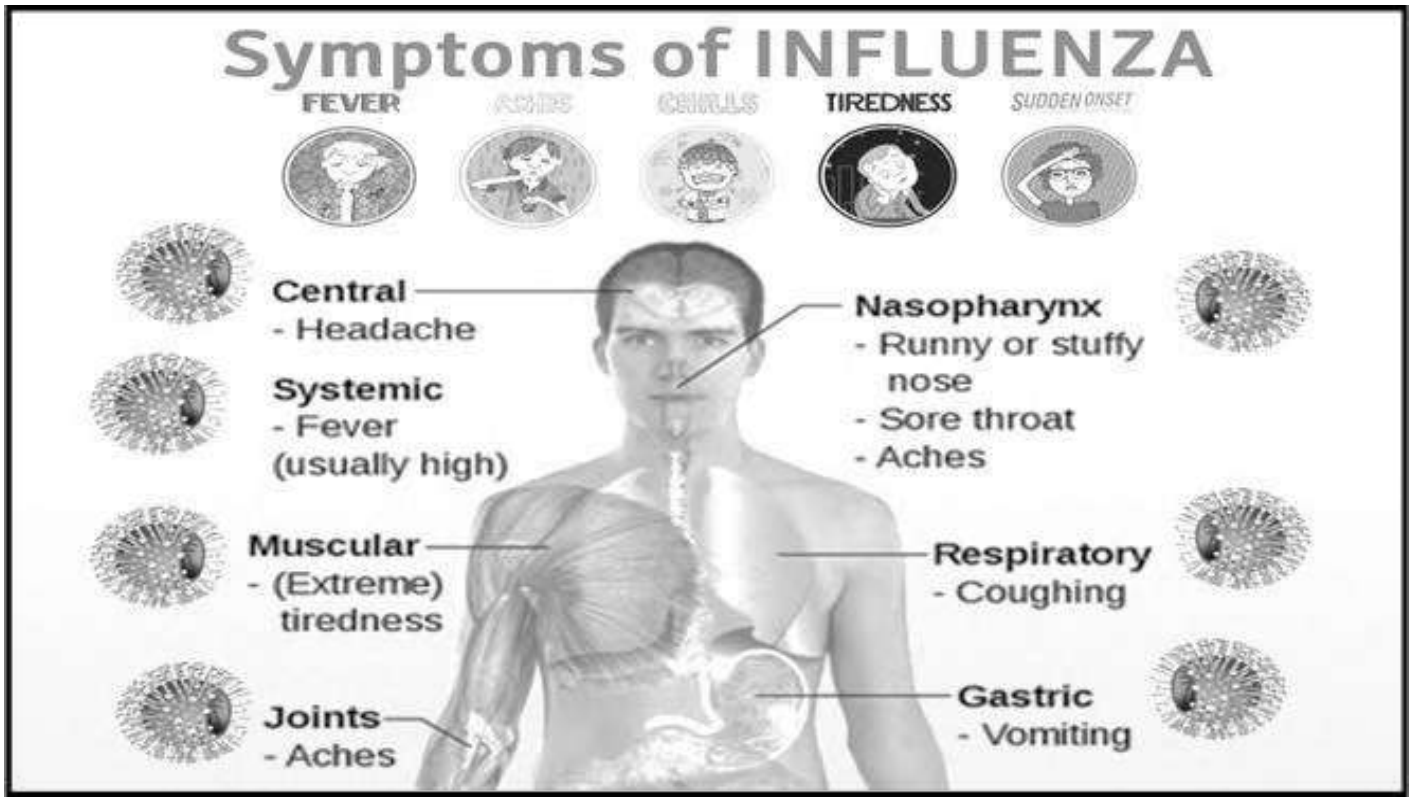
- Infectionoccurs by **inhalationof airbornedroplets** The **incubation period** is **short 1- 4 days**.
- **Influenza (flu)** can cause mild to severe illness, and **at times** can **lead to death.**
- **Flu symptoms** usually come on suddenly. **People who have flu** often feel some
- or all of these symptoms:

✓ **fever** or feeling feverish/chills
✓ **cough**
✓ sore throat
✓ **runny or stuffy** nose
✓ **muscleorbody** aches
✓ headaches
✓ **fatigue** (tiredness)
✓ **somepeople** may have vomiting and diarrhea, though this is **more common** in
✓ **children than adults.**
✓ **Mostpeople** who get flu will **recover** in a few days to less than two weeks

Some people will develop complications such as:

- pneumonia (can **result** from either flu virus infection alone or from co-infection of flu virus and bacteria)
- Sinus and ear infections
- Other **possible serious complications** triggered by flu can include **inflammation of the heart** (myocarditis), **brain** (encephalitis) or **muscle tissues** (myositis, rhabdomyolysis), and **multi-organ failure** (for example, respiratory and kidney failure) **viremia** is **very rare**.

• **IgG** and **IgA** are important in **protection against reinfection.**
• **Antibody** to the HA protein is **most important** since this can **neutralize** the **virus** and **prevent** the **virus initiating** the infection.
• **Antibody** to the NA protein has **some protective effect** since it **seems to slow** the **spread of the virus**.
• **IgG persists longer** than **IgA** and so **plays** a more important role in **long term immunity.**



(Symptoms of influenza)

Diagnosis:

Sample: nasal aspirate, throat swab

1. **Direct detection** of viral **antigens**: **ELISA**,
2. **Detection** of viral **RNA**: by **probes, RT-PCR**
3. **Isolation** of the virus on **embryonated eggs**, **cell culture**

Treatment:

1. **Zanamivir (Relenza) and Oseltamivir (Tamiflu):** They are neuraminidase inhibitors. They are **active against** both influenza A and influenza B and can **reduce** the **duration of uncomplicated influenza** (by **approximately 1 day** in about **70-90% of adults**) **if taken within** two days of the **onset of illness.** However, **oseltamivir resistance** has **been seen** in **some circulating strains recently**
2. **Rimantadine and Amantadine: block virus entry** and also **interfere with virus release.**

Prevention:

Vaccines: An **annual flu vaccine** is the **best way** to **reduce** your **chances of getting seasonal flu** and **spreading** it to others. **CDC recommends** that everyone6months of **age and older** receive an annual flu vaccine.

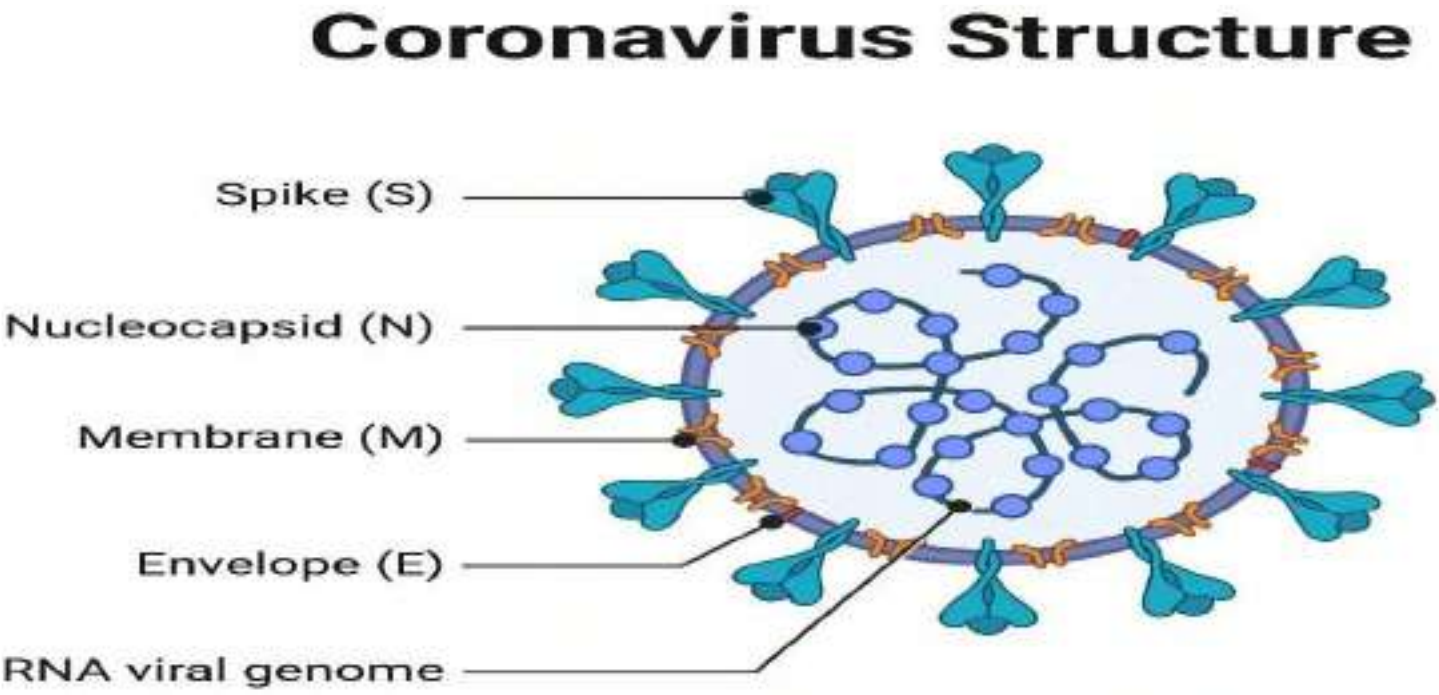
	1-Trivalent Inactivated Vaccine (TIV)	2-Living attenuated influenza vaccine (LAIV)
Composition	• Subvirion andpurified **surface antigen** of **2 A** and **one B** influenza virus strains **antigenically equivalent** to the **annually recommended** strains predicted to be circulating during the influenza season.	• Itcontains**temperaturesensitive mutants**of**2influenza A** and one influenza B virus strains **antigenicallyequivalent**tothe **annuallyrecommended**strains predictedtobecirculatingduring the influenzaseason.Thesecan **replicate**inthe**cooler(33ºC) nasal mucosa**wheretheyinduce**IgA**, but **not** in the**warmer(37ºC)lower respiratory tract**.
Route of administration	• **I.M injections**	• **Nasal spray**
Indications	• **All** persons above 6 months including those that are **healthy** or those **with chronic illnesses** and **pregnant women.**	• **Non-pregnant**healthy**persons 2-49 years old.**
Booster doses	• **Yearly boosters** are recommended **before** the **flu season in October.** • **All children 6 months to 8 years old**, who have **not been previously vaccinated**, should **receive 2 doses** of the age appropriate vaccinein the **same season** and a **single dose** during **subsequent seasons.**	

3. **The quadrivalent flu vaccine** protects against **two** influenza A viruses and **two** influenza B viruses. to give **broader protection** against circulating flu viruses.

Corona viruses

• **Corona viruses** are **enveloped**, **120- to 160-nm** particles that contain an **unsegmented genome** of **single-stranded positive sense RNA** (**27–32 kb**), the **largest genome among RNA viruses**. **Isolated genomic RNA** is **infectious**. The **helical nucleocapsid** is 9–11 nm in diameter.

- There are 20-nm-long club- or petal shaped **projections** that are **widely spaced** on the outer surface of the envelope, **suggestive** of a **solar corona.**
- **Coronaviridae** are best known as the **second most frequent** cause of the **common cold.**
- **Corona viruses** are **zoonotic**, meaning they are **transmitted between animals and people.**
- **Common signs of infection** include **respiratory symptoms**, **fever**, **cough**, **shortness of breath and breathing difficulties.** In **more severe cases**, infection can **cause pneumonia, severe acute respiratory syndrome**, **kidney failure and even death.**



(Corona virus structure)

Coronavirusesarethecauseof:

1- **SARS-CoV** was **transmitted** from civet cats to humans.
2- **MERS-CoV** from camels to humans.
3- Corona virusdisease(COVID-2019)

Diagnosis:

1. **Molecular testing: Real-time reverse-transcription polymerase chain reaction** (**rRT-PCR**) assays to **detect** viral RNA in clinical samples.
2. **Serological testing:** Serology testing is used to **detect** previous infection (**antibodies** in people who may **have been exposed to the virus**).

COVID-19 Vaccines:

COVID-19 vaccine recommendations have been **updated** as of September, 2023

- **Vaccine recommendations** are **based** on age, time since last dose.
- **People** who are **moderately or severely immunocompromised** have **specific** recommendations for **COVID-19 vaccines**.

Vaccines authorized by the U.S. Food and Drug Administration (FDA) currently include:

- **Pfizer-BioNTech** and **Moderna** COVID-19 vaccines which are mRNA vaccines **teach our cells how to make a protein** that **triggers** an immune response **inside our bodies.**
- **Novavax** COVID-19 vaccine which is a protein subunit vaccine
- **J&J/Janssen** COVID-19 vaccine has **expired** and is **no longer available** for use in the **United States** as of **May 6, 2023.** It is a **Viral vector vaccines**

•These **vaccines** are **given** as a **shot** in the **muscle of the upper arm** or in **the thigh of a young child.**

Other Previous commonly used vaccines for COVID-19:

- **The Sinopharm vaccine :** itis inactivated vaccine, therefore, **suitable** forthose with a **compromised immune system.**
- **AstraZeneca:** Recombinant,adenovirus vector **encoding** the **SARS-CoV-2 Spike(S) glycoprotein.**

Side Effects of COVID-19 vaccines:

- Side effects that **happen within 7 days** of **getting vaccinated** are common but are mostly mild. such as fever, chills, tiredness, and headache. They are **more common** after the **second dose** of a **Pfizer-BioNTech, Moderna, or Novavax COVID-19 vaccine.**
- **Severe allergic reactions** to vaccines are **rare** but can happen.
- There is a **rare risk of myocarditis and pericarditis** associated with mRNA COVID-19 vaccination. The **rare risk** may be further **reduced** with a **longer interval** between the first and second dose.
- There was a **relationship between** the **J&J/Janssen COVID-19 vaccine** and a **rare and serious adverse event**. It was **blood clots with low platelets** (**thrombosis with thrombocytopenia syndrome**, or **TTS**). **TTS** occured at a rate of about **4 cases per million J&J/Janssen doses.**

• **N.B. None** of the COVID-19 vaccines **affect or interact with our DNA.**

3 Community and Hospital acquired pneumonia

ILOs

- Define Community and Hospital acquired pneumonia.
- Enlist their risk factors.
- Enlist the etiologic agents and its microbiologic features.
- Discuss the pathogenesis and clinical manifestations of HAP and CAP.
- Discuss the laboratory diagnosis for HAP and CAP.
- Discuss the preventive measures for HAP and CAP.

Definitions

- Pneumonia is an inflammation of the lung parenchyma caused by various microorganisms, including bacteria, fungi and viruses.
  Lung parenchyma consists of the portion of the lung involved in gas transfer (the alveoli, alveolar ducts and respiratory bronchioles).

Pneumonia is frequently categorized based on site of acquisition

- Pneumonia is frequently categorized based on site of acquisition:
- Community-acquired pneumonia (CAP) refers to an acute infection of the pulmonary parenchyma acquired outside of hospitals or extended-care facilities.
- Nosocomial pneumonia refers to an acute infection of the pulmonary parenchyma acquired in hospital settings and encompasses both hospital acquired pneumonia (HAP) and ventilator associated pneumonia (VAP).
- HAP refers to pneumonia acquired ≥ 48 hours after hospital admission.
- VAP refers to pneumonia acquired ≥ 48 hours after endotracheal intubation.

Risk factors of Community acquired pneumonia

- Older age
- Chronic comorbidities as; chronic obstructive pulmonary disease (COPD), chronic lung disease (e.g. bronchiectasis, asthma), congestive heart failure, diabetes mellitus, malnutrition, and immunocompromising conditions.
- Viral respiratory tract infection especially influenza virus infection.

- Impaired airway protection conditions that increase risk of aspiration of stomach contents and/or aspiration of upper airway secretions, such as alteration in consciousness e.g. due to stroke, seizure, anesthesia, drug or alcohol use) or dysphagia due to esophageal lesions or dysmotility.
- Smoking and alcohol overuse.
- Crowded living conditions (e.g. prisons, homeless shelters), residence in low income settings.

Risk factors of Hospital acquired pneumonia

- Previous intravenous antibiotic use (defined as antibiotic use within 90 days) is a significant risk factor for HAP caused by either methicillin- resistant *Staphylococcus aureus* (MRSA) or *P. aeruginosa*.
- Additional risk factors for *P. aeruginosa* HAP include a history of chronic obstructive pulmonary disease (COPD), and prior use of carbapenems, broad spectrum cephalosporins or fluoroquinolones.
- Patients who are immunocompromised (HIV or having chemotherapy).
- Patients with co morbidities (asthma or cystic fibrosis).
- Older age.
- Have recently had major surgery (e.g. abdominal or chest surgery)
- Other hospital patients and staff with poor infection control.

Causative organisms of CAP

1) Typical Bacteria	2) Atypical bacteria
• *Streptococcus pneumoniae* (most common bacterial cause) • *Haemophilus influenzae*,*Moraxella catarrhalis*,*Staphylococcus aureus* , Group A *streptococci*, • Aerobic Gram-negative bacteria (e.g. Enterobacteriaceae such as *Klebsiella* Spp or *Escherichia coli)*, microaerophilic bacteria and anaerobes (associated with aspiration).	• "Atypical" refers to their inability to be visualized on Gram stain or cultured using traditional techniques. ▪ Legionella pneumophila ▪ Chlamydia psittaci ▪ Coxiella burnetii ▪ Chlamydia pneumoniae ▪ Mycoplasma pneumonia

3)Respiratory viruses

- Influenza A and B viruses • Respiratory syncytial virus
- Rhinoviruses • Human metapneumovirus
- Parainfluenza viruses • Adenoviruses
- Corona viruses (e.g. Middle East respiratory syndrome coronavirus).

Causative organisms of HAP

- The most common bacterial pathogens are: Pseudomonas aeruginosa,E. coli, Klebsiella pneumoniae and Acinetobacter.
- Staph aureus and methicillin-resistant Staphylococcus aureus (MRSA)
- Polymicrobial infections can occur especially with acute respiratory distress syndrome (ARDS).
- Viral or fungal pathogens are rare in immunocompetent patients.

Pathogenesis of pneumonia

- Respiratory pathogens are transmitted from person to person via droplets. Following inhalation, the pathogen colonizes the nasopharynx and then reaches the lung alveoli.
- When the inoculum size is sufficient and/or host immune defenses are impaired, infection results. Replication of the pathogen, the productionof virulence factors, lead to inflammation and damage of the lung parenchyma, resulting in pneumonia.

Clinical presentation of pneumonia

➢ Pulmonary signs and symptoms:

- Symptoms: Cough (with or without sputum production), dyspnea, and pleuritic chest pain are among the most common symptoms.
- Signs of pneumonia: on physical examination include tachypnea, increased work of breathing, adventitious breath sounds, including rales/crackles and rhonchi, and dullness to percussion.

- These signs and symptoms result from the accumulation of white blood cells, fluid, and proteins in the alveolar space. Hypoxemia can result from the subsequent impairment of alveolar gas exchange. On chest radiograph, accumulation of WBCs and fluid within the alveoli appears as pulmonary opacities.

➢ Systemic signs and symptoms:

- Fever, chills, fatigue, malaise, chest pain and anorexia. Tachycardia and leukocytosis.
- Inflammatory markers, such as the erythrocyte sedimentation rate (ESR), C-reactive protein (CRP) and procalcitonin may rise.

NOTE

Pneumonia is a leading cause of sepsis; thus, the initial presentation may be hypotension, altered mental status, and organ dysfunction as renal or liver dysfunction..

Preventive measures for CAP

- Smoking cessation.
- Influenza vaccination
- Pneumococcal vaccination

Preventive measures for HAP

There are some interventions that healthcare workers can implement to prevent HAP. These include:

- Infection control measures as good hand hygiene for health care professionals.
- Head-of-bed elevation
- Oral care
- Increase patient mobility

Assessment and diagnostic findings

- History taking particularly recent respiratory tract infection
- Physical examination the number of breaths per minute and breath sounds
- Chest x ray: Identifies structural distribution (e.g. lobar, bronchial) multiple abscesses/infiltrates, empyema (staphylococcus), scattered or localized infiltration (bacterial).
- Fiberoptic bronchoscopy both diagnostic (qualitative cultures) and therapeutic (re-expansion of lung segment)
- Gram stain, cultures of sputum, blood culture
- Inflammatory markers: ESR, CRP, and procalcitonin may rise
- Serologic studies e.g. viral or Legionella titers, cold agglutinins
- Pulmonary function studies.

For diagnosis of HAP

- HAP is suspected: the presence of new or progressive radiographic infiltrate plus at least 2 or 3 clinical features (fever 38 ^{o}c, leukocytosis or leucopenia and purulent secretions).
- Lower respiratory tract sample for microscopy and culture ideally before antibiotic started.

4 Streptococcus pneumoniae

PNEUMOCOCCI

Morphology

- Gram positive lancet-shaped cocci (i.e. oval with pointed ends) arranged in pairs (diplococci) or short chains. They are capsulated. Capsules may appear as unstained halos around the organism.

Cultural characters

- They grow on blood agar producing α- haemolysis similar to viridans streptococci. Since both are found in sputum and their cultures and morphology are hard to differentiate and since pneumococci cause respiratory infections while viridans streptococci do not; the growth on blood agar should be differentiated by:

	Pneumococci	ViridansStreptococci
Solubility in bile	• Soluble	• Not soluble
Inulin fermentation	• Fermented	• Not fermented
Sensitivity to optochin	• Sensitive	• Not sensitive
Pathogenicity to mice	• Pathogenic	• Not pathogenic
Quellung test	• Positive	• Negative

Antigenic structure

- There are at least 91 serotypes of pneumococci based on the specificity of the capsular polysaccharide. Types 1-8 are responsible for 75% of cases of
- pneumococcal pneumonia in adults and for more than half of the fatalities in pneumococcal bacteraemia.

- Serotyping can be done by agglutination or capsule swelling 'quellung test'.
  - Mix bacterial suspension with known antiserum
  - Methylene blue is added.

- In the presence of specific antiserum, the capsule is sharply outlined and appear swollen.

Pathogenesis and virulence

- **The capsule** is the **most important virulence factor**. It interferes with phagocytosis and favours invasiveness.
- **IgA1 protease**, enhances its ability to colonize the mucosa of the upper respiratory tract.
- **Autolysin, pneumolysin and pilli** may contribute to pathogenesis.
- **Peptidoglycan and teichoic acid** in the cell wall, activate the complement, induce cytokine production and contribute to the inflammatory response and to the septic shock that occurs in some immunocompromised patients.

Predisposing factors:

- Predisposition to disease occurs if the local or general resistance is lowered by:
  - Viral or other respiratory infections, bronchial obstruction and respiratory tract injury.
  - Alcoholism or drug intoxication.
  - Abnormal circulatory dynamics e.g. heart failure and pulmonary congestion.
  - Certain chronic diseases e.g. sickle cell anaemia, nephrosis, diabetes. 5- Splenectomyzed and the elderly persons.

Diagnosis of pneumonia:

- Direct microscopic examination of gram stained sputum smears will show the prevalent organism to be pneumococci among pus cells.
- Positive quellung reaction. This can be done for rapid identification.
- Sputum is cultured on blood agar in 5-10% CO_2 (which enhances the growth). Alpha haemolytic colonies should be differentiated from viridans streptococci.
- Intraperitoneal injection of sputum into mice. Animals die in 24-48 hrs. The organism can be seen in tissue smears and pure culture can be obtained from heart blood. The test is rarely used.

- Blood cultures are positive in 15-25% of pneumococcal infections.

Treatment

- Penicillin, Erythromycin, Azithromycin.
- Third generation cephalosporins (e.g. ceftriaxone) and vancomycin are the drugs of choice for treatment of invasive infections with penicillin resistant pneumococci.

Prophylaxis

1. Pneumococcal polysaccharide vaccine containing 23 types (PPV-23)

- For immunosuppressed individuals e.g. elderly, asplenic patients, HIV patients or those under immunosuppressive therapy

2. Pneumococcal conjugate vaccine (PCV13)

- Containing the capsular polysaccharide of 13 pneumococcal serotypes coupled to a carrier protein**.** Four doses are recommended for all children at 2,4 and 6 months and at 12-15 months.

5 Mycobacteria

- It is convenient to divide mycobacteria of clinical interest into:
  - o M. tuberculosis complex (MTC) which includes M. tuberculosis, M. bovis, M. Africanum and M. microti. These are associated with tuberculosis.
  - o Non-tuberculous mycobacteria (NTM) may be associated with human diseases.
  - o M. leprae causing leprosy. Mycobacterium tuberculosis M. tuberculosis and M. bovis are the cause of human tuberculosis.

Morphology

- Theyarethin straight or slightly curved rods. They cannot be stained by simple stainsdueto the high lipid (mycolic acid) content of the cell wall. They can be stainedbyZiehl-Neelsen stain (Z.N.) and appear as thin pink rods arranged singly or in smallgroups in a contrasting blue background. They can be stained by fluorochrome stains (e.g. aura amine, rhodamine) and appear as yellow orange fluorescentbacilli in a black background by fluorescent microscopy.

Constituents of tubercle bacilli

1- Its cell wall contains several complex lipids:

a) **Mycolic acids,** which contribute to the organism's acid-fastness, resistance to phagocytosis and intracellular destruction, resistance to acids, alkalis and antibiotics.

b) **Phosphatides,** which play a role in caseation necrosis.

c) **Wax D;** the active component in Freund's adjuvant, used to enhance the immune response to many antigens in experimental animals.

Cord factor (trehalose-6, 6'-dimycolate) is correlated with virulence.

2- The organism contains several proteins which

a) when combined with

waxes elicit delayed hypersensitivity. These proteins are the antigens in the PPD (purified protein derivative) used in the tuberculin skin test.
3-The organism contains a variety of polysaccharides.

Cultural characters

- They are strictly aerobic; however, better growth occurs in presence of 5-10% CO2. They grow on egg-enriched media such as Lowenstein-Jensen (L-J) medium. They are slow growers; no growth appears before 2-4 weeks incubation at 35-37°C. The growth on L-J medium is irregular, dry and off-white. Other media used include Middlebrook's 7H10,7H11 agar and 7H9, 7H12 broth. Virulent strains grow in a characteristic serpentine cord-like pattern in fluid media due to cord factor.

Biochemical activities and resistance

- M. tuberculosis produces niacin and reduces nitrates while other mycobacteria do not.They are relatively resistant to acids, alkalis and dryness. They can survive in dried expectorated sputum, which is a factor in their transmission by aerosol. They are killed by sunlight and by pasteurization of milk.

6 Tuberculosis

- Infection with M. tuberculosis is airborne while infection with M. bovis occurs by ingestion of milk from infected cattle and causes intestinal tuberculosis which is rare.

Pathogenesis

- The organism produces disease exclusively by invasion of tissues. It is non-toxigenic. The organism preferentially infects macrophages and other reticuloendothelial cells. It resists the destructive properties of normal macrophages by inhibiting phagolysosome formation. It then multiplies within the cellular vacuole or phagosome. As it multiplies intracellularly, it stimulates cell mediated immunity (CMI) and hypersensitivity which leads to tissue damage.
- The disease is either primary or due to reinfection or reactivation of latent infection.

The Difference between Latent TB Infection and Active TB

3) Latent TB Infection	2) TB disease in lungs
• TB present; alive but inactive	• TB present; active
• Tuberculin skin test positive	• Tuberculin skin test positive
• Chest X-ray normal	• Chest X-ray usually reveals lesion
• Sputum smears and cultures negative	• Sputum smears and cultures positive
• No symptoms	• Symptoms such as cough, fever, weight loss
• Not infectious	• Often infectious before treatment
• Not defined as a case of TB	• Defined as a case of TB

Diagnosis of tuberculous infections

- Diagnosis depends on detection, identification and isolation of the tubercle bacilli (TB) from pathologic specimens. Specimen: Tuberculosis can affect virtually every tissue in the body; thus, the specimen can be; sputum or morning

gastric aspirate, urine, CSF, stools and tissue biopsy. The following steps are made for detection or isolation of TB from sputum. Three morning sputum samples collected on three consecutive days should be examined as follows:

- Direct smears are made from the specimen and stained with Ziehl- Neelsen stain. The detection of acid alcohol-fast bacilli in sputum gives a fairly strong indication of pulmonary tuberculosis. However, they are detected only if they are present in large numbers. "The smear may be stained by fluorochrome stains, auramine, rhodamine and examined by fluorescent microscopy for yellow orange fluorescing bacilli in a black background.This staining method is more sensitive than Z.N. staining.

- **Decontamination and concentration**: Contaminated or tenacious specimens e.g. pus, sputum, stools...etc. are subjected to decontaminationand concentration before further processing. Specimens are mixed with an equal volume of N-acetyl-L-eystiene-2% NaOH solution. The mixture is held for 15 min. at room temperature, then diluted with buffer pH 6.8 or distilled water and centrifuged at 3000 rpm for 15 min. Supernatant is decanted and the sediment is processed. The technique leads to;

  a- Liquefaction of the specimen, thus releasing the tubercle bacilli,
  b- Destruction of all bacteria other than tubercle bacilli by NaOH.
  c- By centrifugation, the tubercle bacilli are concentrated in a small Volume.

N.B:

Specimens from sterile sites e.g. CSF do not need decontamination.

Concentrated sediments are processed as follows

- Smears are made from the sediment and stained with Z.N. stain.
- Sediments are cultured on L-J medium (which is the conventional method) and incubated at 35-37°C. Growth appears after 2-3 Weeks. Cultures should not be discarded as negative before 8 weeks. Isolated organisms could be identified by being, niacin positive and nitrate reduction positive and PCR.
- The sediment may be cultured on fluid media using Middlebrook broth as the base for rapid diagnosis by the following tests:

- o The BACTEC radiometric method in which the sediment is inoculated on liquid medium containing radioactive metabolites and incubated at 35-37°C. Growth is detected by production of radioactive carbon dioxide within two weeks.
- o Mycobacteria growth indicator tube (MGIT) in which the sediment is inoculated on liquid medium containing a fluorescence sensor that fluoresces when oxygen is depleted from the medium due to bacterial growth in 7-14 days. Fluorescence is detected by exposure to UV light.

- Moleculdar techniques for rapid direct detection of M. tuberculosis in clinical specimens e.g.
  - o 1- The amplified M. tuberculosis direct test uses enzymes that rapidly make copies of M. tuberculosis 16s ribosomal RNA, which can be detected by genetic probes.
  - o PCR for the detection of chromosomal DNA and for detection of drug resistance genes for rifampicin and others. Commercial PCR kits can confirm the diagnosis in 8hrs.

N.B

The conventional culture method on L-J should be done at the same time with any of the above methods since it is essential to isolate the organism on solid media to be identified and its antibiotic sensitivity tested.

- The Mantoux tuberculin skin test (TST) i
  - o s a delayed hypersensitivity skin test that determines whether a person is infected with M. tuberculosis. The test is performed by intradermal injection of 0.1 ml of the purified protein derivative (PPD), containing 5 tuberculin units, in the forearm. A positive test is read as an induration that appears after 48-72 hours. The induration is due to accumulation of antigen specific CD4 T cells that are activated and release cytokines that attract macophages and polymorphs to the site of injection.

- **Groups that should be screened with TST include;**
  - HIV patients, close contacts of patients with active TB, low-income populations, alcoholics and IV drug users, prison inmates and individuals from countries with a high incidence of tuberculosis.

- Interpretation of TST A positive test cannot distinguish between latent infection and active disease.
  - The diameter of the induration required to judge the test as positive varies depending upon the status of the individual being tested:-

✓ **Induration > 5 mm:**

is considered positive in;

- o Recent contacts of a case of TB.
- o Patients with fibrosis on chest X-ray from prior TB.
- o Immunosuppressed persons e.g. HIV patients and solid organ transplant recipients.

✓ **Induration > 10 mm**

isconsidered positive in;

- o Unvaccinated children under 4 years and children exposed to high risk adults.
- o Persons with high risk factors, such as the homeless, IV drug users, nursing home residents, hospital and laboratory personnel.
- o Patients with high risk clinical conditions e.g. diabetes, silicosis, chronic kidney disease and cancer.

✓ **Induration > 15 mm i**

- o s considered positive in any person, including persons with no risk factors for TB or those who received BCG vaccine.

✓ **False negative reaction**

- o may occur in presence of tuberculous infection when "anergy" develops due to overwhelming tuberculosis, measles, Hodgkin's disease, AIDS, sarcoidosis or immunosuppression.

✓ **False positive reaction**

- may occur due to infection with NTM or previous BCG vaccination. However, BCG vaccination effect wanes in 5 years. The QFTG test mentioned below overcomes these false positive reactions.

✓ **QuantiFERON-TB Gold (QFTG):**

- is a whole-blood test used for diagnosis of latent TB infections (LTBI) or TB disease. The test measures the amount of IFN-gamma released from the patient's white blood cells when mixed with two M. tuberculosis protein antigens (ESAT-6 and CFP-10, that are not present in BCG or in the NTM) and incubated for 16-24 hours. The amount of IFN- garnma released is measured by an ELISA test. IFN gamma release assays IGRAs is another name for QFTG and related tests.

▪ **Advantages of QFTG:**

- The tests require only a single blood specimen and results are read after 24 hrs only. The patient does not need to return for a reading. There is no reader bias. The result is not affected by prior BCG vaccination.

▪ **Disadvantages of QFTG:**

- The tests are costly and the blood should be tested within 12 hrs. They should not be used in severely immunocompromised hosts or in children younger than 5 years. The tests are susceptible to biologic variation in the immune response.

Prevention and Control

➢ Treatment of active tuberculosis patients:

- Prompt adequate treatment of patients is the most important measure to prevent spread of infection.
- The policy in treatment of tuberculosis requires:

A- Prolonged course;

- treatmentshould be continued for 6-12 months as there is a slow response for treatment.

This is due to:

a. Most bacilli are intracellular.

b. The caseous material interferes with penetration of the drug,

c. chronic lesions tubercle bacilli are not dividing i.e. "metabolically inactive" and need a long exposure to more than one drug.

B- Combination of drugs;

- 2-3ormore drugs are given at the same time and the combination is changed every few months; to reduce the drug toxicity and prevent emergence of drug resistant strains.
- The first line of treatment drugs are; isoniazid (INH), rifampicin (RMP), ethambutol, streptomycin and pyrazinamide. In infections with multiple drug resistant strains, second line drugs should be used. These include para-amino-salicylic acid, ethionamide, cycloserine, capreomycin, amikacin kanamycin and the fluorquinolones; ciprofloxacin and levofloxacin.

Multi-drug resistant TB (MDR-TB)

- is defined as those resistant to INH and rifampicin. MDR in tuberculosis is a serious problem that results from the spontaneous and random occurrence of chromosomal mutations. They are treated by second line drugs. The most important causes for development of resistance are; 1- Patients' non-compliance. 2- Inadequate treatment regimens.

Extreme drug resistant TB (XDR-TB)

- results from inadequate treatment of MDR-TB. It is defined as MDR + resistance to fluorquinolones and one of the injectable drugs (kanamycin, amikacin or capreomycin).
- Because drug resistance is a problem, antibiotic sensitivity testing should be performed for all isolated organisms.

DOTS (directly observed therapy short-course)

- is recommended by the WHO to prevent patients' non-compliance in which the health care worker directly observes the patient taking the medication.
- It is a condensed course, consists of 2 months treatment with rifampicin, INH, pyrazinamide and ethambutol. This is followed by rifampicin and INH for 4-6

months. Ideally, the drugs are given daily but to ensure compliance by supervising the administration of the drug they may be given 3 times weekly.

➢ Treatment of latent tuberculosis,

- to reduce the risk of progression to active tuberculosis. Treatment should not begin until active TB has been excluded by clinical, radiological and laboratory methods. Variable protocols are in use. Some use INH alone others use INH with RMP. Periods of treatment may vary from, 12 weeks to 9 months.

  The treatment is required for:

  - o Individuals whose TST or QFTG test has recently converted to positive.
  - o Children who are close contacts of a case of infectious tuberculosis.
  - o Persons with a positive TST or QFTG test who are at high risk of reactivation including; a- Immunosuppressed persons e.g. HIV patients and organ transplant recipients, b- Patients with high risk clinical conditions e.g. diabetes, c- The homeless, IV drug users, hospital and laboratory personnel.

➢ Specific prophylaxis = BCG vaccine:

- The currently used vaccine for immunization against tuberculosis is a living attenuated *M. bovis* strain called BCG (bacillus, Calmette-Guerin).
- The vaccine creates a controlled focus of infection which stimulates cell mediated immunity. It is given by intradermal injection of 0.1 ml in the deltoid region.
- In Egypt, the vaccine is given to newborns during the first month. In USA, it is given to children who are in contact with patients not complying to treatment or those who have MDR tuberculosis. It should be given to health care workers who are TST negative and are working in areas where TB cannot be properly contained.

▪ Advantages of BCG vaccination:

- o It protects against complications of TB e.g. milliary TB and tuberculous meningitis.
- o BCG is used in immunotherapy e.g. in treatment of bladder cancer. The vaccine is instilled into the bladder. It non-specifically stimulates cell mediated immunity, which can inhibit the growth of carcinoma cells.

▪ Disadvantages of BCG vaccination:

- o It does not prevent infection. Its effectiveness varies from 0%- 70%.
- o It gives a false sense of security.
- o It makes the tuberculin test useless in diagnosis especially in latent T.B.

o The vaccine should not be given to immunocompromised children as it may cause disseminated infection, which requires treatment for 6 months.

IV -Eradication of TB in cattle and pasteurization of milk to reduce *M. bovis* infection.

V- Reducing overcrowding and improving socioeconomic conditions.

7 Immunology of the respiratory tract

ILOs

- Hypersensitivity reactions of the respiratory tract.
- Definition of bronchial asthma.
- The differences between extrinsic and intrinsic asthma.
- Types of allergens.
- Pathogenesis of bronchial asthma.
- Role of cytokines in pathogenesis of bronchial asthma.

Hypersensitivity reactions

- Definition of hypersensitivity reactions:
  - Hypersensitivity reactions are exaggerated or inappropriate immunologic responses occurring in response to an antigen or allergen leading to tissue damage.

Hypersensitivity reactions of the respiratory tract

The most common hypersensitivity reactions of the respiratory tract

- Hay fever (allergic rhinitis):
  - It is caused by an allergic response to a harmless outdoor or indoor substance the body identifies as harmful (allergen). It causes cold-like symptoms. These may include a runny nose, itchy watery eyes, nasal congestion, nasal itching, frequent sneezing and sinus pressure.
- Bronchial asthma.
- Hypersensitivity pneumonitis
  - is an allergic reaction that causes inflammation in and around the air sacs (alveoli) of the lungs and the smallest airways (bronchioles).

Definition of bronchial asthma

- Bronchial asthma is a chronic inflammatory disease of the respiratory passages (most especially the bronchi and bronchioles), occurring with the participation of mast cells, eosinophils and T-lymphocytes, the release of a large number of inflammatory mediators.
- Inflammation of the respiratory passages causes their hyperreactivity, bronchial obstruction, and respiratory symptoms such as wheezing, shortness of breath, chest tightness, and coughing.

Types of bronchial asthma

- Types of bronchial asthma according to the cause:..
  - **Extrinsic (allergic asthma)**: Triggered by an allergen as molds, pollens and pet dander.
  - **Intrinsic (non-allergic asthma)**: Triggered by inhaled irritants, chemicals, a respiratory infection, cold weather, or tobacco smoke
  - **Occupational asthma**: Caused by inhaled chemicals or industrial dust in the workplace
  - **Exercise-induced asthma**: Triggered by exercise and physical

Types of bronchial asthma according to the severity of

Intermittent	•This type of asthma comes and goes so the person can feel normal in between asthma flares.
Persistent	•Persistent asthma means that the symptoms are present much of the time. Symptoms can be mild, moderate or severe.

Extrinsic asthma

- **Extrinsic asthma** also known as **allergic asthma or atopic asthma**, is caused by an immune response to environmental allergens such as pollen, dust mites, animal dander, and mold.
- In people with extrinsic asthma, the immune system overreacts to the presence of allergens, leading to the release of histamine and other chemicals that cause inflammation and narrowing of the airways.
- As this is a type of allergic reaction, the immune system goes into overdrive, and an oversupply of immunoglobulin E antibodies is released into the blood.
- Extrinsic asthma is the more common (and typically less serious) form of asthma.
- Symptoms of extrinsic asthma typically include itching and redness in the eyes, runny nose, and cough.
- This type of asthma is more common in individuals who have a family history of allergies or have a personal history of atopic conditions such as hay fever or eczema.

Intrinsic asthma

- **Intrinsic asthma**, **non-allergic asthma** or **non-atopic asthma** on the other hand, is caused by non-allergic triggers such as viral infections, stress, exercise, and exposure to cold air.
- Unlike extrinsic asthma, intrinsic asthma is not associated with an immune response to allergens and can occur in individuals without a personal or family history of allergies.
- In intrinsic asthma, the airways become narrow and inflamed in response to triggers, leading to symptoms such as wheezing, chest tightness, and shortness of breath.
- Intrinsic asthma is more common in adults and in women.
- It is often associated with other underlying medical conditions such as gastroesophageal reflux disease (GERD) and obesity.
- Intrinsic asthma is provoked by factors other than allergens. Here, there is only a small and local release of immunoglobulin E antibodies within the airway passages.

Etiology of Bronchial Asthma

- The etiology of asthma is multifactorial.
- Although the etiology of bronchial asthma remains unknown, its pathogenesis is well believed to be affected by both genetic and environmental factors.

Asthma triggers

- Common asthma triggers include:
- **Allergens: as**
    - **Inhalants** e.g. Pollens, fungal spores, animal dander, dust, mites, chemicals in the air (pollution), Tobacco or other inhalation that is smoked, etc…
    - **Ingestants** e.g. milk, fish, egg, chocolate strawberries, certain medicines (aspirin and other NSAIDs) etc….
    - **Contactants** e.g. wool, nylon etc...
    - **Injectants** including drugs e.g. penicillins, serum injections e.g. antitetanic sera, anesthesia or insect venom.
- **Changes in weather** (most often cold weather).
- **Physical activity.**
- **Respiratory infections,** such as the common cold and many other viruses.
- **Strong emotions (stress).**
- Substances in some workplaces can also trigger asthma symptoms, leading to **occupational asthma.** The most common triggers are wood dust, grain dust, animal dander, fungi, or chemicals.

Pathogenesis of bronchial asthma

➢ Pathogenesis of type I hypersensitivity reaction

- On first exposure to an allergen:
  - The antigen presenting cells (APCs) mainly DCs and macrophages capture the antigen, process it and present its epitopes to TH2 which is activated and produce IL-4 that in turn stimulates class switching of B cells to produce IgE.
  - IgE is cytotropic (i.e. tissue fixing) it binds to receptor for Fc portion of IgE (FcERI) on basophils and mast cells.
- On re-exposure to the same antigen:
  - It causes cross linking of bound IgE which leads to mast cell activation, degranulation and release of its mediators.
  - Mediators derived from mast cells fall into 3 classes:

Preformed mediators

- These mediators are stored in the granules and include histamine.

Newly synthesized

- They include leukotrienes, prostaglandins and platlet activating factor (PAF).

Cytokines

- Secreted by activated mast cells (and basophiles) e.g TNF, IL-3, IL-4, IL- 5 and chemokines. TH2 recruited to the site of reaction also secrete cytokines.

- Preformed mediators (histamine) mediate the immediate manifestations of hypersensitivity:
  - Contraction of smooth muscles leading to bronchospasm.
  - Increased mucus secretion.
  - Vasodilatation and increased vascular permeability leading to edema & hypotension.

- Newly synthesized mediators and cytokines mediate a late phase inflammatory reaction that occurs 2-4 hours after exposure, reaches maximum severity within 24 hours then it subsides gradually. They mediate this inflammatory reaction via:
  - Recruitment of inflammatory cells (granulocytes and TH2) to the site of reaction.
  - Increasing IgE production.
  - Activation of eosinophiles.
  - Tissue damage.

- Mechanisms of airway obstruction in bronchial asthma
  - **Airwayobstruction inbronchialasthmaismainlycausedbythe following four mechanisms:**
    - Contraction of bronchial smooth muscle.
    - Oedema and inflammation of the airway walls.
    - Mucous plugging of the bronchioles.
    - Irreversible changes in the lungs (“remodeling”).

Role of cytokines in pathogenesis of bronchial asthma

- Chemokines
  - these mediators are vital in inducing inflammatory cells into airways and are present abundantly in epithelial cells.

- Cytokines

  - these mediators are involved in the response to asthma and the amount of these cytokines determines how severe the response is to this reaction.
    - **IL-5** is necessary for the differentiation of eosinophil and is derived from Th-2.
    - **IL-4** is also derived from Th-2 and is involved in the differentiation of Th-2 cells.
    - **IL-13** is neededto produceIgE.

➢ Others as tumor necrosis factor, interleukin-1b and granulocyte-macrophage colony-stimulating factor.

- Leukotrienes

  - in their role as bronchoconstrictors, they are unique in that their suppression is directly related to an enhanced lung function and overall clinical presentation.

8 Respiratory fungal infections

ILOs

- The fungal agents causing respiratory tract infections and its classification
- The microbiologic features of opportunistic fungal pathogens
- The route of transmission, clinical manifestations and diagnosis of opportunistic infections
- The microbiologic features for systemic fungal pathogens
- The route of transmission, clinical manifestations and diagnosis of systemic fungal pathogen infection

Classification of fungal agents causing respiratory tract

1-Opportunistic fungal pathogen

- Opportunistic fungi fail to induce disease in most normal persons. But can do so in those with impaired host defenses
- Opportunistic mycoses are caused by saprophytic (i.e. from the environment) or endogenous (i.e. commensal) fungi e.g. systemic candidiasis, cryptococcosis, aspergillosis, mucor mycosis and Pneumocystis infections

2- Systemic fungal pathogens

- These are primary pulmonary lesions that may disseminate to any organ mainly in immunocompromised patients. They are caused by the dimorphic fungi as Histoplasma, Blastomyces and Coccidioides

Opportunistic fungal pathogens

(1) Candida

- Candida albicans, is the most important speciesof Candida. Other species include **C. tropicalis**, **C. parapsilosis**, **C. krusei** and others C. albicans are **Gram positive oval**
- **budding yeasts** which produce pseudohyphae. It is part of the normal flora of mucous membranesof the upper respiratory, gastrointestinal and female genital tract. In these sites, it may predominate and cause superinfection

Pathogenesis and clinical findings

- Predisposing factors to Candida infections are diabetes mellitus, general debility, immunodeficiency, indwelling urinary catheters, intravenous drug abuse, prolonged treatment with broad-spectrum antibiotics, and corticosteroids

Clinical affections include

1-In the mouth, overgrowth of C. albicans produces white patches i.e. oral thrush or moniliasis

2-Vulvovaginitis with itching and discharge which is favoured by prolonged use of antibiotics and diabetes

3-Skin invasion occurs in warm moist areas, which become red and weeping such as the axilla, intergluteal folds, or inframammary foldsmost common in obese and diabetics

4-Nails become involved when repeatedly immersed in water; as in persons involved in dish washing. Painful redness and swelling of nail folds, thickening and loss of nail i.e. paronychia

5-Systemic candidaisis and chronic mucocutaneous candidiasis may occur in debilitated children, diabetics, immunosuppressed patients or drug addicts. They are usually associated with deficient CMI

Systemic candidiasis

➢ mayoccuralmostanywhere in the body

- Candidemias orblood-borneinfectionsoccurmostcommonly in patients with indwelling intravenous catheters
- Endocarditis occurs in patients who have manipulated or damaged valves or in intravenous drug abusers
- Bronchopulmonary disease (usually manifested by persistent cough) occurs in patients with chronic lung disease

Laboratory diagnosis

I-Superficial candidiasis

- Diagnosis is mainly clinical and confirmed by detection of yeast cells in Gram stained film or KOH mount of the lesion.

II-Systemic candidiases

• Samples differ according to the site of infection and include urine, sputum, bronchial washings, cerebrospinal fluid, pleural fluid, blood and tissue biopsies from various visceral organs

1. **Direct microscopic examination** of smear or exudates from the lesions

• shows large oval Gram-positive budding yeast cells with pseudohyphae

2. **Cultures** are done on nutrient agar, corn meal agar and SDA

• Colonies are soft, cream-coloured with a yeasty odour. They are also identified by;

A. Morphology; oval budding Gram positive yeast cells
B. Germ tube formation in serum incubated at 37°C for 30min -2hrs
C. Chlamydospore formation on corn meal agar
D. Biochemical reactions: C. albicans ferments glucose and maltose with acid and gas production. The last three properties (b, c, d) differentiate C. albicans from other candida

(2)Cryptococcus

- C. neoformans are yeast cells with a gelatinous capsule. It is found in soil contaminated with the excreta of birds specially pigeons' faeces
- It is an opportunistic pathogen affecting mainly immunosuppressed individuals specially AIDS patients
- Infection occurs by inhalation where it causes subclinical lung affection or pneumonia
- It may spread systemically to the CNS causing meningoencephalitis or to, the skin, adrenals, bone or eyes

Laboratory diagnosis

1. **Direct microscopic examination** of sputum and CSF after staining with India ink, reveals large gelatinous capsule around budding yeast cells
2. **Cultures** done on SDA without actidione, at 20-37°C show mucoid colonies which are identified by India ink staining, biochemical reactions (urease positive), pathogenicity to mice and DNA probes
3. **Capsular antigens** are detected in CSF, serum or urine by using anticapsular antibodies in a latex agglutination test or enzyme immunoassays
4. **Antibodies** are detected in patients' sera

(3)Aspergillosis

- Aspergillus species are wide spread saprophyticmoulds
- Aspergillus fumigatus is the most common humanpathogen
- A. flavus and A. niger may be involved
- Infection occurs by inhalation of the spores (conidia)
- Aspergillus species can invade any part of the body but commonly cause disease in the respiratory system

There are 3 form of pulmonary aspergillosis

1. Allergic bronchopulmonary aspergillosis: due to allergy to inhaled fungal spores
2. Pulmonary aspergilloma: A fungus ball in the lungs (commonly in the cavities of previous TB infection)
3. Invasive aspergillosis: propagation of the fungus through the lung parenchyma. It is seen in immunocompromised individuals particularly children

Laboratory diagnosis

1.**Microscopic examination**: Aspergillus species are reliably demonstrated by silver stains as Gomori methenamine-silver. Aspergillus hyphae are branching at acute angles of about 45°

2. **Culture**: Aspergillus grows on most media at room temperature. At 25°c colonies appear after 2-3 days with yellow or greenish coloration. Wet mount of the isolated organism shows that colonies are arranged in radiating columns
3. **Serology:** Detection of rising specific IgG in patients with aspergilloma. Detection of IgE in patients with allergic bronchopulmonary aspergillosis
4. **Skin test:** aspergillin test.

(4)Zygomycosis (mucormycosis)

Etiology

- It is an opportunistic infection caused by **Zygomycetes**
- Mainly three genera are inolved **Rhizopus**, **Absidia** and **Mucor** Thes fungi have a
- world wide distribution and commonly isolated from soil, food, organic debris, decaying vegetables and moldy bread

Clinical manifestations

Infection is commonly seen in patients with uncontrolled diabetes mellitus and usually begins in the paranasal sinuses following the inhalation of sporangio-spores

Spread of infection to the brain can occur through the cribriform plate. Once infection reaches the brain it is

Theinfecting fungi have a predilection for invading vessels of the arterial system, causing embolization and subsequent necrosis of surrounding

Other forms of zygomycosis can also present e.g. pulmonary, intestinal, cutaneous and disseminated zygomycosis

Pulmonary mucormycosis is a rapidly progressive infection that occurs after inhalation of spores into the bronchioles and alveoli. Pneumonia with infarction and necrosis results, and the infection can spread to contiguous structures, such as the mediastinum and heart, or disseminate hematogenously to other organs

Diagnosis

- It is mainly clinical to allow immediate start of treatment Typically, there is cotton like
- growth on the roof of the mouth and nares in uncontrolled diabetic patient

Confirmation of diagnosis can be done by

- **KOH mount of the lesion:** shows mold with non septate ribbon like hyphae
- **Culture**: the organism is rapidly growing (24-48 hours) and it is identified by its morphology in the culture
- Immunodiffusion test

(5)Pneumocystis jiroveci

P. jiroveci is the human species (previously known as Pneumocystis carinii). It is a yeast-like fungus commonly found in the lungs of healthy people, but being a source of opportunistic infection in

It causes interstitial pneumonia in malnourished infants and immunosuppressed patients (AIDS, corticosteroid therapy, transplant recipients...etc)

It was a major cause of pneumonia and death in AIDS patients until the use of chemoprophylactic regimens. Cell mediated immunity plays a major role in resistance to disease

It is transmitted by inhalation; growth in the lungs is limited to the surfactant layer above the alveolar epithelium causing interstitial pneumonitis and frothy exudates that blocks oxygen exchange.

P. jiroveci has two morphologicaly distinct forms; thin-walled trophozoites and cysts, which are thick-walled, spherical to elliptical and contain 4-8 nuclei. In clinical specimens, the trophozoites and cysts are present in a tight mass

Clinical manifestations

- Symptoms of PCP include fever, non-productive cough (because sputum is too viscous to become productive unless the patient has an additional bacterial infection), shortness of breath (especially on exertion), weight loss and night sweats
- The fungus can invade other visceral organs, such as the liver, spleen and kidney, but only in a minority of cases. Pneumothorax is a well- known complication of PCP
- Extra-pulmonary infections occur in the late stages of AIDS and affect the liver, spleen, lymph nodes and bone marrow. Mortality is 100% in untreated cases

Laboratory diagnosis

1. ***P. jiroveci*** can not be cultivated on artificial media so diagnosis is carried out by identification of the causative organism in the specimen
2. Bronchoalveolar lavage, lung biopsy or induced sputum are stained by Giemsa, toluidine blue, methenamine silver or calcofluor white and examined for the presence of trophozoites or cysts. The characteristic cysts resemble crushed ping-pong balls and are present in aggregates of 2 to 8
3. Direct immunofluorescent staining using monoclonal antibodies and PCR are used for diagnosis

Systemic mycoses

- These infections result from inhalation of the sporesof dimorphic fungi that have their saprophytic filamentous forms in the soil. Within thelungs, the spores differentiate into yeasts or other specialized forms
- Most lung infections are asymptomatic and self-limited. However, in some persons, mainly irrmiunocompromised patients, infection may disseminate to other organs and may be fatal
- Infected persons do not transmit the disease to others
- These diseases are endemic in the Americas, rarely in Africa and the Middle East and include; histoplasmosis, coccidioidomycosis and blastomycosis

Histoplasmosis caused by Histoplasma capsulationrespiratory tract

- The infection occurs in the lungs as acute pneumonia or as chronic cavitary lesions. The organism may spread to the spleen and liver inside macrophages causing lymphadenopathy or hepato- splenomegaly

Coccidioidomycosis caused by Coccidioides immitis

- Lesions occur in the lungs and can spread by direct extention or via the blood causing granulomatous lesions in any organ mainly the bone and CNS. Skin test is positive

Blastomycosis caused by Blastomyces dermatitidis

- Respiratory infections are mild or asymptomatic
- Dissemination may result in ulcerated granulomas of the skin, bone or other sites

Diagnosis of systemic mycoses

1. Microscopic examination of biopsy specimens, bone marrow aspirates or blood films, after staining by periodic acid- Schiff or calcofluor white or Giemsa stains
2. Culture of sputum, urine, bone marrow aspirate and tissue biopsy, on SDA at 37°C and at 25°C. Cultures are identified by detection of specific antigens or by DNA probes
3. Serologic tests show a rise in antibody titres when dissemination occurs
4. Skin tests using fungal antigens e.g. coccidioidin or histoplasmin
5. Detection of fungal antigens in specimens by RIA and fungal nucleic acids by DNA probes or PCR.

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Respiratory system

Parasitology

RPS_314

page	lecture
1	loeffler pneumonia
2	Focal lesions in the lungs:
3	Infiltrating diseases of the lungs
5	cryptosporidum

1 Parasites Affecting Pulmonary system

I Loeffler's pneumonia:

1. Cercaria of *Schistosoma* spp.
2. Larva of *Ascaris lumbricoides.*
3. Larva of *Ancylostoma duodenale* (Hookworm)
4. Larva of *Strongyloidesstercoralis.*

II Focal lesions in the lungs:

1. Amoebic lung abscess.
2. Hydatid disease.
3. Cysticercosis of the lungs.

III Infiltrating diseases of the lungs:

1. Visceral Larva Migrans (VLM).
2. *Cryptosporidium*.
3. Sparganosis.

Loeffler's pneumonia

Definition:

It is transient pulmonary illness due to sensitization of the lungs by migrating helminths. It caused by cercaria of ***Schistosoma***, larva of ***Ascaris lumbricoides*** or larvae of ***Ancylostoma* (hookworm)** and ***Strongyloides stercoralis*** during their migratory phase through the lungs.

Clinical manifestations:

- Pneumonitis with cough, wheezes, expectoration of blood-tinged sputum.

Diagnosis:

1. **CBC:** high Eosinophilia
2. **Chest x-ray:** pulmonary infiltrates
3. **Sputum examination:** eosinophils and the causative parasite stage (like larvae of *Ascaris,* hookworms or *Strongyloides*)

(**N.B: Loeffler's triad =** Pulmonary symptoms + abnormal chest radiograph + high eosinophilia)

Management:

1. **Treat the cause:** anti-helminthic drugs
2. **Treat the symptoms & complications:** steroid if needed.

Amoebic lung abscess

II.Focallesions in the lung

It is extra-intestinal amoebiasis caused by ***Entamoeba histolytica*** trophozoite which may reach the lung through:

- √ Direct extension from amoebic liver abscess.
- √ Hematogenous spread.
- √ Rupture of liver abscess into pleural cavity.

Clinical manifestations:

1. **Pulmonary symptoms:** Fever, cough, dyspnea, chest pain.
2. **Complications**:
   - If the abscess ruptures into a bronchus, an "anchovy sauce" fluid will pass with the sputum.
   - Secondary bacterial infection.

Diagnosis:

1. Imaging investigations: X-ray, CT scan and MRI

2.Guided Needle Aspiration biopsy: Reveal "Anchovy sauce" aspirate. Microscopic examination of this fluid will show *E. histolytica* **trophozoites**, RBCs and pus cells. The cyst stage is never present.

3. Immunological tests: ELISA.

4. **Sputum examination** may reveal trophozoite.

Management:

1. **Medical:** **Metronidazole** (tissue amoebicidal) + **antibiotic** (prevent 2nd bacterial infection) + **Paromomycin** (luminal amoebicidal to eradicate intestinal amoebiasis).
2. **Abscess drainage:** if large or complicated abscess.
3. **Surgical intervention:** if severe and resistant cases.

III. Infiltrating diseases of the lungs

1. Visceral larva migrans (VLM) = Toxocariasis

Definition:

Invasion of human tissues with the larvae of ascarids of dogs and cats (***Toxocara canis*** and ***Toxocara cati***). The disease occurs mainly in young children usually playing with or near infected pets.

Mode of infection:

Ingestion of embryonated eggs of ***Toxocara canis*** and ***Toxocara cati.***

Pa thogenesis:

The eggs hatch in the small intestine and the larvae penetrate the intestine and enter the lymphatic or blood vessels. Since man is an unnatural host for dog and cat ascarids, the larvae can't complete their cycle but in this case the larvae migrate through viscera and form granulomatous lesions**.**

Diagnosis:

1. **CBC:** high Eosinophilia.
2. Hypergammaglobulinemia IgE.
3. **Biopsy**: larvae surrounded by eosinophilic granulomatous reaction.
4. **Immunodiagnostic methods:** ELISA.

Management:

Diethylcarbamazine (DEC) or Thiabendazole + Steroid.



Chest X-rays showing coin lesions in the lungs

2. *Cryptosporidium*

An opportunistic parasite affecting the brush border of the respiratory tract.

Mode of infection: inhalation of **sporulated oocyst.**

Clinical manifestations: **opportunistic parasite**

In healthy persons: no or mild symptoms.

In immunocompromised patients: Fatal pneumonia.

Diagnosis:

Presence of **oocysts** in the sputum after staining with modified Ziehl-Nelsen.



Cryptosporidium oocyst

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Respiratory System

Pharmacology

RPS-314

page	lecture
1	Treatment of lower respiratory tract infections(pneumonia)
8	Drug therapy of tuberculosis
19	Treatment of bronchial asthma (1) & (2)
30	Anti-tussive drugs and mucolytics

1 Lower respiratory tract infection (pneumonia)

ILOS

After the lecture, students should be able to :

- Enumerate the risk factors for infection.
- Identify the preventive and curative measures of pneumonia .

1 Etiological classification

Bacterial pneumonia

Gram +ve ccoci

- Pneumococcal pneumonia (the commonest cause)35% .
- **Staph aureus**: more common in old age and in immunocompromised patient.

Gram -ve bacilli

- Haemophilus influenza, E coli, Pseudomonas, Klebsiella.

Atypial pneumonia (Caused by organisms devoid of cell wall)

- Mycoplasma, Legionella, chlamydia

Anaerobic: Bacteroides

TB

- is a cause of pneumonia its treatment different from other infective agents

Non bacterial pneumonia

Viral

- influnza, Measles, Herpes simplex and Corona like (SARS) .

Fungal

- Pneumocystitis carinii, Histoplasmosis, Aspergliosis and Candida.

Parasitic

- Toxoplasma

2 *Clinical classification*

- **Community acquired pneumonia(CAP):**
  pneumonia occurring in the community.
- **Hospital acquiredpneumonia(HAP):** (Nosocomial pneumonia)
  Pneumonia occur after 2 day of hospital admission.

Etiology of HAP depend on:

- **Time of occurrence** (Early or late onset)
- **Type of hospitilization** (ICU or not, ventilated or not) .
- **Causative organisms**:

Gram -ve bacilli

- Haemophilus influenza, E coli, Pseudomonas, Klebsiella and Staph aureus.

Atypial pneumonia

- Legionella (which may present in air conditioning system).

Risk factors

Age

- especially infants, young children and the elderly.

Lifestyle

- smoking, alcohol.

Preceding viral infections

- influenza predisposing to Streptococcus pneumoniae infection.

Respiratory

- asthma, chronic obstructive pulmonary disease (COPD), malignancy, bronchiectasis, cystic fibrosis.

Immunosuppression

- AIDS, cytotoxic therapy - increased risk of infection with staph. spp., tuberculosis, Gram-negative bacilli and P. jirovecii.

Intravenous drug abuse

- often associated with Staphylococcus aureus infection.

Hospitalization

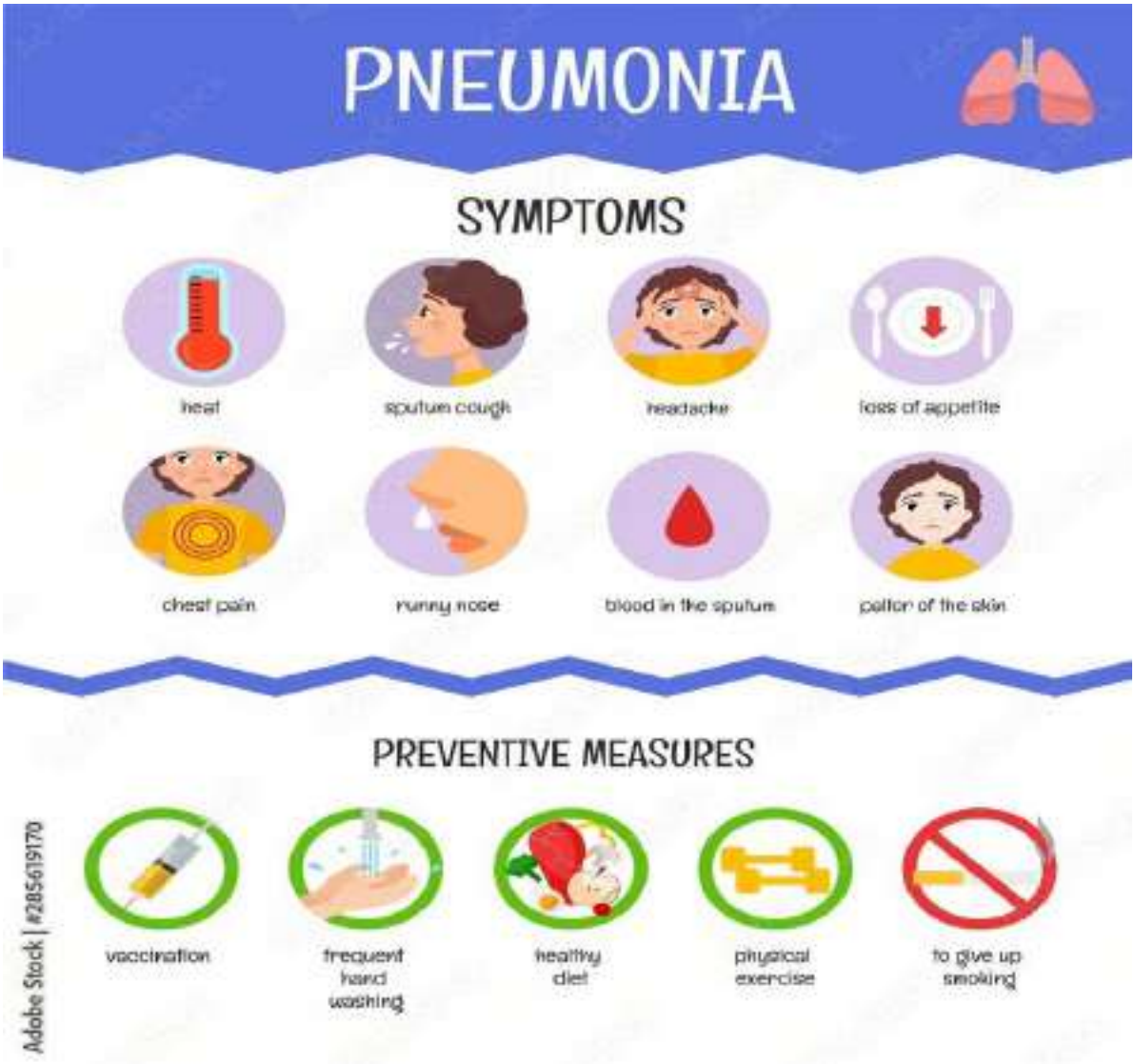
- often Gram-negative organisms.

Aspirationpneumonia

- patients with impaired consciousness or patients with oesophageal obstruction are at risk of aspiration pneumonia which usually affects the right lung and is caused by anaerobes from the oropharynx.

Underlying predisposing disease

- diabetes mellitus, cardiovascular disease



TREATMENT OF PNEUMONIA

i. Specific.
ii. Supportive .
iii. Symptomatic.

1 Specific treatment : Antibiotics.

)A)Community acquired pneumonia (CAP)

1-Mild to moderate CAP in young patients without co-morbidity :

- Benzyl penicillin 1- 2 millionU/6hl. And if the response is inadequate, add one of the following:
  - Azithromycin.
  - Augmentin(Amoxicillin clavulinic acid).
  - Ciprofloxacin.
  - Cephalosporin (2ndgeneration e.g. cefaclor).

2 -Mild to moderate CAP in young patients with co-morbidity or in elderly :

Two of the following :

- Cephalosporin(2"dor3'generation) .
- Augmentin .
- zithromycin.

3-Insevere ill-hospitalized patient with CAP :

- Azithromycinandoneor twoofthefollowing anti-pseudomonal antibiotics:
- Ceftazidime(3dgeneration cephalosporin).
- Piperacillin (anti-pseudomonas pencillin).
- Ciprofloxacin

(B)Hospital acquired pneumonia

1-Non severe HAP :

Type ofpatient :

- Admitted less thans days ago.
- Without co-morbidity (heart, liver or renal failure)
- Recommended antibiotic: one of the following:
  - Cephalosporin (2"" or 3" generation)
  - .Ciprofloxacin

2-severe HAP :

Type ofpatient :

- Admitted more than 5 day ago .
- With co-morbidity.
- Recommended antibiotic: double or triple therapy for 2-3 weeks as follow:
  - Piperacillin/tazabactam
  - Ceftazidime. Imipinim.

    OR
  - Ciprofloxacin. Aminoglycosides (e.g. plus
  - Gentamicin,
  - Amkacin).
  - Vancomycin if resistant staph aureus is suspected.

2 *Supportive treatment*

- Fluid and elctrolytes replacement .
- Total parentral nutrition in severe cases.
- Cardiac support: Cortison and dobbutamine .
- Respiratory support :
  - Oxygen .
  - Mechanical ventilation.

3 *Somptomatic treatment*

Analgesic and antipyretics.

Treatment of Special type of pneumonia

Atypical pneumonia :

(Caused by organisms devoid of cell wall and cannot be detected on ordinary sputum staining and culure) e.g.Mycoplasma pneumonia, Legionellapneumonia,chlamydiapneumoniaand viral pneumonia .

1.Mycoplasma pneumonia:

- Erythromycin or other macrolides e.g. Azithromycin, clarthromycin.
- Ciprofloxacin.
- Tetracycline.

2. Legionella pneumonia:

- Rifampicin, clarthromycin or Ciprofloxacin up to 21 days.

3-viral pneumonia:

- Occurs mainly in immunocompromised patients.
- Influenza, parainfluenza and measles.
- Antiviral and supportive measures.

2 Drug Therapy of Tuberculosis

ILOs

1-Outline the classification of anti-tuberculous drugs.

2-Describe the mechanism of action of different anti-tuberculous drugs.

3-Identify the adverse reactions and contraindications of anti-tuberculous drugs.

4-Outline the therapeutic regimen for treatment of TB case.

Pulmonary disease

It is the most frequent clinical presentation of tuberculosis (TB). Less common clinical manifestations include meningitis, renal disease, vertebral osteomyelitis (Pott's disease), skin involvement (lupus vulgaris), and miliary dissemination.

Mycobacteria are intrinsically **resistant** to most antibiotics because:

1- They grow slowly compared with other bacteria, antibiotics that are most active against growing cells are relatively ineffective.

2- Mycobacterial cells can also be dormant and thus completely resistant to many drugs or killed only very slowly.

3- The lipid-rich mycobacterial cell wall is impermeable to many agents.

4- Mycobacterial species are intracellular pathogens, and organisms residing within macrophages are inaccessible to drugs that penetrate these cells poorly.

5- Mycobacteria are notorious for their ability to develop resistance.

Therefore, combined drug therapy is employed to delay the emergence of this resistance and treatment must include at least two drugs to which the bacteria are sensitive. The response of mycobacterial infections to chemotherapy is slow, and treatment must be administered for months to years, depending on which drugs are used.

Classification of antituberculous drugs

Drugs employed in the treatment of TB can be classified into two categories:

First line drugs	•isoniazid, rifampin, ethambutol, streptomycin and, pyrazinamide.
Second line drugs	•**quinolones, ethionamide, aminosalicylic acid, cycloserine,capreomycin,kanamycinandamikacin.**

First line drugs

They have excellent bactericidal activity for both intracellular and extracellular mycobacteria with an acceptable degree of toxicity.

1) *Isoniazid (INH)*

- It is the most active drug for patients who can tolerate it and whose mycobacteria are sensitive.
- Isoniazid is a prodrug that is converted by mycobacterial catalase-peroxidase into an active metabolite.
- Isoniazid is the most important drug for the treatment of active tuberculosis.
- As a single agent is also indicated for the treatment of latent TB or for prophylaxis.

Mechanism of action

Isoniazid is bactericidal against actively dividing bacilli and bacteriostatic against "resting" bacilli. The primary action of isoniazid is to inhibit the biosynthesis of mycolic acids.

Untoward effects

1- **Hepatitis**: the incidence increases with age and the risk is greater in alcoholics. The drug is only discontinued when there is definite hepatic impairment with transaminases greater than five times the normal value.

2- **Neurological disorders**: mainly peripheral neuritis, others include insomnia, restlessness, muscle twitchings, convulsions and psychotic episodes.
These are dose-related complications as isoniazid increases the excretion of pyridoxine. Concurrent administration of pyridoxine prevents the peripheral neuritis and most other nervous system disorders.

3- Skin rash, fever, arthritic symptoms.

4- Isoniazid can reduce the metabolism of phenytoin increasing its blood level and toxicity.

Contraindications

1- Known hypersensitivity

2- Active hepatic disease

2) *Rifampin (Rimactane)*

Rifampin (Rifampicin) is a semisynthetic derivative of rifamycin B, which belongs to a group of complex macrocyclic antibiotics.

Mechanism of action

Rifampin is bactericidal for gram-positive and gram-negative cocci, some enteric bacteria, mycobacteria, and chlamydia. It inhibits RNA synthesis in bacteria by inhibiting DNA-dependent RNA polymerase.

Untoward effects

1- Occasional side effects include rashes, thrombocytopenia, impaired liver function with jaundice, light chain proteinuria and some impairment of immune response.

2- If given intermittently less than twice weekly or if daily doses of 1200 mg or greater are used, an influenza like syndrome develops. It includes fever, chills, myalgia, headache, acute tubular necrosis, thrombocytopenia, hemolytic anemia and shock.

3- Rifampin is a potent enzyme inducer and increases the metabolism of some drugs as warfarin, digitoxin, oral contraceptives, methadone, ketoconazole, propranolol, sulfonylureas and prednisone. Appropriate increase in dosage is required to compensate for increased drug metabolism.

Co5n-traindications

1- Known hypersensitivity

2- Hepatic dysfunction

3) *Ethambutol*

Mechanism of action

Ethambutol is a bacteriostatic agent. It inhibits mycobacterial arabinosyl transferases which are involved in the polymerization reaction of arabinoglycan, an essential component of the mycobacterial cell wall.

Untoward effects

1- **Optic neuritis**: It is dose-related and manifested by decreased color perception, impaired visual fields or reduced visual acuity. Eye examinations should be done prior to therapy and periodically thereafter. This complication is reversible with early drug withdrawal.

2- Hyperuricemia due to decreased renal excretion of uric acid.

3- Other side effects include GIT upset, headache, dizziness, confusion, hallucinations, peripheral neuritis.

Co6n- t contraindications

1- Known hypersensitivity

2- Pre-existing optic neuritis from any cause

3- Gout

4) *Pyrazinamide*

Pyrazinamide exhibits bactericidal activity against tubercle bacilli.

Mechanism of action

Pyrazinamide is converted to pyrazinoic acid-the active form of the drug by mycobacterial pyrazinamidase. Pyrazinoic acid disrupts mycobacterial cell membrane metabolism and transport functions.

Untoward effects

1- **Hepatotoxicity**: is the major adverse effect.

2- Nausea, vomiting and drug fever can occur.

3- **Hyperuricemia**: It inhibits the excretion of urates resulting in hyperuricemia in nearly all patients; acute episodes of gout have occurred.

Co7n-traindications

1- Known hypersensitivity

2- Hepatic dysfunction

3- Gout

5) *Streptomycin*

- Streptomycin is bactericidal for extracellular mycobacteria.
- It penetrates poorly into living cells and thus cannot kill intracellular microbes.
- In extensive organ tuberculosis, in miliary dissemination or meningitis, three or four drugs are often given and streptomycin is one of them.

Mechanism of action

They are rapidly bactericidal for susceptible organisms by producing irreversible inhibition of protein synthesis.

Untoward effects

1- **Ototoxicity**: All aminoglycosides are capable of affecting both cochlear (hearing) and vestibular (equilibrium) divisions of the eighth cranial nerve.

2- **Nephrotoxicity**: It is usually reversible and may range from mild renal tubular dysfunction to acute tubular necrosis.

3- **Neuromuscular blockade**: This may result with large doses and when combined with curariform drugsresults in respiratory paralysis.

Co8n-traindications

1- Known hypersensitivity

2- Auditory nerve impairment

3- Myasthenia gravis

4- Pregnancy

Second line drugs

➢ These drugs are considered only in:

1- In case of resistance to first-line agents

2- In case of failure of clinical response to conventional therapy

3- In case of serious treatment-limiting adverse drug reactions

➢ -They have lesser efficacy and greater toxicity.

1) Quinolones

- The fluoroquinolones are highly active against M. tuberculosis and are important drugs for multidrug resistant tuberculosis.
- They are also active against atypical mycobacteria.
- Resistance develops rapidly (cross-resistance) if used as a single agent, so must be combined with two or more other agents.
- Agents such as ciprofloxacin, levofloxacin, gatifloxacin and moxifloxacin are used andmoxifloxacin is the most active against M tuberculosis.

Mechanism of action

Quinolones block bacterial DNA synthesis by inhibiting bacterial topoisomerase II (DNA gyrase) and topoisomerase IV.

Untoward effects

1- GIT manifestations: nausea, vomiting, abdominal pain and diarrhea.
2- Headache and dizziness have occasionally been observed.
3- Rashes including photosensitivity reactions can occur.
4- Acute renal failure and anaphylaxis rarely occur.
5- Rare but serious hepatic damage including liver failure with trovafloxacin.
6- Gatifloxacin contraindicated in diabetic patients due to serious reports of hypoglycemia and hyperglycemia.
7- Fluoroquinolones may damage growing cartilage and cause bone and joint damage, so, not used in patients under 18 years and also contraindicated during pregnancy and lactation.

2) *Ethionamide*

Mechanism of action

Chemically related to isoniazid and also blocks the synthesis of mycolic acids.

Untoward effects

1- GIT irritation

2- CNS manifestations

3- Hepatotoxicity

3) *Capreomycin*

Mechanism of action

Peptide protein synthesis inhibitor antibiotic.

Untoward effects

1- Ototoxicity

2- Nephrotoxicity

4) *Cycloserine*

Mechanism of action

An inhibitor of cell wall synthesis.

Untoward effects

1- CNS toxicity

5) *Para-aminosalycylic acid (PAS)*

Mechanism of action

Folate synthesis antagonist that is active almost exclusively against M tuberculosis.

Untoward effects

1- GIT irritation

2- Hypersensitivity reactions

6) Amikacin & Kanamycin

These are aminoglycoside antibiotics.

Amikacin

Folate synthesis antagonist that is active almost exclusively against M tuberculosis.

Kanamycin

Has been used for treatment of streptomycin resistant strains. Availability of less toxic alternatives has rendered it obsolete.

Therapeutic protocol of T.B.

Uncomplicated drug-sensitive pulmonary T.B.

A- Short course therapy for 6 months can be satisfactory. For the first 2 months, isoniazid, rifampin, and pyrazinamide are given, followed by isoniazid and rifampin for the remaining 4 months.

B- Administration of isoniazid in combination with rifampin for 9 months is also effective.

Extensive pulmonaty T.B.

Also applies with extrapulmonary disease, miliary TB, meningitis, or drug resistance:

Another drug is added to the above regimen, which is usually ethambutol or streptomycin.

3 Drugs Used in Bronchial Asthma

Bronchodilators

Bronchodilator drugs relax constricted airway smooth muscle and cause immediate reversal of airway obstruction in asthma. They also prevent bronchoconstriction (and thereby provide bronchoprotection).

Three main classes of bronchodilator are in current clinical use:

- β2 Adrenergic agonists (sympathomimetics)
- Theophylline (a methylxanthine)
- Anticholinergic agents (muscarinic receptor antagonists)

1) β2 Adrenergic agonists

Inhaled β2 is the bronchodilator treatment of choice in asthma because they are the most effective bronchodilator and has minimal side effects:

They relax the airways, smooth muscle by activating β2 receptors in the bronchial smooth muscle

Inhibit the release of bronchoconstricting mediators from mast cells.

Inhibit microvascular leakage.

Increase mucociliary transport by increasing ciliary activity.

As in other tissues, the β agonists stimulate adenylyl cyclase and increase the formation of intracellular cAMP.

Clinical uses

➢ **Short-Acting β2 Agonists; albuterol (salbutamol), levalbuterol, metaproterenol, terbutaline:**

- Inhaled short-acting β2 agonists are the most widely used and effective bronchodilators in the treatment of asthma due to their functional antagonism of bronchoconstriction.
- They are convenient, easy to use, rapid in onset, and without significant systemic side effects.
- Short-acting β2 agonists are the bronchodilators of choice in treating acute severe asthma.
- The nebulized route of administration is easier and safer than intravenous administration and just as effective.
- Inhalation is preferable to oral administration because systemic side effects are less, and inhalation may be more effective.
- Oral β2 agonists are occasionally indicated as an additional bronchodilator. Slow-release preparations may be indicated in nocturnal asthma; however, these agents are less useful than inhaled agonists because of an increased risk of side effects.

Long-Acting Inhaled β2 Agonists; salmeterol, formoterol, and arformoterol:

- proved to be a significant advance in asthma and COPD therapy.
- These drugs have a bronchodilator action of >12 hours
- They are used prophylactically in the treatment of the disease.
- They improve asthma control (when given twice daily) compared with regular treatment with short-acting β2 agonists (four to six times daily).

Untoward effects

Unwanted effects are dose related and due to stimulation of extrapulmonary B receptors.
Side effects are not common with inhaled therapy but quite common with oral or intravenous administration.

- Muscle tremor due to stimulation of B, receptors in skeletal muscle is the most common side effect. It may be more troublesome with elderly patients.
- Tachycardia and palpitations are due to reflex cardiac stimulation secondary to peripheral vasodilation (B2 receptor stimulation) and possibly also from stimulation of myocardial B, receptors as the doses of B2 agonist are increased. These side

effects tend to disappear with continued use of the drug, reflecting the development of tolerance.

- Tolerance continuous treatment with an agonist often leads to tolerance (desensitization, subsensitivity), which may be due to down-regulation of the receptor.

2) *Methylxanthines*

Methylxanthines, such as theophylline, which are related to caffeine, have been used in the treatment of asthma. Theophylline is still widely used in developing countries because it is relatively inexpensive.
However, the frequency of side effects and the relatively low efficacy of theophylline have led to reduced use in many countries, as inhaled beta agonists are far more effective as bronchodilators and inhaled steroids have a greater anti-inflammatory effect.

Mechanism of action

The mechanisms of action of theophylline are still uncertain. In addition to its bronchodilator action, theophylline has many nonbronchodilator effects.

Inhibition of phosphodiesterases:

- At high concentration (minimal at therapeutic dose level.)
- PDE inhibition increases the intracellular of cAMP and cGMP which explain the bronchodilator action of theophylline.
- Inhibition of PDE should lead to synergistic interaction with B,agonists through an increase in cAMP.

Adenosine receptor antagonism:

- Adenosine causes bronchoconstriction in airways by releasing histamine and leukotrienes.
- Theophylline is structurally similar to adenosine and blocks many of its action at therapeutic range concentrations.

- Adenosine antagonism may be responsible for serious side effects of theophylline.

Interleukin-10 release:

- IL-10 has a broad spectrum of anti-inflammatory effects and there is evidence that its secretion is reduced in asthma. IL-10 release is increased by theophylline.

Pharmacodynamics

The methylxanthines have effects on the central nervous system, kidney, and cardiac and skeletal muscle as well as smooth muscle. Of the three agents, theophylline is most selective in its smooth muscle effects, whereas caffeine has the most marked central nervous system effects.

A- Central Nervous System Effects:

In low and moderate doses, the methylxanthines, especially caffeine, cause mild cortical arousal with increased alertness and deferral of fatigue.

The larger doses commonly cause nervousness and tremor in some patients. Very high doses, from accidental or suicidal overdose, cause medullary stimulation and convulsions and may lead to death.

B-Cardiovascular Effects:

- Methylxanthines have positive chronotropic and inotropic effects.
- These agents, in large doses, relax vascular smooth muscle except cerebral blood vessels which are contracted.
- Methylxanthines decrease blood viscosity and may improve blood flow under certain conditions. The mechanism of this action is not well defined, but the effect is exploited in the treatment of intermittent claudication with pentoxifylline, dimethylxanthine agent.

C- Effects on Gastrointestinal Tract:

The methylxanthines stimulate secretion of both gastric acid and digestive enzymes.

D- Effects on Kidney:

The methylxanthines, especially theophylline, are weak diuretics. This effect may involve both increased glomerular filtration and reduced tubular sodium reabsorption. The diuresis is not of sufficient magnitude to be therapeutically useful.

E- Effects on Smooth Muscle:

The bronchodilation produced by the methylxanthines is the major therapeutic action in asthma. Tolerance does not develop, but adverse effects, especially in the central nervous system, may limit the dose.

F- Effects on Skeletal Muscle:

The respiratory actions of the methylxanthines may not be confined to the airways; they also strengthen the contractions of isolated skeletal muscle and improve contractility and reverse fatigue of the diaphragm in patients with COPD.

Clinical uses

1- Bronchial asthma:

Of the xanthines, theophylline is the most effective bronchodilator. Theophylline should be used only where methods to measure theophylline blood levels is available because it has a narrow therapeutic index, and its therapeutic and toxic effects are related to its blood level.

Intravenous aminophylline has been used for acute severe asthma.

NOTE

Nebulized B2 agonists are now preferred over intravenous aminophylline for acute exacerbations of asthma and COPD.

2- Apnea in a premature infant:

In premature infants, episodes of prolonged apnea lasting more than 15 seconds frequently occur, which lead to hypoxia and neurological damage. Oral or IV use of caffeine or theophylline
They can stimulate respiration and eliminate these episodes of apnea.

3- Headache and migraine:

Caffeine is sometimes included as an additional ingredient in some aspirin and paracetamol preparation to potentiate their effects for relieve of headache. And with ergotamine to enhance its absorption for treatment of migraine.

Untoward effects

- Common side effects are headache, nausea, and vomiting, abdominal discomfort and restlessness.
- At high concentrations, cardiac arrhythmias may occur.
- At very high concentrations, seizures may occur.

Use of low doses of theophylline, targeting plasma concentrations of 5-10 mg/L, largely avoids side effects and drug interactions and makes it less necessary to monitor plasma concentrations.

3) *Muscarinic cholinergic antagonists*

Atropine was introduced for treating asthma but due to its significant side effects, particularly drying of secretions, less soluble quaternary compounds, such as atropine methylnitrate and ipratropium bromide, have been developed. These compounds are topically active and are not significantly absorbed from the respiratory or GI tracts.

Mechanism of action

These agents are competitive antagonists of ACh binding to muscarinic cholinergic receptors; thus these drugs block the effects of endogenous ACh at muscarinic receptors, including the direct constrictor effect on bronchial smooth muscle.

Clinical uses

- In asthmatic patients, anticholinergic drugs are less effective bronchodilators than β2 agonists.

- In the acute and chronic treatment of asthma, anticholinergic drugs may have an additive effect with β2 agonists and should therefore be considered when control of asthma is not adequate with nebulized β2 agonists.
- InCOPD, anticholinergic drugs may be as effective as or even superior to β2 agonists.

Therapeutic choices

- Ipratropium bromide: The onset of bronchodilation is relatively slow. It is usually given three to four times daily.
- Oxitropium bromide: is available in higher doses by inhalation and may therefore have a more prolonged effect. Thus, it may be useful in some patients with nocturnal asthma.
- Tiotropium bromide: is a long-acting anticholinergic drug that is suitable for once-daily dosing.

Untoward effects

- Inhaled anticholinergic drugs are generally well tolerated. Systemic side effects are uncommon during normal clinical use because there is little systemic absorption.
- A significant unwanted effect is the **unpleasant bitter taste** of inhaled ipratropium, which may contribute to poor compliance.
- Nebulized ipratropium bromide may precipitate **glaucoma** in elderly patients due to a direct effect of the nebulized drug on the eye. This may be prevented by nebulization with a mouthpiece rather than a face mask.
- **Paradoxical bronchoconstriction** with ipratropium bromide due to blockade of prejunctional M, receptors on airway cholinergic nerves that normally inhibit acetylcholine release; this problem has not been described with tiotropium.
- Tiotropium causes **dryness of the mouth** in 10-15% of patients, but this usually disappears during continued therapy.
- **Urinary retention** is occasionally seen in elderly patients.

2) Anti-inflammatory Agents

A – Corticosteroids

- **Corticosteroids** are used in the **treatment** of several **inflammatory** lung diseases.The **mechanisms of action** of corticosteroids in asthma are **still poorly understood**,but their efficacy is most likely **related** to their anti-inflammatory properties.

1- **Corticosteroids** have **widespread** effects on **gene transcription, increasing** the transcription of several anti-inflammatory genes and **suppressing** transcription of many inflammatory genes.

2- Steroids **potently inhibit** the formation of cytokines that are **secreted** in asthma by **T-lymphocytes, macrophages, and mast cells.**

3- Corticosteroids **prevent** the **increase** in **vascular permeability** and leadto **resolution of airway edema.**

4- Corticosteroids **increase β adrenergic** responsiveness. Steroids **potentiate** the effects of **β agonists** on **bronchial smooth muscle** and **prevent and reverse β receptor desensitization** in airways. **B2 Agonists** also **enhance** the **action of glucocorticoid receptors**, This suggests that **β 2 agonists** and **corticosteroids enhance** each other's beneficial effects in **asthma therapy.**

Routes of Administration and Dosing

❖ Inhaled Corticosteroids in Asthma:-

- Inhaled corticosteroids (**beclomethasone, budesonide, ciclesonide, fluticasone and triamcinolone**) are recommended as **first-line therapy** for all patients with **persistent asthma.** They should be **started** in any patient who **needs** to use a β2 agonist inhaler for symptom control **more than twice weekly.** Effective in **mild, moderate, and severe asthma** and in children as well as adults

- Aerosol treatment is the **most effective** way to **avoid** the systemic adverse effects of corticosteroid therapy.

Side effects inhaled corticosteroids:

1- **Oropharyngeal candidiasis:** the **risk** of this complication can be **reduced** by having **patients gargle water**and **spit** after **each inhaled treatment.**

2- **Hoarseness** can also **result** from a direct local effect of inhaled corticosteroids on the **vocal cords.**

3- These agents may **increase** the risks of **osteoporosis** and **cataracts** over the **long term use.**

❖ Systemic Steroids:-

1- **Intravenous steroids** are indicated in **severe** symptoms as in **status asthmaticus**; where the **attack is severe, prolonged** and **refractory** to other **conventional bronchodilators.**
Hydrocortisone is the **steroid of choice** because it has the **most rapid onset** (**5-6 hours after administration**). **Methylprednisolone** is also **available** for intravenous use.
Intravenous therapy is usually **given** until a **satisfactory response** is obtained, and then **oral prednisolone** may be substituted.

2- **Oral steroids** such as **prednisolone and prednisone** are the **most commonly** used. **Short courses** of oral steroids (**30-40 mg prednisolone daily for 1-2 weeks**) are indicated for **acute exacerbations** of asthma; the dose may be **tapered over 1 week** after the exacerbation is **resolved** (the **taper is not strictly necessary** after a **short course** of therapy, but patients find it **reassuring**).

N.B.
Oralsteroids are usually given as a **single dose** in the **morning** because this **coincides** withthe **normal diurnal increase** in plasma cortisol and **produces less adrenal suppression** than if given in **divided doses** or at **night Alternate-day** treatment has the **advantage of less adrenal suppression**, although in many patients asthma control is not **not optimal** on this regimen.

A high dose of inhaled fluticasone propionate is as **effective** as a **course** of **oral prednisolone** in controlling **acute exacerbations of asthma**
Although this route of delivery is more expensive.

B - Cromones

- **Cromolyn sodium** acts by **stabilizing** the **mast cell membrane** inhumanlung **preventing** the **release of histamine and SRS-A** as a consequence of **IgE mediated allergic resoponse**. It therefore, **prevents** the **immediate and late asthmatic** response to **allergen without any bronchodilator effect.** It can be

effective when taken **before exposure to allergens or exercise**, but **not useful in managing** an **acute asthma attack.**

- A **structurally related drug, nedocromil sodium,** which has a **similar** pharmacological profile to **cromolyn**, was subsequently **developed**.Although **cromolyn** was **popular in the past** because of its **good safety profile**,itsuse has **sharply declined** with the **more widespread** use of the **more effective inhaled steroid.**

C - Anti-Leukotrienes

- **Cysteinyl-leukotrienes (LTs)** are produced in **asthma** and they have **potent** effects on **airway function**, inducing **bronchoconstriction**, **airway hyperresponsiveness, plasma exudation, mucus secretion, and eosinophilic inflammation. Blocking** the leukotriene pathways by **5- lipoxygenase enzyme inhibitors (zileuton)** and **leukotriene-receptor antagonist; montelukast, zafirlukast, and pranlukast** (**LTD4 receptor antagonist**) could be useful in the treatment of asthma.

- They are given **orally** in the **prophylaxis** of **mild to moderate asthma, antileukotrienes** cause a significant improvement in **lung function** and **asthma symptoms**. **Anti-leukotrienes** are considerably less effective than inhaled corticosteroids in the **treatment of mild asthma** and **cannot** be considered the **treatment of first choice. Theoretically**, anti-leukotrienes should be of **particular value** in patients with **aspirin-sensitive asthma.**

- One of the **major advantages** of anti-leukotrienes is their **effectiveness** in tablet form
This may **increase compliance** with **chronic therapy** and **makes treatment** of **children easier. Montelukast** is **effective** as a **once-daily preparation** (10 mg in adults, 5 mg in children). In addition, **oral administration** may treat concomitant **allergic rhinitis.**

- **Adverse Effects: Zileuton, zafirlukast,** and **montelukast** are **all associated** with **rare cases** of hepatic dysfunction.
Thus, **liver-associated enzymes** should be monitored. **Several cases** of Churg-Strauss syndrome has been associated with the use of **zafirlukast and montelukast**. Churg-Strauss syndrome is a **very rare vasculitis** that may **affect** the heart, peripheral nerves, and kidney and is associated with **increased** circulating eosinophils and asthma.

D - Immunomodulatory Therapies

Anti-IgE Receptor Therapy:

Omalizumab is a **humanized monoclonal antibody** that binds to **free circulating IgE** to form a **complex** that **inhibit** the **binding of IgE** to its **receptors** on **mast cell** and **eosinophils**, thus **prevent** release of mediators that cause **bronchospasm**.

Clinical Use:

Omalizumab is used for the treatment of patients with **severe asthma**.
The **antibody** is administered by **subcutaneous injection every 2-4 weeks**, and the dose is determined by the **titer of circulating IgE**.
Omalizumab **reduces** the **requirement** for **oral and inhaled corticosteroids** and **markedly reduces asthma exacerbations**. It is also **beneficial** in treating allergic rhinitis.
The **major side effect** of omalizumab is an **anaphylactic response**, which is uncommon.

MANAGEMENT OF ACUTE ASTHMA

Severe attacks and status asthmaticus, require:

1) Patient is **hospitalized** and should **receive oxygen.**
2) **Inhaled or parenteral sympathomimetics**
3) **I.V** treatment with **corticosteroid**.
4) **Antibiotic** for airway **infection**.
5) **I.V aminophylline** may be added.

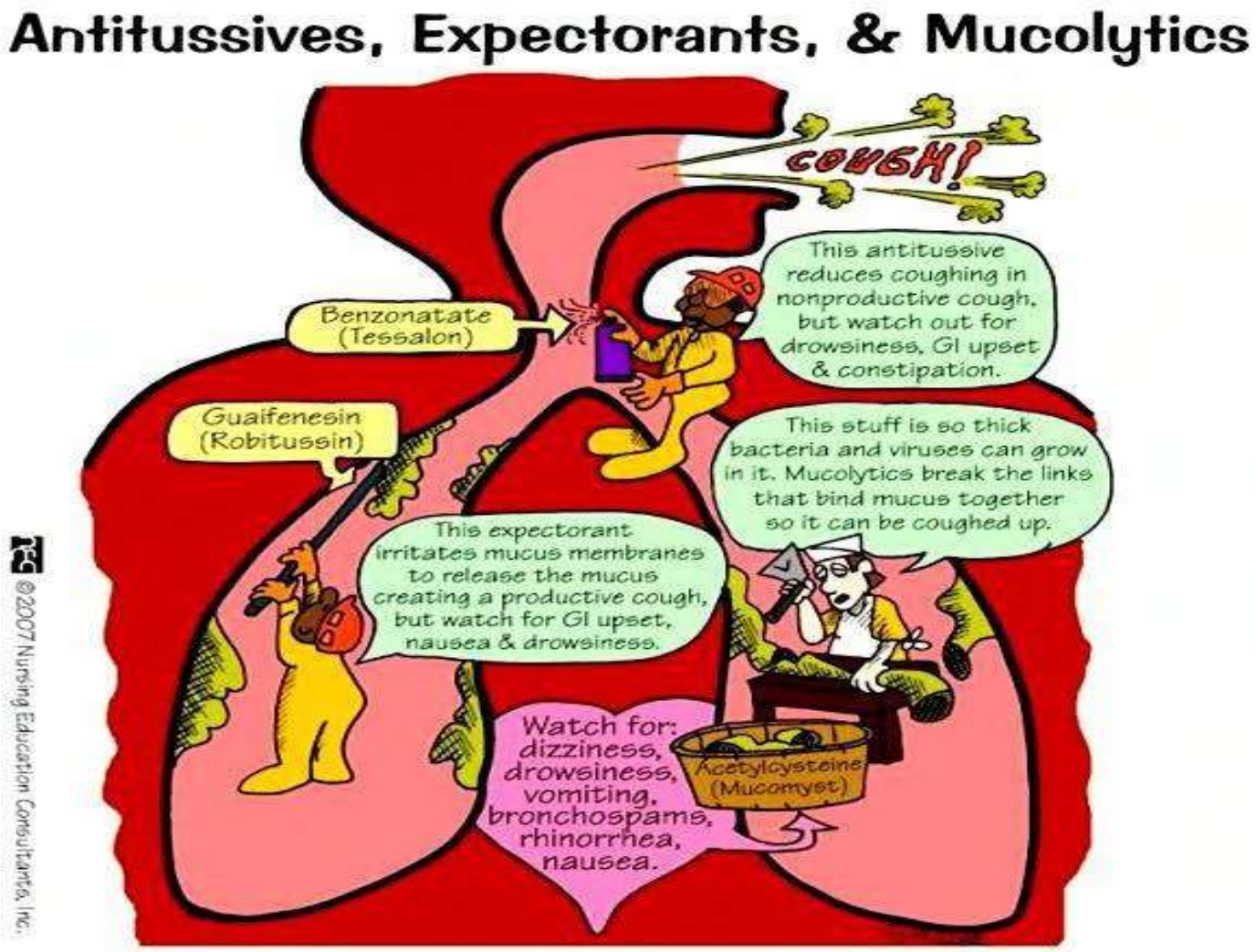
Even this **aggressive** treatment is **not invariably effective,** and **patients must** be **watched closely** for **signs of deterioration**. **General anesthesia**, **intubation**, and **mechanical ventilation** of **asthmatic patients** cannot be **undertaken** lightlybut may be **lifesaving** if respiratory failure supervenes.

4 Antitussive drugs and mucolytics

ILOS

After the lecture, students should be able to:

- Classify the anti-tussive drugs.
- Mention mechanism of actions, indications and adverse effects of anti-tussive drugs.



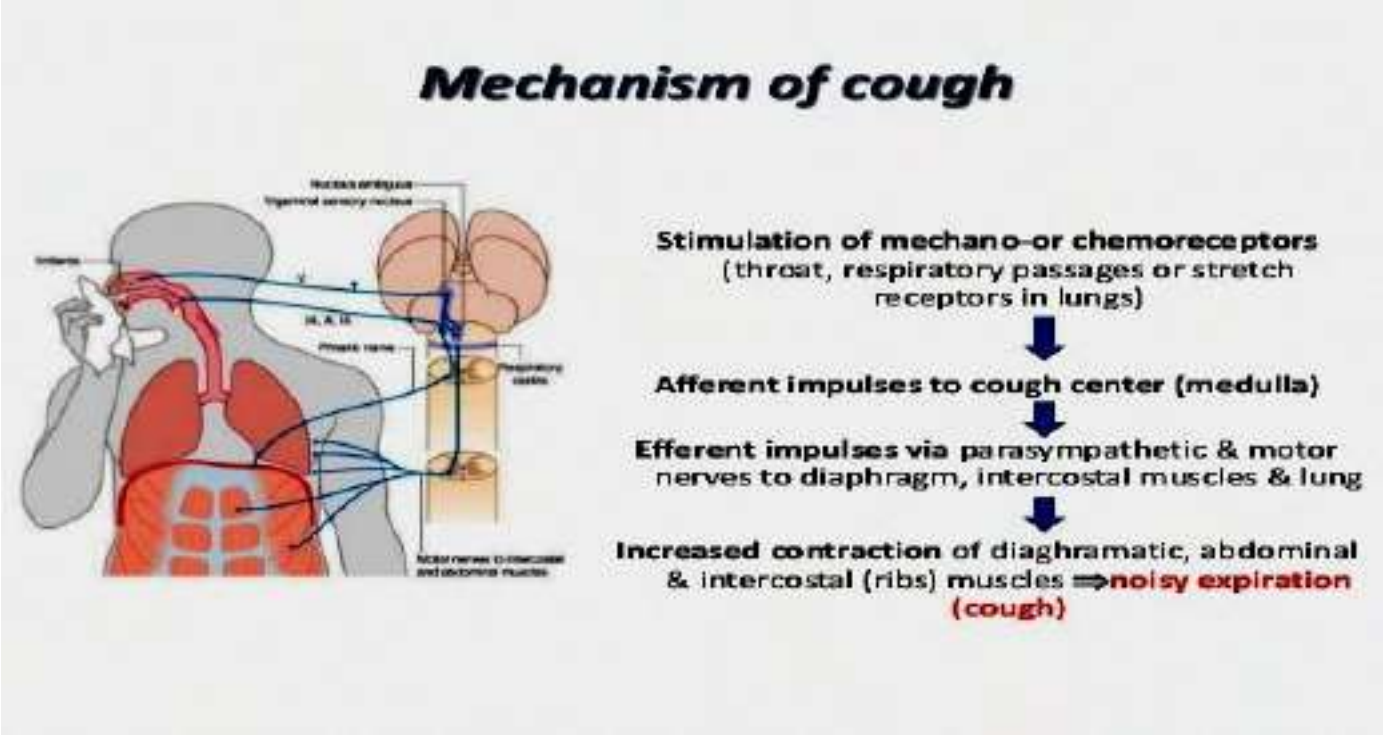
- **Antitussive:** Drugs used to **prevent or relieve a cough.**
- **Expectorant:** Drugs **promotes** the **secretion of sputum** by the **air passage**, used **especially** to treatcough.
- **Mucolytic:** Drugs that acts to **break down thick mucus.**

1-Antitussive

• **Cough** is a **protective reflex,** its **purpose** being **expulsion** of respiratory secretions or foreign particles from the **lungs and upper airway passages.**

• **Cough** may be:

1) **Useful** (**productive**:**sputum** is **coughed up/drain the airway**) or
2) **useless**(**non-productive/dry**). **Useless** cough should be **suppressed**.



1-Antitusive drugs:

Agents used to relieve or suprpress coughing.

These drugs reduce frequency or intensity of coughing.

Benificial in suppressing cough and in decresing morbidity in respiratory disease.

Indicated when **cough** is painful, unproductive, distressing, exhausting and exacerbate lung damage.

Classification of antitussive drugs:

- **Supressing** the cough center in medulla.
  Supressing the cough receptors in throat, lungs.
  Narcotic and non-narcotic.
- Local anesthetics.
- Lozengers.

1. Antitussives

Definition: these drugs reduce the frequency or intensity of coughing.

Classification:



1-Opiods (narcotic antitussive):

A-Codeine:

- Anopium alkaloid; **less potent** than morphine. **Codeine, hydrocodone,** and **hydromorphone** decrease the **sensitivity of the cough center** to peripheral stimuli and decrease **mucosal secretion**. **Suppresses** cough center (**for 6 h**).
- The **antitussive action** is **blocked** by **naloxone**.
- **Abuse liability** is **low**, but present; **constipation** is the **chief drawback.**
- **Higher doses** of respiratory **depression** and **drowsiness** can occur, driving may be impaired.

B-Pholcodeine:

- **similar** inefficacy as **codeine**.
- **long acting** than codeine (**12h**).

C-Ethylmorphine:

- **similar** to **codeine**.

2-Non opiods (non-narcotic antitussive):

A-Noscapine:

- **An opium alkaloid** of **benzoisoquinoline**.
- It **depresses** cough but **not narcotic, analgesic, or dependence** including properties.
- **An equipotent antitussive** as **codeine** and **useful** in spasmodic cough
- **Side effects:** Headache, nausea. It can **produce** bronchoconstriction by **stimulating histamine release.**

B-Dextromethorphane:

- **synthetic compound**;**d-isomer** has **selective** antitussiveaction; **l-isomer** has **analgesic** property.
- **Effective** as **codeine** doesn't **depress** the **mucociliary function** of air way mucosa.
- **Antitussive effect** is **not blocked** by **naloxone**.
- **Side effect:** dizziness,nausea, drowsiness and ataxia.

C-Oxeladin:

- **centrallyacting antitussive.**

2-Demulcent and expectorants

2-Expectorants :

increase the bronchial secretion or **reduce** its viscosity, facilitating its **removal** by coughing.

A-Sodium citrate or acetate(0.3-1g):

- **Increase** bronchial **secretion** by **salt action.**

B-Potassium iodide(0.2-0.3):

- **Act directly** on the bronchial **secretory cells** and are **excreted** into the **respiratory tract.**
- It is an irritant and increasethe **volume of secretion.**
- The irritant action is **not desirable** if bronchial mucosa is **acutely inflamed.**
- **Dangerous** in patients **sensitive to iodine** and **interferes** with **thyroid function test .**
- **prolonged use** can **induce goiter and hypothyroidism.**

C-Ammonium salt (0.3-1g):

- Gastric irritants.
- **Reflexly** enhance bronchial **secretion and sweating.**
- **Nauseating** because of unpleasant taste.

3-Mucolytics

A-Bromohexine/Ambroxol:

- **Derivative** of the **alkaloid** vasicine obtained from **Adhantoda vasica.**
- **potent mucolytic and mucokinetic**, capable of **inducing thin copious** bronchial secretion.
- **Dissolving hard phlegm/mucus plugs.**
- **Side effect:** Rhinorrhoea, lacrymation, gastric irritation, hypersensitivity.

B-Acetylcysteine:

- **Derivatives** of **cysteine**.
- **reduce/open** the **disulfiede bridges** in mucoproteins present in sputum.
- These drugs also acts as **antioxidants** and may therefore **reduce** airway **inflammation**.
- **Side effects:**Not popular one.

Other drugs may used in cough:

1-Bronchodilators:

Bronchospasm induce or aggravate cough.

Mechanism of action:

Surface velocity of air flow during cough.

Clear secretions of airway.

Not used routinely for every type of cough.

Use **only** when **bronchoconstriction is present** (bronchial asthma).

Examples; **β2-agonist (salbutamol,terbutaline).**

2-Antihistamines:

Add to **antitussive/expectorant** formulation.

Relief in cough due to **sedative and anticholinergic** actions.

Lacks selectivity for cough center.
No expectorant action.
Anticholinergic effect
Suitable for **allergic cough** but **not for asthma.**
Example: **chlorpheniramine, diphenhydramine and promethazine.**

3-Local anesthetic:

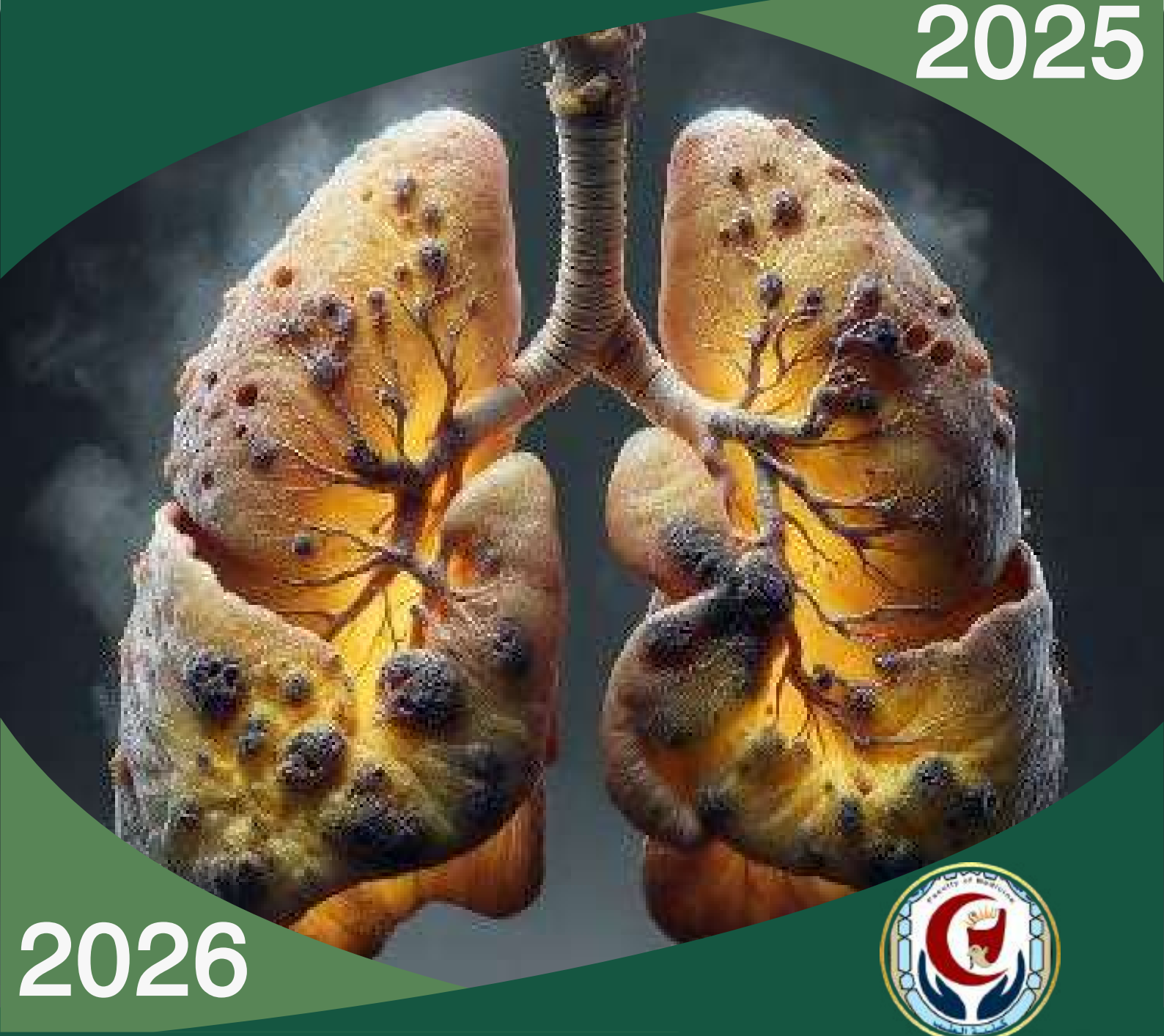
Lingocaine.

- For **intractable** cough of **bronchial carcinoma.** Administer by **nebulizer**.
- To **prevent** cough during **endoscopy**.
- **Instill topically** in airway.

Specific treatment approach to cough

Etiology of cough	Treatment approach
1.Upper/lower respiratory tract infections	Appropriate antibiotics
2. Smoking/chronic bronchitis	Cessation of smoking
3. Pulmonary tuberculosis	Antitubercular drug
4. Asthmatic cough	Inhaled β2-agonists/iprat-ropium/corticosteroid
5. Postnasal drip (sinusitis)	Antibiotics, nasal decongestants Antihistamine
6.Gastroesophageal reflux	PPI,diet modification
7.ACE inhibitor	Losartan

2025



2026

SOHAG UNIVERSITY
FACULITY OF
MEDICINE

Respiratory system

Pathology

RPS_314

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2025

Diseases of the upper respiratory tract

Diseases of the nose

Rhinitis:

Inflammation of mucous membrane of the nose

Acute rhinitis: Catarrhal, allergic

Chronicrhinitis:May bespecificornon specific

1- Chronic specific rhinitis (granuloma): Including syphilis,tuberculosis,leprosy and rhinoscleroma

2- Chronic non specific rhinitis: Appear in one of two forms:

1- Chronic hypertrophic rhinitis: 2- Chronic atrophic rhinitis

ACUTE CATARRHAL RHINITIS (common cold)

Aetiology: A common disease caused by a virus. Exposure to cold is a predisposing factor.

Pathology: Catarrhal inflammation

ComplicationS: (1) Direct spread of infection causing otitis media, nasopharyngitis, laryngitis, bronchitis and bronchopneUmOflia.

(2) Sinusitis specially in the maxillary sinus.

Chronic non specific rhinitis: (1) Hypertrophic chronic rhinitis: The submucosa shows the presence of chronic inflammatory cells, distended vascular spaces and fibrosis. The mucosa is thickened and polypoid.

(2) **ATROPHIC RHINITIS**

Aetiology: A rare disease which is more common in females.

Pathology:

(1) The **nasal mucosa** is thin, atrophic and covered by crusts of dried discharge which have offensive odour.

(2) Atrophy of the nasal bones causing **widening** of the nasal cavity.

(3) Atrophy of the nerve endings causing loss of smell (anosmia).

SCLEROMA (chronic specific (RHINOSCLEROMA) rhinitis)

Definition: A granulomatous inflammation of the upper respiratory tract

- caused by **Kiebsiella rhinoscieromatis.**
- It commonly affects the nose **(rhinoseleroma).**
- The disease is endemic in Egypt.

Gross Picture:

(1) The lesion appears as multiple small hard nodular masses or a single large nodular mass filling the nasal cavity.

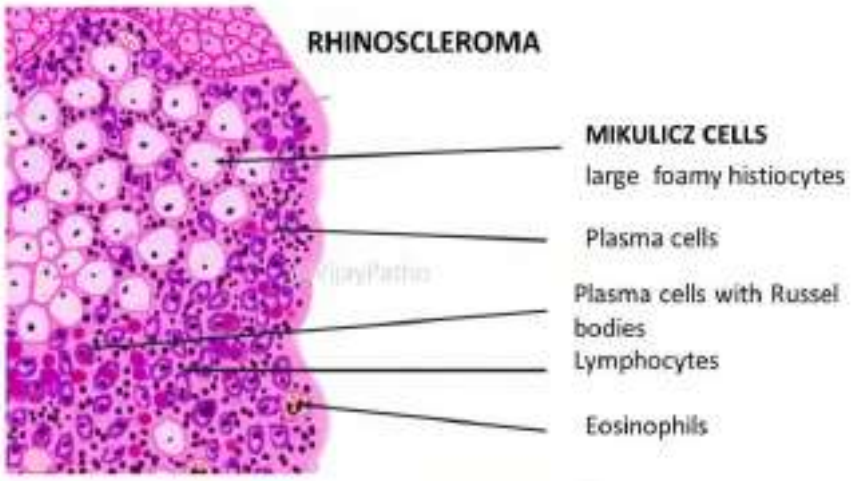
(2) The lesion may extend to the nasal sinuses, anteriorly to the upper lip or posteriorly to the nasopharynx, pharynx (pharyngoscieroma), larynx (laryngoscleroma) and trachea.

Microscopic Picture: The submucosa shows a chronic inflammatory cells. The inflammatory cells arc Mickulicz cells, plasma cells, Russel bodies and lymphocytes. Later the lesion shows proliferation of fibrous tissue.

The characteristic cells in rhinoscieroma are:

(a) **Mickulicz cells:** Are the predominant cells. They are macrophage with hydropic degeneration. The cells are large, sharply outlined with foamy cytoplasm and small pyknotic central or eccentric nuclei,

(b) **Russel bodies**: Plasma cells with hyaline degeneration. They appea round or oval with bright red cytoplasm. Some cells show a pyknoti eccentric nucleus, others are anuclear.



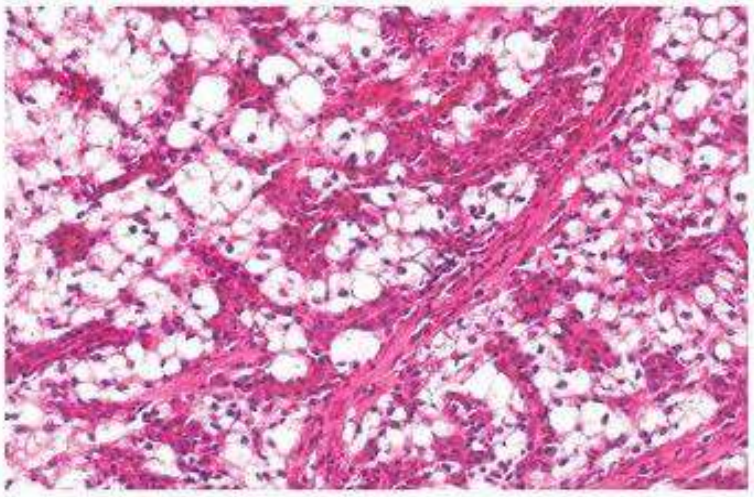


FIGURE 21.6 Nasal rhinoscleroma contains a large number of foamy histiocytes in a plasma cell–rich inflammatory background.



(I) Nasal obstruction and deformity.

(2) Ulceration and secondary infection.

Complications:

SINUSITIS

Definition: Inflammation of the mucosa lining the sinuses.

Acute Sinusitis: Occurs with catarrhal and allergic rhinitis

Chronic Sinusitis:

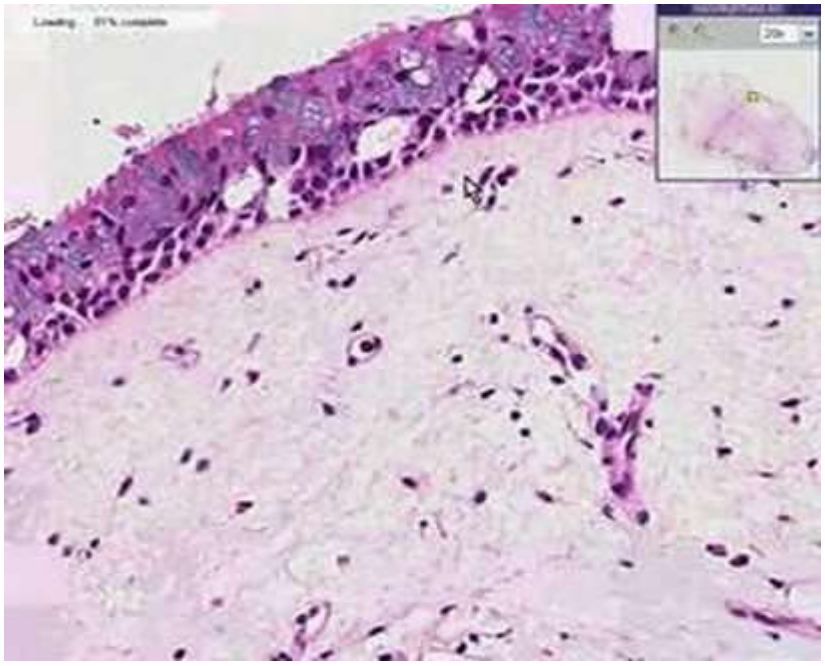
Causes: (1) A complication of repeated attacks of acute sinusitis
(2) Infection of the maxillary sinus from the root of the teeth.

Pathology of chronic sinusitis:



The sinus submucosa shows inflammatory oedema, chronic inflammatory cells, granulation tissue and fibrosis. The oedematous mucosa of the sinus may project through the sinus ostium forming variable sized pink or grey soft masses called nasal polyps.

Microscopically the polyp consists of a core of oedematous connectiv tissue showing chronic inflammatory cells with excess eosinophils, granula tion tissue and proliferating mucous glands.



Epistaxis

It denotes bleeding from the nose

Causes:

- Local causes: Such as
- All types of rhinitis (acute and chronic, specific and non specific)
- All benign and malignant tumours of the nose
- Trauma,

B-General causes:

- Hypertension
- Infectious fevers as typhoid and measles
- Scurvy(vit C deficiency)
- Vitamin K deficiency
- Blood diseases as haemophilia,purpura and leukaemia

TUMOURS OF THE NOSE AND SINUSES

I. Benign Tumours:

- Epithelial: Squamous cell papilloma, transitional cell papilloma, inverted papilloma and pleomorphic salivary adenoma.
- Mesodermal: Angioma and osteoma are the common, but fibroma, chondroma, myxoma, lipoma, neurofibroma and osteoclastoma may occur

II. Malignant Tumours:

(1) Epithelial: Transitional cell carcinoma, squamous cell carcinoma, lymphoepithelioma and adenocarcinoma.
(2) Mesodermal: Fibrosarcoma, osteosarcoma, chondrosarcoma, malignant melanoma and malignant lymphomas.

DISEASES OF THE LARYNX LARYNGITIS

Acute Laryngitis: bacterial as in diphtheria or viral as in common cold
Chronic Laryngitis
(a) Chronic specific: T.B,SYPHILIS
(b) Chronic NON specific : due to smoking or on top of acute inflammation or abuse of voice
laryngeal nodule
Vocal cord polyp

LARYNGEAL NODULE

- Develops on the true vocal cords in heavy smokers or in individuals who impose great strain on their vocal cords as singers and teachers.

- Pathology: Small smooth rounded nodule either sessile or
- peduncujated Microscopically a core of loose myxoid connective tissue showing

vascular channels and fibrosis. The core is covered by squamous epithelium which may become keratotic, hyperplastic or ulcerated.





TUMOURS OF THE LARYNX

Squamous Cell Papilloma:

(1) In **children**: multiple papillomata are caused by a **viral infection.** They have a tendency to recur after surgical removal, but do not turn malignant.

(2) In **adults:** the tumour appears as a single small sessile papilloma on the vocal cords. The tumour lacks the tendency to recurrence, but sometimes it becomes malignant. **Multiple papillomas are rare.**

Squamous Cell Carcinoma:

Mainly affects **males** in the fifth decade of life or beyond.

The precancerous lesions are chronic laryngitis caused by tobacco smoking, **leukoplakia** and squamous cell papilloma.

Pathological features: Grossly the tumour appears as a pink to gray mass

1- 4 cm in diameter often **ulcerated**.

Microscopically: **squamous cell carcinoma** graded into well, moderately and poorly differentiated types.

A rare highly differentiated type of squamous cell carcinoma appears as a polypoid growth and is called verrucous carcinoma. This type is superficially invasive and never metastasize, so the prognosis is good.

Types: (1) Glottic (65%): Arises from the true vocal cords commonly from the anterior third. Microscopically, the tumour is well to moderately differentiated. The prognosis is good because of early diagnosis (causes hoarseness of voice), slow growth, easy surgical removal, radiosensitivity and rare nodal metastasis.

(2) Supraglottic (30%): Arises from false cord, ventricle and epiglottis.

The tumour is moderately to poorly differentiated.

The prognosis is bad because of delayed diagnosis, less differentiation, rapid growth, difficult surgical removal and common nodal metastasis.

(3) Infraglottic (subglottic): Less than 5% of cases.

PulmonaryInfections

Pneumonia & Pneumonitis

- **Definition: Pneumonia/pneumonitis** is an inflammation of the lung
- parenchyma distal to the terminal bronchioles.
  Lung inflammation is more accurately referred to as **pneumonitis;**
- whereas pneumonia refers to pneumonitis due to infection and sometimes non-infectious conditions that have **pulmonary** consolidation.

Types of Pneumonitis/Pneumonia

- Pneumonitis are classified according to:
- Onset and duration:
- Acute pneumonitis/pneumonia
- Chronic pneumonitis/pneumonia
- Morphological types:
- Lobar pneumonitis/pneumonia
- Lobular/Broncho-pneumonitis/pneumonia
- Interstitial pneumonitis/pneumonia
- Etiological types:
- Infective pneumonitis/pneumonia
    - Bacterial pneumonia
    - Viral pneumonia
    - Fungal pneumonia
    - Protozoal pneumonia
    - Tuberculous pneumonia
- **Non-infective pneumonitis**
    - Toxins-induced pneumonitis
    - Chemical pneumonitis
    - Aspiration pneumonitis
    - Auto-immune disease associated pneumonitis

Clinical types: Primary/ secondary pneumonitis/pneumonia depending on the **mode of infection.**

- Typical/ atypical pneumonitis/pneumonia depending on the
- presenting symptoms. Community acquired/ hospital acquired
- pneumonitis/pneumonia depending on where it was acquired. Pneumonitis/pneumonia in **immuno-compromised** patients.

- Currently, combined clinical classification put pneumonia in seven distinct clinical settings “pneumonia syndromes”.

Pneumonia Syndromes

- Community-Acquired Acute Pneumonia(CAP)
- Health Care-Associated Pneumonia
- Hospital-Acquired Pneumonia
- Aspiration Pneumonia
- Chronic Pneumonia
- Necrotizing Pneumonia and Lung Abscess
- Pneumonia in Immunocompromised Host

Lobar Pneumonia

- Definition:
- Acute bacterial infection of a part of a lobe, the entire lobe, or two lobes of one or both lungs.

Etiology of Lobar Pneumonia

- Pneumococcal pneumonia:
- Streptococcus pneumoniae in 90% of cases.
- The majority of cases are community-acquired (CAP).

- Staphylococcal pneumonia:
- It is due to hematogenous spread from another focus or after viral infections.

- Streptococcal pneumonia:
- Causes pneumonia in children after measles or influenza, or in severely debilitated elderly patients and in diabetics.

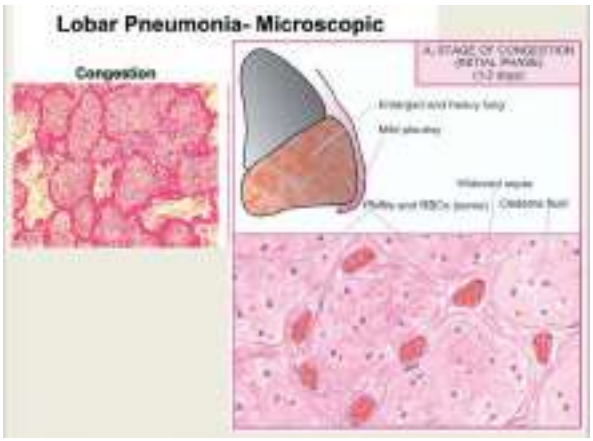
- Pneumonia by gram-negative aerobic bacteria.
- H influenzae, K pneumoniae, Pseudomonas, Proteus and E coli.
- H. influenzae causes pneumonia in children below 3 years of age after a preceding viral infection.

Morphology of Lobar Pneumonia

- Four stages of the inflammatory response:
- Stage of congestion (initial phase):

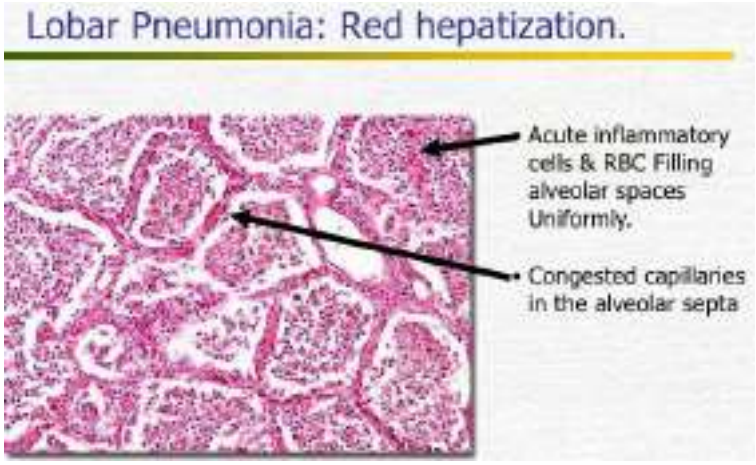
- Initial phase corresponds to the onset of disease.
- It is the early acute inflammatory response to bacterial infection and lasts for 1-2 days.
- Clinically:
- High grade fever, coughing rusty sputum.
- Grossly:
- The affected lobe is enlarged, heavy, and dark red.
- The consistency is wet sponge, and the cut surface exudes frothy blood-stained serous exudate.

- **Microscopically:**
- Dilatation and congestion of the capillaries in the alveolar walls.
- Pale eosinophilic edema fluid in the air spaces.
- A few red cells and neutrophils in the intra-alveolar fluid.
- Numerous bacteria demonstrated in the alveolar fluid by Gram's staining.

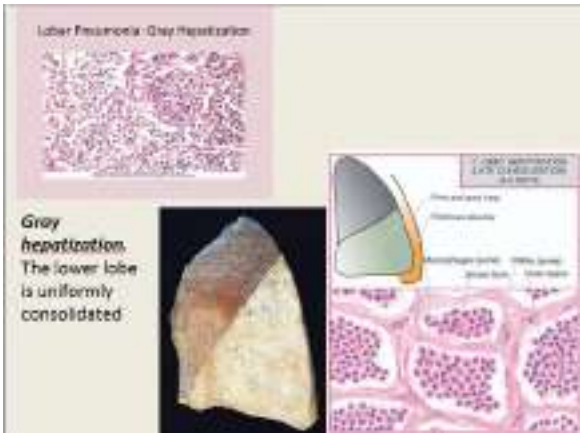


- Stage of red hepatization (earlyconsolidation):
- This phase lasts for 2-4 days.
- Hepatisation in pneumonia refersto liver-like consistency of the affected lobe on cut section.
- Gross examination:
- The lobe is red, firm, and airless (consolidated),with liver-like consistency.
- The cut surface is dark red, dry, and granular.
- The bronchi contain purulent exudate.
- The pleura show serofibrinous pleurisy.
- The hilar lymph nodes are enlarged.

- Microscopic picture:
- The edema fluid of the preceding stage is replaced by strands of fibrin.
- There is marked cellular exudate of neutrophils and extravasation of RBCs.
- Many neutrophils show ingested bacteria.
- Alveolar septa are less prominent than in the first stage due to cellular exudation.



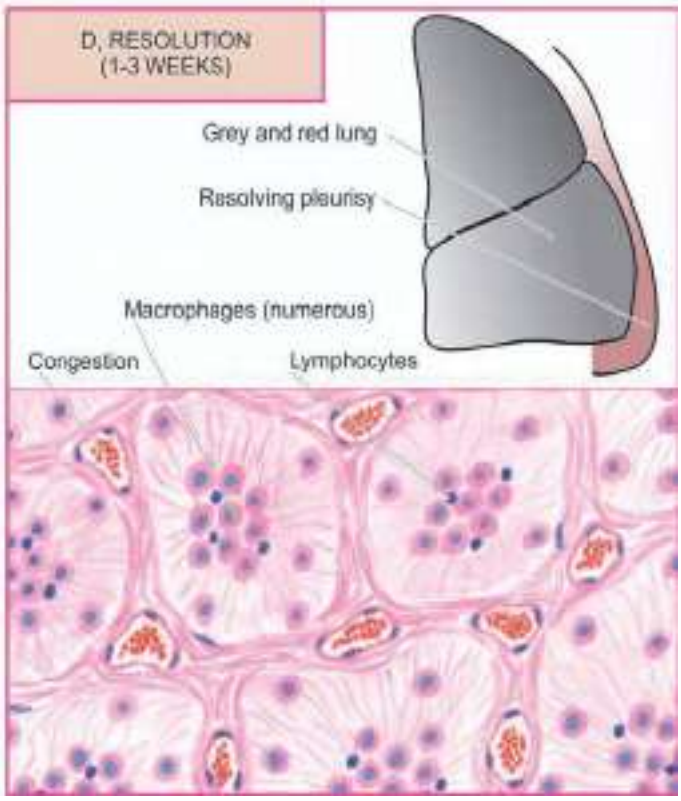
- Stage of gray hepatization (late consolidation):
- This phase lasts for 4-8 days. It is marked by progressive disintegration of RBCs and persistence of a **fibrino-suppurative** exudate, resulting in acolor change to **grayish-brown.** Grossly:
- The affected lobe is firm and heavy.
- The cut surface is dry, granular and grey in appearance with **liver-**
- like consistency.
- The change in color from red to **grayish brown** begins at the hilum and spreads towards the periphery.
  Fibrinous pleurisy is ***prominent.***



Histologically:

- There is less marked thickening of the alveolar wall and less capillary congestion.
- Fibrin strands are dense, more numerous and clumped in the center of the alveoli.
- Cellular exudate of neutrophils is reduced due to their disintegration.
- RBCs are also fewer. Macrophages begin to appear in the exudate.
- Cellular exudate is often separated from the septal walls by a thin clear space.
- Organisms are **less numerous** and degenerated.

- Stage of resolution (final stage):
  This stage begins by 8th to 9th day with no treatment and is completed in 1-3 weeks.
- Antibiotic therapy induces resolution on about the 3rd day.
- Resolution proceeds in a progressive manner.



- Grossly: The exudate within the alveolar spacesisbroken down
- by enzymatic digestion to produce granular,semifluid debris that is resorbed, ingested by macrophages, expectorated, restoring the normal aeration in the affected lobe. The process of softening begins centrally and spreads to the periphery.
- Cut surface is grey-red or dirty brown and frothy, yellow, creamy
- fluid exudes on pressing.
- Pleuritis shows resolution but may undergo organization → fibrous
- thickening and permanent adhesions.

- **Histologically:**

- Macrophages are the predominant cells in the alveolar spaces,
- while less neutrophils.
- Many of the macrophages contain engulfed neutrophils and debris.
- Fragmented strands of fibrin in the alveolar spaces are seen due to progressive enzymatic digestion.
- Alveolar capillaries are engorged.
- There is progressive removal of fluid content and cellular exudate

from air spaces, partly by expectoration, but mainly by lymphatics, resulting in restoration of normal lung parenchyma with aeration.

Clinical Features of Lobar Pneumonia

- Classically, the onset of lobar pneumonia is sudden.
- The major symptoms are: shaking chills, fever, malaise with pleuritic chest pain, dyspnea and cough with expectoration of mucoid, purulent or even bloody sputum.
- Physical findings are high temperature, tachycardia, tachypnea, and cyanosis if the patient is severely hypoxemic. Blood picture shows
- marked neutrophil leucocytosis.

Complications of Lobar Pneumonia

- Organization:
- In about 3% of cases, resolution of the exudate does not occur, it undergoes organization and fibrosis.
- Ingrowth of fibroblasts from the alveolar septa results in fibrosed,
- tough, airless leathery lung.
- Post-pneumonic fibrosis is called carnification.
- Pleural effusion:
- Develop in 5% of treated cases.
  Pleural effusion usually resolves but sometimes undergoes organization with fibrous adhesions between visceral and parietal pleura.

- Empyema:
- Less than 1% of treated cases develop encysted pus in the pleural cavity termed empyema.
- Lung abscess:
- Rare complication especially when there is secondary infection by other organisms.
- Bacteremic dissemination:
- Infection extend → pericarditis, endocarditis and myocarditis.
- Metastatic infection → otitis media, mastoiditis, meningitis, brain abscess and purulent arthritis.

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Bronchopneumonia (Lobular Pneumonia)

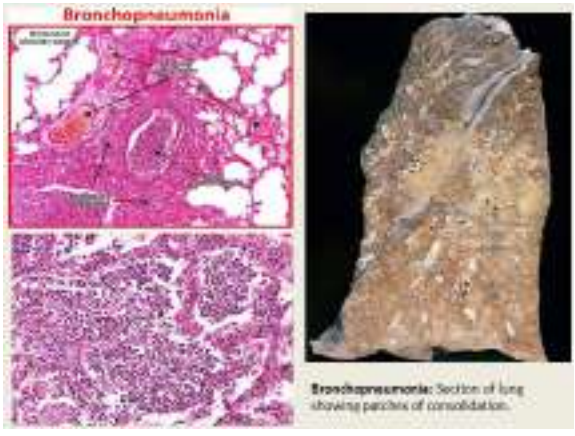
- Definition: Bronchopneumonia or lobular pneumonia is
- infection of the terminal bronchioles that extends into the surrounding alveoli
  resulting in patchy consolidation of the lung.
- Frequent at the extremes of life as a terminal event in chronic debilitating diseases and as a secondary infection following viral respiratory infections such as influenza, measles.

Etiology of lobular pneumonia

- The most common organisms are staphylococci, streptococci, pneumococci, K pneumoniae, H influenzae, and gram-negative bacilli like Pseudomonas and coliform bacteria.

Morphology of Lobular Pneumonia

- Grossly:
- Bronchopneumonia is identified by patchy areas of red or grey consolidation affecting one or more lobes, frequently found bilaterally and more often in the lower zones of the lungs due to gravitation of secretions.
- On cut surface, patchy consolidated lesions are dry, granular, firm, red or grey in color, 3-4 cm in diameter, slightly elevated over the surface and are centered around a bronchiole.



Bronchopneumonia: Section of lung showing patches of consolidation.

Histologically:

- Features of acute bronchitis.
- Suppurative exudate, consisting chiefly of neutrophils, in the peri-bronchiolar alveoli.
- A neutrophil-rich exudate fills the bronchi, bronchioles, and

adjacent alveolar spaces.
- Thickening of the alveolar septa by congested capillaries and leucocytic infiltration.
- Less involved alveoli contain edema fluid.

Clinical Features of Lobular Pneumonia

- Patients are generally infants or elderly individuals.
- History of preceding bed-ridden illness, chronic debilitating disease, aspiration of gastric contents or URT infection.
- For initial 2-3 days, there are features of acute bronchitis but subsequently signs and symptoms similar to those of lobar pneumonia appear.
- Blood picture shows a neutrophilic leucocytosis.
- Chest radiograph shows mottled, focal opacities in both lungs, chiefly in the lower zones.

Complications of Lobular Pneumonia

- Complications of lobar pneumonia may occur in bronchopneumonia as well.
- However, complete resolution of bronchopneumonia is uncommon.
- There is some degree of destruction of bronchioles → foci of bronchiolar fibrosis → bronchiectasis.

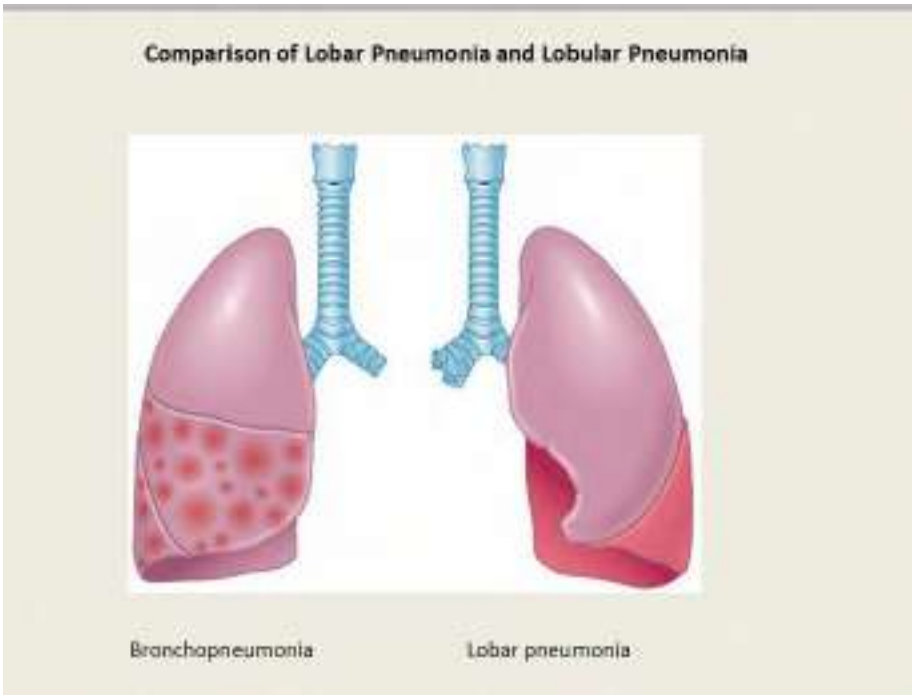
- Tissue destruction and necrosis → abscess formation (common with Klebsiella infections);
- Spread of infection to the pleural cavity, → empyema; and
- Bacteremic dissemination → metastatic abscesses, endocarditis, meningitis, or suppurative arthritis.

Comparison of lobar pneumonia and lobular pneumonia

Comparison of lobar pneumonia and lobular pneumonia

Features	Lobar Pneumonia	Lobular Pneumonia
Definition	Acute bacterial infection of a part of a lobe of one or both lungs, or the entire lobe/s	Acute bacterial infection of the terminal bronchioles extending into adjoining alveoli
Age group	More common in adults	Commoner at extremes of age; infants and old age
Predisposing factors	More often affects healthy individuals	Preexisting diseases e.g. chronic debility, terminal illness, flu, measles
Etiological agents	*Pneumococci, K pneumoniae, Staph, Strept*	Staph, Strept, Pseudomonas, *H influenze*
Type of inflammation	Fibrinous inflammation	Suppurative inflammation

Features	Lobar Pneumonia	Lobular Pneumonia
Pathologic features	Typical case passes through stages of congestion (1-2 days), early consolidation (2-4 days) and late consolidation (4-8 days), followed by resolution (1-3 weeks)	Patchy consolidation with central granularity, alveolar exudation, thickened septa
Healing	Resolution (crisis) in most cases	Lysis (common complications)
Investigations	Neutrophilic leucocytosis, positive blood culture, X-ray shows consolidation	Neutrophilic leucocytosis, positive blood culture, X-ray shows mottled focal opacities
Prognosis	Better response to treatment, resolution, prognosis good	Response to treatment variable, organization may occur, poor prognosis
Complications	Less common; pleural effusion, empyema, lung abscess, organization	Bronchiectasis may occur; other complications are the same as for lobar pneumonia



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Interstitial Pneumonia

- **Etiology**:
- The **most common organism** is **RSV.**
- Others are Mycoplasma pneumoniae, influenza, parainfluenza, CMV, adenoviruses, rhinoviruses.
- Occasionally, psittacosis (Chlamydia) and Q fever (Coxiella) → interstitial pneumonitis.

Morphological features of Interstitial Pneumonia

- Irrespective of the etiology, pathological changes are similar in all cases.
- Grossly patchy to massive widespread consolidation of one or both lungs.
- The lungs are heavy, congested.
- Cut surface of the lung exudes small amount of frothy or bloody fluid.
- Pleural reaction is infrequent and mild.

- ***Histologically***:
- The hallmark of viral pneumonias is the interstitial nature of the inflammatory reaction.
- Interstitial inflammation:

- Thickening of alveolar walls due to congestion, edema and mononuclear inflammatory infiltrate composed of lymphocytes, macrophages and some plasma cells. Necrotising bronchiolitis:
- Foci of necrosis of the bronchiolar epithelium, inspissated
- secretions
  in the lumina and mononuclear infiltrate in the walls and lumina.

- **Reactive changes:**
- Lining epithelial cells of bronchioles and alveoli proliferate and may form multinucleate giant cells in the bronchiolar and alveolar walls.
- Viral inclusions are found (intranuclear and/or intracytoplasmic), especially in pneumonitis caused by CMV.
- **Alveolar changes:**
- In severe cases, the alveolar lumina may contain edema fluid, fibrin, scanty inflammatory exudate and coating of alveolar walls by pink, hyaline membrane.

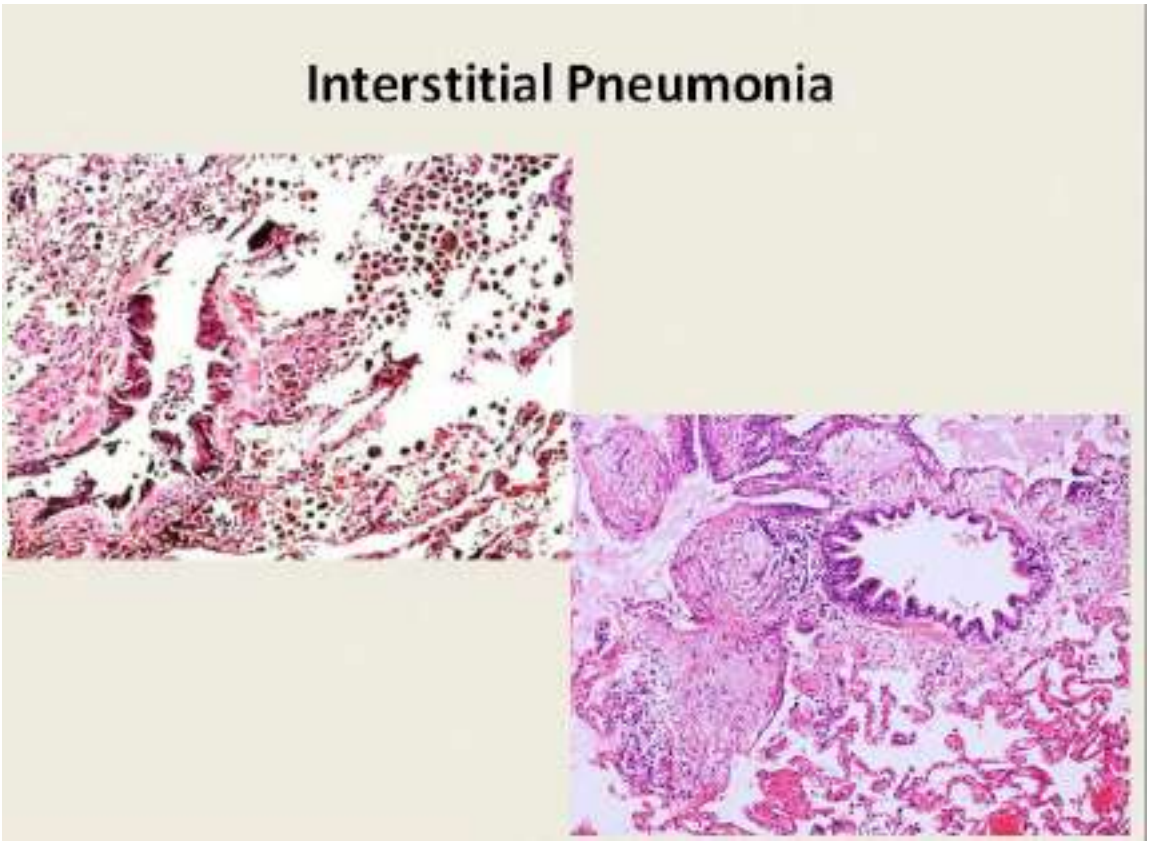
Interstitial Pneumonia

Clinical features of Interstitial Pneumonia

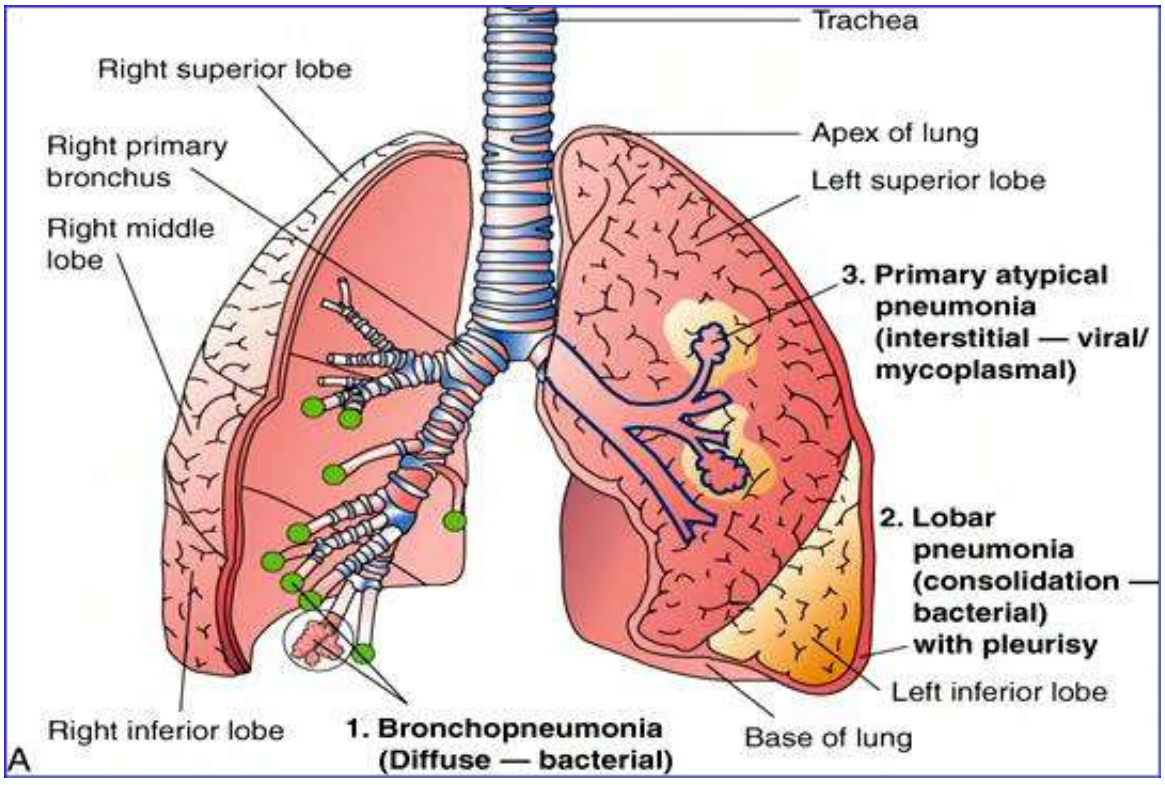
- The majority of cases initially have URT infection with fever, headache and muscle-aches.
- A few days later there is dry, hacking, non-productive cough with retrosternal burning due to tracheitis and bronchitis.
- Blood picture shows lymphocytic leucocytosis.
- Chest radiograph may show patchy or diffuse consolidation.

Complications of Interstitial Pneumonia

- Superimposed bacterial infection and its complications.
- Most cases recover completely.
- In more severe cases, there may be interstitial fibrosis and permanent damage.



Comparing Types of Pneumonia



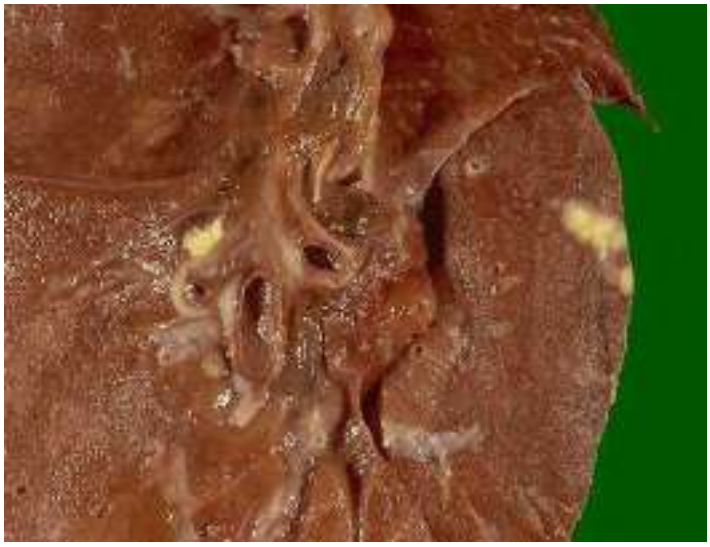
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Pulmonary Tuberculosis

- The classical and most common example of chronic infection of the lungs is pulmonary tuberculosis which may be:
  - Primary pulmonary tuberculosis
  - Secondary pulmonary tuberculosis
- Pulmonary lesions caused by Mycobacterium tuberculosis → granulomatous inflammations.

Primary Pulmonary Tuberculosis

- Etiology: Follows the first infection of the lung with T.B. bacilli.
- Infection is by inhalation of human T.B. bacilli.
- Pathological features:
- The primary complex consists of:
  - Ghon's focus (small subpleural lesion)
  - Tuberculous lymphangitis
  - Tuberculous lymphadenitis
- **Fate:**
  - Healing
  - Spread of infection
  - Encapsulation with later reactivation.



Secondary Pulmonary Tuberculosis

- Etiology: Infection may be:

- Exogenous by inhalation of T.B. bacilli
- Endogenous by reactivation of a capsulated lesion
- Pathological features:
- Chronic Fibro-Caseous Pulmonary Tuberculosis: Consists of:
- Apical cavitation
- Acinar lesions: Part of caseous material is aspirated into distal bronchioles leading to caseating lesions.
- Nodal lesion: LNs are small as the bacilli reaching them are scanty, they are destroyed in the draining LNs.

- Tuberculous Bronchopneumonia
- Rare fatal condition develops as a complication of primary or secondary pulmonary tuberculosis.
- It is caused by a large dose of bacteria in a patient with high hypersensitivity and low immunity.
- The lung shows multiple scattered small rounded caseous lesions in relation to the small bronchi.
- The hilar lymph nodes are enlarged and caseous.

- **Complications:**
- Spread of infection either by:
  - Blood causing miliary tuberculosis or isolated organ tuberculosis.
  - Direct to the pericardium and mediastinum.
  - Direct to the pleura causing sero-fibrinous pleurisy or tuberculous empyema.
- The cavity may rupture → pneumothorax or pyopneumothorax.

- Erosion of a vessel in a cavity causes severe hemoptysis.
- Expectoration of infected sputum → tuberculosis of the, larynx,
- tonsils or tongue.
- Swallowing infected sputum causes intestinal tuberculosis.
- Extensive lung fibrosis leads to right sided heart failure.
- Continuous break down of the lung tissue causes secondary amyloidosis.

Miliary TB

Lung Abscess

- Definition: It is a local suppurative process that produces necrosis of lung tissue.
- Etiology and Pathogenesis of Lung Abscess:
- Common organisms include aerobic and anaerobic streptococci, Staph. aureus, and gram-negative organisms.
- Mixed infections often occur due to inhalation of foreign material.



- ***The causative organisms are introduced by the following mechanisms:***
- Aspiration of infective material (the most frequent cause):
- Common in coma, anesthesia, sinusitis, and debilitation due to depression of cough reflexes.
- Aspiration first causes pneumonia which progresses to tissue necrosis and formation of lung abscess.

- Antecedent primary lung infection:
- Post-pneumonic abscess and Post-transplant.

- Septic embolism: Infected emboli from thrombophlebitis in
- systemic veins or from vegetation on the right side of the heart in acute infective bacterial endocarditis. Neoplasia:
  Secondary infection in bronchopulmonary segment obstructed
- by
- a primary or secondary tumor. Miscellaneous: Direct traumatic penetrations of the lungs; spread
- of infections from a neighboring organ e.g., esophagus, spine, subphrenic space, or pleural cavity; and hematogenous spread of
  pyogenic infection.

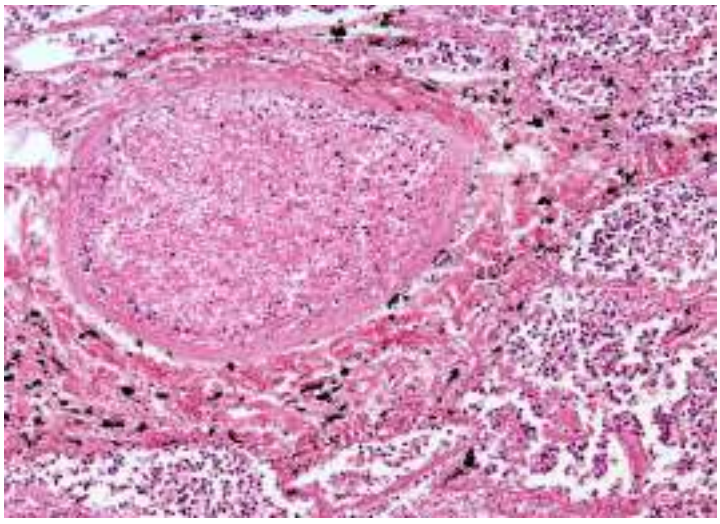
Gross Pathology of Lung Abscess

- Few millimeters to large cavities of 5-6 cm. Affect any part of the lung.
- Single or multiple.
- Pulmonary abscesses due to aspiration are more common on the
- right (because of the more vertical right main bronchus) and are most often single.
  Abscesses due to pneumonia are usually multiple, basal, and
- diffusely scattered.
  Septic emboli → pyemic abscesses which are multiple and affect any
- region of the lungs.
  The cardinal change in all abscesses is suppurative destruction
- of the lung parenchyma within the central area of cavitation.
  Abscess cavity is filled with pus, and if there is communication with
- an air passage, it is partially drained to create an air-containing cavity.
  Continued infection leads to large, poorly demarcated, fetid, green-
- black, multilocular cavities designated gangrene of the lung.
  In chronic cases fibroblastic proliferation produces fibrous wall.
-

- **Lung Abscess** **Clinical Course of Lung Abscess**

- Fever, cough with copious amounts of foul-smelling purulent or sanguineous
- sputum.
- Chest pain, and **weight loss are common.**
- Clubbing of the fingers and toes may appear.

Diagnosis clinical findings must be confirmed by radiology.
Complications: extension of infection into the pleural cavity, development of brain abscesses or meningitis from **septic emboli,** and **secondary amyloidosis.**



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MCQs

- Inflammation in bronchopneumonia is:
  A. Suppurative
  B. Catarrhal
  C. Fibrmous
  D. Serous
  E. Non-bacterial
- The lung tissue has a liver-like consistency in:
  A. Lobar pneumonia
  B. Bronchopneumonia
  C. Bronchiectasis
  D. Bronchogenic carcinoma E. Bronchial asthma

CHRONIC OBSTRUCTIVE PULMONARY DISEASES

ILOs

After the lecture, students should be able to:

Define bronchial asthma and describe the pathophysiology, morphology and pathological consequences of bronchial asthma.

Discuss chronic bronchitis

Define emphysema and enumerate its types, distinguish between two main types with particular reference to the pathogenic mechanism underlying each.

Define bronchiectasis, outlines the main causes, pathogenesis, morphological changes and complication.

Chronic Obstructive Lung Diseases

Chronic obstructive pulmonary diseases (COPD): is a commonly used clinical terms for a group of pathological conditions in which there is chronic, partial or complete, obstruction to the airflow at any level from trachea to the smallest airways resulting in functional disability of the lungs.

The major diffuse obstructive disorders are asthma, chronic bronchitis, emphysema, and bronchiectasis.

BRONCHIAL ASTHMA

Definition:

Asthma is a chronic inflammatory airways disease that manifested clinically by paroxysms and recurrent episodes of wheezing, breathlessness, chest tightness, and cough, particularly at night and/or early in the morning.

Types of Asthma

- **Atopic, extrinsic, or allergic**

- This is the most common type of asthma.
- It usually begins in childhood or in early adult life.
- Most patients have personal and/or family history of preceding allergic diseases such as rhinitis, urticaria or infantile eczema.
- Hypersensitivity to various extrinsic antigenic substances or „allergens" is usually present in these cases.

- Non-atopic, intrinsic or idiosyncratic asthma

- This type of asthma develops later in adult life with negative personal or family history of allergy, negative skin test and normal serum levels of IgE. Most of these patients develop typical symptoms after pulmonary infection by viruses e.g., rhinovirus, para- influenza virus and inhaled air pollutants e.g., sulfur dioxide, ozone, nitrogen dioxide.

Pathophysiology OF bronchial ASTHMA

The major etiologic factors of asthma are:

- Genetic predisposition to type I hypersensitivity (atopy),
- Acute and chronic airway inflammation, and
- Bronchial hyper-responsiveness to a variety of stimuli.

- The inflammation involves many cell types and numerous inflammatory mediators, but the role of type 2 helper T (TH2) cells may be critical to the pathogenesis of asthma.
- The classic atopic form of asthma is associated with an excessive TH2 reaction against environmental antigens. Cytokines produced by TH2 cells account for most of the features of asthma. IL-4 stimulates IgE production, IL-5 activates eosinophils, and IL-13 stimulates mucus production and also promotes IgE production by B cells.
- IgE coats submucosal mast cells, which, on exposure to allergen, release granule contents.
- This induces two waves of reaction: an early (immediate) phase and a late phase. The early reaction is dominated by broncho-constriction, increased mucus production and variable vasodilatation.
- Bronchoconstriction is triggered by direct stimulation of subepithelial vagal receptors.
- The late-phase reaction consists of inflammation, with activation of eosinophils, neutrophils, and T cells.
- In addition, **epithelial cells** are activated to produce chemokines that promote recruitment of more TH2 cells and eosinophils , as well as other leukocytes, thus amplifying the inflammatory reaction.
- Repeated bouts of inflammation lead to structural changes in the bronchial wall, collectively referred to as airway remodeling.
- These changes include hypertrophy of bronchial smooth

muscle and mucus glands, and increased vascularity and deposition of subepithelial collagen, which may occur as early as several years before initiation of symptoms.

- Asthma is a complex genetic disorder in which multiple susceptibility genes interact with environmental factors to initiate the pathological reaction.

Morphology of bronchial asthma

Grossly:

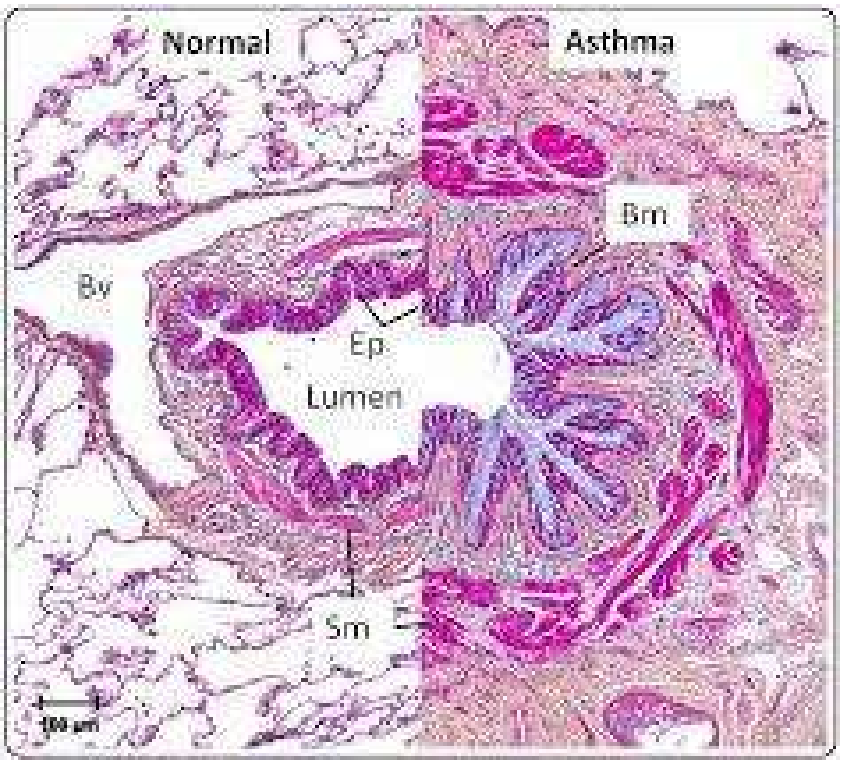
- The lungs are over-distended due to overinflation.
- The most striking macroscopic finding is occlusion of bronchi and bronchioles by thick, viscid and tenacious mucous plugs seen in the cut surface.
- There may be small areas of atelectasis.

Microscopically:

- The sputum contains numerous eosinophils and diamond-shaped crystals derived from eosinophils called Charcot-Leyden crystals.
- The mucus plugs contain normal or degenerated respiratory epithelium forming twisted strips called (Curschmann spirals).
- The bronchial wall shows thickened basement membrane of the bronchial epithelium, submucosal edema and inflammatory infiltrate consisting of lymphocytes and plasma cells with prominence of ~~eosinophils~~.
- There is hypertrophy of submucosal glands as well as of the bronchial smooth muscle.

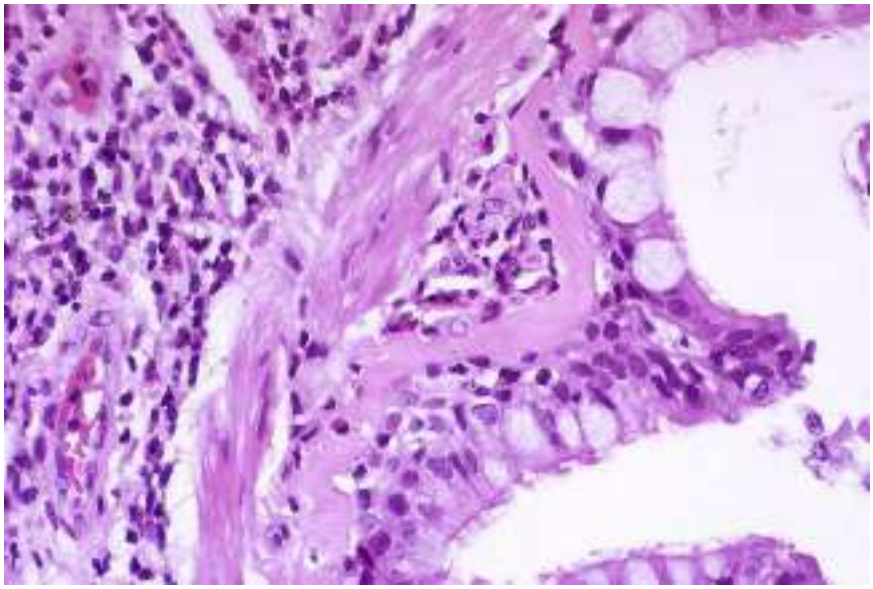
The following are characteristic morphologic changes occurring in asthma, and collectively called “airway remodeling”:

- Thickening of airway wall
- Sub-basement membrane fibrosis
- Increased vascularity in submucosa
- An increase in size of the submucosal glands and goblet cell metaplasia of the airway epithelium
- Hypertrophy and/or hyperplasia of the bronchial muscle.



Complication of bronchial asthma

- Secondary infection
- Chronic bronchitis
- Emphysema



CHRONIC BRONCHITIS

- Definition: the presence of persistent productive cough for at least 3 consecutive months in at least 2 consecutive years.
- The condition is more common in middle-aged males than females.
- Approximately 20% of adult men and 5% of adult women have chronic bronchitis, but only a minority of them develop serious disabling COPD.

- Etiology
- The two most important etiological factors responsible for the majority of cases of chronic bronchitis are: the cigarette smoking and the atmospheric pollution.
- Other contributory factors are occupation, infection, familial and genetic factors.

- **Smoking:**

- The most commonly identified factor implicated in causation of chronic bronchitis.
- Heavy cigarette smokers have 4 to 10 times higher chance to develop chronic bronchitis.

- Prolonged cigarette smoking appears to act on the lungs in a number of ways:

- It impairs ciliary movement.
- It inhibits the function of alveolar macrophages.
- It leads to hypertrophy and hyperplasia of mucus-secreting glands.
- It causes considerable obstruction of small airways.
- It stimulates the vagus and causes broncho-constriction.

- Atmospheric pollution:

- The incidence of chronic bronchitis is higher in industrialized urban areas where air is polluted.

- Occupation.

- Workers engaged in certain occupations such as in cotton mills (byssinosis), plastic factories etc. are exposed to various organic or inorganic dusts which contribute to disabling chronic bronchitis in such individuals.

- Infection:

- Bacterial, viral and mycoplasma infections do not initiate chronic bronchitis but usually occur secondary to bronchitis.
- Cigarette smoke, however, predisposes to infection responsible for acute exacerbation in chronic bronchitis

- Familial and genetic factors:

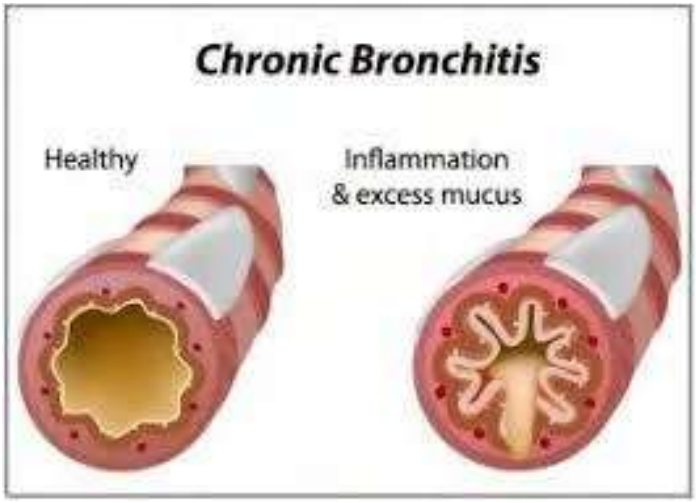
- There is poorly-defined familial tendency and genetic predisposition to develop chronic bronchitis.

- Nonsmoker family members who remain in air-pollution of the home are significantly exposed to smoke (passive smoking) and hence have increased blood levels of carbon monoxide.

Morphology of chronic bronchitis

o Grossly:

- The mucosal lining of the larger airways usually is thickened, hyperemic and edematous.
- It often is covered by a layer of mucinous or muco-purulent secretions.
- Lumina of the bronchi and bronchioles may contain mucus plugs and purulent exudate.
- The smaller bronchi and bronchioles also may be filled with similar secretions.



o Microscopically:

- The diagnostic feature of chronic bronchitis in the trachea and larger bronchi is enlargement of the mucus-secreting glands.
- Largely chronic inflammatory cell infiltrate, mixed with neutrophils, are present in variable density in the bronchial mucosa.

- The bronchial epithelium may show squamous metaplasia and dysplasia.
- The non-cartilage containing small airways show goblet cell hyperplasia and intra-luminal and peri-bronchial fibrosis.

Clinical Features of Chronic Bronchitis

- Some important features of „predominant bronchitis" are :

- Persistent cough with copious expectoration of long duration; initially beginning in a heavy smoker with „morning catarrh" or „throat clearing" which worsens in winter.
- Recurrent respiratory infections are common.
- Dyspnea is generally not prominent at rest but is more on exertion.
- Patients are called „blue bloaters" due to cyanosis and edema.
- Features of right heart failure (cor-pulmonale) are common.
- Chest X-ray shows enlarged heart with prominent vessels.

Complication

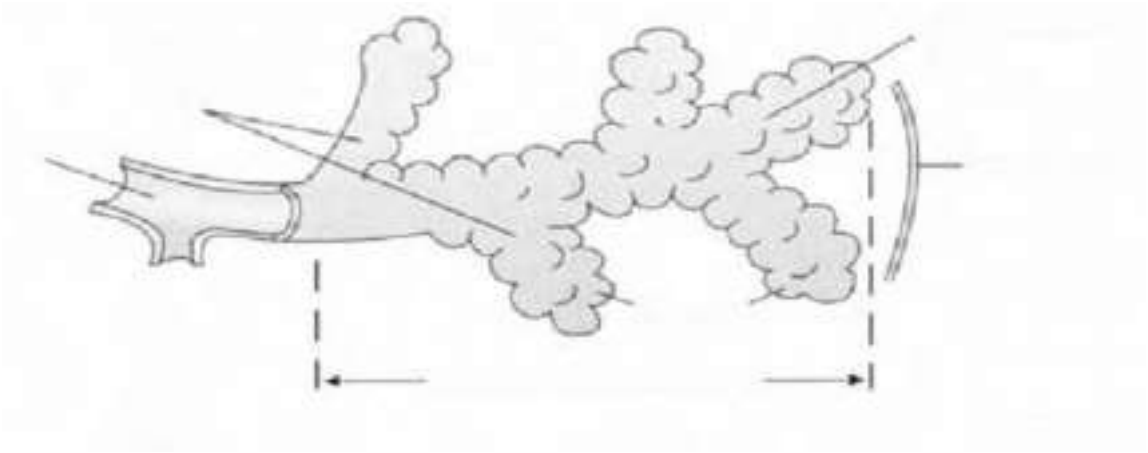
- With progression, chronic bronchitis is complicated by pulmonary hypertension and cardiac failure.
- Recurrent infections and respiratory failure are constant threats.

PULMONARY EMPHYSEMA

Definition:

Emphysema means abnormalpermanentoverdistentionofthe air spaces distal to the terminalbronchioleswithdestruction of their walls and of the alveolarsepta(containingalveolar capillaries) without significantfibrosis.

A terminal bronchiole leads to respiratory bronchioles, alveolarducts and alveoli. The portion of thelung tissue formed by the branching of a single terminal bronchioleis called respiratory acinus.



Pathogenesis of Emphysema:

(1) Protease-antiprotease theory: Destruction of elastic tissue in the lung is carried by a proteolytic enzyme –elastase||.

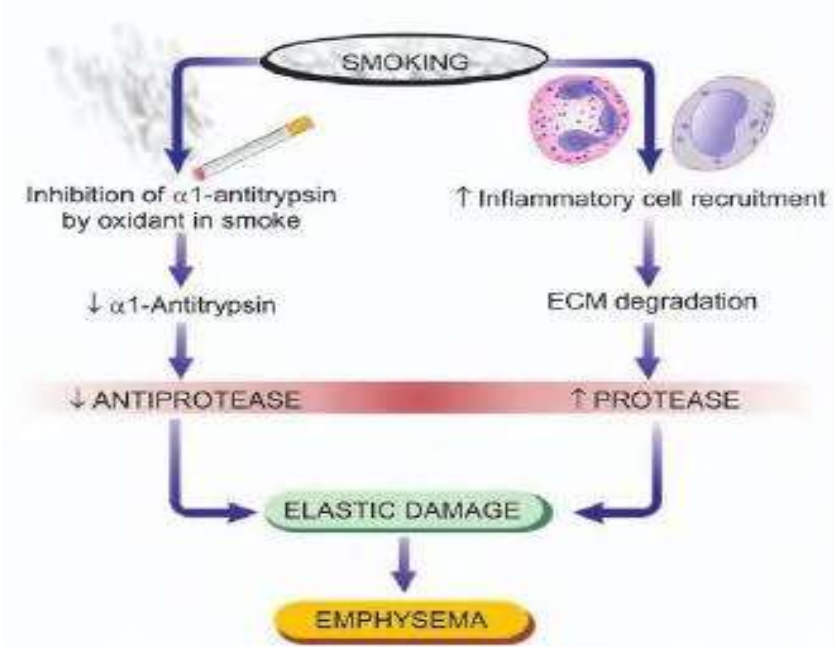
Elastase is derived from neutrophils, macrophages, mast cells and bacteria in the lung. Elastase activity is inhibited mainly by the anti-elastase, alpha1 antitrypsin formed by the liver and present in the serum and interstitial tissue. So emphysema results from:

(a) Increased elastase activity in the lung, or

(b) Decreased anti-elastase activity as in alpha 1 antitrypsin deficiency.

- **(2) Cigarette smoking: Have a role in the genesis of** centriacinar emphysema.

(a) Smokers have greater number of neutrophils and macrophages in their alveoli.
(h) Smoking stimulates release of elastase from neutrophils.
(c) Smoking enhances elastase activity in macrophages. (d) Oxidants in cigarette smoke inhibit cx 1—AT. bronchitis.
(e) Cigarette smoking causes chronic



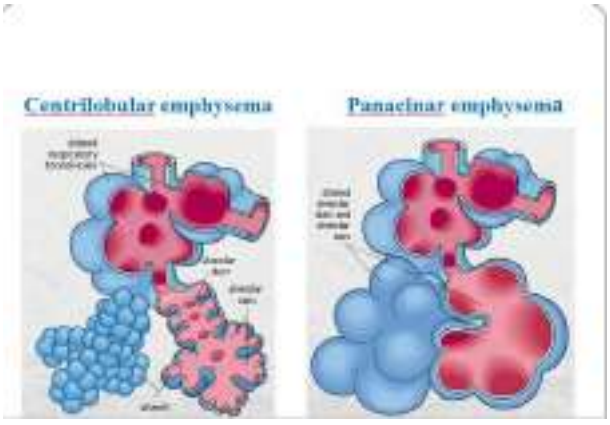
(3) Increased intra-luminal pressure: In chronic bronchitis the small bronchi and bronchioles are narrowed during expiration mainly due to mucus accumulation. Part of the inspired air get entrapped, gradually

- Types of Pulmonary Emphysema:

(1) Centriacinar (centrilobular or bronchiolar)

emphysema: Over- distention affects the central or proximal parts of the acini, formed by respiratory bronchioles, while distal alveoli are spared. The lesions are more common and more severe in the upper lobes. Inflammation is present around the bronchi and bronchioles. Smoking plays an important role in the genesis of this type.

(2) Panacinar „panlobular or alveolar) emphysema: The acini are uniformly over-distended frorn the level of the respiratory bronchiole to the terminal blind alveoli. Panacinar emphysema is more common in the lower lobes and anterior margins. It is the type associated with alpha1 antitrypsin deficiency.



(3) Paraseptal emphysema: The proximal portion of the acinus is normal but the distal part is over-distended. Bullae formation is common.

(4) Irregular emphysema: The acinus is irregularly over-distended. The affected acini are usually adjacent to areas of lung scars.

- Morphology of emphysema
- Gross Picture:

(1) Voluminous lungs covering the heart completely.

(2) The lungs do not collapse on opening the chest. (3) The surface is dry and pale due to wide separation of the septa carrying the blood vessels and carbon particles. The lung surface is indented at the sites of ribs.

(4) Bullae up to few cm in diameter are **present specially along the anterior border, apex and base.**

(5) Cut surface is spongy showing empty spaces 1-2 mms in diameter.

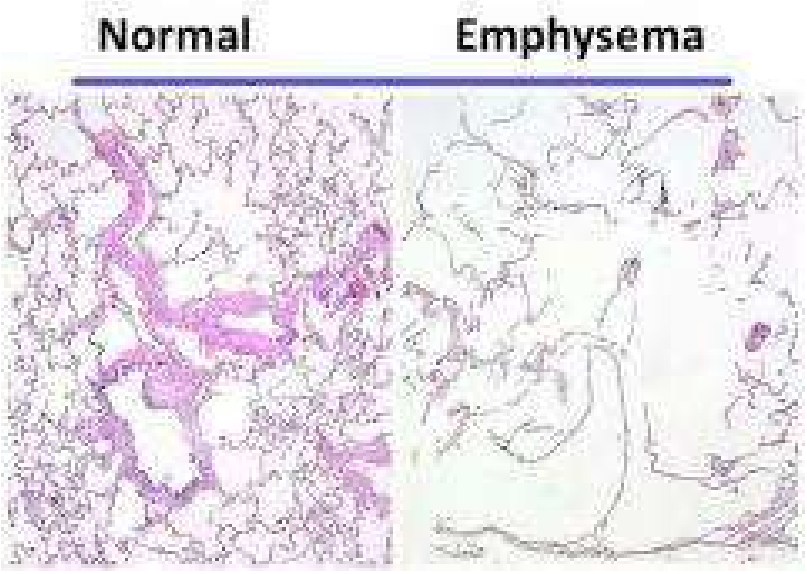
- **Microscopic Picture:**

(1) The alveolar spaces widened, communicating and have thin atrophic walls with projecting blunt spurs (broken-down alveolar septa).

(2) The alveolar capillaries, are compressed and obliterated.

(3) Elastic tissue stains show decrease in the elastic tissue.

(4) Overdistention affects only the central parts of the bronchopulmonary lobules in centrilobular emphysema and whole lobules in panlobular emphysema.





- **Effects:**

(1) Barrel shaped chest: The chest wall acquires an exaggerated inspiratory position. The ribs are horizontal, the subcostal angle is obtuse, the intercostal spaces widen, the sternum protrudes forward and the anteroposterior diameter increases and becomes nearly equal to the transverse one. The costal cartilages are calcified.

(2) Increased residual air: Causes cyanosis in advanced cases.

(3) Pulmonary hypertension: Occurs due to mechanical obstruction of the pulmonary capillaries. The condition may terminate in right-sided heart failure.

- Other Types of Emphysema:

(1) ATROPHIC EMPHYSEMA (SENILE)

Causes: Occurs in old age due to senile atrophy of the alveolar walls and decrease in the elastic tissue.

Gross picture: The lungs are airy and small as they loose air when the thorax is opened. Bullae are seldom seen.

Microscopic picture: Decrease in the number of alveoli. The alveoli are enlarged and their walls are thin and show loss of the

elastic tissue. The capillaries are reduced in number and obliterated.

(2) COMPENSATORY EMPHYSEMA

Localized areas of emphysema near-by areas of lung consolidation, collapse or fibrosis as a compensatory mechanism. Also it occurs after surgical removal of one lung or a lobe in the remaining lung tissue.

(3) INTERSTITIAL EMPHYSEM

Accumulation of air in the interstitial tissues of the lung from ruptured alveoli. It is a rare condition which follows fracture of ribs, penetrating wounds of the lung or overdistention of the alveoli as in whooping cough and straining. In severe cases the air reaches the interstitial tissue in the mediastinum or even the subcutaneous tissues of the neck and chest. Commonly recovery occurs due to absorption of air. Rarely death results due to pneumothorax or compression of the blood vessels.

- **Complications:**

(1) Rupture of the bullae causes pneumothorax.

(2) Rupture of the alveolar septa a causes slight haemoptysis.

(3) Right-sided heart failure.

BRONCHIECTASIS

- **Definition:**

- Bronchiectasis is defined as irreversible permanent dilatation of the bronchi and bronchioles developing secondary to inflammatory weakening of the bronchial walls caused by destruction of the muscle and the supporting elastic tissue, resulting chronic necrotizing infections.

- **The most characteristic clinical manifestation of**

bronchiectasis is persistent cough with expectoration of copious amounts of foul-smelling, purulent sputum.

- Post-infectious cases commonly develop in childhood and in early adult life.
- It is not a primary disease but rather secondary to persisting infection or obstruction caused by a variety of conditions.

Etiology & Pathogenesis of Bronchiectasis:

(1) Bronchial infection:

Causative Organisms: Staphylococcus, streptococcus, pneumococcus, bacillus coli and Haemophilus influenzae

- Bronchitis in association with whooping cough, measles, influenza and pneumonia in early life causes:

(a) Destruction of the lining bronchial epithelium. This interferes with the cough reflex causing retention of secretions which allows further infection.

(b) Destruction of the musculo-elastic tissue in the bronchial wall with replacement fibrosis. This causes weakness in the wall.

(c) Infection spreads from the bronchi to the surrounding lung causing pneumonitis. Pneumonitis heals by fibrosis. The fibrous tissue pulls on the weak bronchial wall causing its dilatation.

(2) **Bronchial obstruction**: By mucous plugs, foreign bodies, bronchial tumours or pressure on the bronchi from outside by enlarged tuberculous lymph node is a factor in some cases.

Bronchial obstruction causes:

(a) **Retention of secretions** favouring infection.

(b) Collapse of the surrounding lung parenchyma. Thus the bronchi lose the support of the surrounding alveoli and become

exposed directly to the increased intrapleural negative pressure and so dilate. (3) Congenital abnormality in the development of the muscle,

elastic tissue and cartilage of the bronchi.

- Morphology of bronchiectasis The disease characteristically affects distal bronchi and bronchioles, particularly those air passages that are most vertical.
- The airways may be dilated to as much as four times their usual diameter.
- Usually, the most severe involvement is found in the more distal bronchi and bronchioles.

- Grossly: The lungs may be involved diffusely or segmentally. Bilateral involvement of lower lobes occurs most frequently. More vertical air passages of left lower lobe are more often involved than the right.
- The pleura is usually fibrotic and thickened with adhesions to

the chest wall.

- When caused by tumors or aspiration of foreign bodies the

involvement may be sharply localized to a single segment of

thelungs.

- The dilated airways, depending upon their gross or bronchographic appearance, have been sub-classified into the following different types:

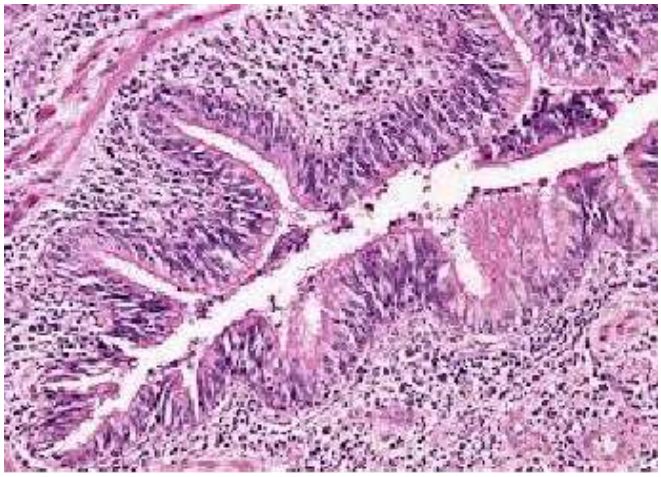
- **Cylindrical:** They are the most common type. It is characterized by tube-like bronchial dilatation.
- Fusiform: They have spindle-shaped bronchial dilatation.
- Saccular: They have rounded sac-like bronchial distension.
- Varicose: They have irregular bronchial enlargements.

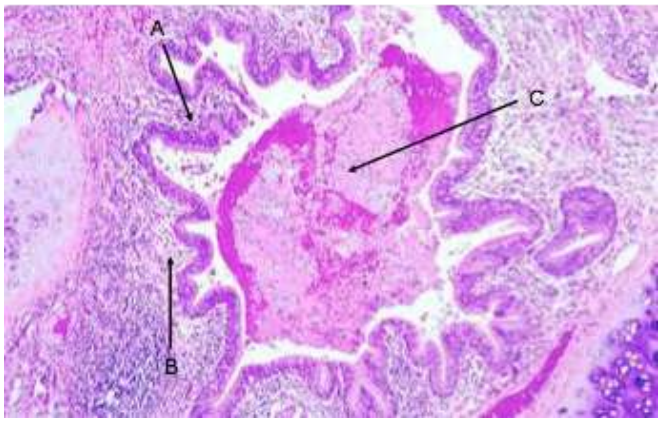
- Cut surface of the affected lobes, generally the lower zones, shows characteristic honey-combed appearance.
- The bronchi are extensively dilated nearly to the pleura.
- Their walls are thickened and the lumina are filled with mucus or muco-pus.
- The intervening lung parenchyma is reduced and fibrotic.

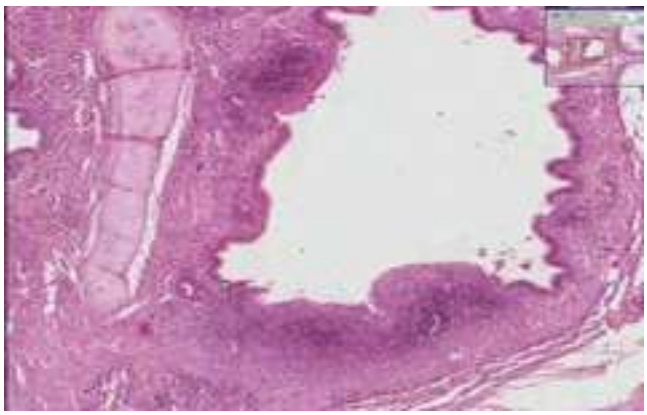
- Microscopically:
- The histological findings vary with the activity and chronicity of the disease.

- The bronchial epithelium may be normal, ulcerated or may show squamous metaplasia.
- The bronchial wall shows infiltration by acute and chronic inflammatory cells and destruction of normal muscle and elastic tissue with replacement by fibrosis.
- The intervening lung parenchyma shows fibrosis, while the surrounding lung tissue shows changes of interstitial pneumonia.

- The pleura in the affected area is adherent and shows bands of fibrous tissue between the bronchus and the pleura.
- In the full-blown active case, an intense acute and chronic inflammatory exudate within the walls of the bronchi and bronchioles and the desquamation of lining epithelium cause extensive areas of ulceration.







The clinical manifestations of bronchiectasis:

- typically consist of chronic cough with foul-smelling sputum production, hemoptysis and recurrent pneumonia.
- Symptoms are episodic and are precipitated by upper respiratory tract infections or introduction of new pathogenic agents.
- Clubbing of the fingers may develop.
- Sinusitis is commonly accompanying diffuse bronchiectasis.

Complications:

(1) Haemoptysis is common.

(2) Lung abscess and gangrene.

(3) Local spread of infection causes empyema, pyopneumothorax, suppurative pericarditis and suppurative mediastinitis.

(4) Septic thrombophlebitis in the bronchial wall causes pyaemic abscesses mainly in the brain and spinal cord.

(5)Secondary amyloidosis.

(6) Lung fibrosis causes right-sided heart failure.

(7) Squamous cell carcinoma may develop in the epithelium ofthe affected bronchi due to chronic irritation.

Chronic obstructive lung diseases include all of the following except:

Bronchial asthma

Emphysema

Lung abscess

Bronchiectasis

2. The prominent inflammatory cells e in bronchial asthma is:

A. Lymphocytes

B. Plasma cells

C. Eosinophils

D. Polymorphs

E. Histiocytes

MCQ

3. The histologic features of bronchiectasis include all EXCEPT:

A. Dilation of the bronchi and bronchioles

B. Destruction of the bronchial wall

C. The bronchial lumen is filled with pus

D. The bronchial wall is infiltrated by many eosinophils

E. The intervening lung tissue shows fibrosis

4. α 1 antitrypsin deficiency plays a role in the pathogenesis of:

A. Lung abscess

B. Pulmonary emphysema

C. Pulmonary collapse

D. All of the above

Restrictive Lung Diseases and Vascular Diseases of the Lung

By the end of this lecture; you should able to: Define

restrictive lung diseases

Classify restrictive lung diseases

Discuss ARDS and NRDS

Define and enumerate causes of pneumoconiosis.

Define idiopathic pulmonary fibrosis

Identify causes and pathology of pulmonary embolism, pulmonary hypertension and pulmonary edema

Part (I)

Restrictive Lung Diseases

Definition

- Group of lung diseases characterized by reduced expansion of lung parenchyma with decreased total lung capacity.

- It is category of extra pulmonary, pleural, or parenchymal respiratory diseases that restrict lung expansion, resulting in:

- Decreased lung volume
- Increased work of breathing
- Inadequate ventilation and/or oxygenation.
- Pulmonary function test demonstrates a decrease in the forced vital capacity.

Clinical picture

- Dyspnea

- Tachypnea
- Cyanosis

Etiology Restrictive lung disease includes 2 types of conditions:

- Restriction due to chest wall disorder
- Restriction due to interstitial and infiltrative diseases.

Etiology

Restrictive lung disease includes 2 types of conditions:

A. Restriction due to chest wall disorder:

- Conditions cause restriction to expansion of lungs
- Due to alterations in chest wall, pleura and neuromuscular apparatus
- But are primarily not lung parenchymal diseases.

1. Kyphoscoliosis
2. Poliomyelitis
3. Severe obesity
4. Pleural diseases

B. Restriction due to interstitial and infiltrative diseases.

- Commonly called as —interstitial lung diseases —(ILDs)
- Are diseases characterized by non-infectious diffuse

- parenchymal involvement of the lung . Diffuse lung parenchymal involvement may be primary, or it may be involved secondary as a part of some other multi organ disease process.

1. Pneumoconiosis
2. Immunological lung diseases
3. Collagenvascular

N.B

- The term „infiltrative" is used here to denote the radiologic appearance of lungs which show characteristic diffuse interstitial ground-glass opacities. i,e diffuse infiltration by nodules or shadows



Pneumoconiosis

Definition

- Lung diseases caused by inhalation of dust, mostly at work (pneumo = lung; conis = dust in Greek).
- Pneumoconiosis is a term originally coined to describe the non-neoplastic lung reaction to inhalation of mineral dusts.
- The term has been broadened to include diseases induced

by organic as well as inorganic particulates, chemical fume- and vapor-induced non-neoplastic lung diseases.

Synonyms

- Dust diseases
- Occupational lung diseases

Causes



- 1. **Silicosis:**

- **Definition: due to inhalation of silica particles (as in** miners) leading to granulomas Grossly: both lungs
- showed numerous hard small grayish nodules. Microscopic: multiple foreign body granulomas
- composed of silica particles surrounded by macrophages, foreign body
  giant cells, lymphocytes and fibrosis
- Complications
- Pulmonary fibrosis…..Rt.sided heart failure
- Bronchogenic carcinoma Pulmonary TB
-

2. Asbestosis

Definition: due to inhalation of asbestos particles

- Grossly: both lungs showed numerous hard small grayish nodules.
- Microscopic: Asbestos fibers or asbestos bodies accompanied by diffuse interstitial fibrosis with chronic inflammation
- Complications
  - Pulmonary fibrosis…Rt. Sided Heart Failure
  - Mesothelioma
  - Bronchogenic carcinoma



3. Anthracosis: inhalation of carbon particles



4. Byssinosis: inhalation ofcotton fibers….lung fibrosis



5. Bagassosis: inhalationof sugar fibers….lung fibrosis



Acute respiratory distress syndrome (ARDS)

Definition

- Sudden, progressive form of respiratory failure characterized by severe dyspnea, refractory hypoxia and diffuse bilateral lung infiltrate

CLINICAL FEATURES AND CONSEQUENCES.

- Adult respiratory distress syndrome (ARDS)
- Newborn respiratory distress syndrome (NRDS)

Neonatal ARDS

- occurring in newborn infants
- Begins with dyspnea within a few hours after birth

- Associated with tachypnea, hypoxia and cyanosis
- Insevere cases death may occur within a few hours.

Adult ARDS

- is known by various synonyms such as (shock-lung syndrome, diffuse alveolar damage, acute alveolar injury and post-traumatic respiratory insufficiency).

- Adult ARDS also presents clinically by sudden and severe respiratory distress, tachypnea, tachycardia, cyanosis and severe hypoxemia.

ETIOLOGY.

Neonatal ARDS

is primarily initiated by hypoxia, either shortly before birth or immediately afterward.

1. Preterm infants
2. Infants born to diabetic mothers
3. Caesarean section
4. Excessive sedation of the mother
6. Birth asphyxia from various causes such as coils of umbilical cord around the neck
8. Many cases of neonatal ARDS remain idiopathic.

- Adult ARDS

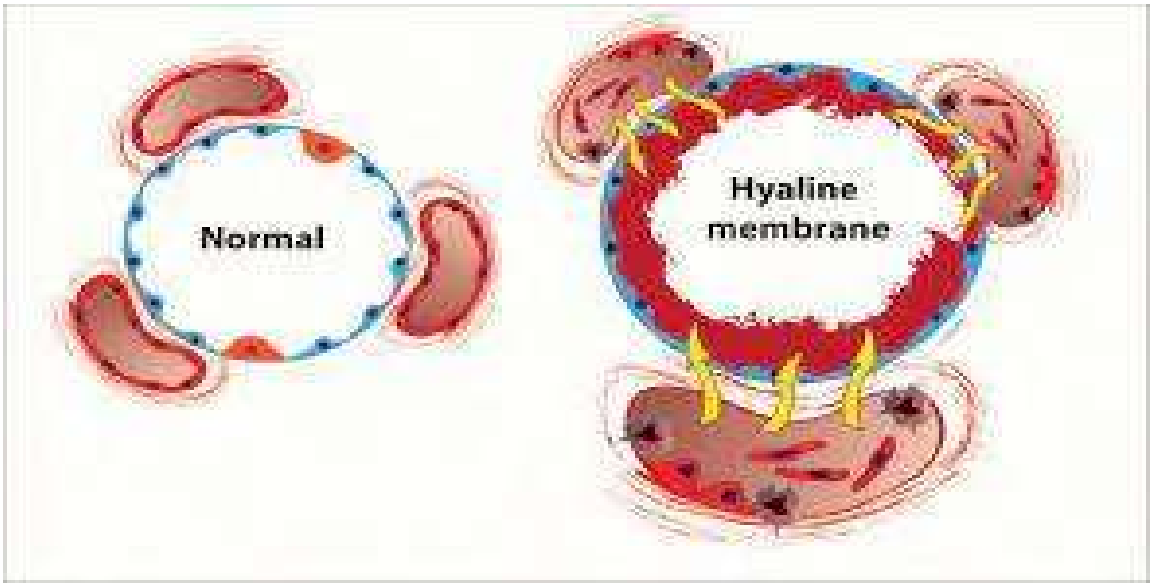
1. Shock
2. Diffuse pulmonary infections, chiefly viral pneumonia
3.Oxygen toxicity
5. Inhalation of toxins and irritants e.g. smoke, war gases, nitrogen dioxide.
6. Narcotic overdose
7. Drugs e.g. salicylates, colchicine
8. Aspiration pneumonitis
9. Fat embolism

PATHOGENESIS

- In both neonatal and adult type ARDS, there is damage to alveolocapillary wall triggered by etiologic factors listed above, and the final pathologic consequence of formation of hyaline membrane is also similar. However, how it occurs is different in the neonates than in adults.

Neonatal ARDS.

- The basic defect in neonatal ARDS is a deficiency of pulmonary surfactant (normally synthesized by type II alveolar cells).



- The production of surfactant is normally increased shortly before birth but in prematurity and in neonatal hypoxia from any of the foregoing causes, its synthesis is decreased.
- The main function of alveolar surfactant being lowering of alveolar surface tension, its deficiency leads to increased alveolar surface tension which in turn causes atelectasis.
- Atelectasis of the lungs results in hypoventilation, pulmonary hypo-perfusion and ischemic damage to capillary endothelium.
- This results in ischemic necrosis of the alveolocapillary wall, exudation of plasma proteins including fibrinogen into the alveoli and eventually formation of hyaline membrane on the alveolar surface containing largely fibrin

Adult ARDS.

- The mechanism of acute injury by etiologic agents listed

above Activated pulmonary macrophages release
- proinflammatory cytokines such as IL8, IL1, and TNF
Number of neutrophils in the alveoli is increased in
- acute
injury. Activated neutrophils on release products which cause active tissue injury e.g. proteases, platelet activating
factor, oxidants and leukotrienes.

- A few fibrogenic cytokines such as transforming growth factor-α (TGF-α) play a role in repair process by stimulation of proliferation of fibroblast and collagen.

- Injury to the capillary endothelium leads to increased vascular permeability while injured pneumocytes, especially type 1, undergo necrosis.

- The net effect of injury to both capillary endothelium and alveolar epithelium is interstitial and intra-alveolar edema, congestion, fibrin deposition and formation of hyaline
membranes.
- As a result of coating of the alveoli with hyaline membranes, there is loss of surfactant causing collapse called ‗stiff lung'. There is an attempt at regeneration of alveolar cells by proliferation of type II alveolar cells so as to increase the secretion of surfactant.

MORPHOLOGIC FEATURES.

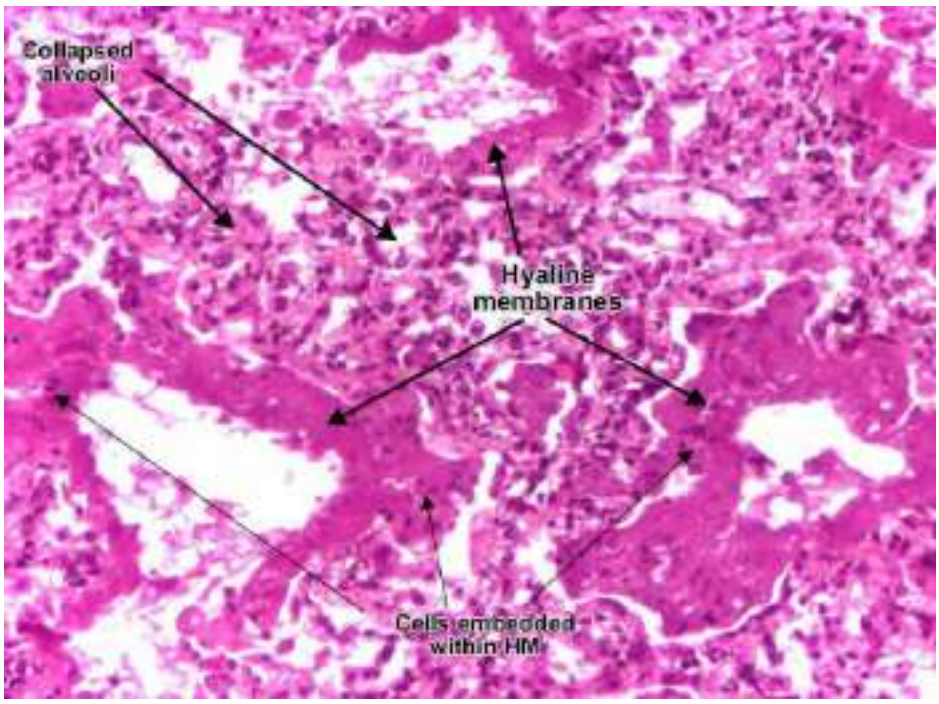
- Grossly, the lungs are normal in size. They are characteristically stiff, congested, heavy and airless so that they sink in water.



- Microscopically,
- the important features are as follows

- There is presence of collapsed alveoli (atelectasis) alternating with dilated alveoli.
- Necrosis of alveolar epithelial cells
- Formation of characteristic eosinophilic **hyaline membranes lining the respiratory bronchioles, alveolar ducts and the proximal alveoli.**
- The membrane is largely composed of fibrin admixed with cell debris derived from necrotic alveolar cells.
- Interstitial and intra-alveolar oedema, congestion and intra-alveolar haemorrhages.

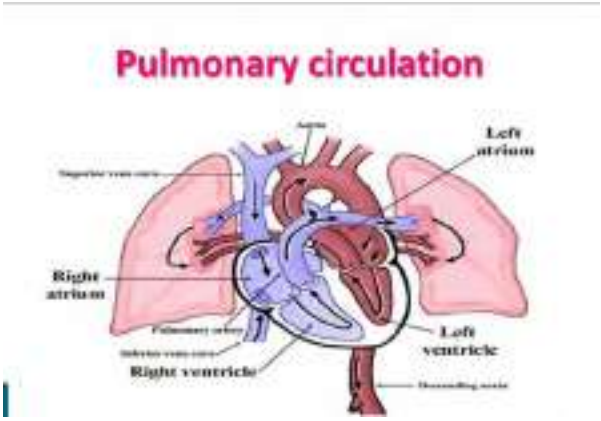
- In organising stage, there is interstitial fibrosis obliterating alveolar spaces.



Vascular Lung Diseases

PULMONARY HYPERTENSION

- Normally, the pulmonary arterial circulation is high-flow and low-pressure with much lower blood pressure than the systemic blood pressure
- Pulmonary arterial pressure does not exceed 30/15 mmHg



Definition

- Pulmonary hypertension is defined as a systolic blood pressure in the pulmonary arterial circulation above 30 mmHg.

Classification

Pulmonary hypertension is broadly classified into 2 groups:

- Primary (idiopathic)
- Secondary

(1) Primary (Idiopathic) Pulmonary Hypertension

- Itisan uncommon condition of unknown cause.
- The diagnosis can be established only after a thorough search for the usual causes of secondary pulmonary hypertension
- The patients are usually young females between the age of

20 and 40 years, or children around 5 years of age.

Secondary Pulmonary Hypertension

- When pulmonary hypertension occurs secondary to a recognized lesion in the heart or lungs, it is termed as
- It is the more common type
- Occurs at any age, but more frequently over the age of 50

ETIOPATHOGENESIS

Based on the underlying mechanism, causes of secondary pulmonary hypertension are divided into the following 3 groups:

(1) Passive pulmonary hypertension.

- The commonest
- Is produced by diseases raising pressure in the pulmonary veins e.g.

1. Mitral stenosis.
2. Chronic left ventricular failure

(2) Hyperkinetic (Reactive) pulmonary hypertension.

- Theblood enters the pulmonary arteries in greater volume orata higher pressure. Forexample: PDA,
- ASD&VSD

(3) Vaso-occlusive pulmonary hypertension.

- All conditions which produce progressive diminution of the vascular bed in the lungs are included in this group.

- sub-divided into 3 types:

Obstructive type (there is block in the pulmonary circ.) e.g. Multiple emboli or thrombi

Obliterative type: (there is reduction of pulmonary vascular bed by chronic parenchymal lung diseases) e.g. Pulmonary tuberculosis & Pneumoconiosis Vasoconstrictive type (there is widespread and sustained hypoxic vasoconstriction and alveolar hyperventilation leading to pulmonary hypertension)

- e.g. Pathologic obesity, neuromuscular diseases such as poliomyelitis &Severe kyphoscoliosis.

Pulmonary Edema

- It is pathological accumulation of transudate in the interstitial tissues and alveolar spaces of the lung.

Signs and symptoms

- The most common symptom is dyspnea
- other symptoms such as cough and excessive sweating may be present.

- Shortness of breath manifest as orthopnea or paroxysmal nocturnal dyspnea

- Pulmonary edema may be associated with peripheral edema of the "pitting" variety, raised jugular venous pressure and hepatomegaly

Types of Pulmonary Edema

- According to the etiology

- Cardiogenic pulmonary edema: It is due to failure of the left ventricle to remove blood adequately from the pulmonary circulation. It is a cardinal feature of congestive heart failure.
- Non-cardiogenic pulmonary edema: or injury to lung parenchyma or lung vasculature

 According to course

- Acute pulmonary edema
- Chronic pulmonary edema

1.Cardiogenic pulmonary edema:

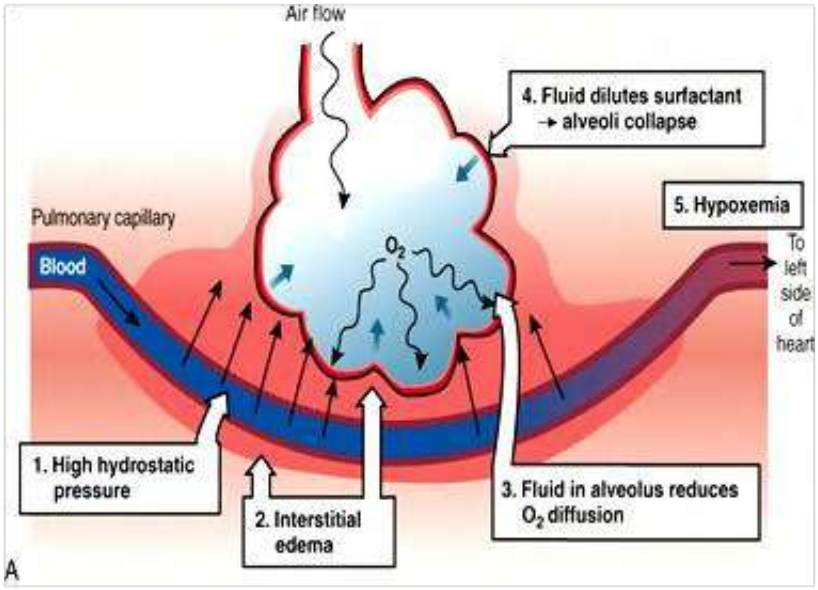
This may be due to left ventricular failure

2. Non-cardiogenic pulmonary edema:

in which there is a significant negative pressure in the chest or due to Injury to the lung vasculature and/or lung parenchyma.

Such as

- Upper airway obstruction
- Neurogenic causes
  Acute respiratory distress syndrome.
- Inhalation of hot or toxic gases.
- Aspiration, e.g., gastric fluid.



Acute pulmonary edema

Causes:

- Acute left ventricular failure
- Inhalation of irritant gases
- Excess intravenous fluid administration
- Viral lung infection

Gross:

bulky lungs, pale and pit on pressure, the cut surface oozes frothy fluid on pressure

Microscopic: the alveolar vessels are inconspicuous, the interstitial tissue , air spaces and bronchi contain pale pink homogenous substances which is

coagulated protein of the edema fluid stained with eosin.

sever acute pulmonary edema is fatal

Chronic pulmonary edema

Causes

- Chronic lung congestion
- As a part of generalized edema

Gross & microscopic:

as acute pulmonary edema

Complications:

Secondary bacterial infection

Pulmonary embolism

Occlusion of pulmonary artery by embolism Sources:

- Recent thrombus in the veins of the lower limbs……. after prolonged bed rest due to bone fracture
- Thrombi in iliac and pelvic veins…….after labor and pelvic operations
  Thrombi in right side of the heart ….RT. sided HF

Effects:

- Big embolus

occlude the pulmonary artery or one of main branches. Serotonin released in big amounts from platelets of the embolus causes bilateral vasoconstriction of pulmonary arterioles.

- death within seconds or minutes due to acute Rt. sided HF,

there is no time for infarction to occur

- Medium sized embolus
- healthy lung ..... No effect as the lung has a double blood supply
- lung with CVC ..... lung infarction
- Small embolus ..... no effect

MCQ

All are causes of acute pulmonary edema except:

- Acute left ventricular failure
- Inhalation of irritant gases
- Viral lung infection
- As part of generalized edema

All are true about the effect of big pulmonary embolus except:

- Complicated by acute sided heart failure Leads to death
- within minutes Occludes the pulmonary artery or one of its
- main branches Lead to lung infarct

- Asbestosis is complicated by Bronchogenic carcinoma
- Mesothelioma

- Cor-pulmonale
- All of above

All are true about NRDS except

- Characterized by hyaline membrane formation
- Affects neonates
- More common in premature infants
- Caused mainly due to increased pulmonary surfactants

Update on pulmonary tumors

Carcinoma of the Lung

6.5 % of all deaths #1 cause of cancer

deaths in males & females 25% of male

cancer deaths, UK 21% of female cancer

deaths, UK

Causes of Lung Cancer

85-95% **smoking**

Rare arsenic, chromium, mustard gas, nickel, vinyl chloride, bis (chloromethyl)
Air pollution

Genetic factors:

C/P of bronchogenic carcinoma:

-Chronic cough + Heamoptysis

-Metastasis

-Paraneoplastic syndrome

Classification of lung tumor

Lung secondary (Metastasis)

Spread:

via blood

via lympatics

direct

Grossly:

multiplesmallnodules(variable sized)

largesize **X-ray:Cannon ball appearance**

renal carcinoma
seminoma

Microscopic:

resemblesthe primary

Primary Lung tumors

SCLC -

Smallcellcarcinoma -

Presentedbylatestage/un operable -

Treatedbychemotherapy -

*Adenocarcinoma

*Squamous cell carcinoma

*Large cell carcinoma

- Operable

- Treated surgery± radiotherapy

- Respond poorly for chemotherapy

NSCLC

Incidence of different histological subtypes Lung Carcinoma (Major Types)

Squamous cell carcinoma35-50%

Adenocarcinoma 15-35%

Large cell carcinoma Small cell 10-15%

carcinoma Classification of 20-25%

Lung IASLC/ATS/ERS

Adenocarcinoma

Frequency: 30%

Smoking: X 3 (increased risk)

Males < females

Preinvasive lesions

Atypical adenomatous hyperplasia

Adenocarcinoma in situ

(Nonmucinous, Mucinous, Mixed mucinous/nonmucinous)

Minimally invasive adenocarcinoma

Nonmucinous

Mucinous

Mixed mucinous/nonmucinous

Invasive adenocarcinoma

Lepidic predominant

Acinar predominant Papillary predominant

Micropapillary predominant Solid

predominant with mucin production

Variants of invasive adenocarcinoma

Invasive mucinous adenocarcinoma

Colloid

Fetal

Enteric

-

NSCLC

Invasive adenocarcinoma

Lepidic predominant Adenocarcinoma

Acinar adenocarcinoma

Papillary adenocarcinoma

Micropapillary adenocarcinoma

Solid adenocarcinoma Invasive

mucinous adenocarcinoma

NSCLC

Squamous cell carcinoma

Frequency: 35%

Smoking: X 25 (increased risk)

Males > females

Arises in bronchial squamous metaplasia

Centrally located

May cavitate

Microscopic pic:

Squamous cell carcinoma

Well Moderately Poorly differentiat.

Small cell carcinoma

Frequency: 25 % Smoking:

95% of patients

Males >> females

Highly malignant& highly aggressive

Early metastatic

Incurable by surgery

Microscopic:

Sheets of small cells

scanty cytoplasm

finely granular chromatin

Mitosis……frequent

Necrosis……..massive

EM……neurosecretory granules

Carcinoid tumor

Locally malignant tumor???

Low grade malignant

Typical Vs atypical

Neuroendocrine cells of bronchial mucosa